KFT™

MINOT AFB FIRE DEPT, ND

HELO FIRETRAINER® O-100

HOSE LINE FIRETRAINER® O-100

MULTI CONTROL SYSTEM FIRETRAINER® O-100

OPERATION & MAINTENANCE MANUALS

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1	HELICOPTER FireTrainer O-100 O&M Manual
2	FireTrainer O-100 O&M Manual
3	Hose Line FireTrainer O-100 Multi Control System Manual
4	O-100 Wireless Deadman Data Sheets

Helicopter FireTrainer® O-100

Operation and Maintenance Manual

KFT™

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1.0 INTRODUCTION

1.1 GENERAL

This manual provides all of the necessary information to operate the:

Helicopter FIRETRAINER® O-100

MANUFACTURED BY:

Kidde Fire Trainers

If you have any questions regarding the operation or maintenance of this trainer that are not answered in this manual contact the Kidde Fire Trainers Customer Service Group at:

KIDDE FIRE TRAINERS

17 PHILIPS PARKWAY

MONTVALE, NJ 07645

USA

TEL: 1(201) 300 8100

FAX: 1(201) 300 8101

CUSTOMER SERVICE TEL: 1 800-899-1701 EXT 502

TO REQUEST SERVICE: HTTPS://KIDDEFT.BLUEFOLDER.COM/PORTAL/

Utilization of these Helicopter FIRETRAINER® O-100 for live-fire training exercises brings a level of versatility and realism that cannot be duplicated using conventional fire fighting training methods. The operator of the trainer will be able to vary the difficulty, complexity or scale of the fires to challenge the skills of the trainees.

To maximize training effectiveness and use the full potential of the system, a thorough understanding of the trainer system and controls is necessary. This section describes the layout of the manual and the Safety Guidelines for the Trainer. Read and understand the Safety Guidelines and system operation procedures in this manual before operating the Trainer. Additionally, refer to the manufacturers safety and operation information located in Section 6 of this manual.

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1.2 HOW TO USE THIS MANUAL

The Operation Manual for the **TRAINER** is presented in the following order:

Section 1	**Introduction**	This section provides a brief functional description how to use the manual and relevant safety guidelines and instructions.
Section 2	**System Description**	This section provides an equipment overview and a functional description of the major equipment components that are part of the site.
Section 3	**Operation**	Detailed operating instructions.
Section 4	**Care and Maintenance**	This section provides Information to keep the trainer functional and operationally effective.
Section 5	**Parts List**	This section provides figures and parts list tables for each of the Trainer major assemblies.
Section 6	**Drawings & Reference Materials**	This section contains interconnection drawings and manufacturers documentation to support the trainer

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1.3 SAFETY

Normal operation of the training system presents potentially hazardous situations to student firefighting personnel. It is of the utmost importance that students read, understand, and follow the safety precautions and guidelines detailed in the following chapters. Failure to follow these safety guidelines could place training area personnel at undue risk of personal injury or death.

1.3.1 SAFETY SYMBOLS

Various symbols are used throughout the manual to alert the reader to a potentially dangerous situation.

WARNING

THE WARNING SYMBOL IS USED WHENEVER A CONDITION EXISTS THAT COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

CAUTION

THE CAUTION SYMBOL IS USED WHENEVER A CONDITION EXISTS THAT MAY CAUSE INJURY OR DAMAGE TO ASSOCIATED EQUIPMENT.

1.3.2 TRAINER PERSONNEL SAFETY RESPONSIBILITIES

Trainer Personnel (instructors, safety officers, system operators and maintenance personnel) are ultimately responsible for the safety of all personnel entering the training area and compliance to all applicable NFPA, OSHA, national, state and local codes and standards.

The safety responsibilities of Trainer Personnel are summarized below:

1. Read and fully understand all provided trainer manuals and this safety handbook.
2. Ensure that the trainer is operated and maintained in accordance with all trainer manuals.
3. Discontinue trainer use if potentially hazardous conditions or anomalies occur.
4. Be familiar with each fireplace's behavior in all modes of operation under all conditions, and insure the safety of personnel with respect to potentially hazardous behaviors.
5. Ensure that personnel are properly trained in the use of self contained breathing apparatus (SCBA).
6. All personnel entering the training area shall wear full firefighting gear. This gear shall include: full-face mask, self-contained breathing apparatus (SCBA) and appropriate protective clothing. As in any fire situation, the exposure of personnel to smoke (real or synthetic), fumes, or other combustion products should be avoided.
7. Ensure that personnel are in a safe position prior to, and during, flame initiation.

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8. Ensure that the operator and instructors have a full knowledge and understanding of the training scenario prior to initiation. This should include all fireplace parameter settings as well as reflash capabilities or simulated flame effects.
9. At no time during live-fire operation should the trainer be left unmanned.
10. Ensure that only water is used as the extinguishing agent in the trainer. Use of extinguishing agents other than water can result in diminished trainer life and performance and adversely impact safe trainer operation.
11. Ensure that at the completion of a training exercise, personnel without adequate protective clothing and SCBA do not enter a training structure until the ventilation system has been operated at maximum capacity for a minimum of 3 minutes to remove combustion by-products and smoke.
12. Ensure that proper communication is established and in operation between instructors and safety officers before and throughout each training exercise.

1.3.3 STUDENT SAFETY GUIDELINES

Prior to conducting training, Trainer Personnel must inform students of the dangers involved in live fire training. The following is a list of safety guidelines that must be presented to instructors and students before use of the trainer.

Normal operation of the training system presents potentially hazardous situations to student firefighting personnel. It is of the utmost importance that students read, understand, and follow the safety precautions and guidelines detailed on this page. Failure to follow these safety guidelines could place training area personnel at undue risk of personal injury or death.

1. Do not enter the training area until authorized to do so by Trainer Personnel.
2. Do not enter the training area without proper protective clothing and self-contained breathing apparatus (SCBA). Keep this equipment on during the entire live fire training exercise.
3. While within the training area, be alert at all times. Treat the live fire-training situation the same as an actual fire emergency. Be prepared for any aspect of flame behavior that is possible with an uncontrolled, fully involved fire.
4. Avoid contact with hot surfaces within the training area before, during, and after the live fire training exercise.
5. Avoid breathing smoke and products of combustion while outside the training area.
6. Minimize or avoid contact with clothing and equipment exposed to smoke, soot, and any other fire training residue. If contact cannot be avoided, clean exposed skin surfaces with soap and water as soon as practical.
7. In addition to the guidelines listed above, listen to and carefully follow all rules and directions provided by Trainer Personnel.

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1.3.4 SYSTEM SAFETY WARNING

HIGH VOLTAGES capable of causing serious injury or death are used in the training system equipment. Unless required for voltage measurement, all power should be turned off when servicing is being performed. When power must be on, use extreme care when working near exposed conductors.

FUEL GAS used in the trainer is capable of forming explosive mixtures with air that can be triggered by open flames or sparks. Even residual gas from piping sections can create a hazardous condition. Work only in well ventilated areas, with no exposed flames or possible ignition source.

OPEN FLAMES generated in the fireplace area during operation of this equipment can cause serious injury or death. Use extreme care when working in this area whenever training fires are in operation.

EXPOSURE to extremes in the environmental conditions such as high or low temperatures, high winds, etc. may occur during a training scenario. Extreme care should be used when training under adverse environmental conditions. Failure to exercise extreme care could result in serious injury or death.

⚠ WARNING

SAFE OPERATION OF THE TRAINING SYSTEM MANDATES THAT THE FOLLOWING SAFETY PRECAUTIONS BE TAKEN PRIOR TO ANY OPERATION OR ACTIVITY. FAILURE TO FOLLOW THESE PROCEDURES MAY RESULT IN SERIOUS INJURY OR DEATH.

1. All personnel entering the training area shall wear full firefighting gear. This gear shall include: full-face mask, self-contained breathing apparatus (SCBA) and appropriate protective clothing. As in any fire situation, the exposure of personnel to smoke (real or synthetic), fumes or other combustion products should be avoided.
2. All live fire training evolutions shall be conducted in accordance with all applicable NFPA Codes (such as 1402,1403, 1404, and 1500); OSHA; national; state; and local codes and standards.
3. If fuel leaks are detected, they must be repaired. Under no circumstance is trainer operation to be performed when a leak is present.
4. Keep flames and ignition sources away from fuel handling components. When ignited, the fuel may cause serious injury or death.
5. Ensure the upstream fuel supply valve is fully closed prior to the purging of any lines to atmosphere. Tag all valves closed to alert other personnel that service is being performed. Failure to comply may result in serious injury or death.
6. Only smoke generators approved by Kidde Fire Trainers for use in flame areas shall be used.
7. Only smoke fluid recommended by Kidde Fire Trainers shall be used in the smoke generator(s).

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1.3.5 TRAILERS

NOTE

PRIOR TO OPERATION REFER TO THE MANUFACTURERS SAFETY INFORMATION LOCATED IN SECTION 6 OF THIS MANUAL.

WARNING

FAILURE TO FOLLOW THE EQUIPMENT SAFETY GUIDELINES COULD PLACE TRAINING AREA PERSONNEL AT UNDUE RISK OF PERSONAL INJURY OR DEATH!

1.3.6 PERIPHERAL EQUIPMENT

NOTE

PRIOR TO OPERATION REFER TO THE MANUFACTURERS SAFETY INFORMATION LOCATED IN SECTION 6 OF THIS MANUAL.

WARNING

FAILURE TO FOLLOW THE EQUIPMENT SAFETY GUIDELINES COULD PLACE TRAINING AREA PERSONNEL AT UNDUE RISK OF PERSONAL INJURY OR DEATH! INFORMATION LOCATED IN SECTION 6 OF THIS MANUAL.

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2.0 SYSTEM DESCRIPTION

2.1 INTRODUCTION

The **Helicopter FIRETRAINER® O-100** is a helicopter live fire training prop that provides a means to conduct realistic training in the orderly, expedient, and effective extinguishment of fires in a simulated environment. This trainer is a system that generates propane liquid/vapor fueled fires under the control of an instructor.

2.2 TRAINER CONFIGURATION

The **Helicopter FIRETRAINER® O-100** consists of the following:

- One (1) helicopter mockup (Trainees will not enter mockup)
- One (1) Cockpit Fire.
- One (1) Cargo Fire.
- One (1) Engine Fire.
- One (1) O-100 Pilot Box Assembly
- One (1) O-100 Fire Pan Assembly
- One (1) O-100 Control Pendant Assembly
- All Required liquid propane and vapor Hoses and Connectors.
- All Required Electrical Cables and Connectors.
- Two (2) 100 gallon Propane Tanks
- One (1) Trailer equipped with a storage boxes, winch and Lift Jack. The trailer is capable of transporting the items listed above.
- One (1) 12 VDC battery.
- One (1) Power Inverter (Converts 12 VDC to 110 VAC)

NOTE

THE OVERALL HEIGHT OF THE TRAILER WITH HELICOPTER MOCKUP IS 13'-2".

Refer to Figure 2-1 for Trainer Layout.

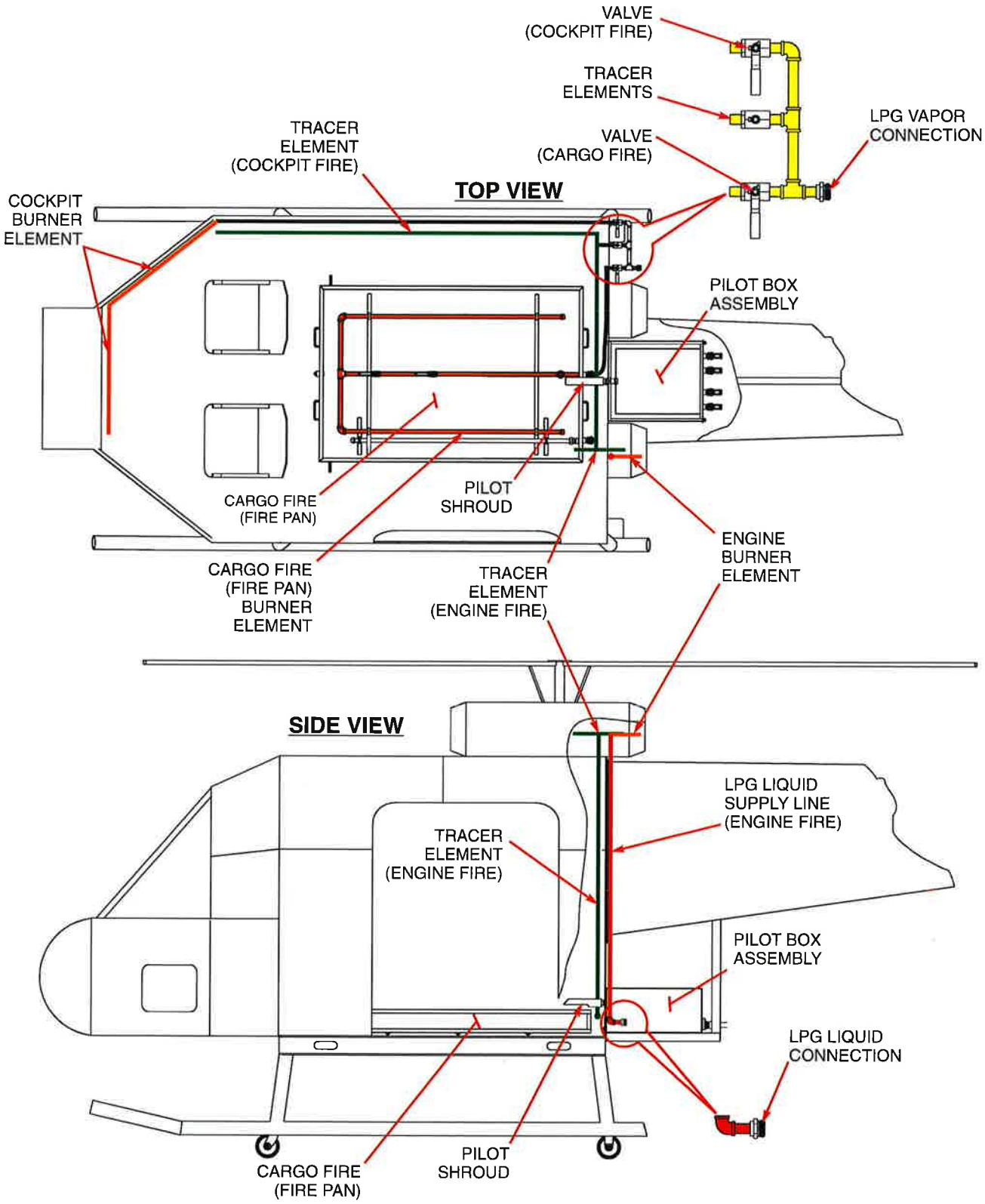

Figure 2-1. Helicopter Fires Trainer Layout

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2.2.1 HELICOPTER MOCK-UP ASSEMBLY

The Helicopter Mock-Up assembly is constructed of carbon steel plate and is equipped with four casters to allow the mock-up to be unloaded from the trailer and moved into positioned for training. The mock-up is also equipped with a LPG vapor connection and LPG liquid connection. The LPG vapor connection supplies the cockpit fire, cargo fire and tracer elements. The LPG liquid connection supplies the engine fire burner element. (Refer to Figures 2-1 and 2-2)

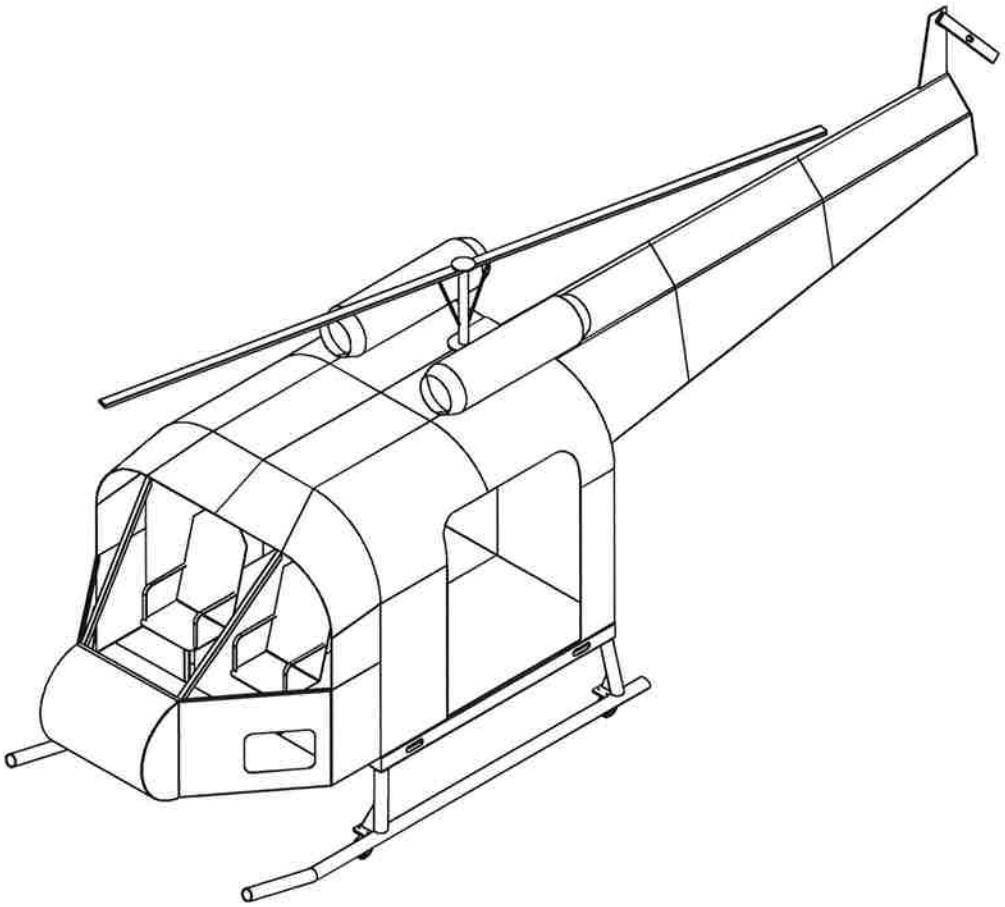

Figure 2-2. Helicopter Mock-Up Assembly

2.2.2 PILOT BOX ASSEMBLY

The Pilot Box Assembly controls the distribution of the liquid and vapor propane and produces the pilot flame. See Figure 2-1 for location and Figure 2-3 for Pilot Box Assembly detail. The pilot flame ignites the cargo fire burner element (fire pan), the cockpit and engine fire tracer elements. The tracer element delivers the ignition flame for the cockpit and engine fires (see Figure 2-1 for locations).

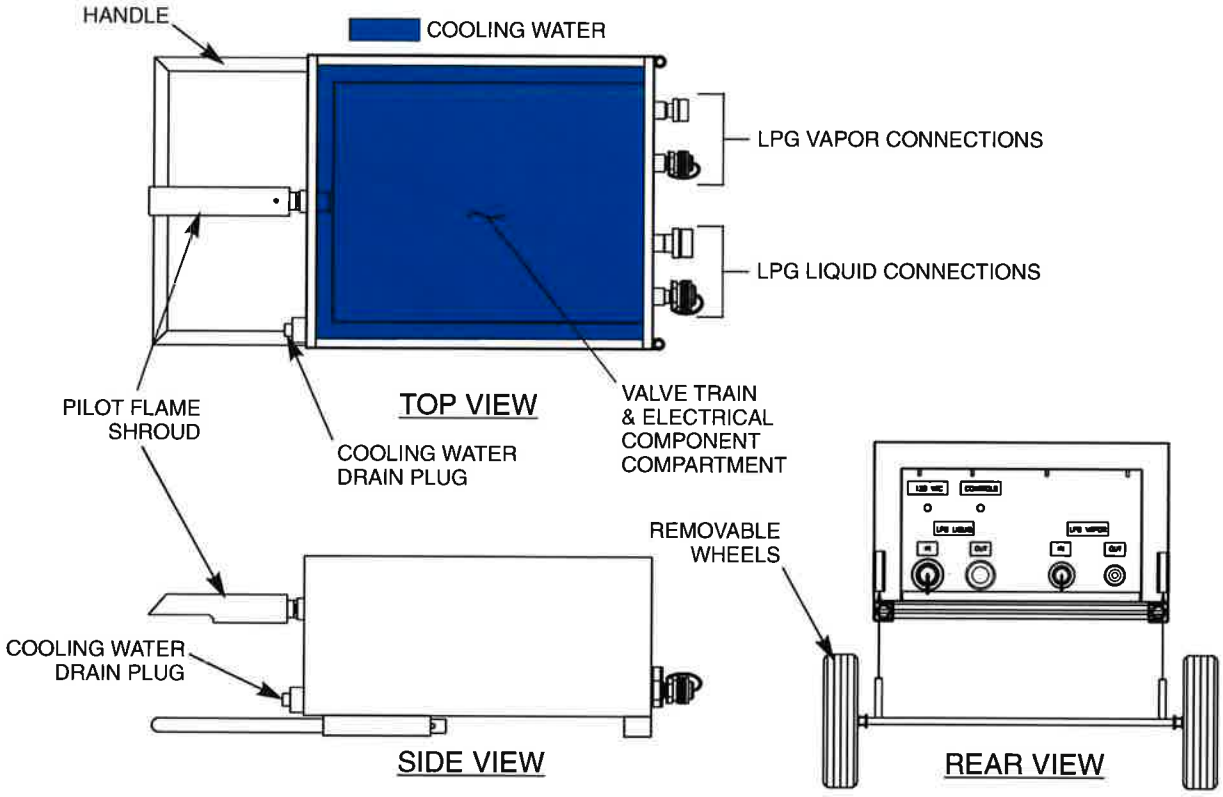

Figure 2-3. Pilot Box Assembly

2.2.3 FIRE PAN ASSEMBLY

The Fire Pan is positioned in the cargo section of the mock-up and used to create a cargo compartment fire. The Fire Pan is designed to create a fire over its 4' x 6' surface. The Fire Pan is constructed of stainless steel. There is an acme connector at the center of the Fire Pan for the hookup of the vapor propane supply hose. The liquid propane connection is not used for the **HELICOPTER FIRETRAINER® O-100**.

The supply gas is distributed through a burner element in the Fire Pan. The burner element has two, quarter turn shut off valves in line. The burn pan is filled with water. The gas bubbles up through the water and is ignited by the standing flame from the pilot igniter located on the Pilot Box Assembly. See Figure 2-1 for the location and Figure 2-4 for the Fire Pan detail.

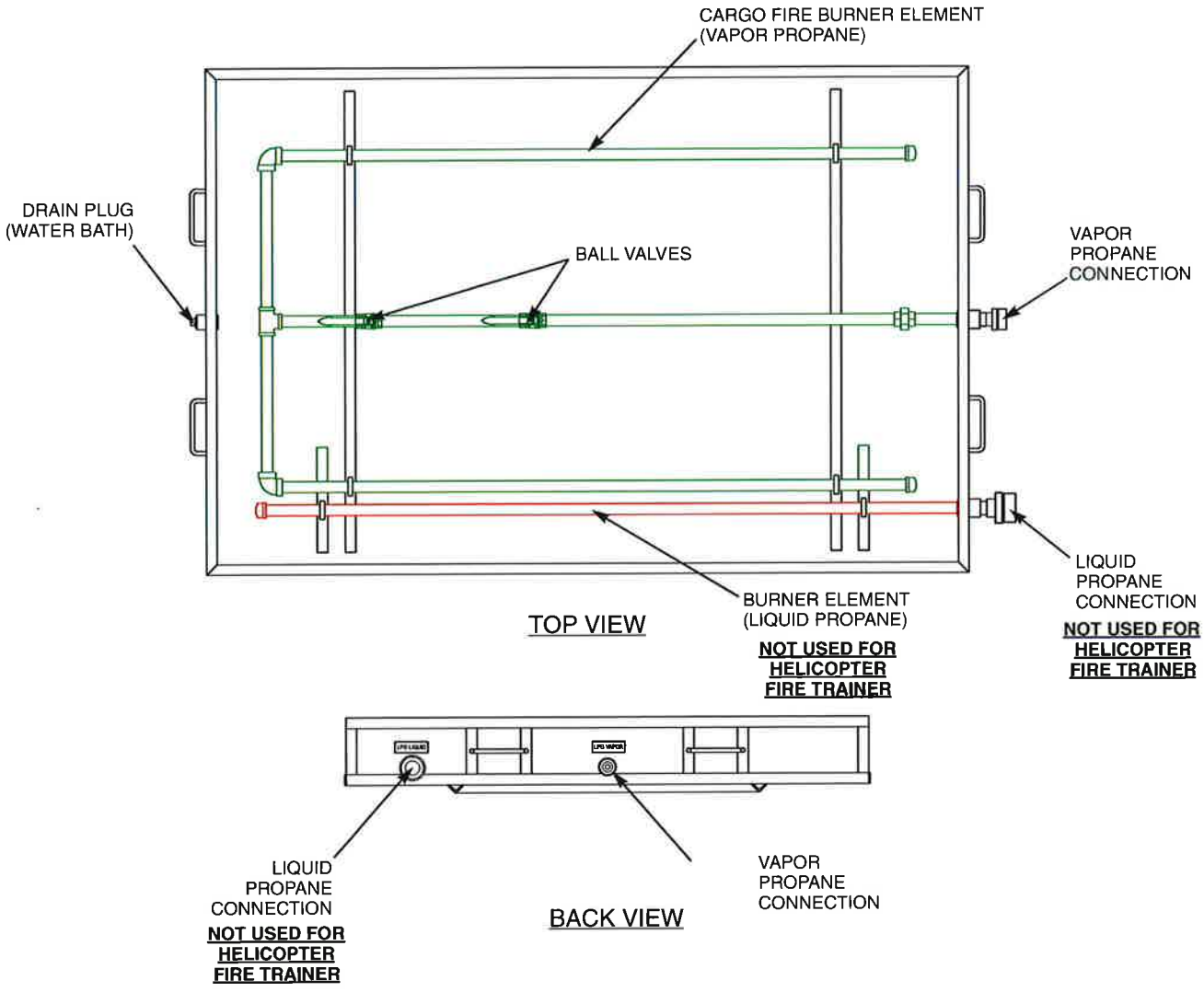

Figure 2-4 – Fire Pan Assembly

2.2.4 TRAILER

The Trailer is used to transport all the components that make-up the **Helicopter FIRETRAINER® O-100** (figure 2-5). The trailer is equipped with two (2) 100 gallon propane storage tanks, a storage compartment located above the propane storage compartment, binding straps, a winch, lift jack and ramps to allow for loading and unloading of the helicopter mock-up. The trailer is also equipped with a 12 VDC battery, used to power the lift jack and winch.

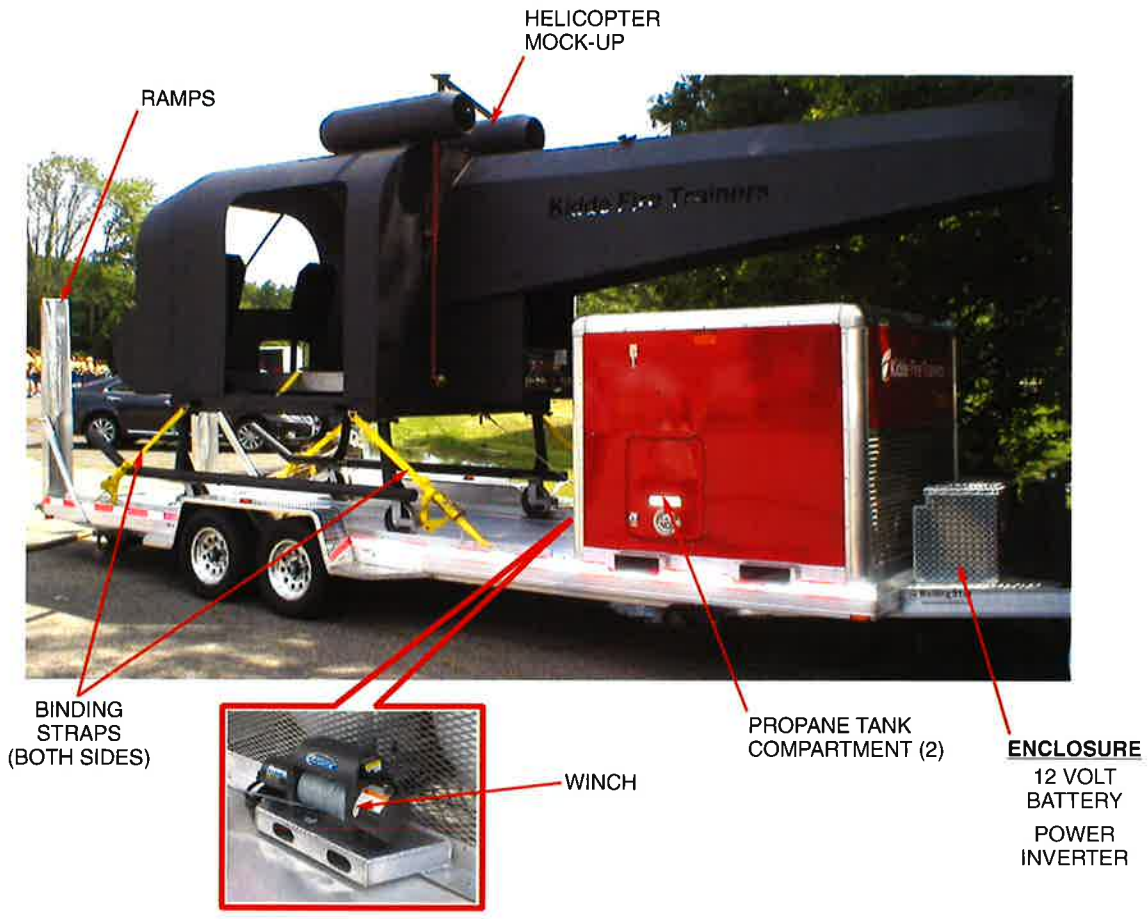

Figure 2-5. Trailer

2.2.4.1 PROPANE SUPPLY TANKS

Two (2) 100 Gallon Propane supply tanks come mounted to the front end of the Trailer. Each tank is furnished with a liquid propane supply shutoff valve, a vapor propane supply shutoff valve, a fill point, tank vent, level gauge & relief valve. See Figure 2-5 for location and 2-6a for details. Both tank supplies run to the liquid and vapor supply connections located externally on the back of the propane supply tank enclosure (Figure 2-6b).

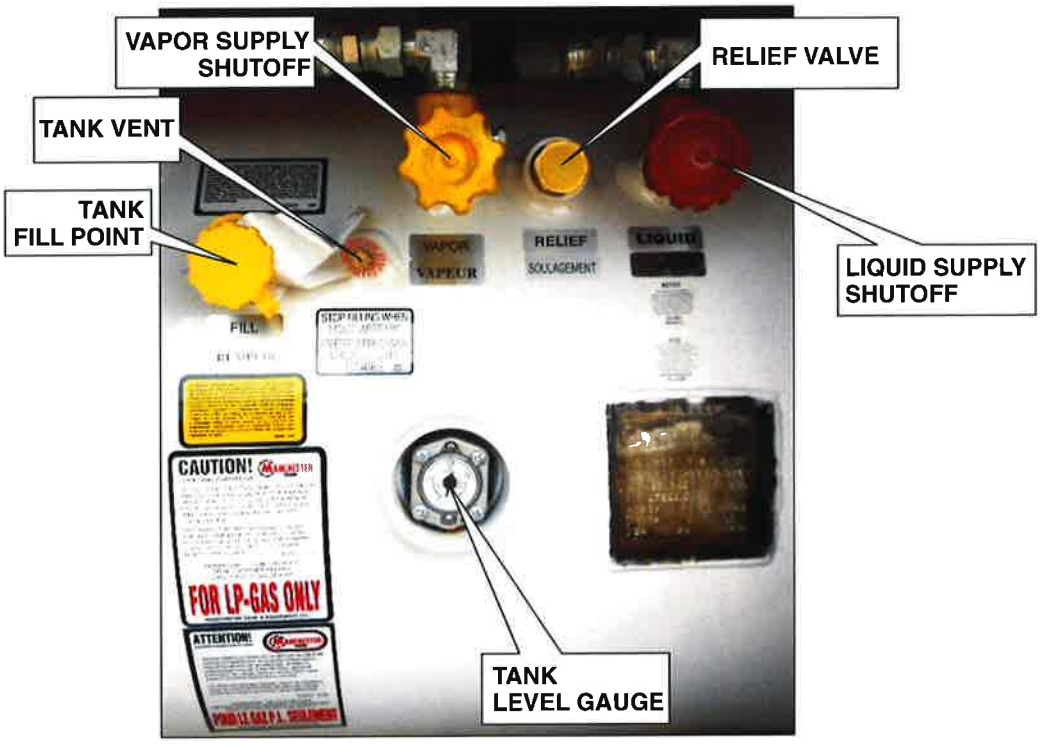

Figure 2-6a – 100 Gallon Propane Storage Tank

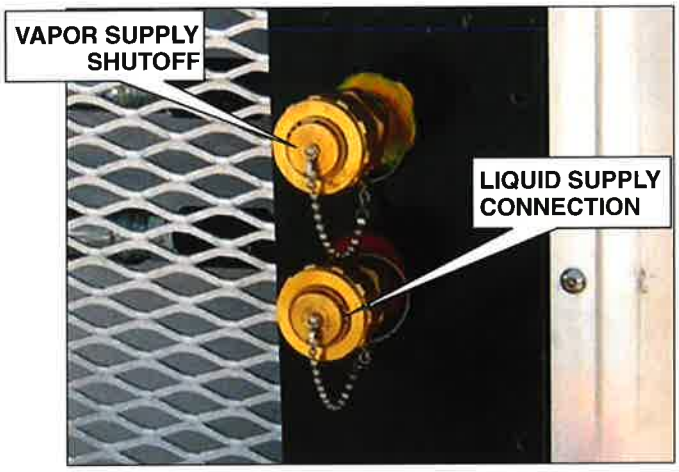

Figure 2-6b –Propane Supply Hose Connections (Exterior of Propane Enclosure)

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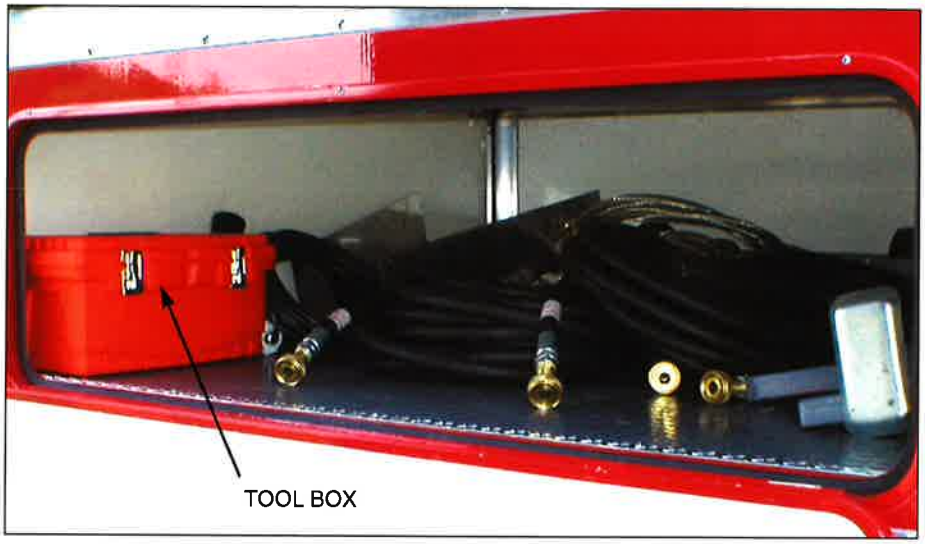
2.2.4.2 STORAGE COMPARTMENT

The trailer is equipped with a storage compartment, located above the propane Supply tank area. This compartment is used to store all the equipment that supports the **Helicopter FIRETRAINER® O-100.** See Figure 2-5 for location and 2-7 for details.



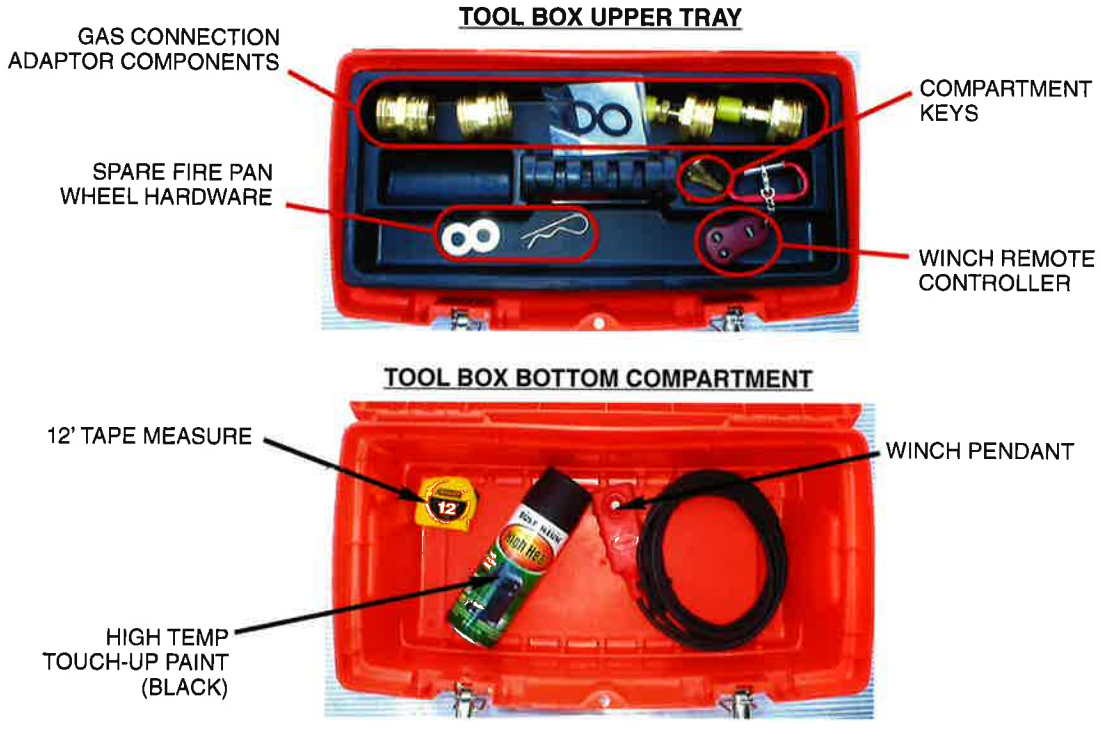


Figure 2-7. Trailer Storage Compartment

FRONT OF TRAILER VIEW

Figure 2-8. Trailer - Front View

2.2.4.3 TRAILER LIFT JACK

The trailer is equipped with a 12 VDC powered lift jack (Figure 2-8). The lift jack is used to raise the trailer coupler off the tow vehicle hitch and to allow for the helicopter mockup to be unloaded from the trailer. The lift jack is operated from a control unit that is mounted to the battery/inverter enclosure. For detailed information about the Trailer Lift Jack, refer to Section 6 of this manual.

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2.2.4.4 12 VDC BATTERY

The trailer is equipped with a 12 VDC battery (Figure 2-9), located in the bottom of the front enclosure (Figure 2-8). The battery is used to power the winch and lift jack. The battery is charged from a solar charging system. The solar charging system consists of solar panels, that are located on the top of the trailer storage compartment and a charger controller that is mounted to the inside of the enclosure cover, located at the front of the trailer (See Figure 2-9). For detailed information about the Solar Charging System, refer to Section 6 of this manual.

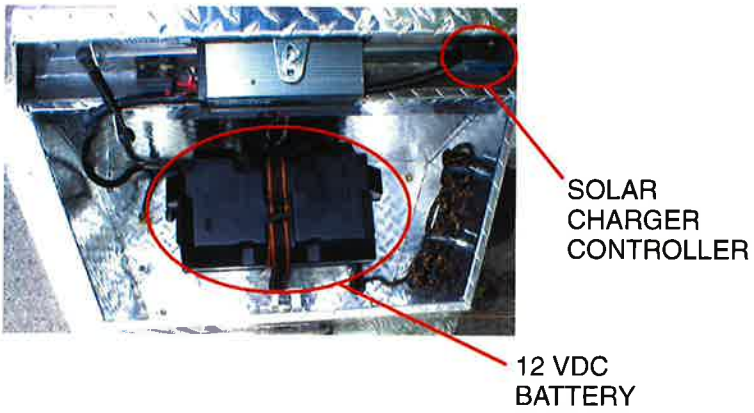

Figure 2-9. Battery & Solar Charger Controller Locations – Front of Trailer

2.2.4.5 POWER INVERTER

The trailer is equipped with a Power Inverter. The Power Inverter is used to convert 12 VDC battery power to 110 VAC power to operate the Pilot Box Assembly. The inverter is mounted to the inside of the enclosure cover, located at the front of the trailer (See Figure 2-10). For detailed information about the Power Inverter, refer to Section 6 of this manual.

Figure 2-10. Power Inverter Location - Front of Trailer

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2.3 CONTROL SYSTEMS

The following describes each of the three control systems used to operate the **Helicopter FIRETRAINER® O-100.**

2.3.1 CONTROL PENDANT

A Control Pendant is used to control the operation of the **Helicopter FIRETRAINER® O-100**. The Control Pendant is a three-button hand held control device used to activate the pilot flame and the propane vapor and liquid flames from a safe distance. See Figures 2-11.

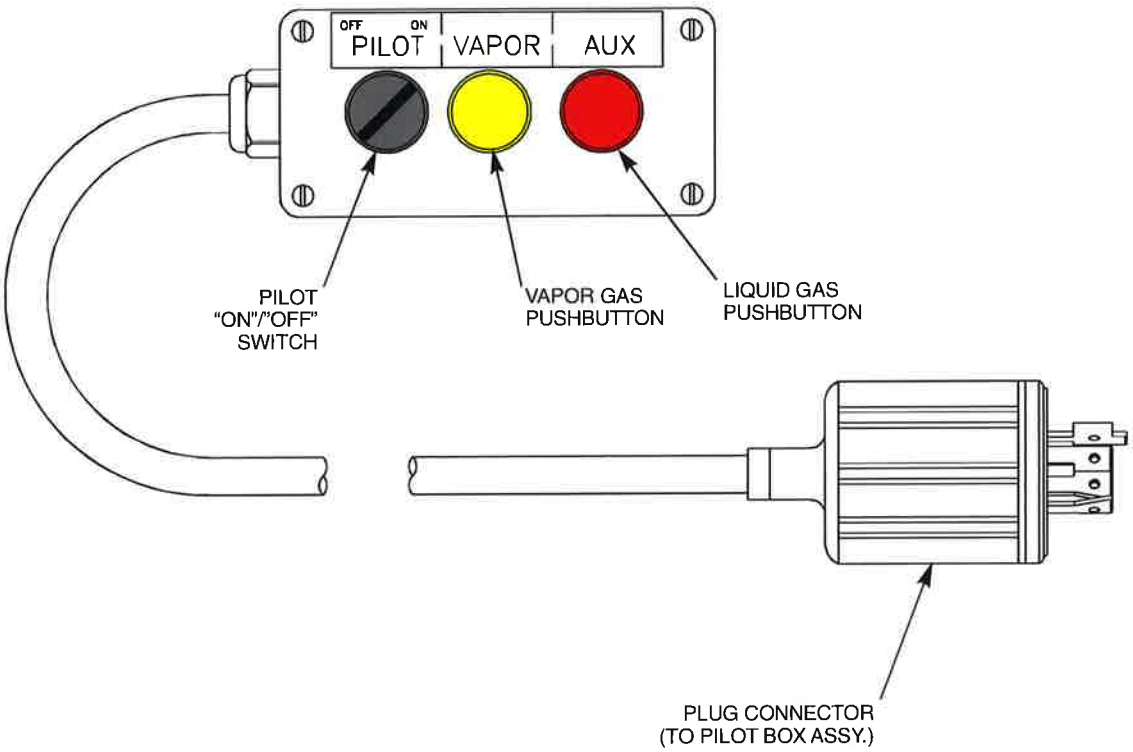

Figure 2-11 – Control Pendant

2.3.2 WIRELESS CONTROL SYSTEM (OPTION)

The wireless control system assembly option allows the instructor to control the operation of the **KIDDE O-100 Helicopter FIRE TRAINER** with the inherent mobility available without a cable connection. The switches on the front panel are used for functional operations with an additional trigger switch on the pistol grip that functions as a "Dead Man" switch for the LPG vapor and liquid burner flames. The trigger switch on the pistol grip must be pressed and held for the LPG vapor and liquid burner flames to operate.

The wireless control system assembly consists of a hand held battery powered weatherproof transmitter, with unique ID code and a receiver Located in a weatherproof enclosure. See Figure 2-12).

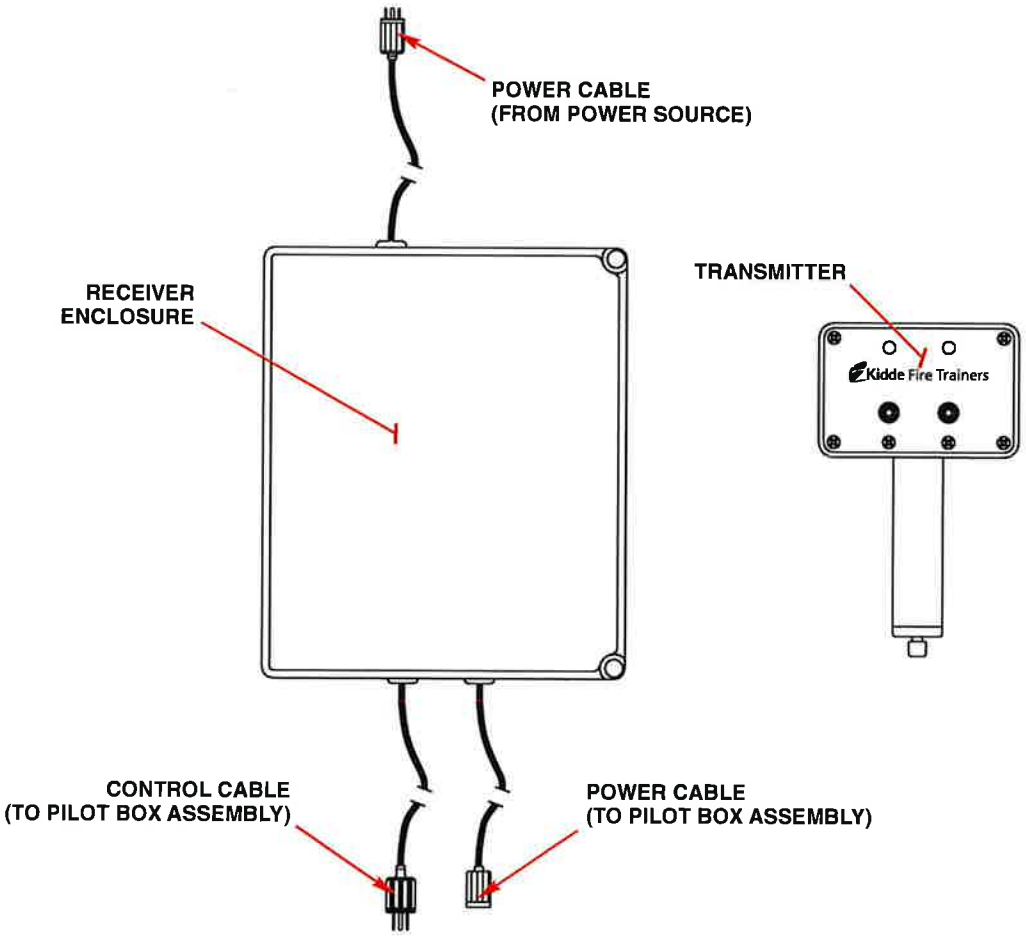

Figure 2-12. Wireless Control System Assembly

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2.3.3 CONTROL STAND ASSEMBLY (OPTION)

The Control Stand Assembly option allows the instructor to control the operation of three **KIDDE O-100 OUTDOOR FIRE TRAINERS** from one location. The Control Stand Assembly is equipped with a Control Enclosure, LPG Vapor/Liquid manifolds and the power/control junction box. See Figure 2-13.

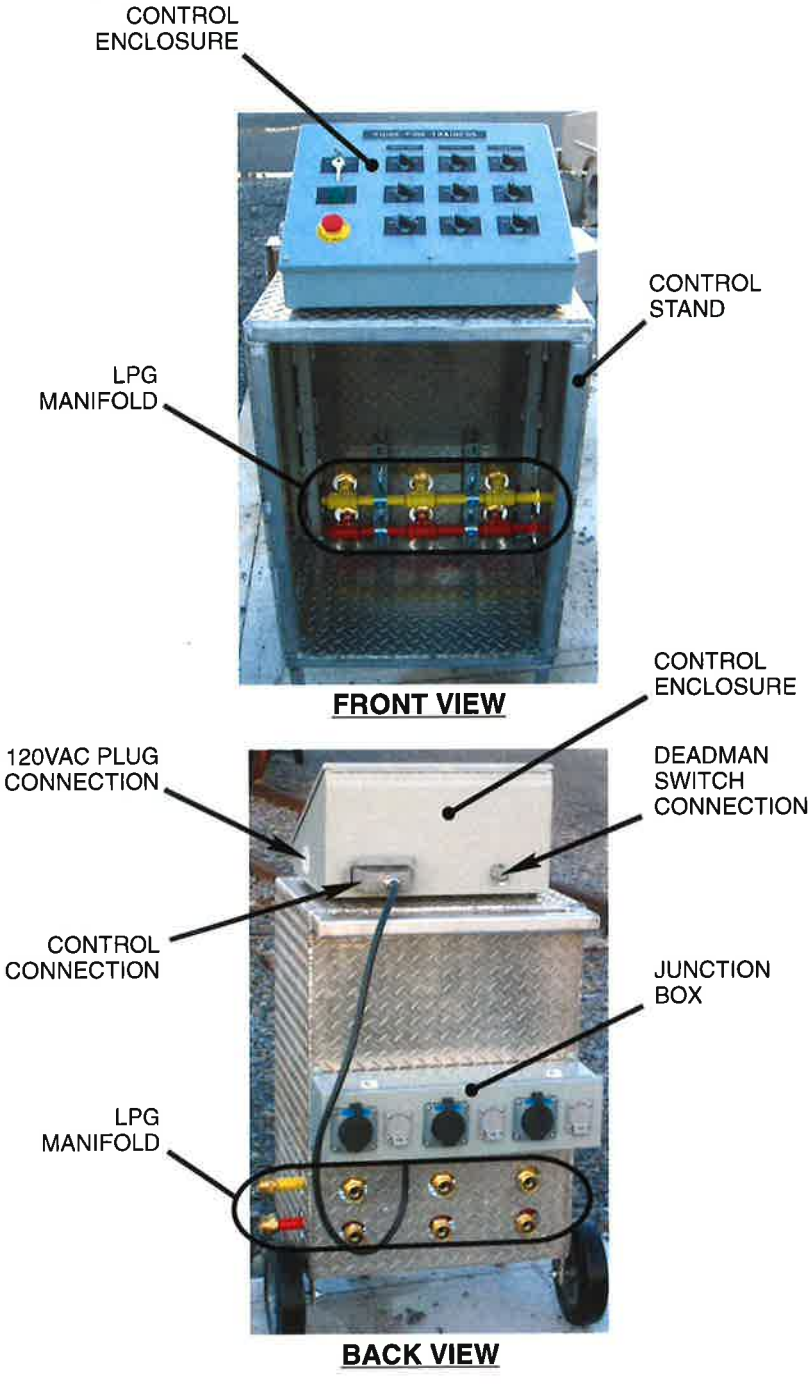

Figure 2-13. Control Stand Assembly

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3.0 OPERATION

3.1 INTRODUCTION

This section describes the necessary procedures to operate the **Helicopter FIRETRAINER® O-100**.

3.2 UNLOADING HELICOPTER MOCKUP

Refer to Figure 3-1 for Components

1. Choose a location for the mock-up outdoors; clear of all ground/aerial obstructions on level ground.
2. Position trailer so that the door side of the helicopter mockup is facing into the wind.
3. Prevent the trailer from rolling when the helicopter mockup is unloaded from the trailer, by blocking the tires on both sides of the trailer.

⚠ WARNING

TIRES ON BOTH SIDES OF TRAILER MUST BE BLOCKED PRIOR TO PROCEEDING TO THE NEXT STEP.

NOTE

IF O-100 FIRE PAN(S) ARE ONBOARD THE TRAILER, REMOVE BINDING STRAPS AND REMOVE PAN(S) AT THIS TIME.

4. Be sure the winch hook is securely connected to the helicopter mockup, Item (1).

⚠ WARNING

THE WINCH HOOK MUST BE SECURELY CONNECTED TO THE HELICOPTER MOCKUP ITEM (1) PRIOR TO PROCEEDING TO THE NEXT STEP.

TWO (2) PEOPLE ARE REQUIRED TO SAFELY LOWER THE REAR RAMPS.

5. While supporting the ramp disconnect the securing bar (2) from the ramp by removing the securing pin and sliding the bar off the ramp connection rod. Carefully lower the ramp to the ground, Item (3).
6. Repeat step 5 for the remaining ramp.
7. Remove the securing bars from both sides of the trailer by removing the lower securing pins. Store both bars in trailer storage compartment. Reinstall all four securing pins on the ramp and trailer.
8. Align both ramps with centerline of helicopter mock-up casters.
9. Disconnect the electrical brake and light connections from the tow vehicle.
10. Disconnect the safety chains from the tow vehicle.
11. Disconnect the breakaway brake switch tether from the tow vehicle.
12. Unlock trailer coupler from tow vehicle hitch.

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13. Release all four of the helicopter mock-up wheel brakes.

⚠ CAUTION

ALL TRAILER COMPONENTS MUST BE DISCONNECTED FROM THE TOW VEHICLE PRIOR TO PROCEEDING TO THE NEXT STEP.

⚠ WARNING

YOU MUST READ AND UNDERSTAND THE EQUALIZER HYDRAULIC TRAILER JACK OPERATION MANUAL, LOCATED IN SECTION 6 OF THIS MANUAL, BEFORE PROCEEDING.

14. At the trailer jack control unit, Item (4), insert the key and turn to the right. Toggle and hold the "TRAILER UP/TRAILER DOWN switch, Item (4a) in the "TRAILER UP" position to raise the front of the trailer. Raise the trailer until trailer coupler is clear of the hitch on the tow vehicle.

15. With the trailer coupler clear of the tow vehicle hitch, move tow vehicle a safe distance from the trailer to avoid damage to vehicle during the unloading of the mockup.

⚠ CAUTION

THE TOW VEHICLE MUST BE MOVED TO A SAFE DISTANCE FROM THE TRAILER, PRIOR TO PROCEEDING TO THE NEXT STEP.

16. At the trailer jack control unit, Item (4a), use the "TRAILER UP/TRAILER DOWN switch until there is a smooth transition at the trailer, ramp and the bottom of the ramp and the ground.

17. At the trailer jack control unit, Item (4a), turn off power to the trailer jack by turning the key to the left and remove key from switch.

18. Remove the two trailer supports from the storage compartment and install one on each side as shown in Figure 3-1, Item (5).

19. With the handle lower each trailer support foot until they are touching the ground.

⚠ WARNING

BE SURE BOTH RAMPS ARE ALIGNED WITH CENTERLINE OF HELICOPTER MOCK-UP CASTERS, BEFORE PROCEEDING.

BE SURE NO ONE IS BEHIND, THE HELICOPTER MOCK-UP OR BETWEEN THE MOCK-UP AND WINCH AT ANY TIME WHEN THE MOCK-UP IS REMOVED FROM THE TRAILER.

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WARNING

YOU MUST READ AND UNDERSTAND THE RAMSEY PATRIOT 9500 WINCH OWNER'S MANUAL, LOCATED IN SECTION 6 OF THIS MANUAL, BEFORE PROCEEDING.

20. At the winch remove the protective rubber cover and connect the end of the control pendant cable to the control switch receptacle located on the winch, Item (6).

21. With the winch pendant, carefully remove the slack from the winch cable by pressing the "IN" button.

WARNING

THE WINCH CABLE MUST BE TENSIONED SO THE HELICOPTER MOCKUP WILL NOT ROLL WHEN THE BINDER STRAPS ARE REMOVED.

WARNING

THE WINCH HOOK MUST BE SECURELY CONNECTED TO THE HELICOPTER MOCKUP ITEM (1) PRIOR TO PROCEEDING TO THE NEXT STEP.

22. With the winch hook still attached to the helicopter mockup, remove all the binding straps that secure the helicopter mockup to the trailer and store binding straps in trailer storage compartment.

WARNING

YOU MUST READ AND UNDERSTAND THE RAMSEY PATRIOT 9500UT WINCH OWNER'S MANUAL, LOCATED IN SECTION 6 OF THIS MANUAL, BEFORE PROCEEDING.

23. With the winch pendant, press the winch "OUT" button to allow the helicopter mockup to be slowly lowered onto the ground, Items (7) & (8).

24. Once all four of the helicopter mockup casters are on the ground continue pressing the winch "OUT" button to release tension on the winch cable. Disconnect the winch cable hook from the helicopter.

25. Press the "IN" button on the winch pendant to roll cable back onto reel. Use the red hook strap to keep the winch cable under tension as it feeds onto the reel.

26. Reinstall both ramp securing bars to the trailer and secure in place with securing pins. Place both trailer ramps in the upright position and secure each in place with securing pins, Item (2).

27. Remove both of the trailer supports, Item (5) and stow in trailer storage compartment.

28. Connect the trailer coupler to the tow vehicle hitch.

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29. Connect the safety chains to the tow vehicle
30. Connect the electrical brake and light connections to the tow vehicle.
31. Connect the breakaway brake switch tether to the tow vehicle.
32. Remove the Pilot Box Assembly from the trailer storage compartment.
33. Remove required hoses and adapters from storage compartment.
34. Remove the electrical power cable and Control Pendent from storage compartment.
35. Drive the tow vehicle and trailer at least 75 ft. upwind of the helicopter trainer.

NOTE

IF THE KIDDE MULTI CONTROL SYSTEM IS USED, THE MULTI CONTROL STAND IS CONSIDERED THE PROPANE SUPPLY. THE MULTI CONTROL STAND IS POSITIONED 75 FEET UPWIND OF THE FIRE TRAINER MOCK-UP(S) AND THE TRAILER IS POSITIONED 40 FEET UPWIND OF THE MULTI CONTROL STAND.

36. Position the helicopter mockup as shown in Figure 3-2.
37. Engage the brakes on all four of the mock-up casters.

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Figure 3-1 Trainer Setup (Sheet 1 of 2)

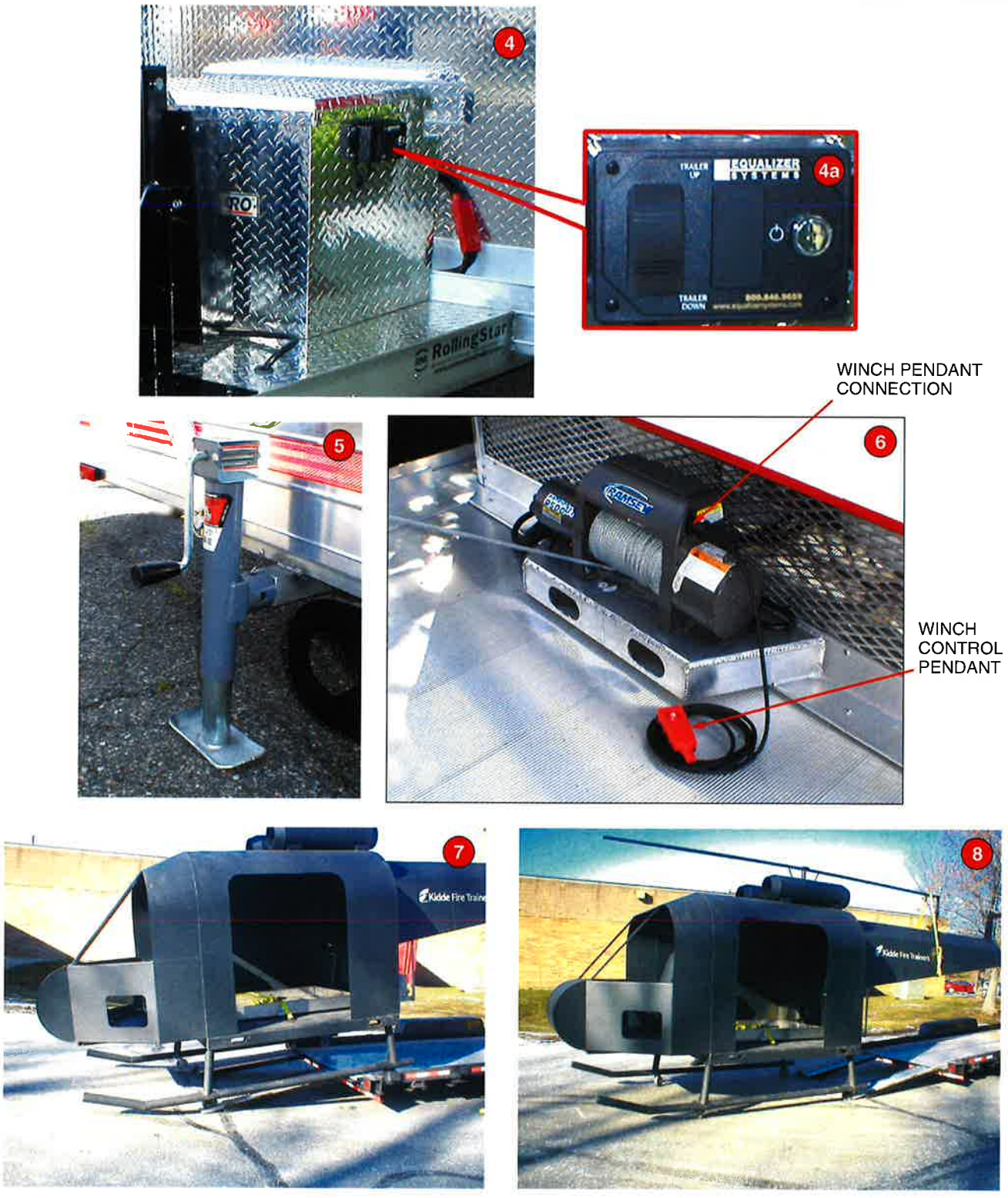

Figure 3-1. Trainer Setup (Sheet 2)

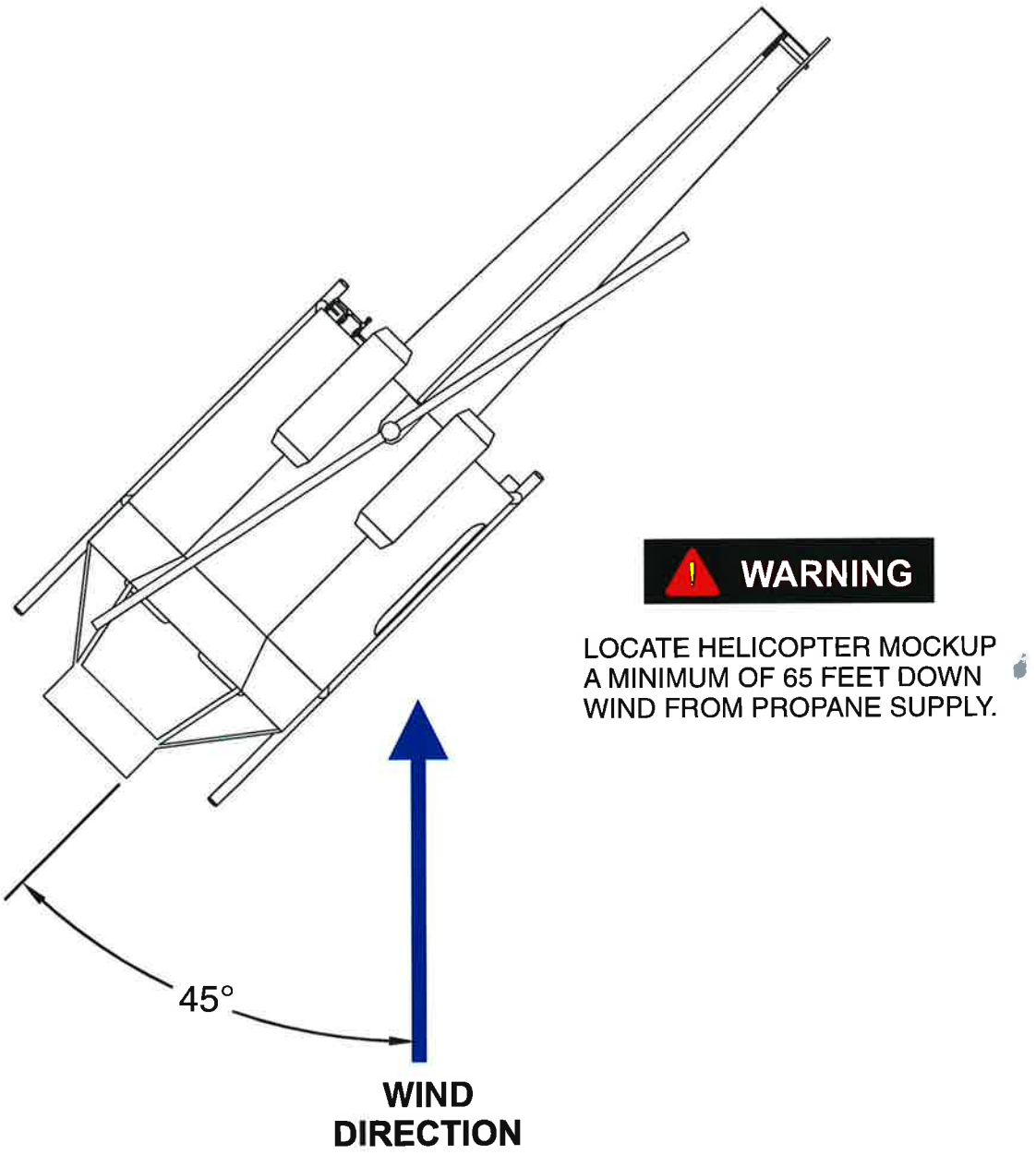

Figure 3-2. Helicopter Mockup Position

3.3 HELICOPTER FIRE TRAINER SETUP

1. At the main rotor support bracket, remove the binding strap that secures the main rotor in place during transit.

⚠ WARNING

THE MAIN ROTOR SUPPORT BRACKET MUST BE SECURED SO IT WILL NOT DROP DOWN WHEN STEP 2. IS PERFORMED.

2. Support the main rotor support bracket. With the bracket supported, from under the tail section of the mock-up, remove the securing pin and lower the bracket down. Reinstall the securing pin.
3. Remove the binding strap that secures the fire pan to the floor of the helicopter mockup (see Figure 3-3).

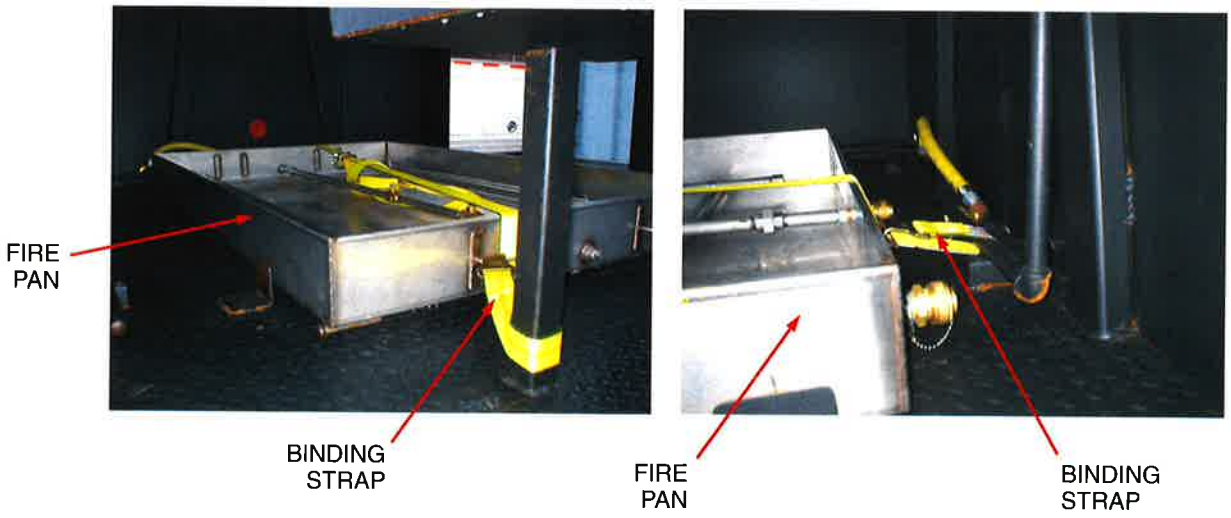

Figure 3-3. Fire Pan Binding Strap

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4. Be sure the 1-1/4" male connection on the flexible stainless steel LPG Vapor supply hose is connected to the 1-1/4" female connection marked **LPG VAPOR** on the Fire Pan Assembly (see Figure 3-4).

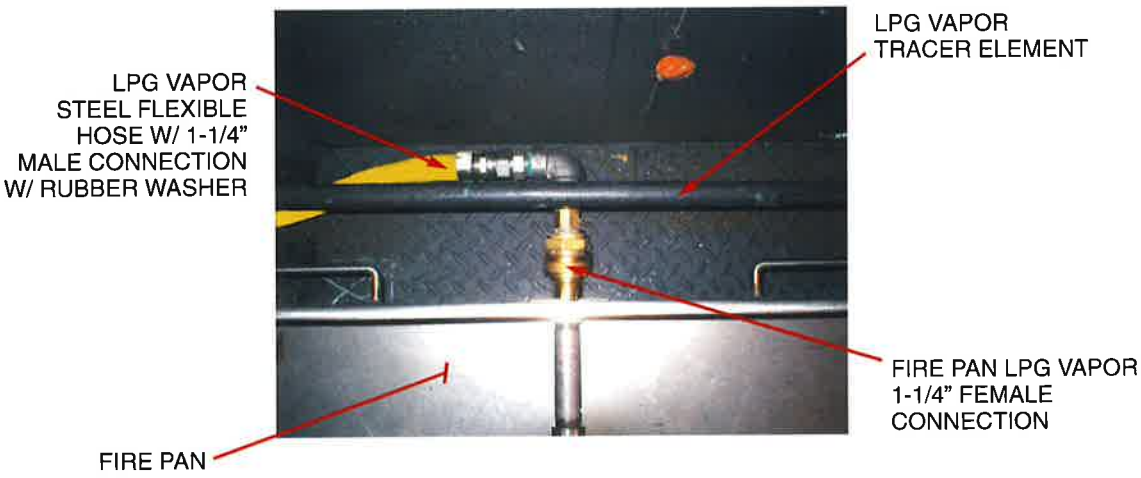

Figure 3-4. Fire Pan LPG Vapor Connection

5. Locate the fire pan against the floor stops located on the right side of the mockup (see Figure 3-5).

Figure 3-5. Fire Pan Location

6. Ensure that drain plug is installed in the front of the fire pan.

7. Ensure that drain plug is installed in the Pilot Box Assembly.

⚠ WARNING

TWO (2) PEOPLE ARE REQUIRED TO SAFELY INSTALL THE PILOT BOX ASSEMBLY.

8. Position the Pilot Box Assembly onto the support frame as shown in Figure 3-6.

9. While aligning the Pilot Box shroud and drain plug with the corresponding through holes in the back wall of the helicopter mockup, slide the Pilot Box assembly into position flush against the mockup wall see Figure 3-6.

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SIDE VIEW

Figure 3-6. Pilot Box Assembly Installation

10. Extend all lines (gas, control, electric) and check for any damage.

WARNING

DO NOT PROCEED ANY FURTHER IF ANY OF THE CONTROL OR ELECTRICAL LINES ARE DAMAGED. DO NOT PROCEED ANY FURTHER IF ANY OF THE GAS HOSES ARE DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

11. At the Pilot Box Assembly remove protective thread cap from the 1-1/4" male **LPG VAPOR** connection marked **"IN".** Make sure the 1-1/4" male **LPG VAPOR** connection is free of debris, damage and that the rubber washer is present (see Figure 3-7).

WARNING

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE LPG VAPOR "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

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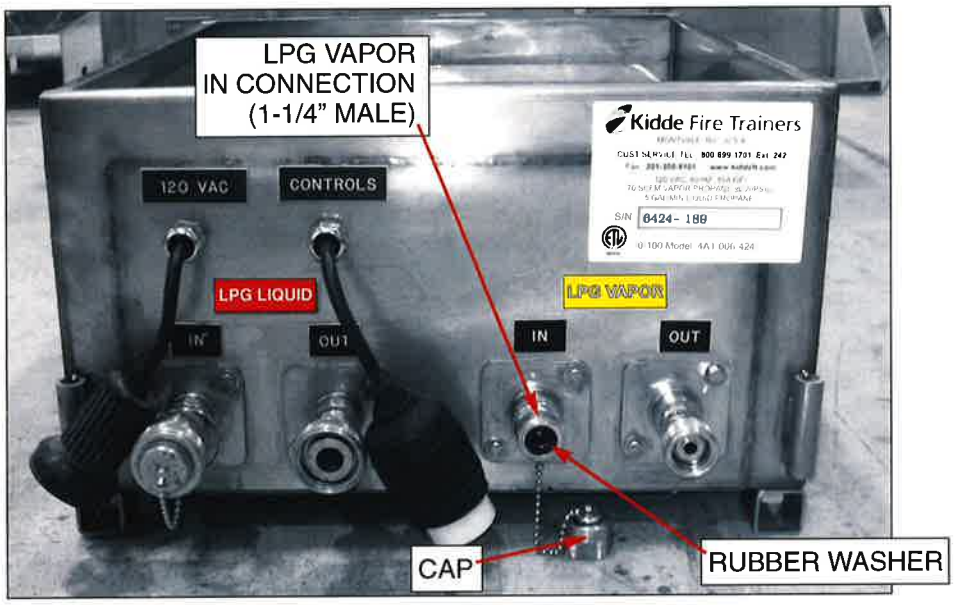

Figure 3-7. Pilot Box Assembly LPG VAPOR "IN" Connection

12. Make sure the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose is free of debris and damage.

⚠ WARNING

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

13. Connect the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose to the 1-1/4" male **LPG VAPOR** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 3-8).

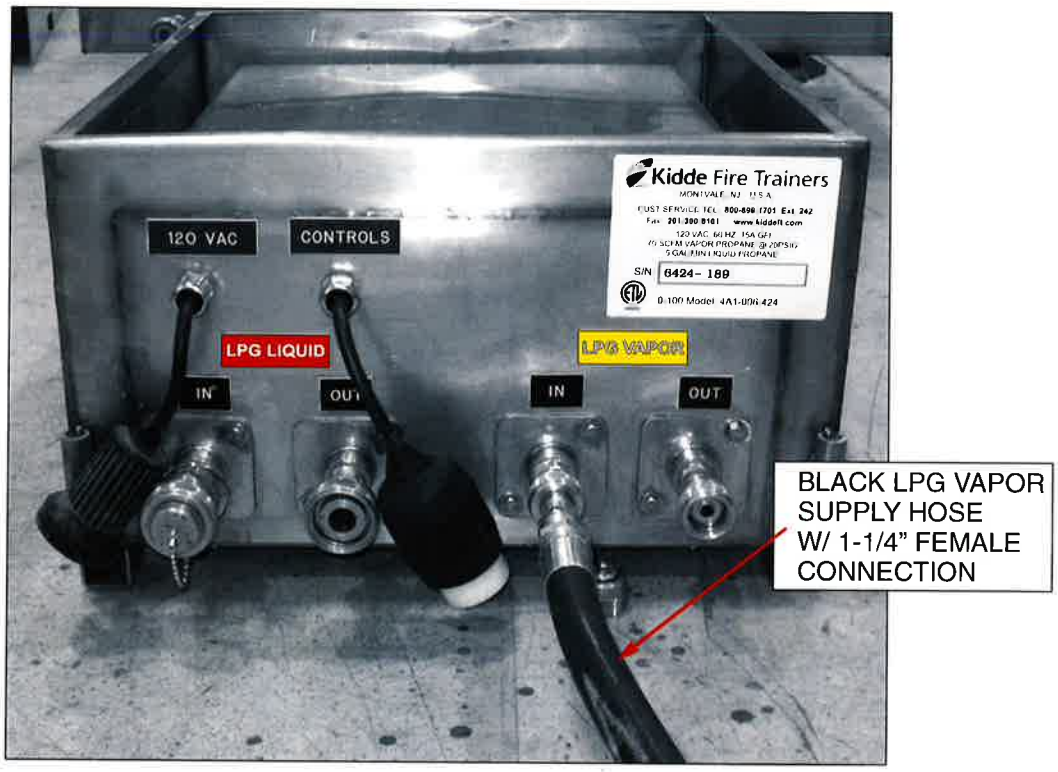

Figure 3-8. Pilot Box Assembly LPG VAPOR "IN" Supply Hose Connection

14. An additional 50' black rubber LPG Vapor supply hose must be added to the end of the existing one so the propane supply is at a safe distance from the fire trainer. Take the Kidde supplied 1-3/4" male connection adaptor and check that it is free of debris, damage and that the two rubber washers are present (see Figure 3-9 to identify adaptor).

⚠ WARNING

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE ADAPTOR CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER(S) IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

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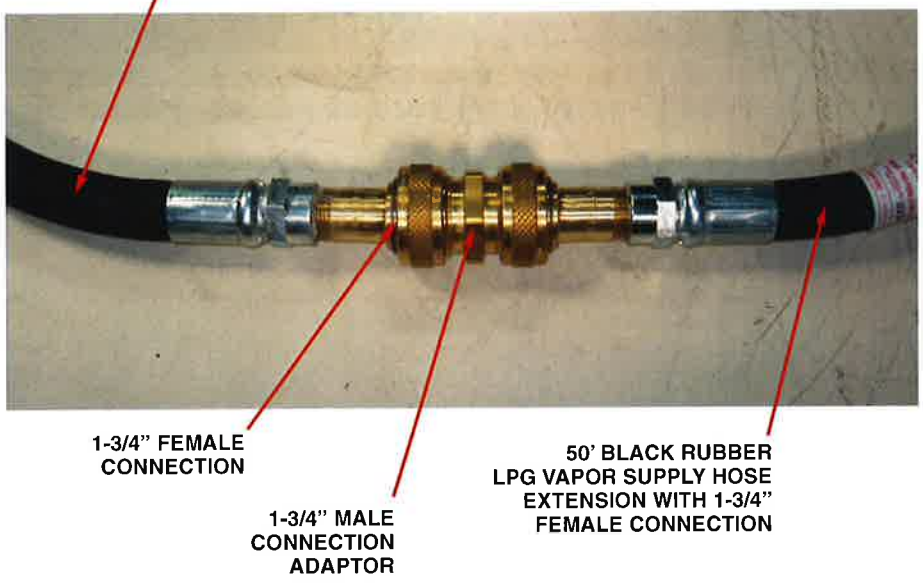

Figure 3-9. LPG Vapor Supply Hose Extension

15. Make sure the 1-3/4" female connection on both 50' black rubber LPG Vapor supply hoses are free of debris and damage (see Figure 3-9 to identify components).

⚠ WARNING

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG VAPOR SUPPLY HOSES HAVE DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

16. Using the Kidde supplied 1-3/4" male connection adaptor, connect the second 50' black rubber LPG Vapor supply hose to the end of the 50' black rubber LPG Vapor supply hose that is connected to the Pilot Box **LPG VAPOR** connection marked **"IN"** (Attach fittings hand tight (see Figure 3-9). Run the hose straight back from the Pilot Box Assembly

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17. At the Pilot Box Assembly remove protective thread cap from the 1-3/4" male **LPG LIQUID** connection marked **"IN".** Make sure the 1-3/4" male **LPG LIQUID** connection is free of debris, damage and that the rubber washer is present (see Figure 3-10).

WARNING

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE LPG LIQUID "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

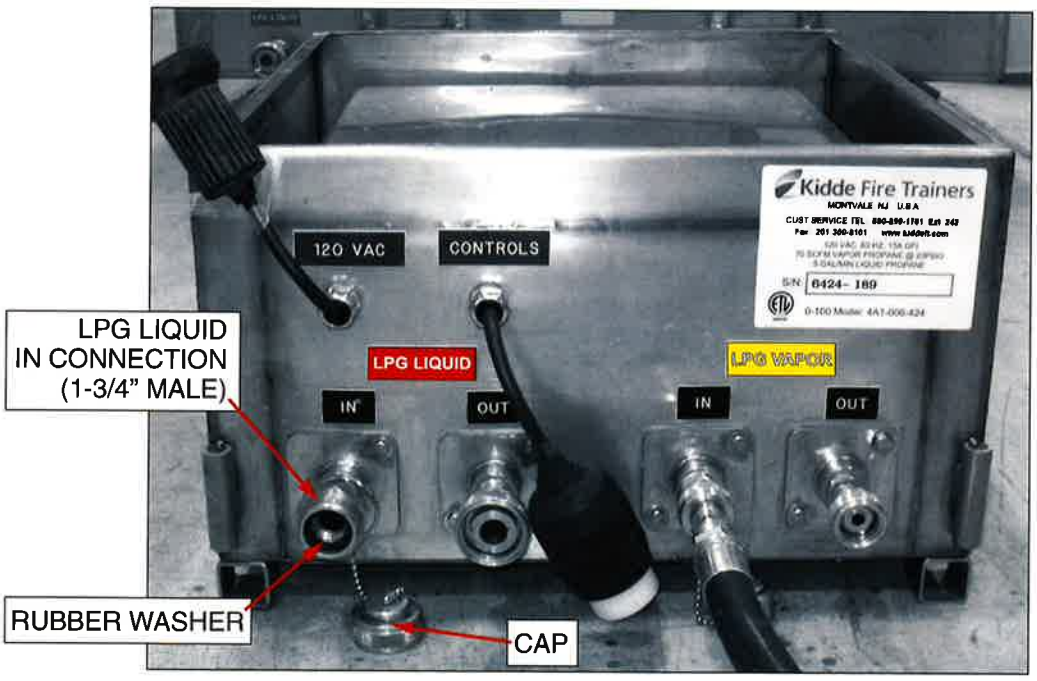

Figure 3-10. Pilot Box Assembly LPG LIQUID "IN" Connection

18. Make sure the 1-3/4" female connection on the 50' black rubber LPG Liquid supply hose is free of debris and damage.

⚠ WARNING

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

19. Connect the 1-3/4" female connection on the 50' black rubber LPG Liquid supply hose to the 1-3/4" male **LPG LIQUID** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 3-11).

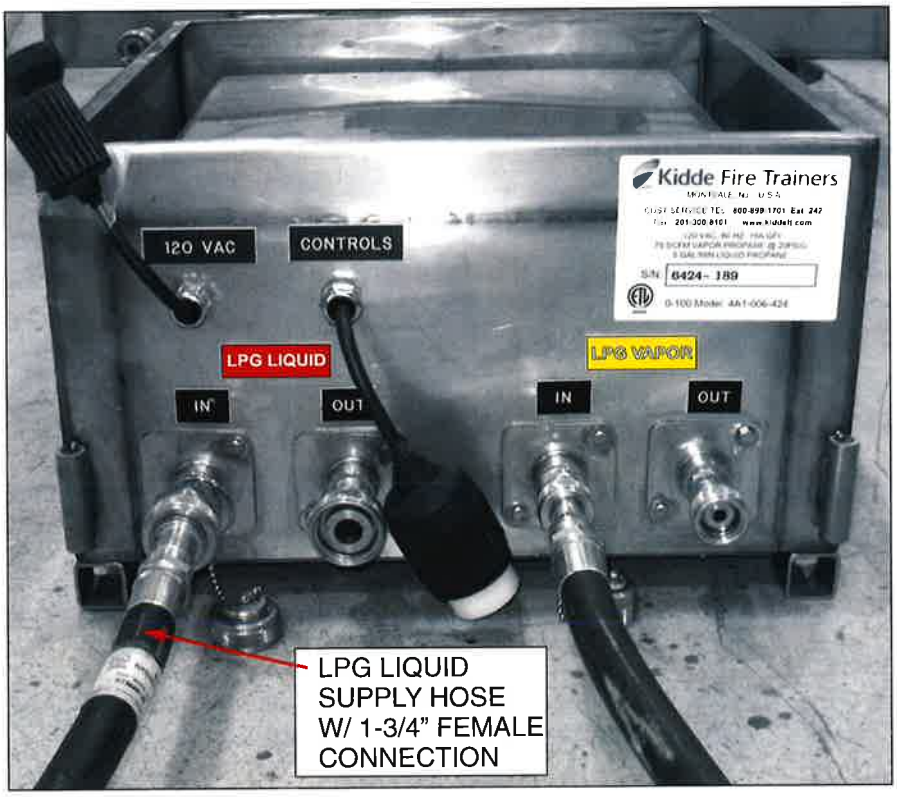

Figure 3-11. Pilot Box Assembly LPG LIQUID "IN" Supply Hose Connection

20. An additional 50' black rubber LPG Liquid supply hose must be added to the end of the existing one so the propane supply is at a safe distance from the fire trainer. Take the Kidde supplied 1-3/4" male connection adaptor and check that it is free of debris, damage and that the two rubber washers are present (see Figure 3-12 to identify adaptor).

WARNING

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE ADAPTOR CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER(S) IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

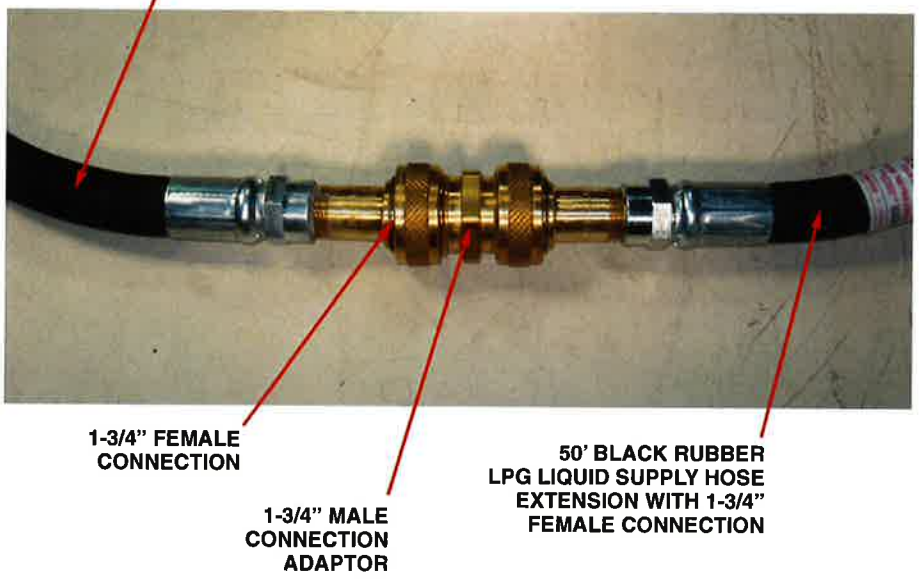

Figure 3-12. LPG Liquid Supply Hose Extension

21. Make sure the 1-3/4" female connection on both 50' black rubber LPG Liquid supply hoses are free of debris and damage (see Figure 3-12 to identify components).

WARNING

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG LIQUID SUPPLY HOSES HAVE DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

22. Using the Kidde supplied 1-3/4" male connection adaptor, connect the second 50' black rubber LPG Liquid supply hose to the end of the 50' black rubber LPG Liquid supply hose that is connected to the Pilot Box **LPG LIQUID** connection marked **"IN"** (Attach fittings hand tight (see Figure 3-12). Run the hose straight back from the Pilot Box Assembly.

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23. At the Pilot Box Assembly make sure the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** is free of debris and damage.

WARNING

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE LPG VAPOR "OUT" CONNECTION HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

24. Remove protective thread cap from the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

25. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 3-13).

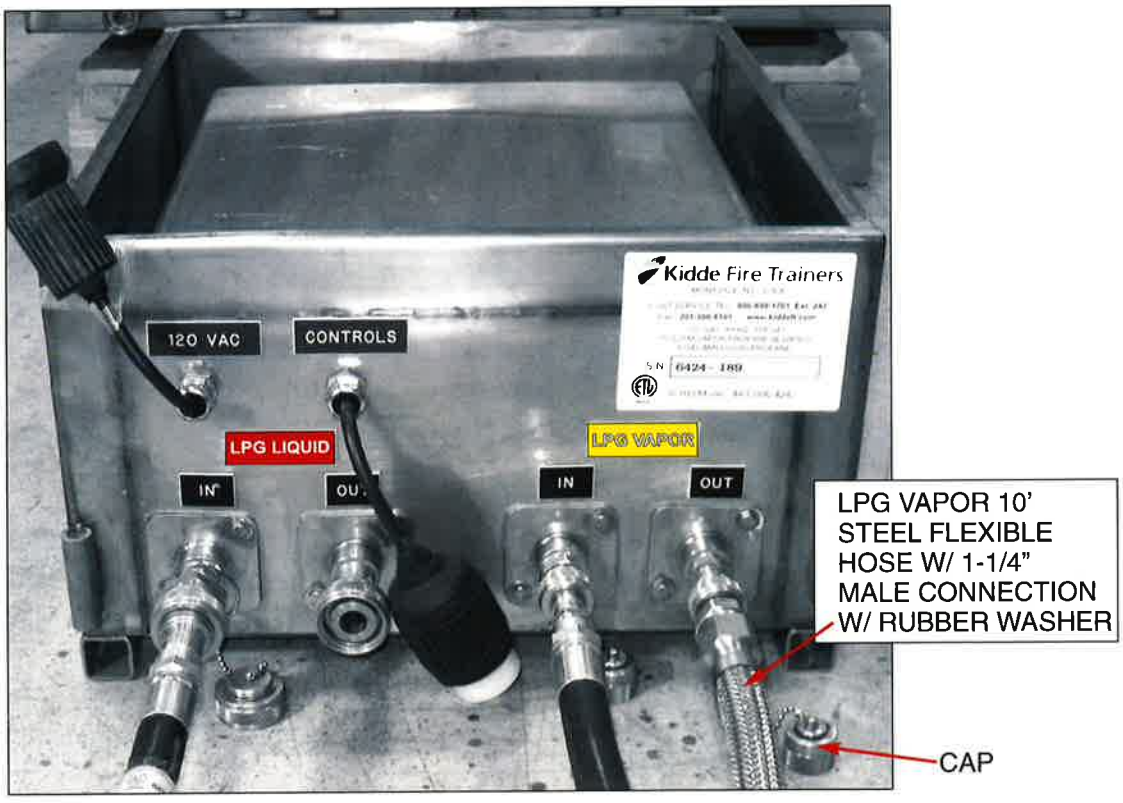

Figure 3-13. Pilot Box Assembly LPG VAPOR "OUT" Hose Connection

26. Remove protective thread cap form the 1-1/4" male connection on the remaining end of the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

⚠ WARNING

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

27. At the yellow LPG Vapor manifold, to the right of the Pilot Box Assembly, make sure the 1-1/4" female LPG Vapor connection is free of debris and damage (see Figure 3-15).

⚠ WARNING

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE LPG VAPOR CONNECTION ON THE YELLOW LPG VAPOR MANIFOLD HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

28. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the 1-1/4" female LPG Vapor connection on the yellow LPG Vapor manifold to the right of the Pilot Box Assembly. Attach fittings hand tight (see Figure 3-15).

29. Remove protective thread cap from the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose. Make sure the 1-3/4" male connection is free of debris, damage and that the rubber washer is present.

⚠ WARNING

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

30. At the Pilot Box Assembly make sure the 1-3/4" female **LPG LIQUID** connection marked **OUT** is free of debris and damage.

⚠ WARNING

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION MARKED LPG LIQUID "OUT" HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

31. Connect the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose to the 1-3/4" female **LPG LIQUID** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 3-14).

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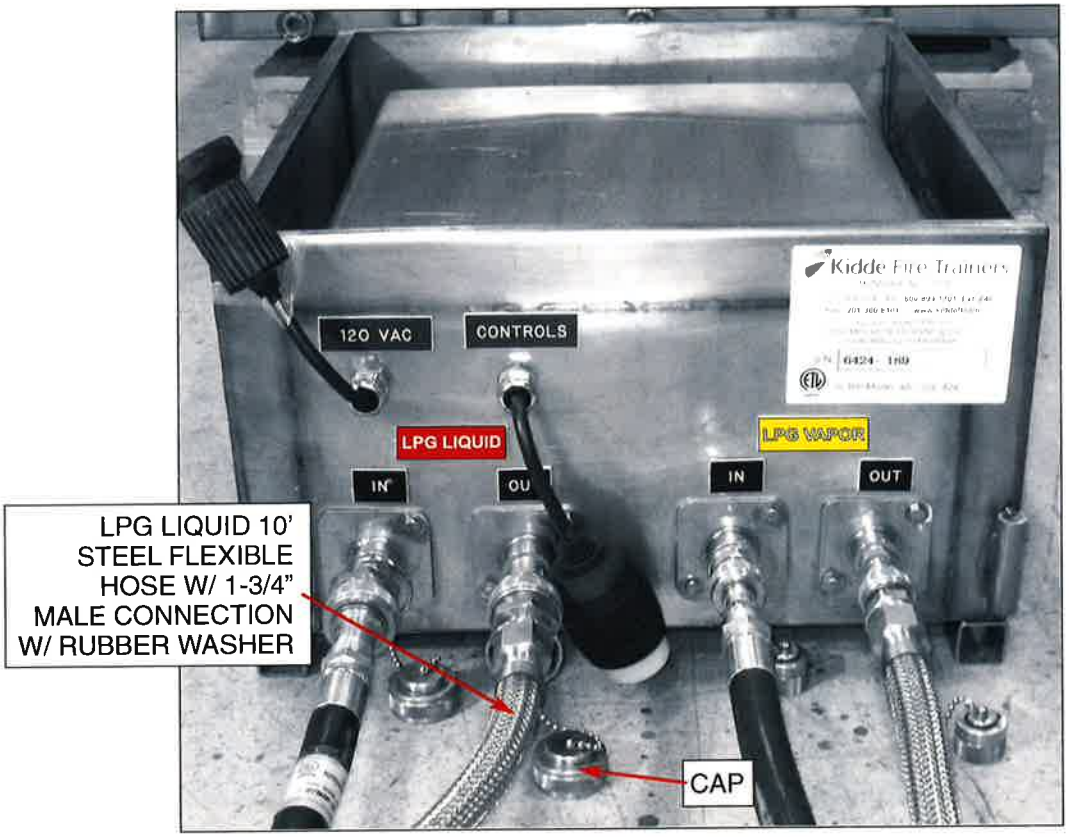

Figure 3-14. Pilot Box Assembly LPG LIQUID "OUT" Hose Connection

32. Remove protective thread cap form the 1-3/4" male connection on the remaining end of the 10' flexible stainless steel LPG Liquid supply hose. Make sure the 1-3/4" male connection is free of debris, damage and that the rubber washer is present.

⚠ WARNING

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

33. At the red LPG Liquid connection, to the left of the Pilot Box Assembly, make sure the 1-3/4" female LPG Liquid connection is free of debris and damage (see Figure 3-15).

⚠ WARNING

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE LPG LIQUID CONNECTION ON THE RED LPG LIQUID LINE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

34. Connect the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose to the 1-3/4" female LPG Liquid connection on the red LPG Liquid line. Attach fittings hand tight (see Figure 3-15).

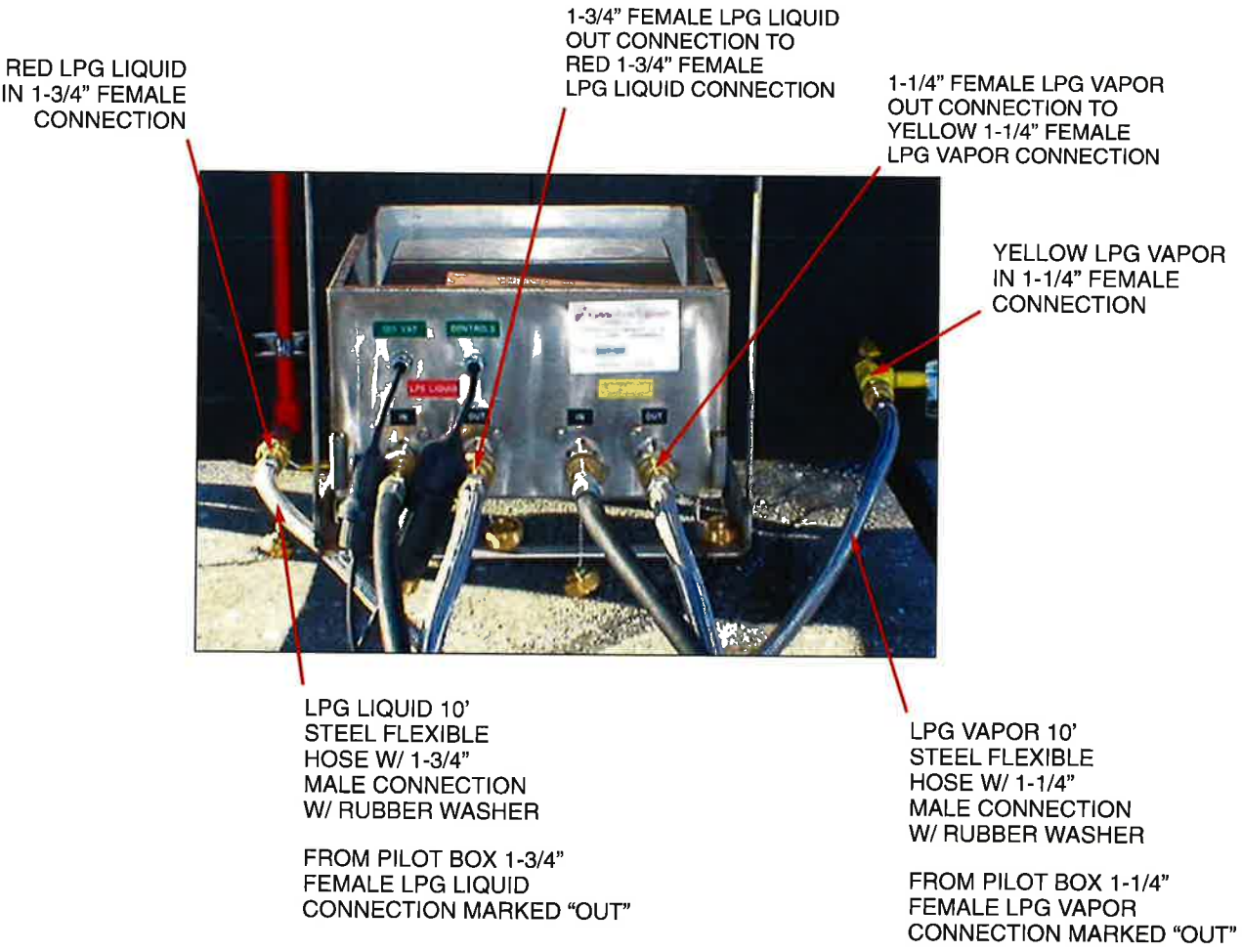

Figure 3-15. Pilot Box LPG Vapor & LPG Liquid Connections to Helo

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35. Attach the Electric Supply cord to the connection marked **"120 VAC"** on Pilot Box Assembly (see Figure 3-16). Run the Electric Supply cord straight back from the Pilot Box Assembly.

NOTE

IF USING THE WIRELESS CONTROL SYSTEM, REFER TO THE WIRELESS CONTROL SYSTEM USERS MANUAL, SECTIONS 3.1 AND 3.2. FOR WIRELESS CONTROL SYSTEM SETUP, THEN RETURN TO THIS MANUAL, STEPS 36 THROUGH 44.

36. Be sure the "PILOT" ON/OFF switch, on the Control Pendant, is in the OFF position. Attach the Control Pendant cable to the connection marked **"CONTROLS"** on Pilot Box Assembly (see Figure 3-16). Run the Control Pendant cable straight back from the Pilot Box Assembly.
37. Open the Fire Pan valves on the LPG Vapor burner element (see Figure 3-17).
38. Ensure that drain plug is installed in the front of the fire pan.
39. Fill Fire Pan with water until water overflows (see Figure 3-17).
40. Ensure that drain plug is installed in the Pilot Box Assembly.
41. Fill Pilot Box Assembly top with water until it overflows (see Figure 3-17).

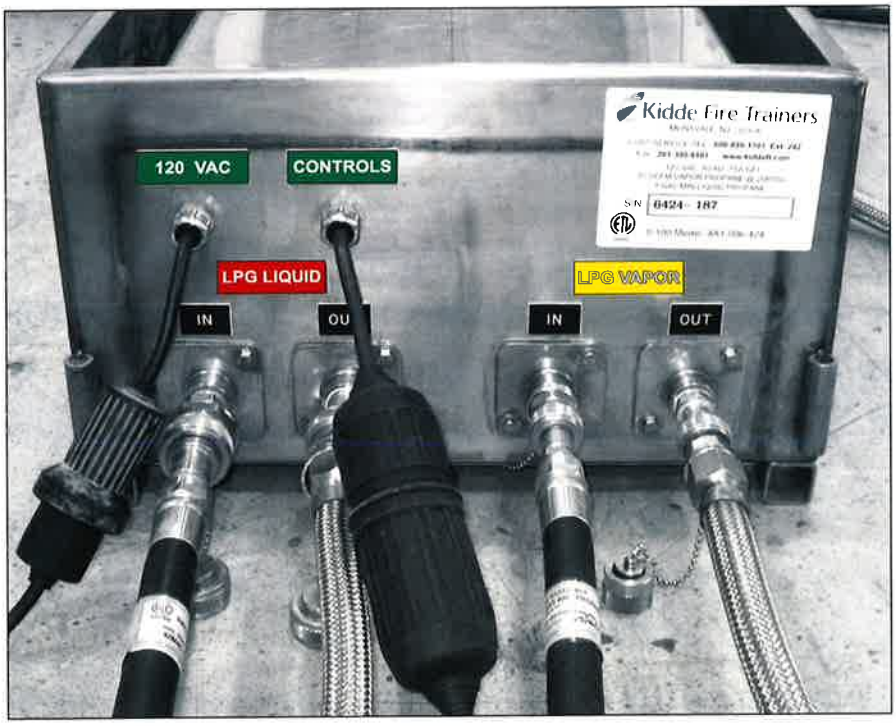

Figure 3-16. Electrical Supply & Control Pendant Connections

NOTE

IF USING THE WIRELESS CONTROL SYSTEM, REFER TO THE WIRELESS CONTROL SYSTEM USERS IF THE O-100 OUTDOOR FIRE TRAINER is to be controlled by the Kidde Multi Control System, refer to the O-100 Multi Control System Users Manual, Section 3.1.

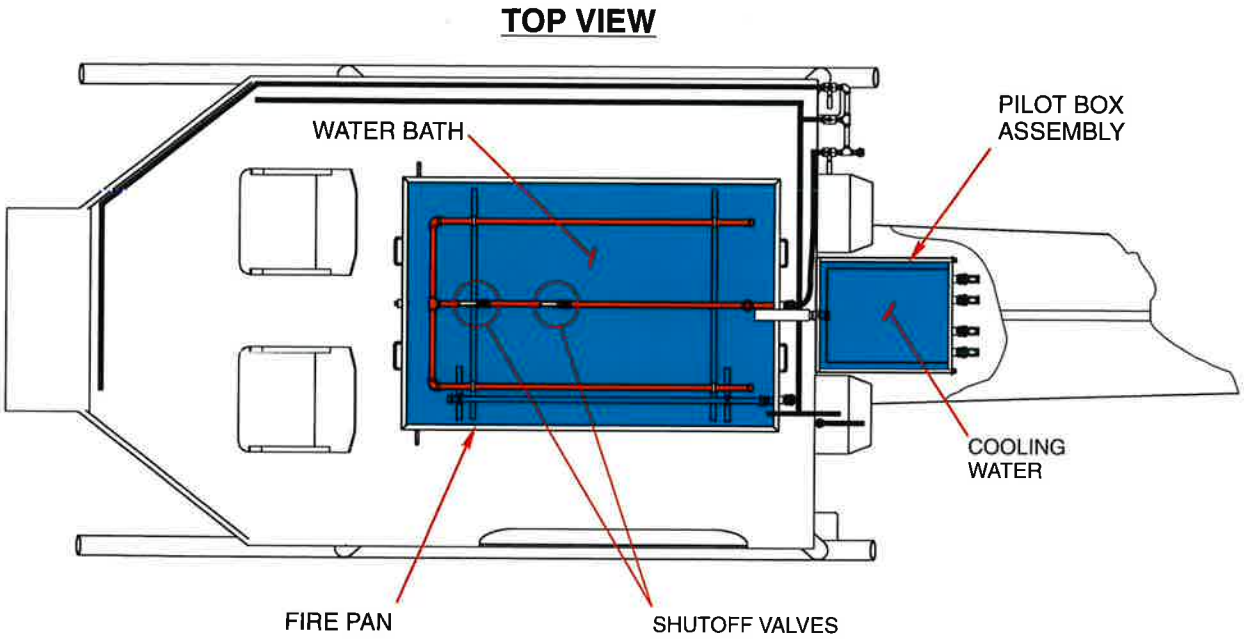

Figure 3-17. Pilot Box / Fire Pan

42. Connect the remaining end of the 100' black rubber LPG vapor supply hose from the Pilot Box Assembly **LPG VAPOR** connection marked **"IN"**, to the yellow **LPG Vapor Outlet** connection located on the exterior of the trailer propane tank and storage compartment (Figure 3-18a). Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male connection. Attach fittings hand tight.

WARNING

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER LPG VAPOR SUPPLY HOSE IS CONNECTED TO THE LPG VAPOR OUTLET ON THE EXTERIOR OF THE TRAILER PROPANE TANK AND STORAGE COMPARTMENT. FAILURE TO DO THIS WILL CAUSE DEATH OR SERIOUS INJURY.

43. Connect the remaining end of the 100' black rubber LPG liquid supply hose, from the Pilot Box Assembly **LPG LIQUID** connection marked **"IN"**, to the red **LPG LIQUID Output** connection located on the exterior of the trailer propane tank and storage compartment (Figure 3-18a). Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male adaptor connection. Attach fittings hand tight.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

⚠ WARNING

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER **LPG LIQUID** SUPPLY HOSE IS CONNECTED TO THE **LPG LIQUID** OUTLET ON THE EXTERIOR OF THE TRAILER PROPANE TANK AND STORAGE COMPARTMENT. FAILURE TO DO THIS WILL CAUSE DEATH OR SERIOUS INJURY.

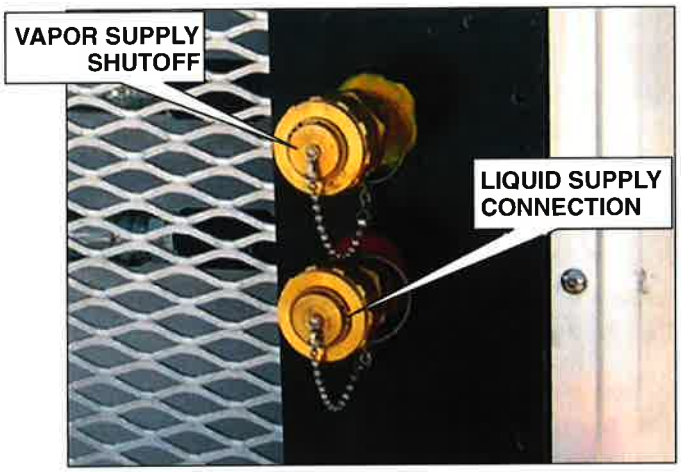

Figure 3-18a. Propane Supply Hose Connections (Exterior of Trailer Enclosure)

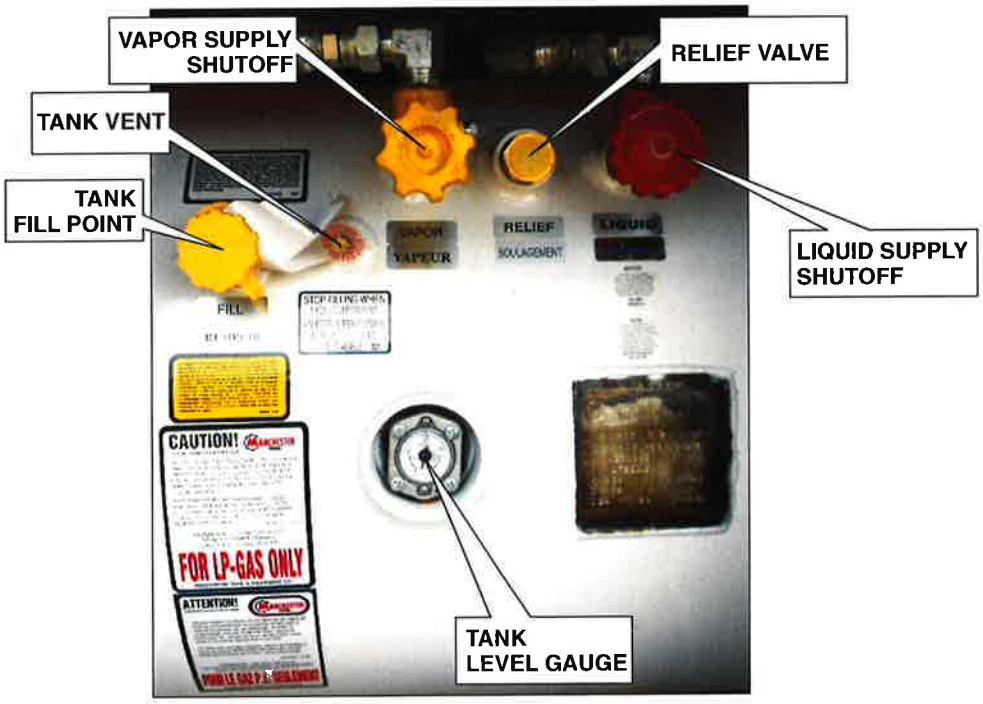

Figure 3-18b. 100 Gallon Propane Storage Tank

3.4 BEFORE STARTUP

3.4.1 PILOT BOX VALVES OPERATIONAL TEST

Prior to operating the trainer the Pilot Box Assembly Pilot, Vapor and Liquid solenoid valves must be operationally tested.

NOTE

WHEN PERFORMING THIS TEST WITH THE KIDDE WIRELESS CONTROL SYSTEM, REFER TO THE WIRELESS CONTROL SYSTEM USERS MANUAL, SECTIONS 3.1 & 3.3.

WHEN PERFORMING THIS TEST WITH THE KIDDE MULTI CONTROL SYSTEM, REFER TO THE O-100 MULTI CONTROL SYSTEM USERS MANUAL, SECTIONS 3.1, 3.2 & 3.3.

⚠ WARNING

YOU MUST READ AND UNDERSTAND THE OWNERS MANUAL THAT CAME WITH THE POWER INVERTER PV700HF, BEFORE PROCEEDING.

1. If inverter is used, connect the power cord from the Pilot Box Assembly into the inverter. Inverter is mounted to the inside of the enclosure cover, located behind the trailer lift jack. Refer to Section 2, paragraph 2.2.6 for location.
2. If shore power is used, attach the electric supply cord from the Pilot Box assembly to an outdoor 120V power GFIC receptacle.
3. Turn the "PILOT" Button on the Control Pendant to the ON position (see Figure 3-19) and listen for the solenoid valve to open.
4. Once it has been confirmed that the Pilot solenoid valve is operating, push and hold the VAPOR button on the Control Pendant (see Figure 3-19) and listen for the solenoid valve to open.
5. Once it has been confirmed that the Vapor solenoid valve is operating, while holding the VAPOR button in, push and hold the AUX button (see Figure 3-19) and listen for the solenoid valve to open.
6. Once it has been confirmed that the Liquid solenoid valve is operating, release the "VAPOR" & "AUX" buttons and turn the "PILOT" Button on the Control Pendant to the OFF position
7. Disconnect Pilot Box Assembly power cord from power source, (Inverter/shore power).
8. Once it has been confirmed that the Pilot, Vapor and Liquid solenoid valves are operational continue to 3.4.2.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

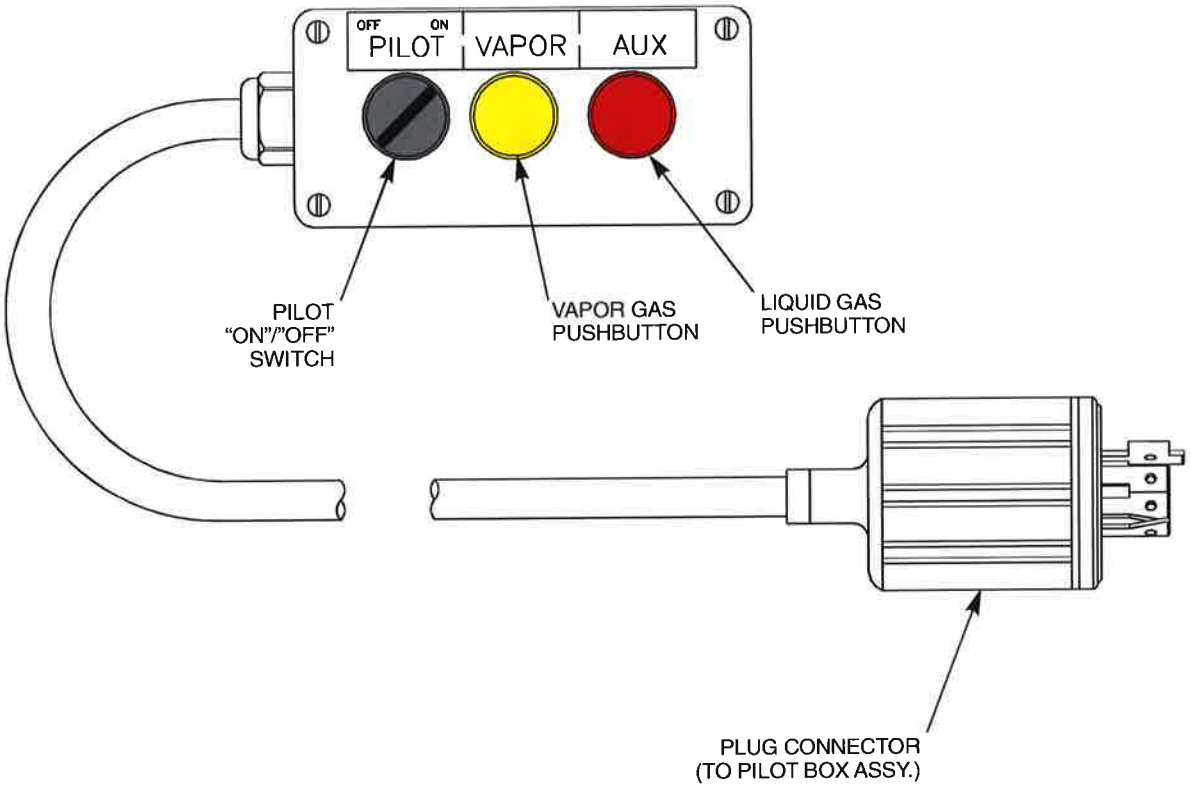

Figure 3-19. Control Pendant Assembly

3.4.2 PILOT FLAME TEST

Only after procedure 3.4.1 has been successfully completed, the Pilot Box Assembly Pilot flame must be operationally tested.

NOTE

WHEN PERFORMING THIS TEST WITH THE KIDDE WIRELESS CONTROL SYSTEM, REFER TO THE WIRELESS CONTROL SYSTEM USERS MANUAL, SECTIONS 3.1 & 3.4.

WHEN PERFORMING THIS TEST WITH THE KIDDE MULTI CONTROL SYSTEM, REFER TO THE O-100 MULTI CONTROL SYSTEM USERS MANUAL, SECTION 3.1, 3.2 & 3.4.

1. At the trailer propane compartment, open the compartment doors. At the propane supply tank, open the LPG vapor supply valve SLOWLY (see Figure 3-18b).
2. Assure there are no leaks in the black 100' LPG vapor supply hose and at all connections.

WARNING

IF A LEAK IS DETECTED TURN PROPANE SUPPLY VALVE OFF IMMEDIATELY. MAKE THE APPROPRIATE REPAIRS. DO NOT USE TRAINER UNTIL ALL LEAKS ARE ELIMINATED.

3. Connect the power cord, from the Pilot Box Assembly, to the power source, (inverter/shore power-120V power GFIC receptacle).

WARNING

WHEN USING THE INVERTER, YOU MUST READ AND UNDERSTAND THE TRIPP-LITE POWER INVERTER PV700HF OWNERS MANUAL, BEFORE PROCEEDING. THIS MANUAL IS LOCATED IN SECTION 6 OF THIS MANUAL.

4. Turn the "PILOT" Button on the Control Pendant to the ON position (see Figure 3-19). The Pilot may take a few seconds to ignite initially because it takes a while for the gas to get through the lines.
5. Once Pilot Flame has ignited Turn the "PILOT" Button on the Control Pendant to the OFF position (see Figure 3-19).
6. At the propane supply tank, close the LPG vapor supply valve SLOWLY (see Figure 3-18b).
7. Disconnect Pilot Box Assembly power cord from power source, (Inverter/shore power).
8. Once it has been confirmed that the Pilot flame is operational continue to 3.4.3.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

3.4.3 LIQUID PROPANE LEAK CHECK

NOTE

PROCEDURES 3.4.1 AND 3.4.2 MUST BE SUCCESSFULLY COMPLETED PRIOR TO PERFORMING THE LIQUID PROPANE LEAK CHECK!

1. At the trailer propane compartment, open the compartment doors. At the propane supply tank, open the LPG liquid supply valve SLOWLY (see Figure 3-18b).
2. Assure there are no leaks in the black 100' LPG liquid supply hose and at all connections.

WARNING

IF A LEAK IS DETECTED TURN PROPANE SUPPLY VALVE OFF IMMEDIATELY. MAKE THE APPROPRIATE REPAIRS. DO NOT USE TRAINER UNTIL ALL LEAKS ARE ELIMINATED.

3. At the propane supply tank, close the LPG liquid supply valve (see Figure 3-18b).

3.5 STARTUP

1. At the helicopter mockup, select the desired internal fire(s) by opening the cockpit fire LPG vapor supply valve, and or the cargo fire LPG vapor supply valve (see Figure 3-20).
2. The fire pan used for the cargo fire has a "U" shaped burner element that has two, quarter turn shut off valves in line. Use the quarter turn shut off valves to adjust the cargo fire flame size (see Figure 3-17).

WARNING

YOU MUST READ AND UNDERSTAND THE OWNERS MANUAL THAT CAME WITH THE POWER INVERTER PV700HF, BEFORE PROCEEDING.

3. Connect the power cord, from the Pilot Box Assembly, to the power source, (inverter/shore power-120V power GFIC receptacle).
4. At the propane supply tank, open the LPG vapor and LPG liquid supply valves SLOWLY (see Figure 3-18b).

3.6 OPERATION

WARNING

BE SURE NO ONE IS IN THE HELICOPTER MOCKUP AND ALL PERSONNEL ARE AT A SAFE DISTANCE BEFORE PROCEEDING.

ONLY WHEN ALL THE ABOVE OPERATIONAL AND LEAK CHECKS HAVE BEEN SUCCESSFULLY PERFORMED YOU ARE READY TO TRAIN.

1. All training scenarios should be explained fully to the students and the explanation should include a description of the vehicle mock-up burn pan and its operation.

NOTE

WHEN USING THE WIRELESS CONTROL SYSTEM, REFER TO THE WIRELESS CONTROL SYSTEM USERS MANUAL, SECTIONS 3.1 & 3.6.

WHEN PERFORMING THIS TEST WITH THE KIDDE MULTI CONTROL SYSTEM, REFER TO THE O-100 MULTI CONTROL SYSTEM USERS MANUAL, SECTION 3.1, 3.2 & 3.6.

2. Once all hookups have been made, the gas lines/connections leak checked, you are ready to train. Turn the "PILOT" Button on the Control Pendant to the ON position (see Figure 3-21). The Pilot may take a few seconds to ignite initially because it takes a while for the gas to get through the lines.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

WARNING

PRIOR TO PERFORMING THE NEXT STEP BE SURE THAT PILOT HAS IGNITED.

3. **SEE ABOVE WARNING PRIOR TO PERFORMING THIS STEP!** Once Pilot Flame has ignited, the vapor main flame for the cargo, cockpit fire and tracer elements can be ignited by pushing and holding the VAPOR button on the Control Pendant (see Figure 3-21).

WARNING

PRIOR TO PERFORMING THE NEXT STEP BE SURE THAT FLAME FROM VAPOR BURNER ELEMENT IS VISIBLE.

4. **SEE ABOVE WARNING PRIOR TO PERFORMING THIS STEP!** Once the vapor flame can be seen in the helicopter fire trainer, the AUX button (see Figure 3-21) can be pressed to ignite the liquid propane that supplies the engine fire.

NOTE

WHEN THE VAPOR/AUX. BUTTON IS RELEASED THE CORRESPONDING SOLENOID VALVE, IN THE PILOT BOX ASSY., CLOSES AND GAS FLOW TO THE BURNER AND TRACER ELEMENTS STOPS. THE FLAMES WILL CONTINUE TO BURN FOR A SHORT PERIOD DUE TO THE RESIDUAL GAS IN THE SUPPLY LINES.

Figure 3-20. Valve Locations

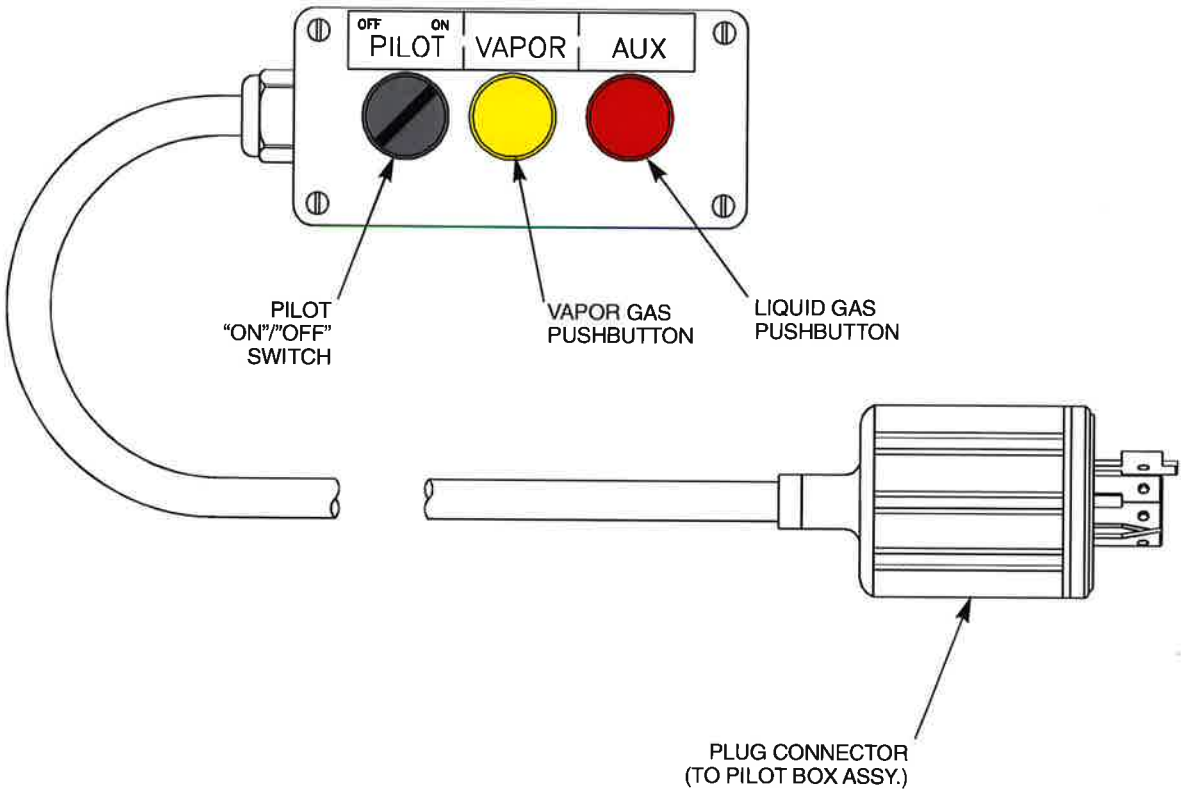

Figure 3-21. Control Pendant Assembly

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

3.7 TRAINER SHUTDOWN

WARNING

RESIDUAL LPG VAPOR AND LIQUID GAS IN THE HOSES AND PILOT BOX ASSEMBLY MUST BE BURNT OFF AT THE END OF A TRAINING SESSION AND PRIOR TO DISASSEMBLY OF THE TRAINER EQUIPMENT.

At the end of a training session there will still be Fuel in the hoses from the Propane Supply Tank to the Pilot Box Assembly. This will have to be burnt off before the equipment is taken apart. It is important that this procedure be followed even if you are not taking equipment apart but are just shutting down at the end of a training session.

NOTE

WHEN PERFORMING THIS GAS BURN OFF PROCEDURE WITH THE WIRELESS CONTROL SYSTEM, REFER TO THE WIRELESS CONTROL SYSTEM USERS MANUAL, SECTIONS 3.1 & 3.7.

WHEN PERFORMING THIS GAS BURN OFF PROCEDURE WITH THE O-100 Multi Control System, REFER TO THE O-100 Multi Control System USERS MANUAL, SECTIONS 3.1 & 3.7.

1. Close the LPG liquid valve at that Propane Storage Tank (see Figure 3-18b).
2. Start the Pilot by turning the "PILOT" switch to the ON position (see Figure 3-21). Verify that Pilot ignited.

WARNING

PRIOR TO PERFORMING THE NEXT STEP BE SURE THAT PILOT HAS IGNITED.

3. **SEE ABOVE WARNING PRIOR TO PERFORMING THIS STEP!** Once Pilot Flame ignites, push and hold "VAPOR" button on the Control Pendant (see Figure 3-21).

WARNING

PRIOR TO PERFORMING THE NEXT STEP BE SURE THAT FLAME FROM VAPOR BURNER ELEMENT IS VISIBLE.

4. **SEE ABOVE WARNING PRIOR TO PERFORMING THIS STEP!** Once the vapor flame is visible press the "AUX" button to ignite the liquid propane that supplies the liquid propane burner element for the engine fire (see Figure 3-21).
5. Leave Pilot and vapor/liquid propane flames on for about 2 minuets, at which time you should see the engine fire flame height reduce, indicating that the residual Liquid gas has burned off.
6. When all the residual Liquid propane has burned off, release "VAPOR"/"AUX" button (see Figure 3-21), turn "PILOT" switch to the OFF position (see Figure 3-21).
7. Close the LPG vapor valve at that Propane Storage Tank (see Figure 3-18b).
8. Start the Pilot by turning the "PILOT" switch to the ON position (see Figure 3-21). Verify that Pilot ignited.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

⚠ WARNING

PRIOR TO PERFORMING THE NEXT STEP BE SURE THAT PILOT HAS IGNITED.

9. **SEE ABOVE WARNING PRIOR TO PERFORMING THIS STEP!** Once Pilot Flame has ignited, push and hold the "VAPOR" button on the Control Pendant (see Figure 3-21).
10. Leave Pilot and Vapor main flame on until the residual Vapor gas is burned off and Pilot and Vapor main flames self-extinguish.
11. When ALL flames have self-extinguished, release "VAPOR" button (see Figure 3-21), turn "PILOT" switch to the OFF position (see Figure 3-10) and allow helicopter mockup and any accessories to cool down before working on them.
12. Disconnect Pilot Box Assembly power cord from power source.
13. Unhook black hose ends from Propane Supply Tank connections and Pilot Box Assembly.
14. Disconnect the 50' black hoses from each other to allow them to be placed in the trailer compartment box. Inspect hoses for damage.
15. At the Propane Supply Tank exterior connections, Install the protective thread caps onto the LPG vapor and liquid connections and close and lock door to propane compartment.
16. At the Pilot Box Assembly, Install the protective thread caps onto the **LPG VAPOR** and **LIQUID** connections marked **"IN"**.
17. Disconnect the electrical and pendant control cables from the Pilot Box Assembly and Inspect for damage.
18. Disconnect the 10" steel flex hose from the Pilot Box Assembly **LPG LIQUID** connection marked **"OUT"** and the **RED** LPG liquid connection located to the left side of the Pilot Box Assembly and Install the protective thread cap onto both ends of the 10" steel flex hose.
19. At the helicopter mockup, close the Cargo and Cockpit fire yellow LPG vapor supply valves located to the right of the Pilot Box Assembly.
20. Disconnect the 10" steel flex hose from the Pilot Box Assembly **LPG VAPOR** connection marked **"OUT"** and the **YELLOW** LPG vapor connection located to the right side of the Pilot Box assembly. Install the protective thread cap onto both ends of the 10" steel flex hose.
21. In the helicopter mockup, drain water from the fire pan by removing the drain plug. Once the pan has drained reinstall drain plug.
22. Drain water from the Pilot Box Assembly by first sliding the Pilot Box back and then remove the drain plug. Once drained, reinstall drain plug.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

CAUTION

WHEN NOT IN USE THE FIRE PAN WATER BATH AND PILOT BOX ASSEMBLY SHOULD BE DRAINED OF WATER AND DURING COLD WEATHER PERIODS, WHEN THE ANTICIPATED AMBIENT TEMPERATURE IS GOING TO BE BELOW 32°F

WHEN TRAINING DURING THE ABOVE MENTIONED CONDITION, FILL FIRE PAN WATER BATH AND TOP OF THE PILOT BOX ASSEMBLY JUST PRIOR TO OPERATION AND DRAIN IMMEDIATELY UPON THE COMPLETION OF TRAINING.

FAILURE TO FOLLOW THESE CAUTIONS MAY RESULT IN SERIOUS DAMAGE TO THE ASSOCIATED EQUIPMENT.

WARNING

TWO (2) PEOPLE ARE REQUIRED TO SAFELY REMOVE THE PILOT BOX ASSEMBLY.

23. Remove the Pilot Box assembly from the helicopter mockup support frame.
24. Wash down Pilot Box assembly and fire pan as required. ONLY USE WATER. Drain any residual water.
25. Store the Pilot Box Assembly, hoses and cables in the trailer storage compartment.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

3.8 LOADING HELICOPTER MOCKUP ONTO TRAILER

1. Align main rotor with support bracket. Raise the main rotor support bracket and secure in place with pin. Secure main rotor to support bracket with binder strap.

WARNING

THE MAIN ROTOR MUST BE SECURED IN PLACE TO THE SUPPORT BRACKET, WITH THE BINDING STRAP PRIOR TO LOADING AND TRANSPORTING THE HELICOPTER MOCKUP.

2. Install binding strap that holds the fire pan in place during transit (see Figure 3-3).
3. Position rear of trailer behind the helicopter mockup. Be sure to leave enough room to lower the two trailer ramps.
4. Prevent the trailer from rolling when the helicopter mockup is loaded onto the trailer, by blocking the tires on both sides of the trailer.

WARNING

THE TIRES ON BOTH SIDES OF THE TRAILER MUST BE BLOCKED PRIOR TO PROCEEDING TO THE NEXT STEP.

5. Disconnect the electrical brake and light connections from the tow vehicle.
6. Disconnect the safety chains from the tow vehicle
7. Disconnect the breakaway brake switch tether from the tow vehicle.
8. Unlock trailer coupler from tow vehicle hitch.

CAUTION

ALL TRAILER COMPONENTS MUST BE DISCONNECTED FROM THE TOW VEHICLE PRIOR TO PROCEEDING TO THE NEXT STEP.

WARNING

TWO (2) PEOPLE ARE REQUIRED TO SAFELY LOWER AND POSITION THE RAMPS.

Refer to Figure 3-22, sheets 1 and 2 for component location

9. While supporting the ramp disconnect the securing bar, Item (1) from the ramp by removing the securing pin and sliding the bar off the ramp connection rod. Carefully lower the ramp to the ground, Item (2).
10. Repeat step 9 for the remaining ramp.
11. Align both ramps with centerline of helicopter mock-up casters.

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CAUTION

ALL TRAILER COMPONENTS MUST BE DISCONNECTED FROM THE TOW VEHICLE PRIOR TO PROCEEDING TO THE NEXT STEP.

WARNING

YOU MUST READ AND UNDERSTAND THE EQUALIZER HYDRAULIC TRAILER JACK OPERATION MANUAL, LOCATED IN SECTION 6 OF THIS MANUAL, BEFORE PROCEEDING.

12. At the trailer jack control unit, Item (4), insert the key and turn to the right. Toggle and hold the "TRAILER UP/TRAILER DOWN switch, Item (4a) in the "TRAILER UP" position to raise the front of the trailer. Raise the trailer until trailer coupler is clear of the hitch on the tow vehicle.
13. With the trailer coupler clear of the tow vehicle hitch, move tow vehicle a safe distance from the trailer to avoid damage to vehicle during the loading of the helicopter mockup.

CAUTION

THE TOW VEHICLE MUST BE MOVED TO A SAFE DISTANCE FROM THE TRAILER, PRIOR TO PROCEEDING TO THE NEXT STEP.

14. At the trailer jack control unit, Item (4a), use the "TRAILER UP/TRAILER DOWN switch until there is a smooth transition at the trailer, ramp and the bottom of the ramp and the ground.
15. At the trailer jack control unit, Item (4a), turn off power to the trailer jack by turning the key to the left and remove key from switch.
16. Remove the two trailer supports from the storage compartment and install one on each side as shown in Figure 3-11, Item (3).
17. With the handle lower each trailer support foot until they are touching the ground.

WARNING

YOU MUST READ AND UNDERSTAND THE RAMSEY PATRIOT 9500UT WINCH OWNER'S MANUAL, LOCATED IN SECTION 6 OF THIS MANUAL, BEFORE PROCEEDING.

18. At the winch remove the protective rubber cover and connect the end of the control pendant cable to the control switch receptacle located on the winch, Item (5).
19. At the winch pendant, press the "OUT" button to pull the cable hook toward the helicopter mockup. Use the red hook strap to keep the winch cable under tension as it feeds off the reel. Secure the hook to the helicopter mockup, Item (6).

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WARNING

THE WINCH HOOK MUST BE SECURELY CONNECTED TO THE HELICOPTER MOCKUP PRIOR TO PROCEEDING TO THE NEXT STEP ITEM (6).

20. Release all four brakes on the mock-up casters and align the casters with the trailer ramps.

WARNING

BE SURE BOTH RAMPS ARE ALIGNED WITH CENTERLINE OF HELICOPTER MOCK-UP CASTERS, BEFORE PROCEEDING.

BE SURE NO ONE IS BEHIND, THE HELICOPTER MOCK-UP OR BETWEEN THE MOCK-UP AND WINCH AT ANY TIME WHEN THE MOCK-UP IS MOVED ONTO THE TRAILER.

21. With the winch pendant, Item (5), press the winch "IN" button to allow the helicopter trainer to be slowly pulled onto the trailer. Stop the winch when all four casters are centered on the black plates located on the trailer guides.

22. Engage the brakes on all four of the mock-up casters.

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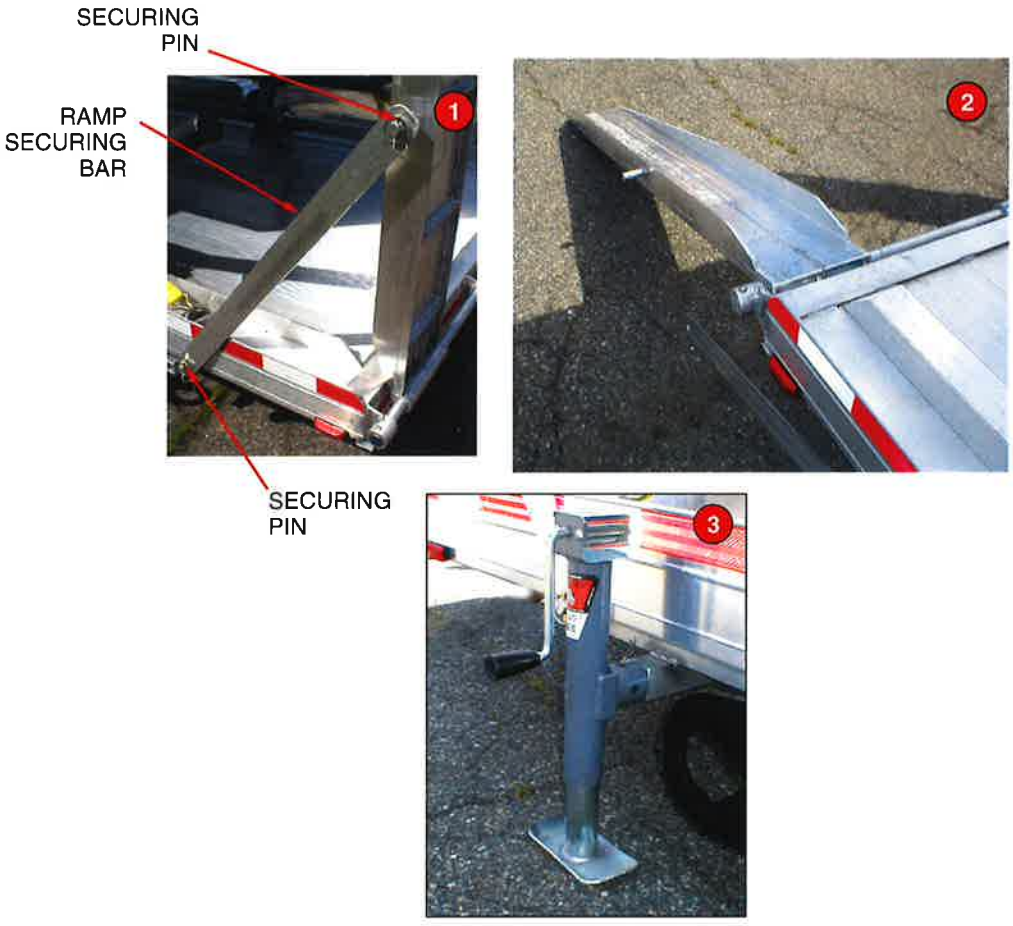

Figure 3-22. Loading Helicopter Mockup onto Trailer (Sheet 1 of 2)

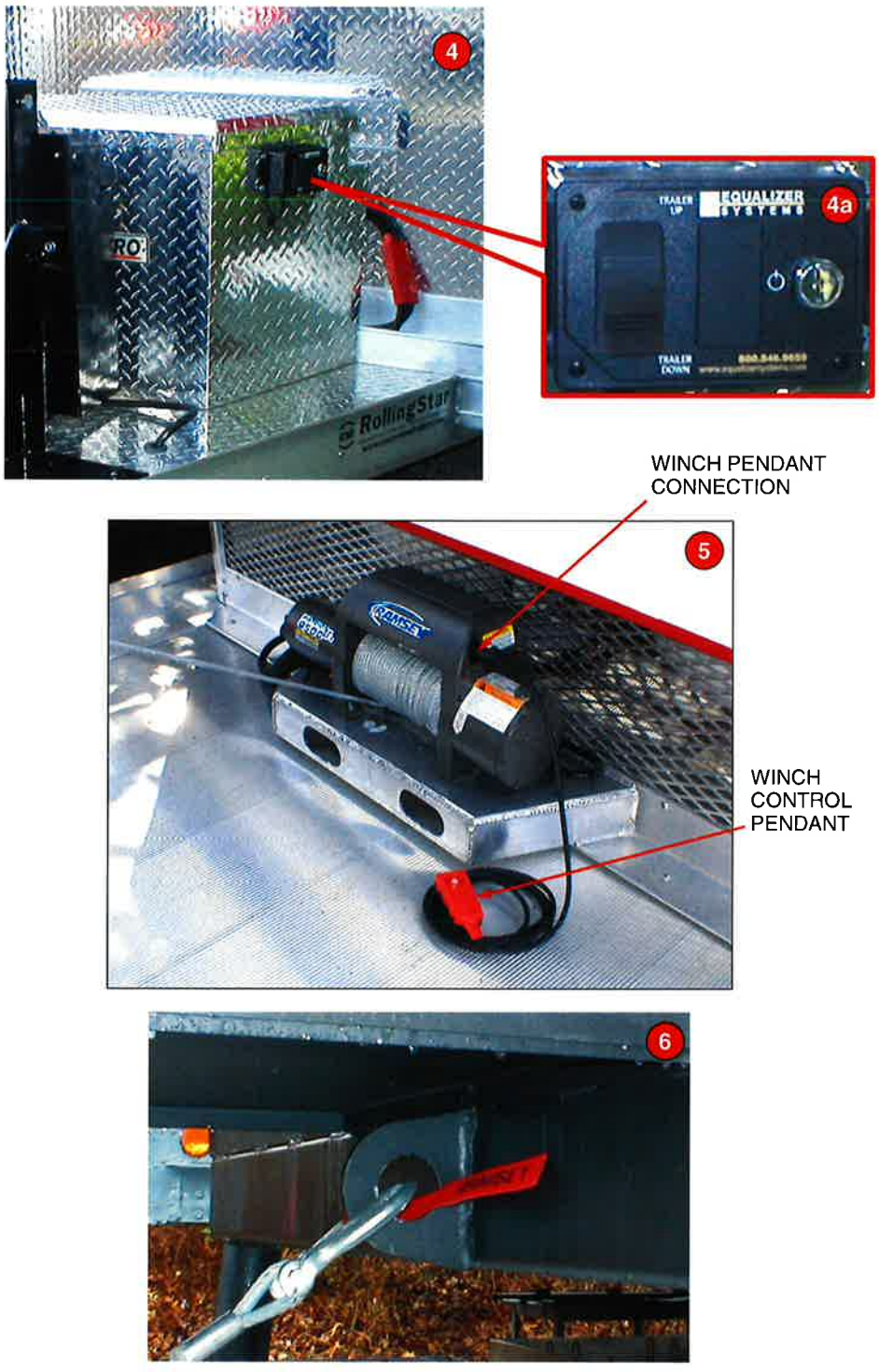

Figure 3-22. Loading Helicopter Mockup Onto Trailer (Sheet 2)

23. Remove both of the trailer supports, Figure 3-22, Item (3) and stow in trailer storage compartment.
24. Connect the trailer coupler to the tow vehicle hitch.
25. Connect the safety chains to the tow vehicle
26. Connect the electrical brake and light connections to the tow vehicle.
27. Connect the breakaway brake switch tether to the tow vehicle.

3.9 SECURING HELICOPTER MOCKUP TO TRAILER

Refer to Figure 3-12 sheets 1 and 2 for component location

There are five (4) binding straps that secure the helicopter mockup to the trailer during transit.

- Two (2) Rear Binding Straps (Figure 3-23, Items 1, 1a & 1b)
- Two (2) Front Binding Straps (Figure 3-23, Items 2, 2a & 2b)

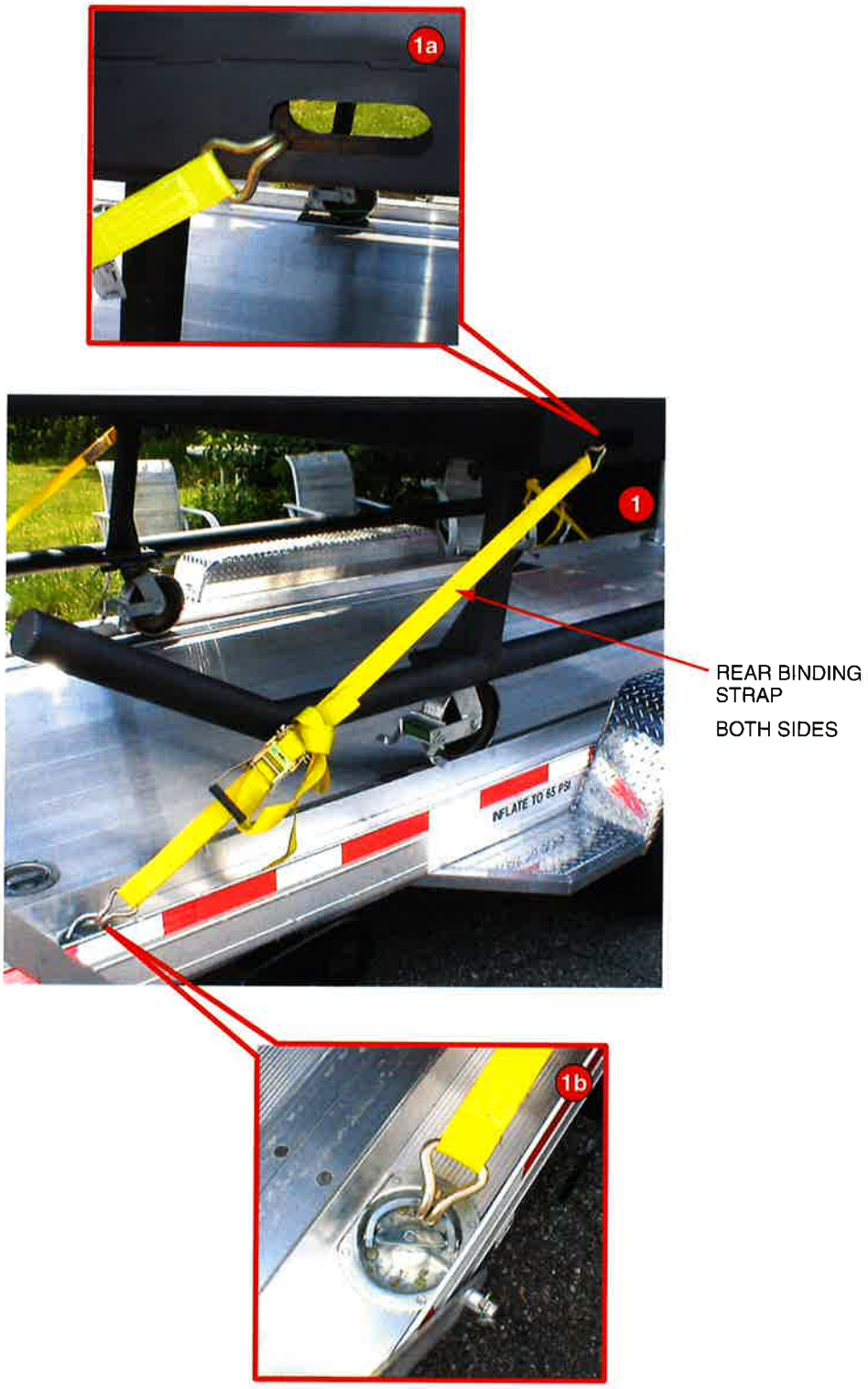

Figure 3-23. Secure Helicopter Mockup To Trailer (Sheet 1 of 2)

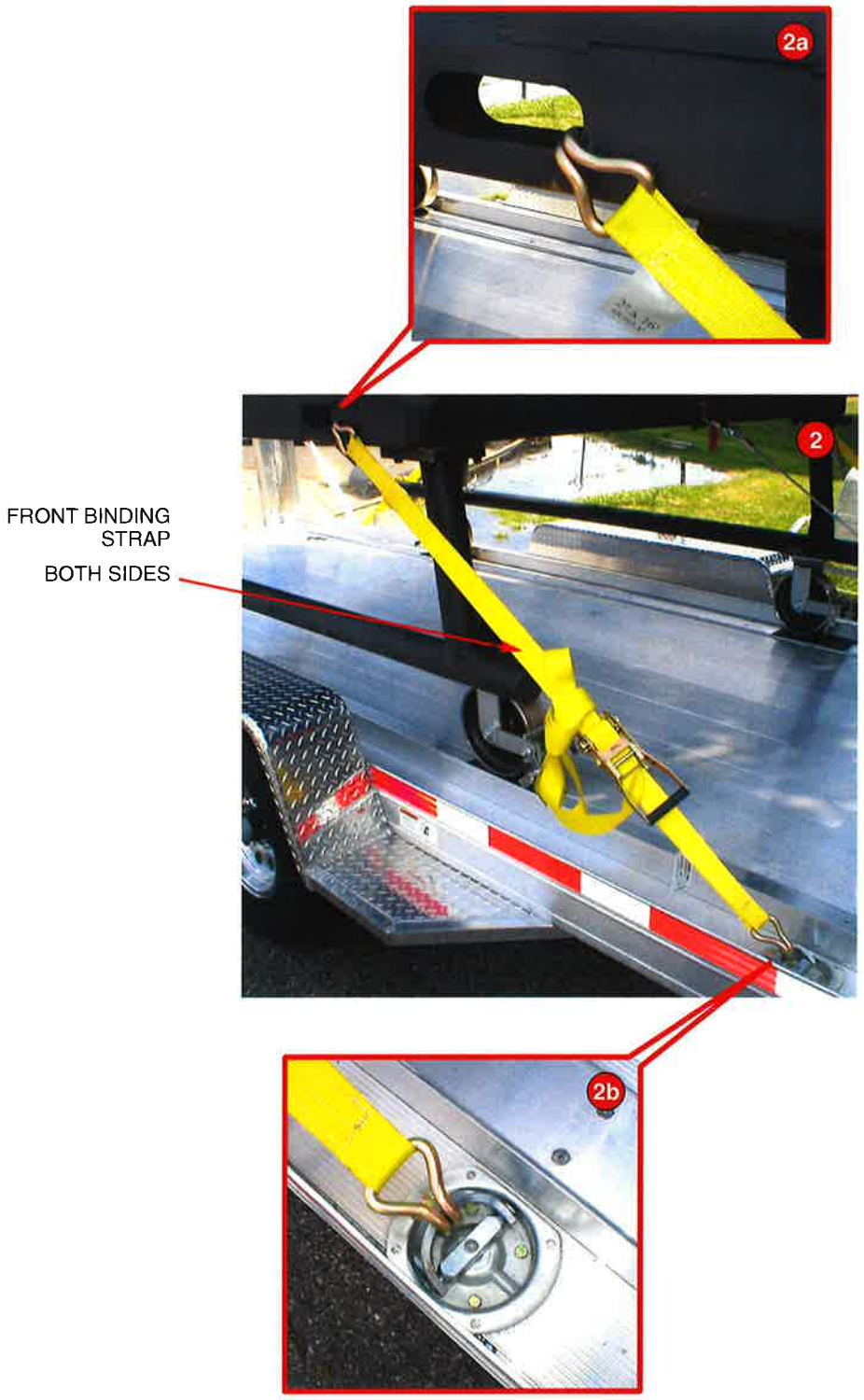

Figure 3-23. Secure Helicopter Mockup To Trailer (Sheet 2)

1. To protect the winch cable from damage during transit tension needs to be removed. With the winch pendant, carefully remove tension from the cable by pressing the "OUT" button.
2. At the winch disconnect the control pendant from the winch connection. Install the protective rubber cover over the connection and store control pendant.

WARNING

TWO (2) PEOPLE ARE REQUIRED TO SAFELY RAISE THE RAMPS.

3. Raise the trailer ramp. Attach securing bar to ramp connection rod and secure bar to ramp with pin.
4. Repeat step 3 for remaining ramp.

NOTE

IF O-100 FIRE PAN(S) ARE TO BE TRANSPORTED PLACE BEHIND THE MOCKUP AND SECURE TO TRAILER WITH BINDING STRAP AS SHOWN IN FIGURE 3-24.

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Figure 3-24. Secure Fire Pan Assemblies To Trailer

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4.0 CARE AND MAINTENANCE

4.1 GENERAL

This section provides Information to keep the trainer functional and operationally effective.

4.2 CARE AND MAINTENANCE

4.2.1 FireTrainer® O-100

The Pilot Box assembly and Burn Pan will be a very functional and operationally effective unit if cleaned and maintained properly.

Replace caps on all Acme fittings and junction box connections before cleaning or storage. Clean surfaces with water only. *Never* use flammable liquids to clean surfaces.

WARNING

CLEAN SURFACES WITH WATER ONLY. NEVER USE FLAMMABLE LIQUIDS TO CLEAN SURFACES. FAILURE TO FOLLOW THESE PROCEDURES MAY RESULT IN SERIOUS INJURY OR DEATH.

4.3.1 PERIPHERAL EQUIPMENT

- Trailer
- Winch
- Solar Charger
- Trailer Lift Jack

NOTE

FOR THE ABOVE EQUIPMENT REFER TO SECTION 6.

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5.0 SPARE PARTS

5.1 General

This section provides a figure and parts list table for each of the **Helicopter FIRETRAINER® O-100** major assemblies.

5.2 Pilot Box Assembly

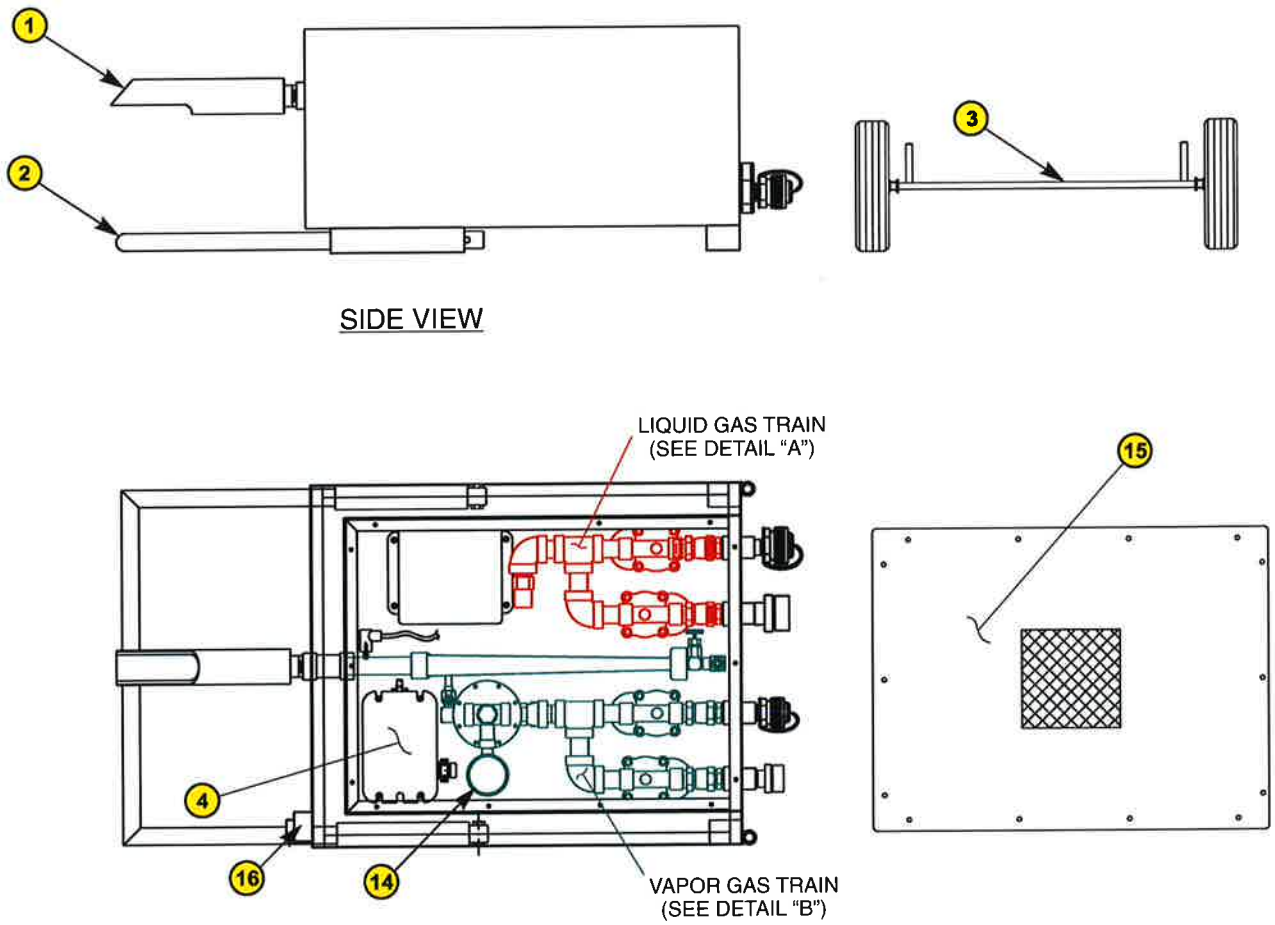

Figure 5–1 Pilot Box Assembly Details (Sheet 1 of 2)

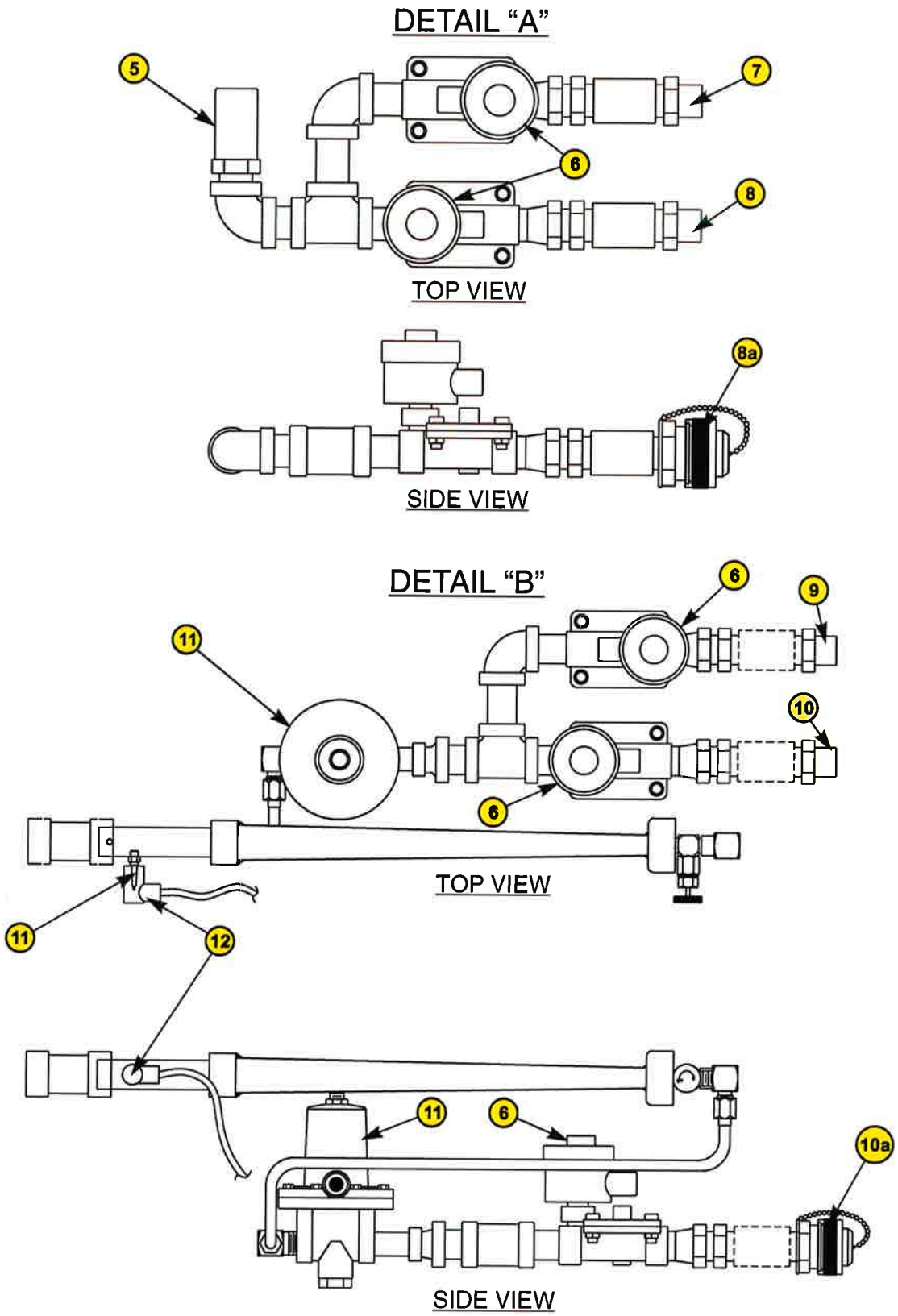

Figure 5–1 Pilot Box Assembly Details (Sheet 2)

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Table 5-1. Pilot Box Assembly Parts Lists			
Figure 5-1 Item No.	**Description**	**Part Number**	**Cage Code**
REF	**Pilot Box Assembly**	**4A1-006-424**	**06003**
1	Pilot Shroud	3D1-003-288	06003
2	Handle	4D1-006-902-002	06003
3	Wheel	ASP102B-1/2	64598
4	Transformer, Igniter, 6000 Volts	A06-SA6	18284
5	Relief Valve, Hydrostatic	SS8022G	2Y894
6	Valve, Solenoid, ¾" NPT, 120VAC	S211GF02K4EG5	0BCX7
7	Coupling, BRS, 1-3/4"F. ACME X ¾"MNPT	ME111	3S251
8	Adaptor, BRS, 1-3/4"M. ACME X ¾"MNPT	ME215	3S251
8a	Cap,Brs, ACME, 1-3/4"F ACME W/ Chain	ME229-1	3S251
9	Coupling, BRS, 1-1/4"F. ACME X 1/2"MNPT	ME101	3S251
10	Adaptor, 1-1/4"M. ACME X ¾"MNPT	ME498-6-3	3S251
10a	Cap,Brs, ACME, W/ split ring & Chain	ME220FP, 92730A12, 90177A223	N/A
11	Regulator, 5-35PSI	64SR/23	3S251
12	Boot Plug, RT Angle	727501	16428
13	Spark Plug	UY6	63477
14	Gauge, Liq Fill, 0-30psi, ¼"NPT, bot	3795K131	39428
15	Bottom Plate, Cover	4D1-006-902-003	06003
16	Plug, Pipe, BRS, 1-1/4" NPT	4429K317	39428
-	LPG Liquid 50ft LG RBR Hose Assy	2A1-004-743-002	06003
-	LPG Vapor 50ft LG RBR Hose Assy	2A1-004-743-001	06003
-	LPG Liquid 10ft LG SS Hose Assy	2A1-004-744-002	06003
-	LPG Vapor 10ft LG SS Hose Assy	2A1-004-744-001	06003

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

5.3 Fire Pan Assembly (O-100)

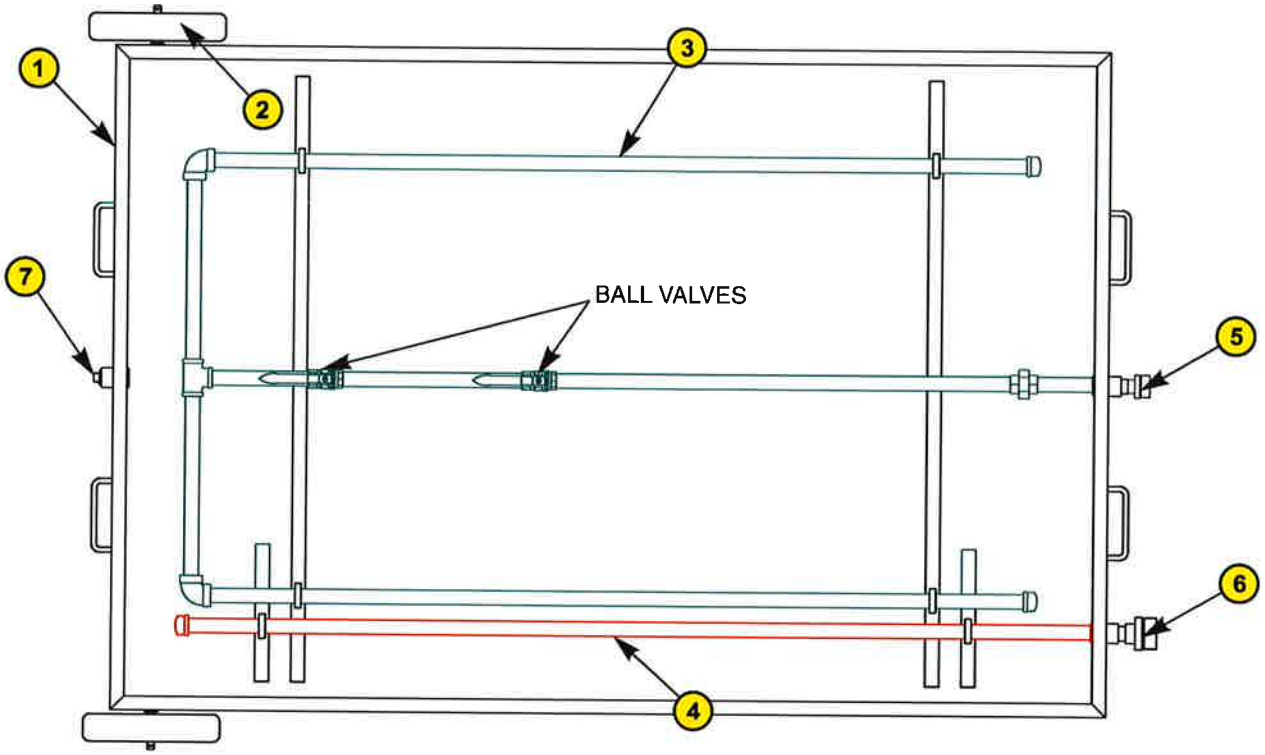

Figure 5-2. Fire Pan Assembly (O-100)

Table 5-2. Fire Pan Assembly (O-100) Parts Lists

Figure 5-2 Item No.	Description	Part Number	Cage Code
REF	**Fire Pan Assembly**	**4A1-006-501**	**06003**
1	802 P4 Pan	802-310	06003
2	Wheel	ASP102B-1/2	64598
-	Hair Pin, Cotter Pin, SS	92375A325	39428
-	Washer, Flat, ½"	MS15795-855	96906
3	Pan Burner Element, Vapor	4A1-006-505-002	06003
4	Pan Burner Element, Liquid	4A1-006-505-001	06003
5	Coupling, BRS, 1-1/4"F. ACME X 1/2"MNPT	ME101	3S251
6	Coupling, BRS, 1-3/4"F. ACME X ¾"MNPT	ME111	3S251
7	Plug,Pipe,SQ-HD,3000#,SS,3/4"NPT	4464K325	39428

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

5.4 Propane Tank Hose Adaptor Kit

(For non Kidde Trailer Propane Supply Tank)

Table 5-3. Propane Tank Hose Adaptor Kit Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
REF	**Propane Tank Hose Adaptor Kit**	**1R1-002-088**	**06003**
1	Adaptor, BRS, 1-3/4"M ACME x ¼" FNPT	ME210	3S251
2	Adaptor, BRS, Soft Nose, MPOL x ¼" MNPT	ME1625	3S251

5.5 Hose Line Extension

Table 5-4. Hose Line Extension Parts Lists			
Item No.	**Description**	**Part Number**	**Cage Code**
REF	**Propane Tank Hose Adaptor Kit**	**1R1-002-111-001**	**06003**
1	Adaptor, BRS, 1-3/4"M ACME x1-3/4"	ME273	3S251
2	Hose Assy,RBR,3/4"FNPT/FNPT x 50' LG	S13600-12	N/A
3	Hose end Coupling,BRS,1-3/4"F ACME x ¾" MNPT	ME111	3S251

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

5.6 Control Pendant

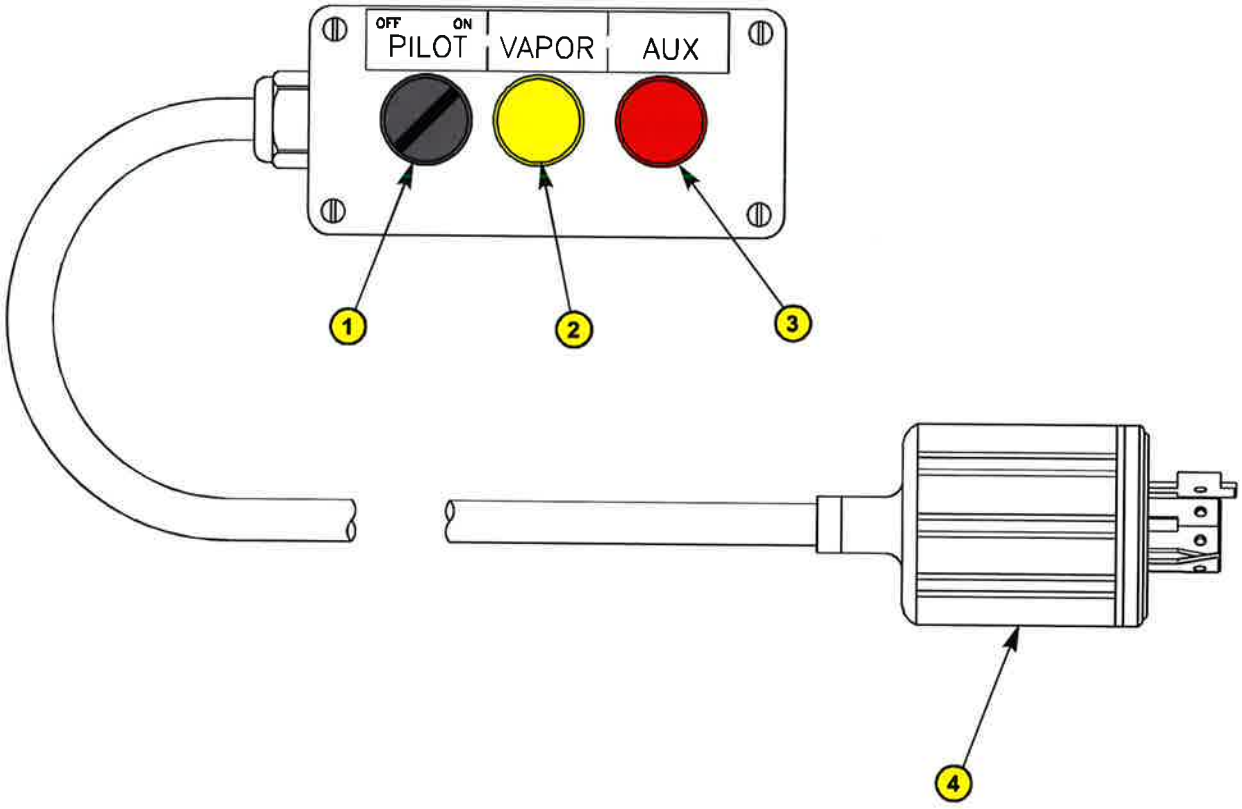

Figure 5-3 – Control Pendant Assembly

Table 5-5. Control Pendant Assembly Parts Lists

Figure 5-3 Item No.	Description	Part Number	Cage Code
REF	**Control Pendant, O-100-Pan**	**3A1-003-285**	**06003**
1	Switch, Sel, 2-MNTD, STD	ZB5AD2	81487
2	Button, Ops, Mom, Non-Illum, Flush, Yel	ZB5AA5	81487
3	Button, Ops, Mom, Non-Illum, Flush, Red	ZB5AA4	81487
-	Contact Block	ZENL1111	81487
4	Plug, 20A, 4P&G	HBL2511	74545
-	Boot, Plug	HBL6036	74545

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5.7 Peripheral Equipment

- RollingStar Trailer
- Ramses Patriot 9500 Winch
- Tripp Lite PV700HF Power Inverter
- Equalizer Lift Jack (Trailer)
- Sunforce Solar Battery Charging Kit

NOTE

REFER TO THE APPROPRIATE MANUFACTURERS PARTS LIST INFORMATION IN SECTION 6 FOR THE ABOVE EQUIPMENT.

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Section 6.0 Drawings & Reference Material

6.1 INTRODUCTION

This section contains reference material that supports the **Helicopter FIRETRAINER® O-100**.

Refer to Table 6-1 for a list of drawings that have been supplied in this section.

Refer to Table 6-2 for a list of the reference materials that have been supplied in this section.

Table 6-1. Drawings

Drawing Number	Title
WD4A1-006-460 (Rev A)	Pilot Box Assy. Electrical Junction Box Wiring Diagram

Table 6-2. Reference Materials

Item	Title
RollingStar	• Trailer Manual
Ramses Patriot 9500 Winch	• Patriot 9500 Owner's Manual • Wireless Remote Operating Instructions
Tripp Lite PV700HF Power Inverter	• Owner's Manual
Equalizer Lift Jack (Trailer)	• Installation & Operation Guide
Sunforce 15W Solar 12 V Battery Charger Kit	• User's Manual

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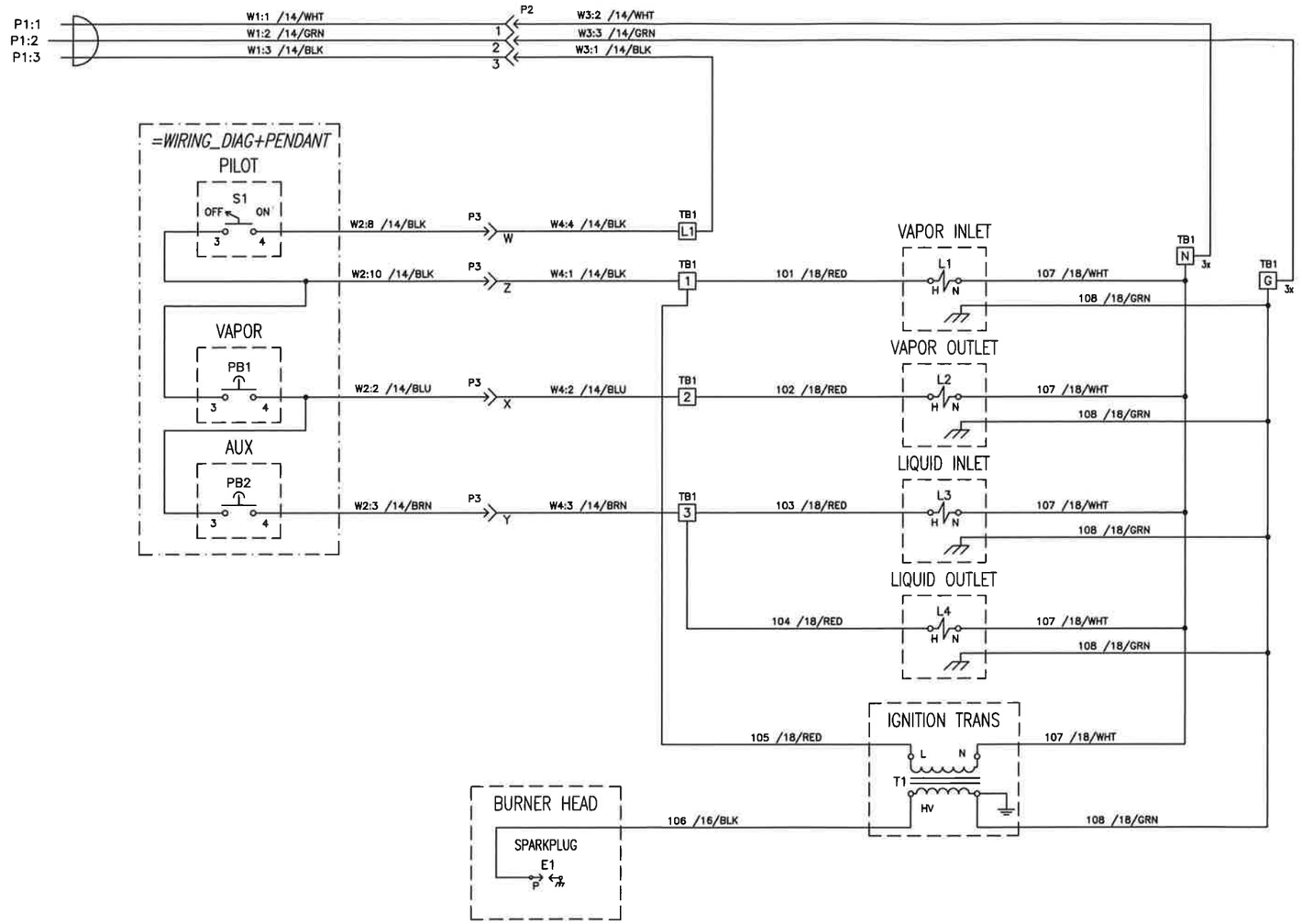
DWG NO WD4A1-006-460 SH 1

REVISIONS			
REV	DRAWING CHANGE RECORD NO	DATE	APVD
A	ECO 30444	6/23/05	BA/DB



PROPRIETARY INFORMATION OF Kidde Fire Trainers
The information contained herein is proprietary to Kidde Fire Trainers, it is not to be copied, disclosed, or disseminated without written permission of Kidde Fire Trainers and then only to the extent authorized.

PROJECT	ICS_0-100	INSTALLATION	WIRING_DIAG	Kidde Fire Trainers	
		LOCATION/S		TITLE	ELECTRICAL JUNCTION BOX 0-100 WIRING DIAGRAM ELECTRICAL ENCL ASSY
4A1-006-460	0-100	DWN	S HUYNH	08/06/04	SIZE D · CAGE CODE 06003 · DWG NO WD4A1-006-460
NEXT ASSY	USED ON	CHKD	S HUYNH	08/20/04	
APPLICATION		APVD	D BRADY	08/24/04	SCALE NA · CAD FILE: WD4A1460A · SHEET 1 OF 1

RMi RollingStar
MANUFACTURING INCORPORATED

OPERATIONS MANUAL

Kidde Fire 24' Custom Flat-Deck Trailer

Table of Contents

Operational Procedure

1. Towing Procedures
2.Propane Tank
3.Winch Operation
4. 7Amp Solar Charge Controller
5.Hydraulic Trailer Jack

OPERATIONAL PROCEDURES

1.) TOWING PROCEDURES

1. Back up to the trailer
2. Hook up the safety chains
3. Set the trailer on the truck ball
4. Lower the trailer on the ball
5. Raise the front jack
6. Make sure all external jacks are stored inside
7. Hook up the breakaway switch cable to the tow vehicle
8. Make sure ramps are properly secured with pins.

2.) PROPANE TANK

This trailer is equipped with two propane tanks from Sleeger's Engineering 24"x60" with a 1 ¼" vapor port.

3.) WINCH

This unit is equipped with a Ramsey Front Mount Electric Winch Patriot 9500UT. This winch features a highly efficient 3 stage planetary gear set which transmits torque from a series wound DC motor. A semi-automatic clutch allows free spooling for quick cable deployment and eliminates unnecessary trips back and forth to the winch. A wireless remote allows the winch to be operated up to 50 feet away. An automatic load holding brake is designed to hold the full rated capacity of the winch.

Please see Owner's Manual for further instructions.

4.) 7Amp Solar Charge Controller

The green "charged" light indicates a full-charged battery. When the battery reaches 14.2 V, the Solar Charge Controller will cut out voltage thereby ensuring no overcharging. The yellow "charging" light indicates a battery that is being charged. When the battery reaches 13 V, the Battery Charge Controller will cut-in and begin charging.

Please see the User's Manual for further instructions.

5.) Hydraulic Trailer Jack

This unit is equipped with an Equalizer Systems Hydraulic Single Leg Trailer Jack.

To raise and lower the jack, simply push and hold the rocker switch in the up or down position. The jack will "hold" a position by releasing the switch at any time. There is no need to take the "weight" off the jack when storing the trailer, as the positive check valves in the system will not allow the jack to "bleed down."

WARNING!!

With ANY hydraulic application, holding any position on a cylinder must be done with safety in mind. Failure in the system may cause the leg to retract or extend on its own.

When working under or near the trailer, always use jack stands of appropriate rating to support the weight of the trailer.

The keyed switch MUST be in the off position and the key must be removed when the jack is not in use and/or when the trailer is in transit. If the unit is NOT supplied with an eyed switch, the +12V battery lead must be fed through a power disconnect switch to fully isolate the system during inactivity or travel. The minimum switch rating must be 60 ampere DC.

Contact Information:

RMi RollingStar
MANUFACTURING INCORPORATED

PO Box 471

125 Liberty Lane

Barneveld, NY 13304

(315) 896-4767

Rstar143@aol.com

www.customizedtrailers.com

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ROLLINGSTAR MANUFACTURING, INC

125 LIBERTY LANE
P.O. BOX 471
BARNEVELD, NEW YORK 13304

LIMITED WARRANTY

Rollingstar Manufacturing, Inc. strives to provide our customers with the highest quality product available. Rollingstar Manufacturing, Inc. here warrants that each Rollingstar Trailer operated by the original purchaser under normal use in the Continental United States or Canada will be free from defects in materials and workmanship for one year following the original purchase with the exceptions noted below.

Electrical components are covered for a period of one (1) year with the following exclusions:

- Accident, abuse, or neglect
- Alteration, additions, or modifications in any form.
- Battery has a two year warranty, except in cases of abuse or neglect

Five Year Limited Warranty

The frame of your Rollingstar trailer is warranted to the original purchaser for a period of five (5) years. The steel or aluminum frame shall be free of defects and failure under normal use conditions. The following limitations apply:

- Failure due to accident, abuse, or overloading
- Paint on steel painted frames
- Alteration or modification in any way without prior written approval from Rollingstar

NOTE REGARDING TOWING VEHICLES: **The payload carrying capacity of any trailer/towing vehicle combination is determined by the GVWR of the trailer, the curb weight of the trailer, the GVWR of the town vehicle and the curb weight of the tow vehicle. It is important that you determine that the trailer and the tow vehicle capacity are suitable to your application.**

NOTE REGARDING TOW VEHICLES WITH EXTRA STIFF SUSPENSIONS: **Tow vehicles equipped with extra-stiff suspensions (including, but not limited to, Freightliner, International, Ford 450-550 Series) can place excessive or abnormal shock loads on trailer frames, resulting in broken welds and flexing sidewalls. This is especially true for lighter weight trailers. Rollingstar Manufacturing highly recommends the use of "Air-Ride" type hitch units on tow vehicles with exceptionally stiff suspensions. Rollingstar Manufacturing reserves the right to void its warranty if the trailer is towed using vehicles with heavy duty suspensions.**

NOTE REGARDING USE OF LOAD LEVELING PRODUCTS:
Use of load distribution hitches on tow vehicles of more than 1 ton capacity can cause damage to trailer frame, truck frame, or hitch voiding the warranty.

EXCLUSION OF COMPONENTS WARRANTED BY OTHER MANUFACTURERS

The following components, purchased and installed by Rollingstar Manufacturing, Inc. are warranted by their manufacturers and are excluded from this limited warranty.

- SIDE WALLS: Crane Cortec offers a five (5) year limited warranty on their FRP Panels (side walls) as stated in their enclosed warranty.
- Consumer is responsible to wash, check, and recaulk sidewalls as needed and to wax twice a year with a quality UV Protectant wax.
- GENERATOR: Onan Gensets are covered with a two (2) year limited warranty in compliance with original manufacturer's warranty. (Refer to Onan's owners' manual). Honda carries a one (1) year limited warranty in compliance with original manufacturer warranty (Refer to Honda's owners' manual).
- AIR CONDITIONER: Coleman air conditioners carry a one (1) year limited warranty in compliance with original manufacturer's warranty (refer to owners' manual).
- AXLES: Dexter Torflex axles come with a five (5) year limited warranty from date of purchase by original owner, exclusive of hubs, drums, brakes, bearings, and seals. Dexter warrants to the original purchaser that the hubs, drums, brakes, bearings, and seals that are part of the torflex axle, will be free from defects for a period one (1) year in compliance with original manufacturer's warranty. (enclosed in this packet)

NORMAL USE, NO REPAIRS OR ALTERATIONS

This limited warranty covers only defects from original components which arise from normal use and does not apply if trailer has been subjected to negligence, accident, abuse, misuse, neglect, or overload or has been repaired or altered without the prior written consent of Rollingstar Manufacturing. Normal use items will not be replaced due to wear.

PRIOR WRITTEN CONSENT REQUIRED

No reimbursement will be made to any dealer or owner for repairs made without the prior written consent of Rollingstar Manufacturing.

LIMITATIONS

The sole responsibility of Rollingstar Manufacturing, Inc. under this limited warranty shall be to repair and replace parts at the Rollingstar factory or at a more convenient place agreed to by both the purchaser and Rollingstar. All other obligations or liabilities, including incidental or consequential damages or contingent liabilities arising out of failure of any parts to operate properly, are hereby excluded, including but not limited to any damages resulting from loss of use, inconvenience, loss of time, commercial loss or any other type of damages, general or specific, foreseen or unforeseen, unless applicable state law provides otherwise.

All defects and other warranty issues must be reported within five (5) days of discovering the problem.

In the even a defect or failure does occur, please immediately call Rollingstar at

(315) 896-4767

The forgoing warranties are in lieu of all other warranties, expressed, or implied. This warranty gives the customer specific legal rights. You may also have other rights that may vary from state to state.

Rollingstar Manufacturing, Inc.
P.O. Box 471
125 Liberty Lane
Barneveld, NY 13304

Reporting Safety Defects:

If you believe that your vehicle has a defect which could cause a crash or could cause injury or death, you should immediately inform the National Highway Traffic Safety Administration (NHTSA) in addition to notifying RollingStar Manufacturing, Inc.

If NHTSA receives similar complaints, it may open an investigation, and if it finds that a safety defect exists in a group of vehicles, it may order a recall and remedy campaign. However, NHTSA cannot become involved in individual problems between you, your dealer, or RollingStar Manufacturing, Inc.

To contact NHTSA, you may either call the Auto Safety Hotline toll-free at 1-800-424-9393 (or 366-0123 in Washington, DC area) or write to:

ADMINISTRATOR
NHTSA
1200 New Jersey Avenue S.E.
Washington, DC 20590

You can also obtain other information about motor vehicle safety from the Hotline or go to http://www.safercar.gov.

1. TIRE SAFETY INFORMATION

This portion of the User's Manual contains tire safety information as required by 49 CFR 575.6.

Section 2.1 contains "Steps for Determining Correct Load Limit - Trailer".

Section 2.2 contains "Steps for Determining Correct Load Limit – Tow Vehicle".

Section 2.3 contains a Glossary of Tire Terminology, including "cold inflation pressure", "maximum inflation pressure", "recommended inflation pressure", and other non-technical terms.

Section 2.4 contains information from the NHTSA brochure entitled "Tire Safety – Everything Rides On It". This brochure This brochure, as well as the preceding subsections, describes the following items;

- Tire labeling, including a description and explanation of each marking on the tires, and information about the DOT Tire Identification Number (TIN).
- Recommended tire inflation pressure, including a description and explanation of:
  A. Cold inflation pressure.
  B. Vehicle Placard and location on the vehicle.
  C. Adverse safety consequences of under inflation (including tire failure).
  D. Measuring and adjusting air pressure for proper inflation.
- Tire Care, including maintenance and safety practices.
- Vehicle load limits, including a description and explanation of the following items:
  A. Locating and understanding the load limit information, total load capacity, and cargo capacity.
  B. Calculating total and cargo capacities with varying seating configurations including quantitative examples showing / illustrating how the vehicles cargo and luggage capacity decreases as combined number and size of occupants' increases. This item is also discussed in Section 3.
  C. Determining compatibility of tire and vehicle load capabilities.
  D. Adverse safety consequences of overloading on handling and stopping on tires.

1.1. STEPS FOR DETERMINING CORRECT LOAD LIMIT – TRAILER

Determining the load limits of a trailer includes more than understanding the load limits of the tires alone. On all trailers there is a Federal certification/VIN label that is located on the forward half of the left (road) side of the unit. This certification/VIN label will indicate the trailer's Gross Vehicle Weight Rating (GVWR). This is the most weight the fully loaded trailer can weigh. It will also provide the Gross Axle Weight Rating (GAWR). This is the most a particular axle can weigh. If there are multiple axles, the GAWR of each axle will be provided.

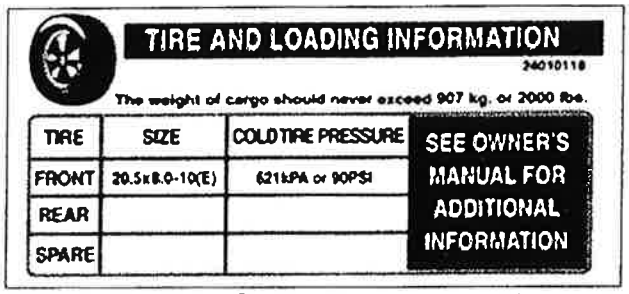
If your trailer has a GVWR of 10,000 pounds or less, there is a vehicle placard located in the same location as the certification label described above. This placard provides tire and loading information. In addition, this placard will show a statement regarding maximum cargo capacity. Cargo can be added to the trailer, up to the maximum weight specified on the placard. The combined weight of the cargo is provided as a single number. In any case, remember: the total weight of a fully loaded trailer can not exceed the stated GVWR.

For trailers with living quarters installed, the weight of water and propane also need to be considered. The weight of fully filled propane containers is considered part of the weight of the trailer before it is loaded with cargo, and is not considered part of the disposable cargo load. Water however, is a disposable cargo weight and is treated as such. If there is a fresh water storage tank of 100 gallons, this tank when filled would weigh about 800 pounds. If more cargo is being transported, water can be off-loaded to keep the total amount of cargo added to the vehicle within the limits of the GVWR so as not to overload the vehicle. Understanding this flexibility will allow you, the owner, to make choices that fit your travel needs.

When loading your cargo, be sure it is distributed evenly to prevent overloading front to back and side to side. Heavy items should be placed low and as close to the axle positions as reasonable. Too many items on one side may overload a tire. The best way to know the actual weight of the vehicle is to weigh it at a public scale. Talk to your dealer to discuss the weighing methods needed to capture the various weights related to the trailer. This would include the weight empty or unloaded, weights per axle, wheel, hitch or king-pin, and total weight.

Excessive loads and/or underinflation cause tire overloading and, as a result, abnormal tire flexing occurs. This situation can generate an excessive amount of heat within the tire. Excessive heat may lead to tire failure. It is the air pressure that enables a tire to support the load, so proper inflation is critical. The proper air pressure may be found on the certification/VIN label and/or on the Tire Placard. This value should never exceed the maximum cold inflation pressure stamped on the tire.

1.1.1. TRAILERS 10,000 POUNDS GVWR OR LESS



Tire and Loading Information Placard - Figure 1-1

1. Locate the statement, "The weight of cargo should never exceed XXX kg or XXX lbs.," on your vehicle's placard. See figure 1-1.
2. This figure equals the available amount of cargo and luggage load capacity.
3. Determine the combined weight of luggage and cargo being loaded on the vehicle. That weight may not safely exceed the available cargo and luggage load capacity.

The trailer's placard refers to the Tire Information Placard attached adjacent to or near the trailer's VIN (Certification) label at the left front of the trailer.

1.1.2. TRAILERS OVER 10,000 POUNDS GVWR (NOTE: THESE TRAILERS ARE NOT REQUIRED TO HAVE A TIRE INFORMATION PLACARD ON THE VEHICLE)

1. Determine the empty weight of your trailer by weighing the trailer using a public scale or other means. This step does not have to be repeated.
2. Locate the GVWR (Gross Vehicle Weight Rating) of the trailer on your trailer's VIN (Certification) label.
3. Subtract the empty weight of your trailer from the GVWR stated on the VIN label. That weight is the maximum available cargo capacity of the trailer and may not be safely exceeded.

1.2. STEPS FOR DETERMINING CORRECT LOAD LIMIT – TOW VEHICLE

1. Locate the statement, "The combined weight of occupants and cargo should never exceed XXX lbs.," on your vehicle's placard.
2. Determine the combined weight of the driver and passengers who will be riding in your vehicle.
3. Subtract the combined weight of the driver and passengers from XXX kilograms or XXX pounds.
4. The resulting figure equals the available amount of cargo and luggage capacity. For example, if the "XXX" amount equals 1400 lbs. and there will be five 150 lb. passengers in your vehicle, the amount of available cargo and luggage capacity is 650 lbs. (1400-750 (5 x 150) = 650 lbs.).
5. Determine the combined weight of luggage and cargo being loaded on the vehicle. That weight may not safely exceed the available cargo and luggage capacity calculated in Step # 4.
6. If your vehicle will be towing a trailer, load from your trailer will be transferred to your vehicle. Consult the tow vehicle's manual to determine how this weight transfer reduces the available cargo and luggage capacity of your vehicle.

1.3. GLOSSARY OF TIRE TERMINOLOGY

Accessory weight
The combined weight (in excess of those standard items which may be replaced) of automatic transmission, power steering, power brakes, power windows, power seats, radio and heater, to the extent that these items are available as factory-installed equipment (whether installed or not).

Bead
The part of the tire that is made of steel wires, wrapped or reinforced by ply cords and that is shaped to fit the rim.

Bead separation
This is the breakdown of the bond between components in the bead.

Bias ply tire
A pneumatic tire in which the ply cords that extend to the beads are laid at alternate angles substantially less than 90 degrees to the centerline of the tread.

Carcass
The tire structure, except tread and sidewall rubber which, when inflated, bears the load.

Chunking
The breaking away of pieces of the tread or sidewall.

Cold inflation pressure
The pressure in the tire before you drive.

Cord
The strands forming the plies in the tire.

Cord separation
The parting of cords from adjacent rubber compounds.

Cracking
Any parting within the tread, sidewall, or inner liner of the tire extending to cord material.

CT
A pneumatic tire with an inverted flange tire and rim system in which the rim is designed with rim flanges pointed radially inward and the tire is designed to fit on the underside of the rim in a manner that encloses the rim flanges inside the air cavity of the tire.

Curb weight
The weight of a motor vehicle with standard equipment including the maximum capacity of fuel, oil, and coolant, and, if so equipped, air conditioning and additional weight optional engine.

Extra load tire
A tire designed to operate at higher loads and at higher inflation pressures than the corresponding standard tire.

Groove
The space between two adjacent tread ribs.

Gross Axle Weight Rating
The maximum weight that any axle can support, as published on the Certification / VIN label on the front left side of the trailer. Actual weight determined by weighing each axle on a public scale, with the trailer attached to the towing vehicle.

Gross Vehicle Weight Rating
The maximum weight of the fully loaded trailer, as published on the Certification / VIN label. Actual weight determined by weighing trailer on a public scale, without being attached to the towing vehicle.

Hitch Weight
The downward force exerted on the hitch ball by the trailer coupler.

Innerliner
The layer(s) forming the inside surface of a tubeless tire that contains the inflating medium within the tire.

Innerliner separation
The parting of the innerliner from cord material in the carcass.

Intended outboard sidewall
The sidewall that contains a white-wall, bears white lettering or bears manufacturer, brand, and/or model name molding that is higher or deeper than the same molding on the other sidewall of the tire or the outward facing sidewall of an asymmetrical tire that has a particular side that must always face outward when mounted on a vehicle.

Light truck (LT) tire
A tire designated by its manufacturer as primarily intended for use on lightweight trucks or multipurpose passenger vehicles.

Load rating
The maximum load that a tire is rated to carry for a given inflation pressure.

Maximum load rating
The load rating for a tire at the maximum permissible inflation pressure for that tire.

Maximum permissible inflation pressure
The maximum cold inflation pressure to which a tire may be inflated.

Maximum loaded vehicle weight
The sum of curb weight, accessory weight, vehicle capacity weight, and production options weight.

Measuring rim
The rim on which a tire is fitted for physical dimension requirements.

Pin Weight
The downward force applied to the 5^{th} wheel or gooseneck ball, by the trailer kingpin or gooseneck coupler.

Non-pneumatic rim
A mechanical device which, when a non-pneumatic tire assembly incorporates a wheel, supports the tire, and attaches, either integrally or separably, to the wheel center member and upon which the tire is attached.

Non-pneumatic spare tire assembly
A non-pneumatic tire assembly intended for temporary use in place of one of the pneumatic tires and rims that are fitted to a passenger car in compliance with the requirements of this standard.

Non-pneumatic tire
A mechanical device which transmits, either directly or through a wheel or wheel center member, the vertical load and tractive forces from the roadway to the vehicle, generates the tractive forces that provide the directional control of the vehicle and does not rely on the containment of any gas or fluid for providing those functions.

Non-pneumatic tire assembly
A non-pneumatic tire, alone or in combination with a wheel or wheel center member, which can be mounted on a vehicle.

Normal occupant weight
This means 68 kilograms (150 lbs.) times the number of occupants specified in the second column of Table I of 49 CFR 571.110.

Occupant distribution
The distribution of occupants in a vehicle as specified in the third column of Table I of 49 CFR 571.110.

Open splice
Any parting at any junction of tread, sidewall, or innerliner that extends to cord material.

Outer diameter
The overall diameter of an inflated new tire.

Overall width

The linear distance between the exteriors of the sidewalls of an inflated tire, including elevations due to labeling, decorations, or protective bands or ribs.

Ply
A layer of rubber-coated parallel cords.

Ply separation
A parting of rubber compound between adjacent plies.

Pneumatic tire
A mechanical device made of rubber, chemicals, fabric and steel or other materials, that, when mounted on an automotive wheel, provides the traction and contains the gas or fluid that sustains the load.

Production options weight
The combined weight of those installed regular production options weighing over 2.3 kilograms (5 lbs.) in excess of those standard items which they replace, not previously considered in curb weight or accessory weight, including heavy duty brakes, ride levelers, roof rack, heavy duty battery, and special trim.

Radial ply tire
A pneumatic tire in which the ply cords that extend to the beads are laid at substantially 90 degrees to the centerline of the tread.

Recommended inflation pressure
This is the inflation pressure provided by the vehicle manufacturer on the Tire Information label and on the Certification / VIN tag.

Reinforced tire
A tire designed to operate at higher loads and at higher inflation pressures than the corresponding standard tire.

Rim
A metal support for a tire or a tire and tube assembly upon which the tire beads are seated.

Rim diameter
This means the nominal diameter of the bead seat.

Rim size designation
This means the rim diameter and width.

Rim type designation
This means the industry of manufacturer's designation for a rim by style or code.

Rim width
This means the nominal distance between rim flanges.

Section width
The linear distance between the exteriors of the sidewalls of an inflated tire, excluding elevations due to labeling, decoration, or protective bands.

Sidewall
That portion of a tire between the tread and bead.

Sidewall separation
The parting of the rubber compound from the cord material in the sidewall.

Special Trailer (ST) tire
The "ST" is an indication the tire is for trailer use only.

Test rim
The rim on which a tire is fitted for testing, and may be any rim listed as appropriate for use with that tire.

Tread

- Tire safety tips.

Use this information to make tire safety a regular part of your vehicle maintenance routine. Recognize that the time you spend is minimal compared with the inconvenience and safety consequences of a flat tire or other tire failure.

1.5. Safety First–Basic Tire Maintenance

Properly maintained tires improve the steering, stopping, traction, and load-carrying capability of your vehicle. Underinflated tires and overloaded vehicles are a major cause of tire failure. Therefore, as mentioned above, to avoid flat tires and other types of tire failure, you should maintain proper tire pressure, observe tire and vehicle load limits, avoid road hazards, and regularly inspect your tires.

1.5.1. Finding Your Vehicle's Recommended Tire Pressure and Load Limits

Tire information placards and vehicle certification labels contain information on tires and load limits. These labels indicate the vehicle manufacturer's information including:

- *Recommended tire size*
- Recommended tire inflation pressure
- Vehicle capacity weight (VCW–the maximum occupant and cargo weight a vehicle is designed to carry)
- Front and rear gross axle weight ratings (GAWR– the maximum weight the axle systems are designed to carry).

Both placards and certification labels are permanently attached to the trailer near the left front.

1.5.2. Understanding Tire Pressure and Load Limits

Tire inflation pressure is the level of air in the tire that provides it with load-carrying capacity and affects the overall performance of the vehicle. The tire inflation pressure is a number that indicates the amount of air pressure– measured in pounds per square inch (psi)–a tire requires to be properly inflated. (You will also find this number on the vehicle information placard expressed in kilopascals (kpa), which is the metric measure used internationally.)

Manufacturers of passenger vehicles and light trucks determine this number based on the vehicle's design load limit, that is, the greatest amount of weight a vehicle can safely carry and the vehicle's tire size. The proper tire pressure for your vehicle is referred to as the "recommended cold inflation pressure." (As you will read below, it is difficult to obtain the recommended tire pressure if your tires are not cold.)
Because tires are designed to be used on more than one type of vehicle, tire manufacturers list the "maximum permissible inflation pressure" on the tire sidewall. This number is the greatest amount of air pressure that should ever be put in the tire under normal driving conditions.

1.5.3. Checking Tire Pressure

It is important to check your vehicle's tire pressure at least once a month for the following reasons:

- Most tires may naturally lose air over time.
- Tires can lose air suddenly if you drive over a pothole or other object or if you strike the curb when parking.
- *With radial tires, it is usually not possible to determine underinflation by visual inspection.*

For convenience, purchase a tire pressure gauge to keep in your vehicle. Gauges can be purchased at tire dealerships, auto supply stores, and other retail outlets.
The recommended tire inflation pressure that vehicle manufacturers provide reflects the proper psi when a tire is cold. The term cold does not relate to the outside temperature. Rather, a cold tire is one that has not been driven on for at least three hours. When you drive, your tires get warmer, causing the air pressure within them to increase. Therefore, to get an accurate tire pressure reading, you must measure tire pressure when the tires are cold or compensate for the extra pressure in warm tires.

1.5.4. Steps for Maintaining Proper Tire Pressure

- Step 1: Locate the recommended tire pressure on the vehicle's tire information placard, certification label, or in the owner's manual.
- Step 2: Record the tire pressure of all tires.
- Step 3: If the tire pressure is too high in any of the tires, slowly release air by gently pressing on the tire valve stem with the edge of your tire gauge until you get to the correct pressure.
- Step 4: If the tire pressure is too low, note the difference between the measured tire pressure and the correct tire pressure. These "missing" pounds of pressure are what you will need to add.
- Step 5: At a service station, add the missing pounds of air pressure to each tire that is underinflated.
- Step 6: Check all the tires to make sure they have the same air pressure (except in cases in which the front and rear tires are supposed to have different amounts of pressure).

If you have been driving your vehicle and think that a tire is underinflated, fill it to the recommended cold inflation pressure indicated on your vehicle's tire information placard or certification label. While your tire may still be slightly underinflated due to the extra pounds of pressure in the warm tire, it is safer to drive with air pressure that is slightly lower than the vehicle manufacturer's recommended cold inflation pressure than to drive with a significantly underinflated tire. Since this is a temporary fix, don't forget to recheck and adjust the tire's pressure when you can obtain a cold reading.

1.5.5. Tire Size

To maintain tire safety, purchase new tires that are the same size as the vehicle's original tires or another size recommended by the manufacturer. Look at the tire information placard, the owner's manual, or the sidewall of the tire you are replacing to find this information. If you have any doubt about the correct size to choose, consult with the tire dealer.

1.5.6. Tire Tread

The tire tread provides the gripping action and traction that prevent your vehicle from slipping or sliding, especially when the road is wet or icy. In general, tires are not safe and should be replaced when the tread is worn down to 1/16 of an inch. Tires have built-in treadwear indicators that let you know when it is time to replace your tires. These indicators are raised sections spaced intermittently in the bottom of the tread grooves. When they appear "even" with the outside of the tread, it is time to replace your tires. Another method for checking tread depth is to place a penny in the tread with Lincoln's head upside down and facing you. If you can see the top of Lincoln's head, you are ready for new tires.

1.5.7. Tire Balance and Wheel Alignment

To avoid vibration or shaking of the vehicle when a tire rotates, the tire must be properly balanced. This balance is achieved by positioning weights on the wheel to counterbalance heavy spots on the wheel-and-tire assembly. A wheel alignment adjusts the angles of the wheels so that they are positioned correctly relative to the vehicle's frame. This adjustment maximizes the life of your tires. These adjustments require special equipment and should be performed by a qualified technician.

1.5.8. Tire Repair

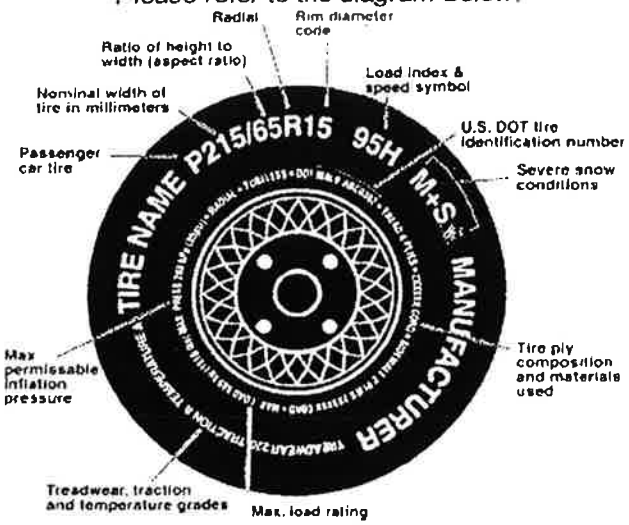
The proper repair of a punctured tire requires a plug for the hole and a patch for the area inside the tire that surrounds the puncture hole. Punctures through the tread can be repaired if they are not too large, but punctures to the sidewall should not be repaired. Tires must be removed from the rim to be properly inspected before being plugged and patched.

1.5.9. Tire Fundamentals

Federal law requires tire manufacturers to place standardized information on the sidewall of all tires. This information identifies and describes the fundamental characteristics of the tire and also provides a tire identification number for safety standard certification and in case of a recall.

1.5.9.1. Information on Passenger Vehicle Tires

Please refer to the diagram below.



P

The "P" indicates the tire is for passenger vehicles.

Next number

This three-digit number gives the width in millimeters of the tire from sidewall edge to sidewall edge. In general, the larger the number, the wider the tire.

Next number

This two-digit number, known as the aspect ratio, gives the tire's ratio of height to width. Numbers of 70 or lower indicate a short sidewall for improved steering response and better overall handling on dry pavement.

R

The "R" stands for radial. Radial ply construction of tires has been the industry standard for the past 20 years.

Next number

This two-digit number is the wheel or rim diameter in inches. If you change your wheel size, you will have to purchase new tires to match the new wheel diameter.

Next number

This two- or three-digit number is the tire's load index. It is a measurement of how much weight each tire can support. You may find this information in your owner's manual. If not, contact a local tire dealer. Note: You may not find this information on all tires because it is not required by law.

M+S

The "M+S" or "M/S" indicates that the tire has some mud and snow capability. Most radial tires have these markings; hence, they have some mud and snow capability.

Speed Rating

The speed rating denotes the speed at which a tire is designed to be driven for extended periods of time. The ratings range from 99 miles per hour (mph) to 186 mph. These ratings are listed below. Note: You may not find this information on all tires because it is not required by law.

Tire Safety Information

Letter Rating	Speed Rating
Q	99 mph
R	106 mph
S	112 mph
T	118 mph
U	124 mph
H	130 mph
V	149 mph
W	168* mph
Y	186* mph

* For tires with a maximum speed capability over 149 mph, tire manufacturers sometimes use the letters ZR. For those with a maximum speed capability over 186 mph, tire manufacturers always use the letters ZR.

U.S. DOT Tire Identification Number
This begins with the letters "DOT" and indicates that the tire meets all federal standards. The next two numbers or letters are the plant code where it was manufactured, and the last four numbers represent the week and year the tire was built. For example, the numbers 3197 means the 31st week of 1997. The other numbers are marketing codes used at the manufacturer's discretion. This information is used to contact consumers if a tire defect requires a recall.

Tire Ply Composition and Materials Used
The number of plies indicates the number of layers of rubber-coated fabric in the tire. In general, the greater the number of plies, the more weight a tire can support. Tire manufacturers also must indicate the materials in the tire, which include steel, nylon, polyester, and others.

Maximum Load Rating
This number indicates the maximum load in kilograms and pounds that can be carried by the tire.

Maximum Permissible Inflation Pressure
This number is the greatest amount of air pressure that should ever be put in the tire under normal driving conditions.

1.5.9.2. UTQGS Information

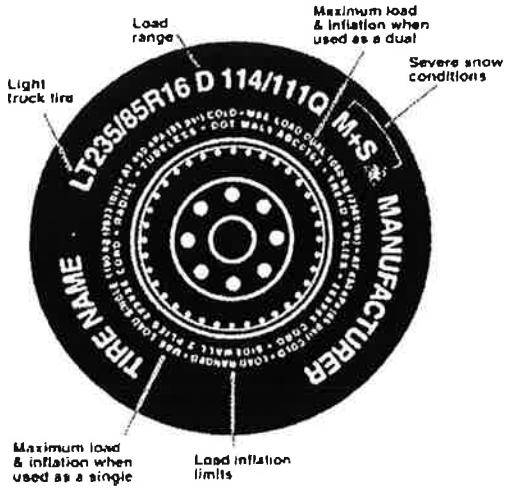
Treadwear Number
This number indicates the tire's wear rate. The higher the treadwear number is, the longer it should take for the tread to wear down. For example, a tire graded 400 should last twice as long as a tire graded 200.

Traction Letter
This letter indicates a tire's ability to stop on wet pavement. A higher graded tire should allow you to stop your car on wet roads in a shorter distance than a tire with a lower grade. Traction is graded from highest to lowest as "AA","A", "B", and "C".

Temperature Letter
This letter indicates a tire's resistance to heat. The temperature grade is for a tire that is inflated properly and not overloaded. Excessive speed, underinflation or excessive loading, either separately or in combination, can cause heat build-up and possible tire failure. From highest to lowest, a tire's resistance to heat is graded as "A", "B", or "C".

1.5.9.3. Additional Information on Light Truck Tires

Please refer to the following diagram.



Tires for light trucks have other markings besides those found on the sidewalls of passenger tires.

LT
The "LT" indicates the tire is for light trucks or trailers.

ST
An "ST" is an indication the tire is for trailer use only.

Max. Load Dual kg (lbs) at kPa (psi) Cold
This information indicates the maximum load and tire pressure when the tire is used as a dual, that is, when four tires are put on each rear axle (a total of six or more tires on the vehicle).

Max. Load Single kg (lbs) at kPa (psi) Cold
This information indicates the maximum load and tire pressure when the tire is used as a single.

Load Range
This information identifies the tire's load-carrying capabilities and its inflation limits.

1.6. TIRE SAFETY TIPS

Preventing Tire Damage

- Slow down if you have to go over a pothole or other object in the road.
- Do not run over curbs or other foreign objects in the roadway, and try not to strike the curb when parking.

Tire Safety Checklist

- Check tire pressure regularly (at least once a month), including the spare.
- Inspect tires for uneven wear patterns on the tread, cracks, foreign objects, or other signs of wear or trauma.
- *Remove bits of glass and foreign objects wedged in the tread.*
- Make sure your tire valves have valve caps.
- Check tire pressure before going on a long trip.
- Do not overload your vehicle. Check the Tire Information and Loading Placard or User's Manual for the maximum recommended load for the vehicle.

12 Volt Wiring Schematic
Kidde Fire Trainer Unit

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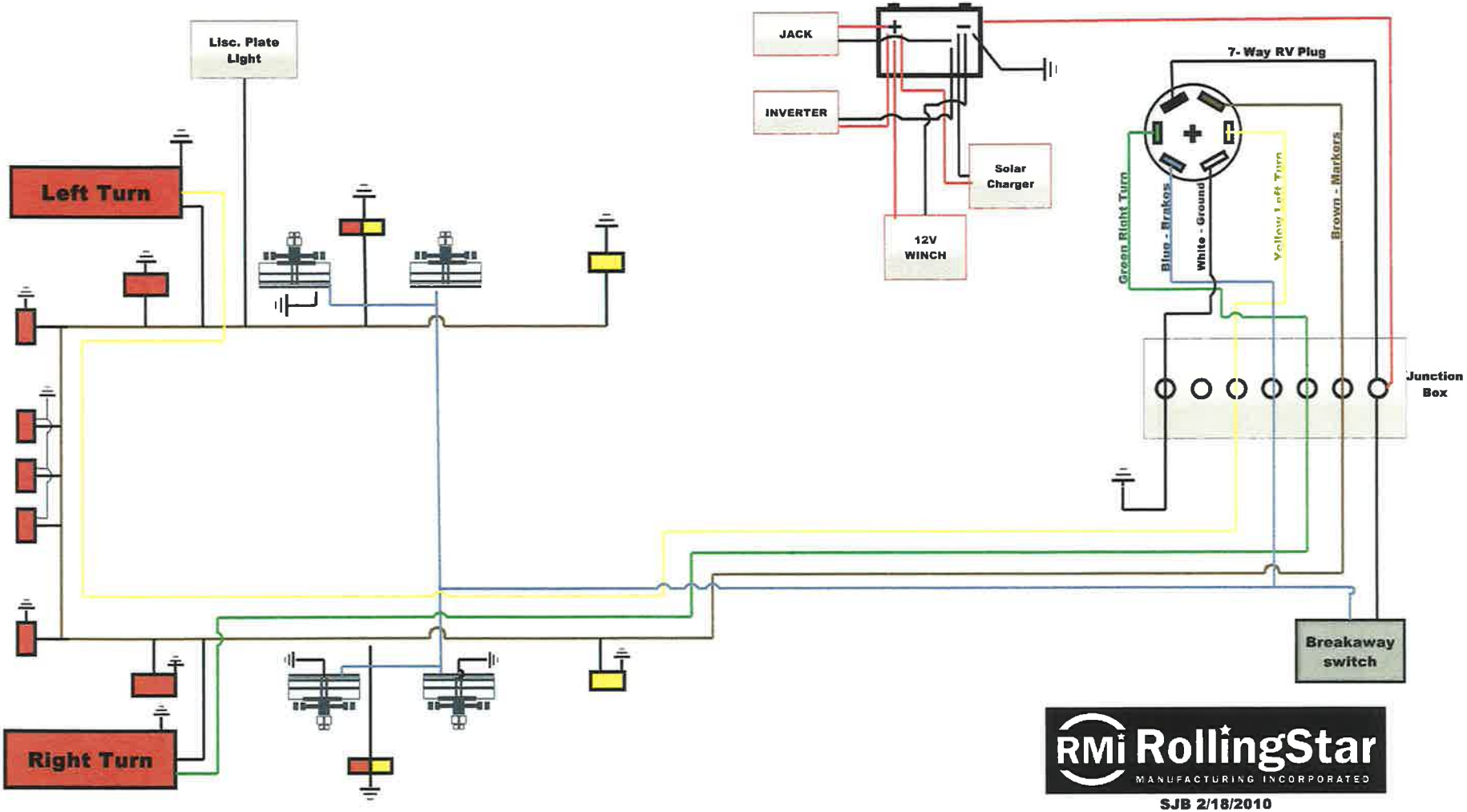

RMi RollingStar
MANUFACTURING INCORPORATED

SJB 2/18/2010

Ramsey Winch Company
Owner's Manual
Front Mount Electric Winch
PATRIOT 9500UT

Layer of Cable		1	2	3	4	5
Rated Line Pull Per Layer	(lbs)	9,500	7,700	6,500	5,700	4,900
	(kg)	4,309	3,480	2,940	2,580	2,210
Cumulative Cable Capacity Per Layer (5/16" – 8mm – dia. Cable)	(ft)*	15	35	60	90	105
	(m)*	4	10	18	27	32

* Depends on cable being uniformly wound onto drum

Line Pull First Layer	(lbs)	NO LOAD	2,000	4,000	6,000	8,000	9,500
	(kg)		900	1,810	2,720	3,620	4,309
Line Speed First Layer	(fpm)	33	14	11	9	7	6
	(mpm)	10.0	4.2	3.3	2.5	2.1	1.8
Amp Draw		95	205	270	340	405	450

Congratulations

You have purchased the finest winch available in its service class. It features a highly efficient 3 stage planetary gear set which transmits torque from a series wound DC motor. A semi-automatic clutch allows free spooling for quick cable deployment and eliminates unnecessary trips back to the winch. A wireless remote allows the winch to be operated while standing up to 50 feet away. An automatic load holding brake is designed to hold the full rated capacity of the winch. It was designed and manufactured to provide you with the utmost in utility. As with any device that combines power and movement in its use, there are dangers if improperly used. At the same time, there are easier ways for getting the job done if certain precautions are taken first.

Please read this manual carefully. It contains useful ideas in obtaining the most efficient operation from your Ramsey Winch and safety procedures you need to know before beginning use. When you follow our guidelines for operation, your Ramsey Winch will give you many years of satisfying service. Thank you for choosing Ramsey. You will be glad you have one working for you.

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Tips for Safe Operation 2
Techniques of Operation 3
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Operating Instructions 5
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Maintenance 5
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Winch Parts List 8-10
Warranty ..... Back Cover

Please note: Ramsey Patriot™ series winches are designed for front mount vehicle use. The winches are not designed for and should not be used in industrial applications (car haulers/carriers, wreckers, hoisting, etc.), and Ramsey does not warrant them to be suitable for such use. Ramsey makes a separate, complete line of winches for industrial/commercial use. Please contact the factory for further information.

CAUTION: Read and understand this manual before installation and operation of winch. See Safety Precautions!

Safety Precautions

To Guard against Possible Injury...
A minimum of five wraps of cable around the drum barrel is necessary to hold the rated load. Cable clamp is not designed to hold the load.

A. Keep yourself and others a safe distance to the side of the cable when pulling under load.

B. Do not step over a cable, or near a cable under load.

C. Use supplied hook strap when handling hook for spooling wire rope.

D. Do not move the vehicle to pull a load on the winch cable. This could result in cable breakage and/or winch damage.

E. Use a heavy rag or gloves to protect hands from burrs when handling winch cable.

F. Apply blocks to wheels when vehicle is on an incline.

G. Winch clutch should be fully engaged when in use.

H. Modification, alteration, or deviation to the winch should only be made by Ramsey Winch Company.

I. Keep the duration of your pulls as short as possible. If the motor becomes uncomfortably hot to the touch, stop and let it cool for a few minutes. Do not pull more than one minute at or near rated load. Do not maintain power to the winch if the motor stalls. Electric winches are for intermittent usage and should not be used in constant duty applications.

J. Disconnect the remote control switch from the winch when not in use. A Ramsey Part No. 282053 safety on-off switch in your vehicle is recommended.

K. NOTE: Do not use winch in hoisting applications due to required hoist safety factors and features.

L. Do not exceed maximum line pull ratings shown in tables. Shock loads must not exceed these ratings.

M. To respool correctly, it is necessary to keep a slight load on the cable. This can be accomplished by (wearing gloves) holding the cable with one hand and the remote control switch with the other, starting as far back and in the center as you can, walking up keeping load on the cable as the winch is powered in. Do not allow the cable to slip through your hand and do not approach the winch too closely. Turn off the winch and repeat the procedure until all the cable except a few feet is in. Disconnect the remote control switch and finish spooling in cable by rotating the drum by hand with clutch disengaged. On hidden winches, spool in cable under power using supplied hook strap.

Tips for Safe Operation

Don't underestimate the potential danger in winching operations. Neither should your fear them. Do learn the basic dangers and avoid them.

Observe the spooling of cable onto drum. Side pulls can cause cable to pileup at one end of the drum. To correct uneven stacking, spool out that section of the cable and move it to the other end of the drum and continue winching. Uneven spooling, which causes cable pileup, can interfere with the solenoid housing causing damage to the winch.

Store the remote control switch inside your vehicle where it will not become damaged. Inspect it before you plug it in.

When ready to begin spooling in, plug in remote control switch with clutch disengaged.

Never connect the hook back to the cable. This causes cable damage. Always use a sling or chain of suitable strength, as shown in the illustration.

Observe your winch while winching, if possible, while standing at a safe distance. If you use vehicle drive to assist, stop and get out every few feet to assure the cable is not piling up in one corner. Jamming cable can break your winch.

Do not attach tow hooks to winch mounting apparatus. They must attach to vehicle frame.

When double lining during stationary winching, the winch hook should be attached to the chassis of the vehicle.

Since the greatest pulling power is achieved on the innermost layer of your winch, it is desirable to pull off as much line as you can for heavy pulls. If this is not practical, use a snatch block and double the arrangement (see illustration). Remember, a minimum of 5 wraps of cable around the drum barrel is necessary to hold the rated load.

Neat, tight spooling avoids cable binding which is caused when a load is applied and the cable is pinched between two others. If this happens, alternately power the winch in and out a few inches. Do not attempt to work a bound cable under load, free by hand.

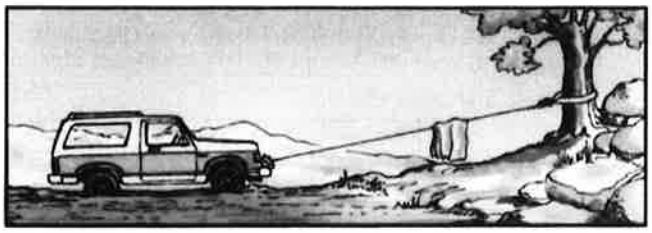
Techniques of Operation

The best way to get acquainted with how your winch operates is to make a few test runs before you actually need to use it. Plan your test in advance. Remember you hear your winch as well as see it operate. Get to recognize the sound of a light steady pull, a heavy pull, and sounds caused by load jerking or shifting. Soon you will gain confidence in operating your winch and its use will become second nature with you.

Your winch will not only pull your vehicle up a steep grade, it will also pull another vehicle or a load while your vehicle is anchored in a stationary position. The sketches on this page show you a few techniques.

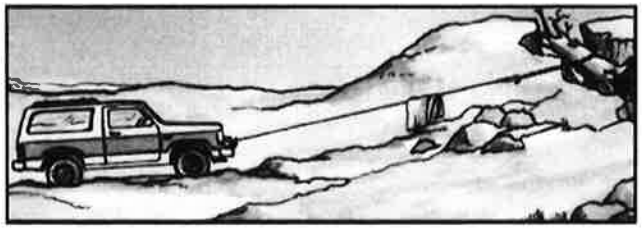
When pulling a heavy load, place a blanket, jacket or tarpaulin over the cable five or six feet from the hook. It will slow the snap back in the event of a broken cable. Also, open the vehicle hood for additional protection.

Use the vehicle wheel power to help the winch, but don't overtake the winch line. Plan your pull. You can't always hook up and pull out in one step. Examine all the areas for anchoring possibilities as well as leverage situations, direction, and goal.

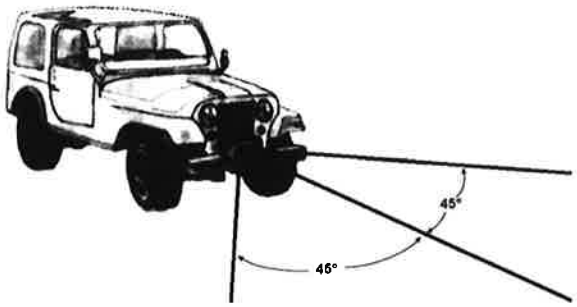


For basic self-recovery, anchor to a tree or heavy rock. When anchoring to a tree, always use a tree trunk protector.

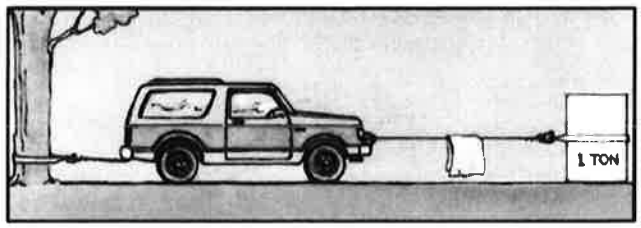
Stakes driven in solid earth and chained together make a good anchor point for self-recovery when no solid anchor point is available.



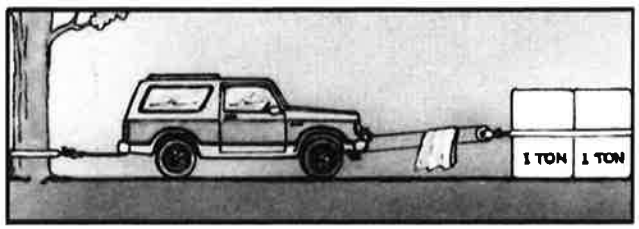
For a solid anchor, bury a log with earth or sand or place it in a deep ravine.



Winches equipped with cable guide fairleads can pull from several directions. Pull from an angle only to straighten up the vehicle-otherwise you can damage structural members or other parts of your vehicle and cause excess cable buildup on one end of the winch drum.



For a direct pull of 2000 lbs., hitch truck to a tree or solid anchor, and take out of gear.



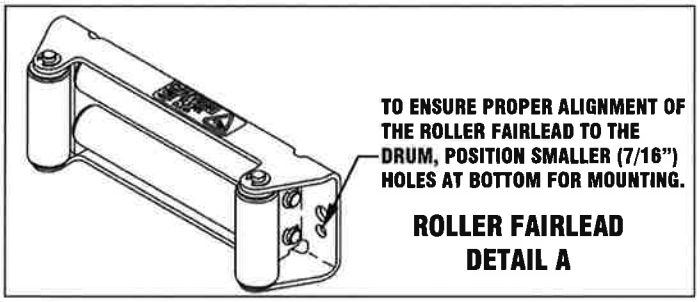
To double the pull, use 2-part line and tie off to chassis. Take out of gear.

Installation

The winch shown in this owner's manual is solely and exclusively designed for vehicle mounted, non-industrial applications. All other applications will void warranty.

It is very important that the winch be mounted on a flat surface so that the three major sections (the motor end, the cable drum, and the gear housing end) are properly aligned. It is recommended that Ramsey kits be used to mount the winch. They are designed to align the winch and distribute up to the full rated load evenly, to avoid possible damage to the winch or vehicle.

NOTE: If recommended mounting is not used, a kit of equal design must be used.



Also available for mounting the Patriot 9500 UT are the following winch mounting channels:

- #251126 short length (23.63") black
- #251127 medium length (30.00") black
- #251128 long length (36.00") black

It is recommended that Ramsey mounting channels be used with all non-Ramsey mounting.

Attach fairlead to channel using hardware furnished with winch. Attach winch to channel. Thread capscrews with lockwashers through mounting holes in channel and into winch feet (see Figure 1).

Substitution of attaching hardware items (bolts, nuts or washers) different from those supplied with your winch and mounting kit can lead to failure causing damage or serious injury (use SAE grade 5 bolts or better and torque to 34 ft.lbs.).

Place end of wire rope through fairlead and attach cable hook. Use clevis pin and cotter pin (see Figure 1).

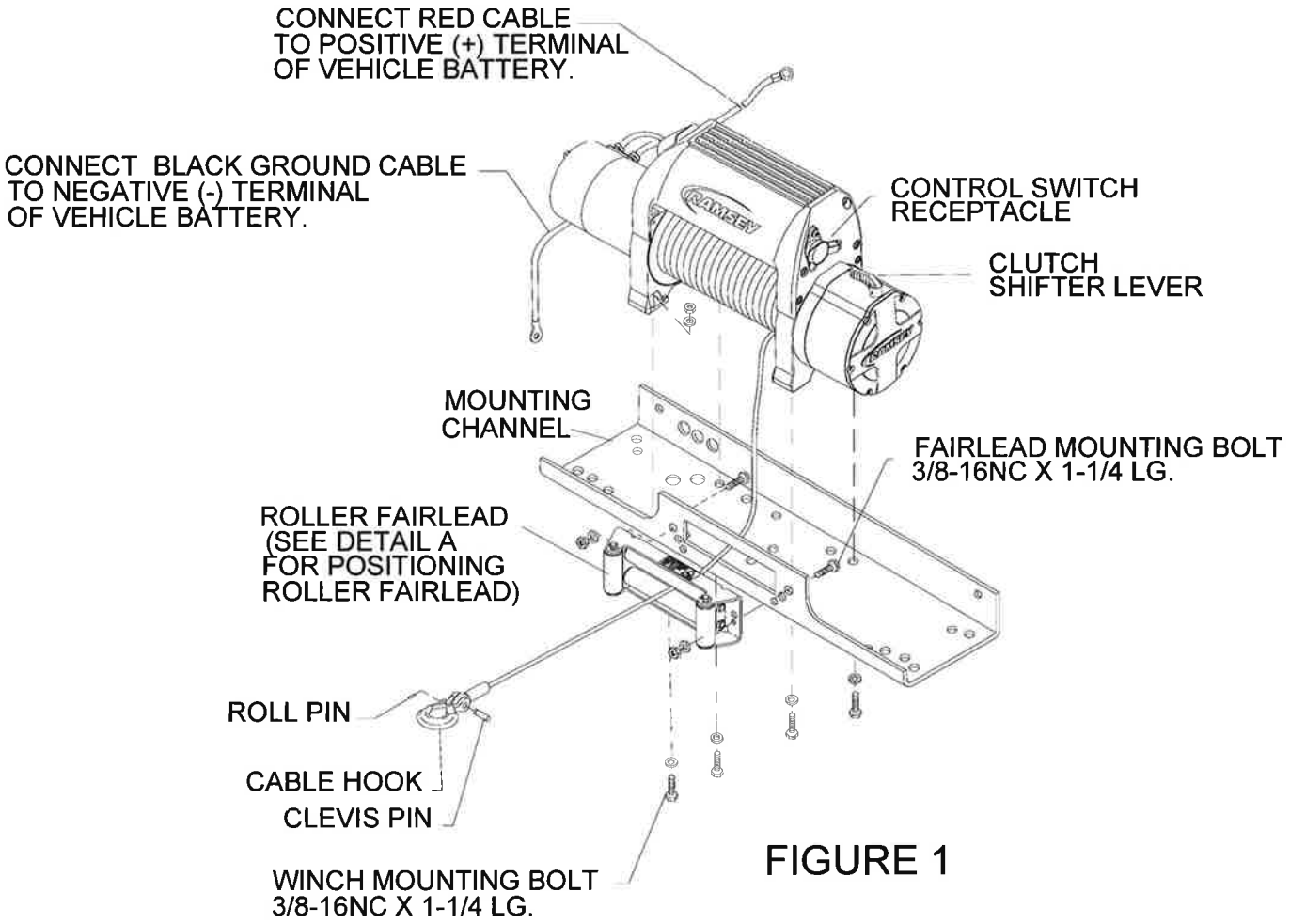


FIGURE 1

Operating Instructions

The winch clutch allows rapid unspooling of the wire rope for hooking onto the load or anchor point. The clutch is operated by the shifter lever located on the gear housing end of the winch as follows:

1. To disengage the clutch, move the clutch shifter lever to the "DISENGAGED" position. Wire rope may now be freespooled off the drum.
2. To engage the clutch, power winch motor to reel in cable. The clutch will automatically engage and the clutch shifter lever will automatically move to the "ENGAGED" position.

   Note: Slight tension on the cable may be necessary to fully engage the clutch.

   Note: Clutch may be manually engaged by manually moving the clutch lever to the "ENGAGED" position. Rotation of the drum may be necessary to fully engage the clutch manually.

   Caution: Insure the clutch fully engages by observing clutch shifter lever position before winching.

   Caution: Stay clear of clutch lever while clutch is being automatically engaged. Disconnect the remote from the winch before operating the clutch manually.

Electrical Connections and Operations

For normal self-recovery work, your existing electrical system is adequate. Your battery must be kept in good condition. A fully charged battery and proper connections are essential. Run the vehicle engine during winching operations to keep battery charged.

Route battery cables up to battery.

Caution: Be sure battery cables are not drawn taut across any surfaces which could possibly damage them.

Connect red cable to positive (+) battery terminal. Connect black ground cable to negative (-) terminal of battery (See Figure 1).

Models Equipped with Push Button Remote Control Switch

The remote control switch is water proof. It has push buttons on either side. Make sure the motor has stopped fully before reversing. To actuate winch simply plug remote control switch into receptacle in cover of winch. Run winch forward and reverse to check directions. Snap appropriate "IN" and "OUT" disc into proper thumb cavity. **Do not leave switch plugged in when winch is not in use.**

Models Equipped with Wireless Remote Control

See Installation and Operating Instructions for Ramsey Universal Remote Control (OM-914057) included with the wireless remote.

Maintenance

All moving parts are permanently lubricated with high temperature lithium grease at the time of assembly. Under normal conditions factory lubrication will suffice.

Lubricate cable periodically using light penetrating oil. Inspect the cable for broken strands and replace if necessary. If the cable becomes worn or damaged, it must be replaced.

Corrosion on electrical connections will reduce performance or may cause a short. Clean all connections especially in remote control switch and receptacle. In salty environments use a silicone sealer to protect from corrosion.

To minimize corrosion of the internal motor components that may occur due to condensation, power the winch in or out periodically. Energizing the motor will generate heat, which will help dissipate any moisture buildup in the motor. This should be performed at periodic intervals (such as with each oil change of your vehicle). **Note:** Refer to the Troubleshooting Guide if the motor has been submerged.

Cable Installation

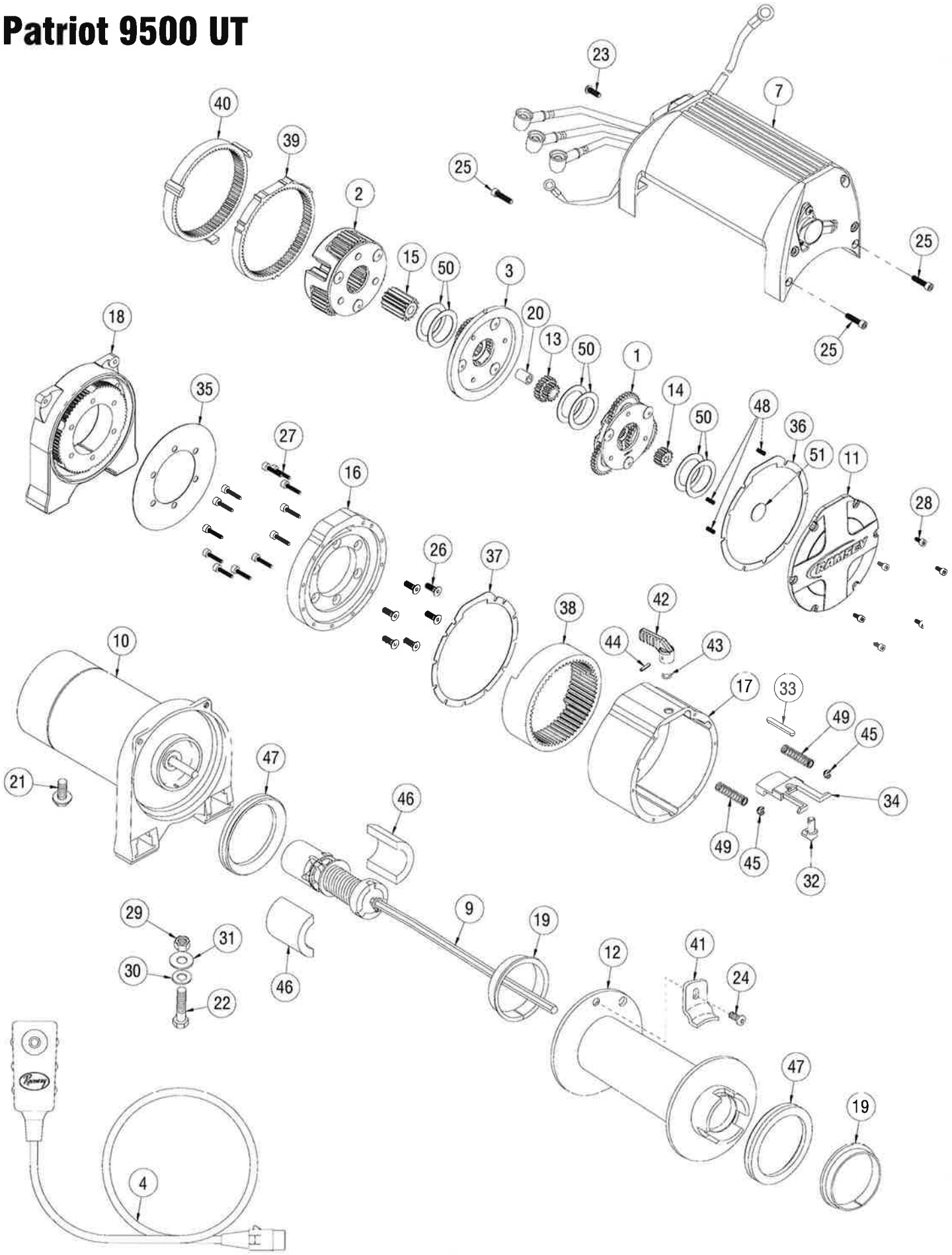
1. Unwind the new cable by rolling it out along the ground to prevent kinking.
2. Remove old cable and observe the manner in which it is attached to the cable drum flange.
3. Before installing the new cable assembly, securely wrap the end of the cable with plastic tape or similar tape to prevent fraying.
4. Position the cable drum so that the large 13/32" diameter hole in the motor end drum flange is approximately on the top.
5. Form a short bend (approximately 1/2" long) in the end of the cable. Insert the bend into the 13/32" hole in the drum flange and then carefully run the winch in the "reel in" direction approximately 3/4 revolution until the 1/4" diameter threaded hole in the drum flange is on top.
6. Secure the cable to the drum flange using cable anchor and capscrew shown in the parts drawing on page 8-9 (Item nos. 41 and 24). Securely tighten the capscrew, but do not over-tighten.
7. Wind 5 wraps of cable onto the drum. Wind on the rest of the cable by pulling in a light load to keep the tension constant. Allow the cable to swivel by using a length of chain or a swivel block between the cable hook and the load.

Troubleshooting Guide

Condition	Possible Cause	Correction
MOTOR RUNS IN ONLY ONE DIRECTION	Defective or stuck solenoid	Jar each solenoid to free contacts. Check each solenoid by applying 12 volts to coil terminal (it should make an audible click when energized).
	Defective remote control switch	Disengage winch clutch, remove remote control switch plug from the socket and jump hot pin and either of the other pins. Motor should run. Jump hot pin and the remaining pin. Motor should run.
MOTOR RUNS EXTREMELY HOT	Long period of operation	Cooling off periods are essential to prevent overheating.
	Insufficient battery	Check battery terminal voltage under load. If 10 volts or less, replace or parallel another battery to it.
MOTOR RUNS, BUT WITH INSUFFICIENT POWER, OR WITH LOW LINE SPEED.	Bad connection	Check battery cable for corrosion; clean and grease.
	Insufficient charging system	Replace with larger capacity charging system.
MOTOR RUNS, BUT DRUM DOES NOT TURN	Motor running in the reel out direction while clutch is not engaged	If motor is running in the reel in direction or clutch is engaged but symptom still exists, it will be necessary to disassemble winch to determine cause and repair.
MOTOR WILL NOT OPERATE	Defective or stuck solenoid	Jar each solenoid to free contacts. Check each solenoid by applying 12 volts to coil terminal (it should make an audible click when energized.
	Defective remote control switch	Disengage winch clutch, remove remote control switch plug from the socket and jump hot pin and either of the other pins. Motor should run. Jump hot pin and the remaining pin. Motor should run.
	Defective motor	If solenoids operate, check for voltage at armature post; replace motor.
	Loose Connections	Tighten connections on bottom side of hood and on motor.
MOTOR WATER DAMAGED	Submerged in water or water from high pressure car wash	Allow dry thoroughly, then run motor without load in short bursts to dry windings.

NOTES

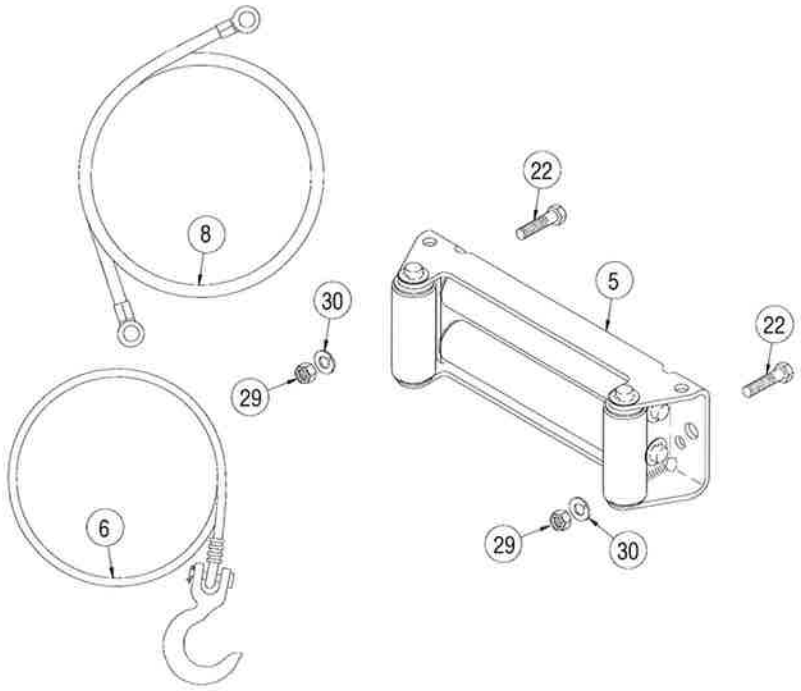
Patriot 9500 UT



Patriot 9500 UT Winch Parts List

Item No.	Qty.	Part No.	Description
1	1	247009	GEAR CARRIER ASSEMBLY – INPUT
2	1	247023	GEAR CARRIER ASSEMBLY – OUTPUT
3	1	247037	GEAR CARRIER ASSY – INTERMEDIATE
4	1	251110	SWITCH ASSEMBLY
5	1	251183	ROLLER FAIRLEAD
6	1	251210	CABLE ASSEMBLY – 5/16 DIA X 105'
7	1	278195	SOLENOID ASSEMBLY – 12V
8	1	289141	CABLE ASSEMBLY - GROUND
9	1	296181	BRAKE/INPUT SHAFT ASSEMBLY
10	1	297040	MOTOR 12V
11	1	328163	COVER-GEAR HOUSING
12	1	332219	DRUM – CABLE
13	1	334147	GEAR – INTERMEDIATE SUN
14	1	334154	GEAR – INPUT SUN
15	1	334197	GEAR – OUTPUT SUN
16	1	338361	GEAR HOUSING – OUTPUT
17	1	338362	GEAR HOUSING – INPUT
18	1	338364	END BEARING – GEAR HOUSING
19	2	412056	BUSHING – DRUM
20	1	412061	BUSHING – SHAFT
21	1	414092	CAPSCREW 5/16-18UNC X 3/8 LG HX
22	6	414316	CAPSCREW 3/8-16NC X 1-1/4 HX HD
23	1	414823	BOLT-1/4-20NC X 3/4 LG BUTTON HD
24	1	414830	BOLT-1/4-20NC X 3/8 BUTTON HD
25	3	414986	BOLT-1/4-20NC X 1 LG SOCKET HD
26	6	414861	BOLT-1/4-20NC X 3/4 LG FLAT SOC HD

Item No.	Qty.	Part No.	Description
27	12	414960	BOLT-#8-32NC X 7/8 LG. SOC HD
28	6	416208	BOLT-#8-32NC X 1/2 LG. SOCKET HD
29	6	418035	NUT 3/8-16NC HEX REG PLATED
30	6	418177	LOCKWASHER – 3/8 ID MED SECT
31	4	418181	WASHER-FLAT 3/8 ID SAE PLATED
32	1	426052	ACTUATOR
33	1	426053	LINK – CLUTCH
34	1	426056	PLUNGER – LOCKING
35	1	442227	GASKET – GEAR HSG/END BEARING
36	1	442228	GASKET – GEAR HSG/COVER
37	1	442229	GASKET – GEAR HSG/GEAR HSG
38	1	444110	GEAR – OUTPUT RING
39	1	444111	GEAR – INPUT RING
40	1	444112	GEAR – INTERMEDIATE RING
41	1	448046	CABLE ANCHOR
42	1	452007	HANDLE – CLUTCH
43	1	462064	O-RING – 5/16 ID X 1/16 THICK
44	1	470050	ROLL PIN – 1/8 DIA X 5/8 LG
45	2	472080	SPRING GUIDE
46	2	477004	RING – HALF
47	2	486082	SEAL – DRUM
48	3	494077	SPRING – DRAG
49	2	494104	SPRING – CLUTCH
50	6	518020	THRUST WASHER
51	1	518027	THRUST DISC



Wireless Remote Kit #251200

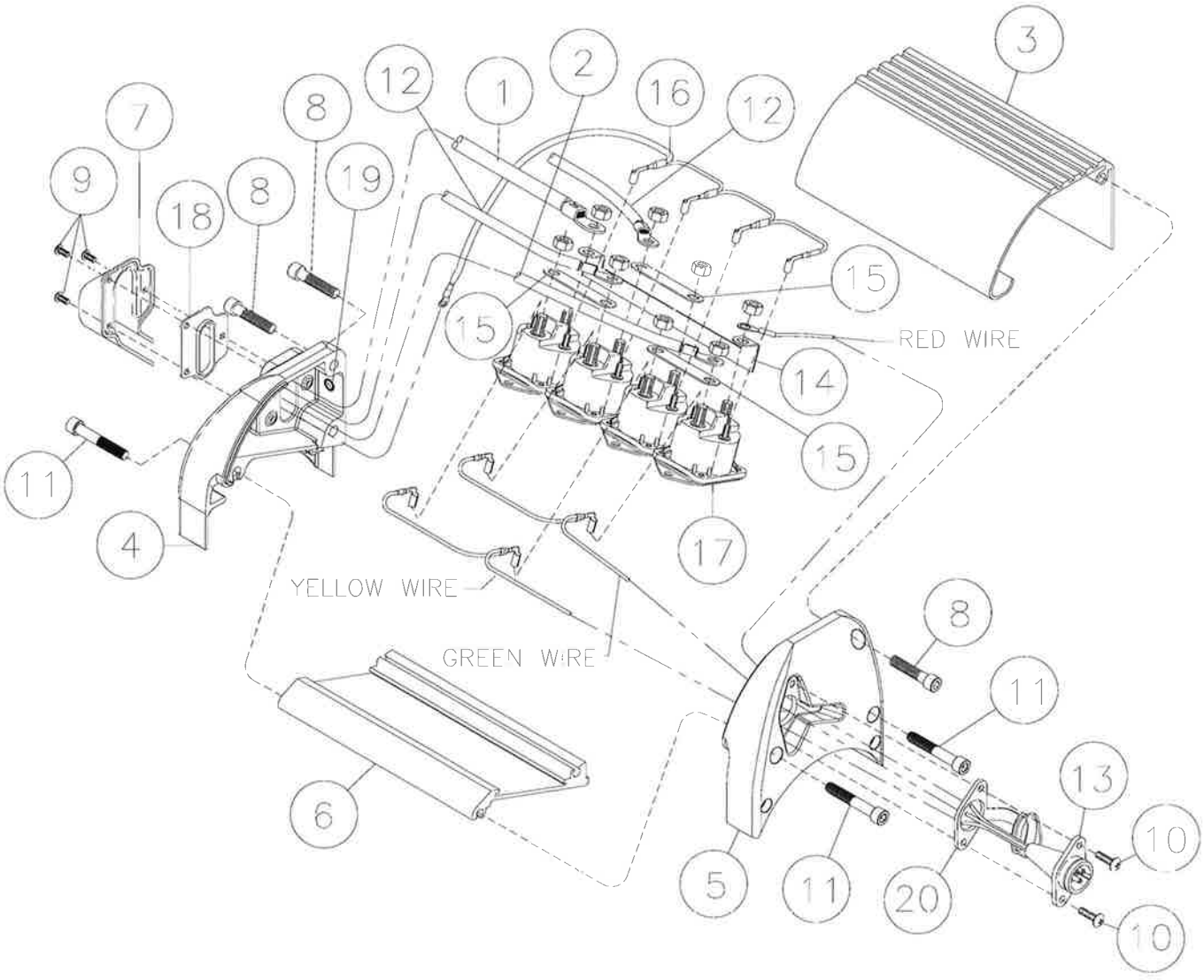
See "Installation And Operating Instructions For Ramsey Universal Front Mount Wireless Remote Control Kits" for wireless remote parts list.

Solenoid Assembly Parts List

278195 12V (PATRIOT 9500 UT)

Item No.	Qty.	Part No.	Description
1	1	289015	WIRE ASSY – BATTERY CABLE
2	1	289195	WIRE ASSY – MOTOR LEADS
3	1	682186	COVER
4	1	316107	CAP – L.H. END
5	1	316106	CAP – R.H. END
6	1	682185	CHANNEL – SOLENOID MOUNTING
7	1	413078	COVER – WIRE
8	3	414880	CAPSCREW 5/16-18NC X 1-1/2 LG SOC HD
9	3	416207	SCREW
10	2	416227	SCREW #10-24NC X ¾ LG TRUSS HD

Item No.	Qty.	Part No.	Description
11	3	414989	CAPSCREW 5/16-18UNC X 2 LG SOC HD
12	2	289196	WIRE ASSY – MOTOR LEAD
13	1	430022	CONNECTOR
14	1	440259	STRAP
15	3	440260	STRAP
16	1	440276	WIRE ASSY – GROUND
17	4	440262	SOLENOID 12V
18	1	442222	GASKET
19	1	472047	GROMMET
20	1	482029	COVER – RECEPTACLE



NOTES

Warranty Information

Ramsey Winches are designed and built to exacting specifications. Care and skill go into every winch we make. If the need should arise, warranty procedure is outlined on the back of your self-addressed postage paid warranty card. Please read and fill out the enclosed warranty card and send it to Ramsey Winch Company. If you have problems with your winch, please follow instructions for proper service on all warranty claims.

Limited Lifetime Warranty

Ramsey Winch offers a limited lifetime warranty for each new Ramsey consumer/RV winch against manufacturing defects in workmanship and materials on all mechanical components.

Warranty registration cards for each winch must be submitted at the time of purchase or within 30 days. Warranty will only be valid for the original purchase of the winch and installed on the vehicles with which they were originally registered.

New cable assemblies are warranted against defects in workmanship and materials. No warranty applies after initial use.

All Ramsey mounting kits and other accessories carry a 1-year limited warranty against defects in material and workmanship.

Chrome finish warranted for one year against manufacturing defects. Cracking, scratching, or corrosion caused by winching not covered by warranty.

This warranty is void if winch is used in commercial/industrial applications other than front mount self-recovery.

Electrical components consisting of motors, solenoids, wiring, wire connectors and associated parts carry a 1-year limited warranty. Battery isolators carry a 90-day limited warranty.

An optional extended 2-year limited warranty for all electrical components may be purchased.

The obligation under this Warranty, statutory or otherwise, is limited to the replacement or repair at the manufacturer's factory, or at a point designated by the manufacturer, upon inspection of such part, to have been defective in material or workmanship. This Warranty does not obligate Ramsey Winch Company to bear the cost of transportation charges in connection with the replacement or repair of defective parts, nor shall it apply to a product upon which repairs or alterations have been made, unless authorized by the manufacturer, or for equipment misused, neglected, or improperly installed.

IMPORTANT NOTICE: To the fullest extent permitted by applicable law, the following are hereby excluded and disclaimed: 1. All warranties of fitness for a particular purpose; 2. All warranties of merchantability; 3. All claims for consequential or incidental damages. There are no warranties that extend beyond the description that appears on the face hereof.

Some states do not allow the above exclusions or disclaimers in consumer transactions and as such this disclaimer/exclusion may not apply to your particular case.

To the extent such warranties of fitness for a particular purpose or merchantability are deemed to apply to this product, they exist for only so long as the express limited warranty elsewhere set forth is in existence.

Ramsey Winch Company makes no warranty in respect to accessories, same being subject to the warranties of their respective manufacturers.

Ramsey Winch Company, whose policy is one of continuous product improvement, reserves the right improve any product through changes in design and materials as it may deem desirable without being obligated to incorporate such changes in products of previous manufacture.

If field service at the request of the buyer is rendered and the fault is found not to be with Ramsey Winch Company's product, the buyer shall pay the time and expense cost of the field representative. Bills for service, labor, or other expenses which have been incurred by the buyer without express approval or authorization by Ramsey Winch Company will not be accepted.

This warranty gives you specific legal rights and you may also have other legal rights which vary from state to state.

RAMSEY WINCH

Ramsey Winch Company

P.O. Box 581510 - Tulsa, OK 74158-1510 USA - Phone: (918) 438-2760 - Fax (918) 438-6688

Visit us at http://www.ramsey.com

OM-914182-0407-B

Warranty Registration: ...ister online today for a chance ...n a FREE Tripp Lite product— ...www.tripplite.com/warranty

Owner's Manual

PowerVerter®

DC-to-AC Inverter

Model: PV700HF

Input	Output
12V DC	120V, 60Hz AC

RoHS COMPLIANT

TRIPP·LITE

1111 W. 35th Street, Chicago, IL 60609 USA
Customer Support: (773) 869-1234
www.tripplite.com

Reliable AC Power Wherever You Need It

Congratulations! You've purchased a high-quality Inverter designed to function as a mobile energy source powered by your automotive battery. PowerVerter Inverters convert 12V DC (battery) power into 120V AC (household) power, allowing you to use equipment you commonly use at home—computers, appliances, electronics, power tools and more—while traveling by automobile or working at remote locations that lack AC power. PowerVerter Inverters include advanced features that protect your equipment, lengthen the service life of your battery and ensure that you'll always have battery power to start your vehicle:

- Automatic Overload Protection
- Automatic Low Battery Protection
- High-Performance DC-to-AC Inversion
- Frequency-Controlled Output Power
- Simple, Maintenance-Free Operation
- Moisture-Resistant Construction*
- 3 AC Outlets

** The inverter is moisture-resistant, not waterproof.*

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Important Safety Instructions

SAVE THESE INSTRUCTIONS!

This manual contains important instructions and warnings that should be followed during the installation, operation and storage of all Tripp Lite Inverters.

Location Warnings

- Install your Inverter (whether for a mobile or stationary application) in a location or compartment that minimizes exposure to heat, dust, direct sunlight and moisture.
- Although your Inverter is moisture resistant, it is NOT waterproof. Flooding the unit with water will cause it to short circuit and could cause personal injury due to electric shock. Never immerse the unit, and avoid any area where standing water might accumulate. Mounting should be in the driest location available.
- Leave a minimum of 2 inches (5 cm) clearance at front and back of the Inverter for proper ventilation. The heavier the load of connected equipment, the more heat will be generated by the unit. Any compartment that contains the Inverter must be properly ventilated with adequate outside airflow to avoid overheating the Inverter.
- Do not install the Inverter directly near magnetic storage media, as this may result in data corruption.
- Do not install near flammable materials, fuel or chemicals.
- **Do not mount unit with its front or rear panel facing down (at any angle). Mounting in this manner will seriously inhibit the unit's internal cooling, eventually causing product damage not covered under warranty.**

Battery Connection Warnings

- Multiple battery systems must be comprised of batteries of identical voltage, age, amp-hour capacity and type.
- Because explosive hydrogen gas can accumulate near batteries if they are not kept well ventilated, your batteries should not be installed (whether for a mobile or stationary application) in a "dead air" compartment. Ideally, any compartment would have some ventilation to outside air.
- Sparks may result during final battery connection. Always observe proper polarity as batteries are connected.
- Do not allow objects to contact the two DC input terminals. Do not short or bridge these terminals together. Serious personal injury or property damage could result.
- Connect the Inverter to the battery with recommended DC fusing (see *Battery Connection*).

Ground Connection Warnings

- Safe operation requires connecting the Inverter's main grounding screw directly to the frame of the vehicle or earth ground.

Equipment Connection Warnings

Use of this equipment in life support applications where failure of this equipment can reasonably be expected to cause the failure of the life support equipment or to significantly affect its safety or effectiveness is not recommended. Do not use this equipment in the presence of a flammable anesthetic mixture with air, oxygen or nitrous oxide.

- You may experience uneven performance results if you connect a surge suppressor, line conditioner or UPS system to the output of the Inverter.

Operation Warnings

- Your Inverter does not require routine maintenance. Do not open the device for any reason. There are no user-serviceable parts inside.
- Potentially lethal voltages exist within the Inverter as long as the battery is connected. During any service work, the battery should therefore be disconnected.
- Do not connect or disconnect batteries while the Inverter is operating. Dangerous arcing may result.

Feature Identification

1 **AC Outlets:** 3 NEMA 5-15R outlets allow you to connect equipment that you would normally plug into a utility outlet.

2 **Illuminated ON/OFF Switch:** When you set the switch to the ON position, the Inverter provides AC power by converting DC power from the connected battery. Set the switch to the OFF position to shut down the Inverter and conserve the battery's charge when you are not using connected equipment. Also set the switch to the OFF position to reset the Inverter if it has shut down due to low battery or overload. See the *Operation* section for more information.

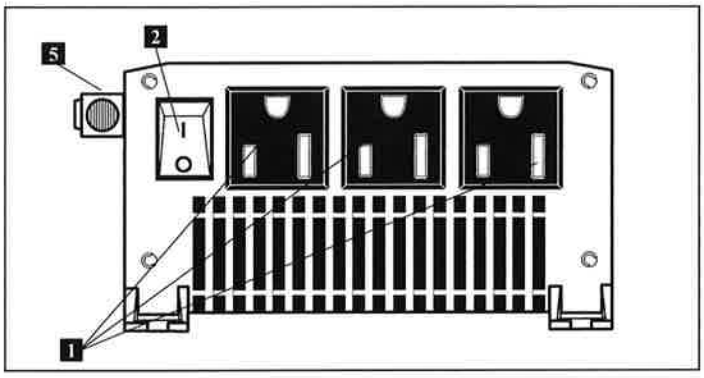
3 **Cooling Fan:** This fan regulates the internal temperature of the Inverter and prolongs service life.

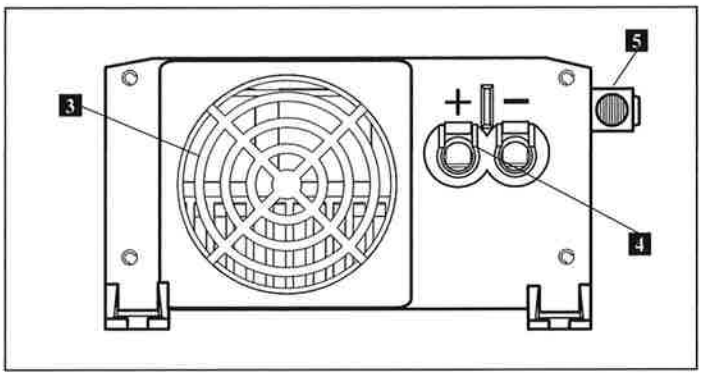
4 **DC Power Terminals:** These positive and negative terminals connect to the battery via user-supplied cabling. See the *Battery Connection* section for instructions.

5 **Main Ground Screw:** Connect to earth ground or to a vehicle or boat grounding system in order to properly ground the Inverter. See the *Battery Connection* section for instructions.

Low Battery Alarm (not shown): An internal circuit automatically detects low battery voltage and shuts down the Inverter to preserve your vehicle's battery. Turn the ON/OFF switch to the OFF position if the alarm sounds. See the *Operation* section for more information.

Overload Alarm (not shown): An internal circuit automatically detects overload conditions and shuts down the Inverter as a protective measure. Turn the ON/OFF switch to the OFF position if the alarm sounds. See the *Operation* section for more information.





Operation

Operating Modes

After mounting and connecting your Inverter according to the instructions in this manual, use the illuminated ON/OFF switch to choose the Inverter's operating mode.

ON: When you set the switch to the ON position, the Inverter provides AC power to connected equipment by converting DC power from your vehicle's battery.

OFF: Set the switch to the OFF position to shut down the Inverter completely, preventing it from drawing power from your vehicle's battery. Also set the switch to the OFF position to reset the Inverter if it has shut down due to low battery or overload.

Resetting the Inverter

The inverter may shut down and cease supplying AC power under certain conditions in order to protect itself, the battery and connected equipment. Follow these instructions to restore normal operation:

- **Low Battery Alarm Reset:** If the Inverter has shut down due to low battery, (1) set the ON/OFF switch to the OFF position, (2) allow the Inverter to cool, (3) start the vehicle's engine to recharge the battery, (4) allow the battery to recharge completely and (5) set the ON/OFF switch to the ON position.
- **Overload Alarm Reset:** If the Inverter has shut down due to overload, (1) set the ON/OFF switch to the OFF position, (2) allow the Inverter to cool, (3) remove the equipment that caused the overload, (4) confirm that any equipment now connected does not exceed the rated wattage of the Inverter and (5) set the ON/OFF switch to the ON position.

Battery Selection

Match Battery Amp-Hour Capacity to Your Application

Select a battery or system of batteries that will provide your Inverter with proper DC voltage and an adequate amp-hour capacity to power your application. Even though Tripp Lite Inverters are highly efficient at DC-to-AC inversion, their rated output capacities are limited by the total amp-hour capacity of connected batteries plus the output of an alternator when one is used.

- **STEP 1) Determine Total Wattage Required**

Add the wattage ratings of all equipment you will connect to your Inverter. Wattage ratings are usually listed in equipment manuals or on nameplates. If your equipment is rated in amps, multiply that number times AC utility voltage to estimate watts. (Example: a drill requires 2.5 amps. 2.5 amps × 120 volts = 300 watts.)

NOTE: Your Inverter will operate at higher efficiencies at about 75% - 80% of nameplate rating.

Example

Tools

300W + 220W + 20W = 540W

Appliances

300W + 140W + 100W = 540W

- **STEP 2) Determine DC Battery Amps Required**

Divide the total wattage required (from Step 1, above) by the battery voltage to determine the DC amps required and multiply by 1.2 to account for conversion losses.

1.2 x 540 watts ÷ 12V = 54 DC Amps

- **STEP 3) Estimate Battery Amp-Hours Required**

Multiply the DC amps required (from Step 2, above) by the number of hours you estimate you will operate your equipment exclusively from battery power before you have to recharge your batteries. This will give you a rough estimate of how many amp-hours of battery power (from one or several batteries) you should connect to your Inverter.

54 DC Amps x 5 Hrs. Runtime = 270 Amp-Hours

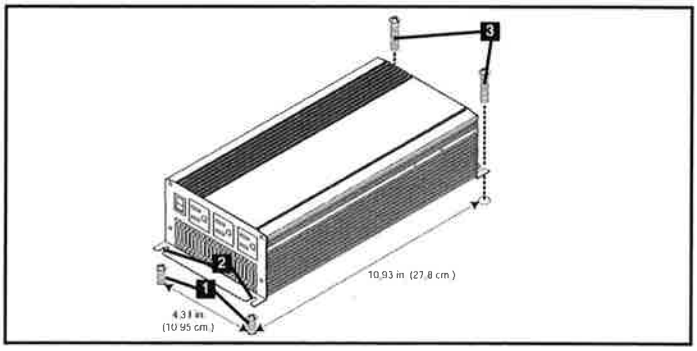
NOTE: Battery amp-hour ratings are usually given for a 20-hour discharge rate. Actual amp-hour capacities are less when batteries are discharged at faster rates. For example, batteries discharged in 55 minutes provide only 50% of their listed amp-hour ratings, while batteries discharged in 9 minutes provide as little as 30% of their amp-hour ratings.

Mounting

WARNING! Mount your Inverter BEFORE DC battery connection. Failure to follow these instructions may lead to personal injury and/or damage to the Inverter and connected systems.

Tripp Lite recommends that you permanently mount your Inverter in the configuration illustrated below. The Inverter features integral mounting brackets on the front and rear of the unit. The user must supply mounting hardware and is responsible for determining if hardware and mounting surface are adequate to support the weight of the unit.

Using the measurements as shown in the diagram install two user-supplied fasteners 1, leaving the heads slightly raised. Slide the unit over the fasteners to engage the mounting bracket slots 2. Tighten fasteners. Install two additional fasteners 3 over the remaining mounting bracket.



Battery Connection

Connect your Inverter to your batteries using the following procedures:

• **Connect DC Wiring:** Though your Inverter is a high-efficiency converter of electricity, its rated output capacity is limited by the length and gauge of the cabling running from the battery to the unit. Use the shortest length and largest diameter cabling (maximum 4 AWG or 5 mm) to fit your Inverter's DC Input terminals. Shorter and heavier gauge cabling reduces DC voltage drop and allows for maximum transfer of current. Your Inverter is capable of delivering peak wattage at up to 200% of its rated continuous wattage output for brief periods of time. Heavier gauge cabling should be used when continuously operating heavy draw equipment under these conditions. Tighten your Inverter and battery terminals to approximately 3.5 Newton-meters of torque to create an efficient connection and to prevent excessive heating at this connection. Insufficient tightening of the terminals could void your warranty. **See Specifications for Recommended Cable Sizing.**

• **Connect Ground:** Using a 12-18 AWG (1-2 mm) wire, directly connect the Main Ground Screw to the vehicle's chassis or earth ground. See the Feature Identification section to locate the Main Ground Screw. All installations must comply with national and local codes and ordinances.

• **Connect Fuse:** Tripp Lite recommends that you connect your Inverter's positive DC Terminal directly to a fuse(s) and fuse block(s) within 45 cm (18 inches) of the battery. The fuse's rating must equal or exceed the Minimum DC Fuse Rating listed in your Inverter's specifications. See Specifications for fuse recommendations. See diagrams below for proper fuse placement.

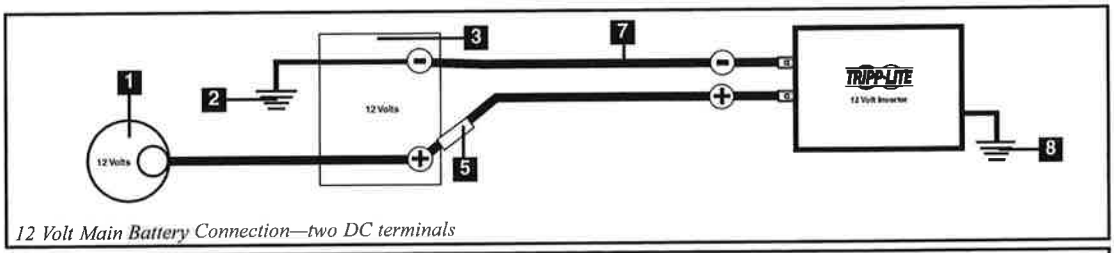
WARNING!

- **Failure to properly ground your Inverter to a vehicle's chassis or earth ground may result in a lethal electrical shock hazard.**
- **Never attempt to operate your Inverter by connecting it directly to output from an alternator rather than a battery or battery bank.**
- **Observe proper polarity with all DC connections.**

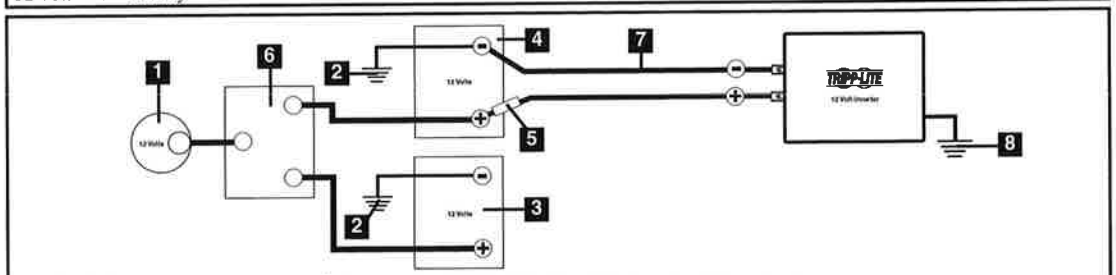
Vehicular Applications

Your Inverter's Nominal DC Input Voltage **must match** the voltage of your battery or batteries—12 Volts in most vehicular applications. It is possible to connect your Inverter to the main battery within your vehicle's electrical system. In many vehicular contexts, the Inverter will be connected to one or more dedicated auxiliary (house) batteries which are isolated from the drive system to prevent possible draining of the main battery.

1. **12 Volt Alternator**
2. **Vehicle Battery Ground**
3. **12 Volt Main Battery**
4. **12 Volt Auxiliary (House) Battery**
5. **UL or CE Approved Fuses & Fuse Blocks (mounted within 18 inches [45 cm] of the battery)**
6. **Battery Isolator**
7. **Large Diameter Cabling, Maximum 4 AWG or 5 mm to Fit Terminals**
8. **12-18 AWG (1-2 mm) Ground Wire to Vehicle Frame or Earth Ground**



12 Volt Main Battery Connection—two DC terminals



12 Volt Main and Auxiliary (House) Battery Connection (Isolated Parallel)—two DC terminals

Service

If you are returning your Inverter to Tripp Lite, please pack it carefully, using the ORIGINAL PACKING MATERIAL that came with the unit. Enclose a letter describing the symptoms of the problem. If the Inverter is within the warranty period, enclose a copy of your sales receipt. To obtain service you must obtain a Returned Material Authorization (RMA) number from Tripp Lite or an authorized Tripp Lite service center.

Maintenance

Your Inverter requires no maintenance and contains no user-serviceable or replaceable parts, but should be kept dry at all times. Periodically check, clean and tighten all cable connections as necessary, both at the unit and at the battery.

Troubleshooting

Please check the following before sending the Inverter in for service:

Symptom	Problem	Correction
Alarm sounds.	Low battery voltage (<10.5 V).	Turn Inverter OFF and run engine to recharge vehicle battery.
	Output overload.	Turn Inverter OFF and remove overload by unplugging high-wattage devices. Load should not exceed your Inverter's maximum continuous output power (see Specifications).
Inverter does not turn ON when power switch is turned ON.	Automatic Inverter shutdown due to low battery voltage (<10V).	Turn Inverter OFF and run engine to recharge vehicle battery.
	Automatic Inverter shutdown due to output overload.	Turn Inverter OFF and remove overload by unplugging high-wattage devices. Load should not exceed your Inverter's maximum continuous output power (see Specifications).
	Blown fuse.	Replace fuse with standard automotive fuse of identical amperage (see Specifications).
Inverter is unable to power connected equipment.	Battery running low. Low battery voltage reduces Inverter power output.	Turn Inverter OFF and run engine to recharge vehicle battery.
	Connected equipment load exceeds Inverter's maximum continuous output power.	Turn Inverter OFF and remove overload by unplugging high-wattage devices. Load should not exceed your Inverter's maximum continuous output power (see Specifications).
Connected equipment experiences audio/video distortion.	Loose connections.	Check and secure all connections.
	Audio/video interference.	Reposition equipment antennas and Inverter.

Specifications

Model:	PV700HF
Output Power (Maximum Continuous Watts)*:	700
Output Power (Peak Watts)**:	1400
Input Voltage (DC):	12V nominal (10-15V)
Output Voltage (AC)/Frequency:	120V / 60 Hz. nominal
Output Waveform:	PWM Sine Wave
Low Battery Voltage Alarm (Volts):	10.5V
Low Battery Voltage Shutdown (Volts):	10V
AC Outlets:	3 (NEMA 5-15R)
Circuit Protection (Minimum DC Fuse Rating):	125-amp fuse (external)
Minimum Battery Cable Size:	6 AWG (4 mm)
Maximum Battery Cable Size:	4 AWG (5 mm)
Ground Cable Size:	12-18 AWG (1-2 mm)
Housing Material:	Extruded Aluminum

** Maximum output power (continuous or peak) only available when vehicle battery is properly charged. Run vehicle engine often to maintain proper charge. ** Peak Output Power is instantaneous.*

Typical Runtimes Before Battery Recharge

Although you can operate your Inverter with your vehicle engine off, best results are usually attained with the engine running. Since the Inverter converts electrical energy (and doesn't produce it), the Inverter's performance is relative to the condition of your vehicle's electrical system (battery, alternator and wiring). If other loads (air conditioner, heater, lights, etc.) are also using power, you may get less runtime.

Model: PV700HF	With Typical *Compact Vehicle (4 cylinder)* Battery, Alternator & Wiring	With Typical *Mid-Size Vehicle (6 cylinder)* Battery, Alternator & Wiring	With Typical *Full-Size Vehicle (8 cylinder)* Battery, Alternator & Wiring
1/2 Load (engine ON)	continuous	continuous	continuous
Full Load* (engine ON)	1-2 hours	2-4 hours	continuous
1/2 Load (engine OFF)	not recommended	1 hours	2 hours
Full Load* (engine OFF)	not recommended	not recommended	1 hour

NOTE: You can maximize output power by following one or more of these steps: (1) connect multiple batteries to the Inverter; (2) keep batteries fully charged by running the vehicle's engine often; (3) install the Inverter as close to the batteries as possible; (4) connect the heaviest gauge cable possible to the Inverter's DC terminals; (5) use the Inverter in a vehicle with a heavy-duty alternator; (6) run the vehicle's engine at higher RPMs.

Battery Wiring Recommendations

For maximum efficiency, keep battery wires as short as possible. For very short runs, 6 AWG (4 mm) wiring may be adequate; for longer runs 4 AWG (5 mm) wiring is recommended. Acceptable output power is directly dependent on wire length: the shorter the wiring, the better the performance.

Limited Warranty and Warranty Registration

Tripp Lite warrants its Inverters to be free from defects in materials and workmanship for a period of 1 year from the date of initial purchase. Tripp Lite's obligation under this warranty is limited to repairing or replacing (at its sole option) any such defective products. To obtain service under this warranty you must obtain a Returned Material Authorization (RMA) number from Tripp Lite or an authorized Tripp Lite service center. Products must be returned to Tripp Lite or an authorized Tripp Lite service center with transportation charges prepaid and must be accompanied by a brief description of the problem encountered and proof of date and place of purchase.

This warranty does not apply to equipment which has been damaged by accident, negligence or misapplication or has been altered or modified in any way, including opening of the unit's casing for any reason. This warranty applies only to the original purchaser who must have properly registered the product within 10 days of purchase.

EXCEPT AS PROVIDED HEREIN, TRIPP LITE MAKES NO WARRANTIES, EXPRESS OR IMPLIED, INCLUDING WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. Some states do not permit limitation or exclusion of implied warranties; therefore, the aforesaid limitation(s) or exclusion(s) may not apply to the purchaser.

EXCEPT AS PROVIDED ABOVE, IN NO EVENT WILL TRIPP LITE BE LIABLE FOR DIRECT, INDIRECT, SPECIAL, INCIDENTAL OR CONSEQUENTIAL DAMAGES ARISING OUT OF THE USE OF THIS PRODUCT, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE. Specifically, Tripp Lite is not liable for any costs, such as lost profits or revenue, loss of equipment, loss of use of equipment, loss of software, loss of data, costs of substitutes, claims by third parties, or otherwise.

Warranty Registration

Visit www.tripplite.com/warranty today to register the warranty for your new Tripp Lite product. You'll be automatically entered into a drawing for a chance to win a FREE Tripp Lite product!*

* No purchase necessary. Void where prohibited. Some restrictions apply. See website for details.

Regulatory Compliance Identification Numbers

For the purpose of regulatory compliance certifications and identification, your Tripp Lite product has been assigned a unique series number. The series number can be found on the product nameplate label, along with all required approval markings and information. When requesting compliance information for this product, always refer to the series number. The series number should not be confused with the marking name or model number of the product.

Tripp Lite follows a policy of continuous improvement. Product specifications are subject to change without notice.

EQUALIZER SYSTEMS

Electric and Hydraulic Lifting Systems

Installation and Operation Guide

March 2008

Hydraulic Trailer Jack

Single Leg: AJ 45 and AJ 70

Installation

Tools Required for Installation

- Ratchet, sockets and wrench set
- Wire cutters/crimpers
- Electric drill and bits
- Screw gun bit
- Welding equipment (if welding leg in place)

Additional Parts Needed for Installation

- #10 gauge power wire (to connect battery +12V to the pump)
- #10 gauge ground wire (to connect battery ground to pump)
- #10 gauge ring terminals
- Loom clips (to secure switch and harness to the trailer)
- Self tapping screws or pop rivets (to secure loom clips)
- Wire ties

Jack Mounting

The AJ style jack is available in "bolt on" or "weld on" configurations (depending on the mounting option ordered for the system). Mounting provisions must be designed with adequate strength to sustain trailer weight and jack lifting capacity. If bolting the jack in place, ensure the use of appropriate size and quantity of mounting hardware. The use of **Grade 8** bolts is recommended. Welding the jack in place requires sound welding practices.

The jack should be mounted so that when the trailer is level (while mounted to the tow vehicle) there is a **minimum** of 10 inches of ground clearance. This is usually achieved by mounting the jack so that the foot is slightly below the bottom edge of trailer. The bottom of the foot pad should not be lower than any other item mounted on the trailer.

Switch Harness and Hydraulic Lines

The AJ jack has been shipped with all necessary switchgear, harnesses and hydraulic lines. These items are specifically engineered to operate your system and should not be altered in any manner. **Modification of any factory-supplied item may result in the denial of all warranty claims**.

Switchgear

The unit may have been provided with a key switch in the switchgear box (depending on the option ordered). If unit is not supplied with a key switch, the +12v battery lead must be fed through a power disconnect switch to fully isolate the system during travel or inactivity. Minimum switch rating must be 60 ampere DC. Switchgear is plugged into the system harness through a weather resistant connector (shown in Fig.1)

Fig.1- Weather Resistant Connector

Battery Connections

Battery Lead (+12volts): Attach a #10-gauge (minimum) wire between the positive 12-volt terminal on the battery and the supplied butt connector (Fig.2) on the power unit "RED" lead.

If circuit protection is required, install a 40 amp (minimum) circuit breaker.

Fig.2- Red Lead= +12V Battery
Black Lead= -12V Battery (Ground)

Pump Ground (-12volts): Attach a #10 gauge (minimum) wire between the negative 12-volt terminal on the battery and the supplied butt connector (Fig.2) on the power unit "BLACK" lead.

It is not acceptable to allow the jack mounting bolts or the weld to be the sole grounding connection.

Reservoir Breather Cap

With the jack installed in the vertical position, remove the 1-1/8 inch round flush mounted fill plug from the top of the reservoir and discard.
Replace with the 1-1/16 inch 6-sided breather cap. Failure to do so will cause the jack to operate erratically. Make sure the reservoir has adequate fluid.

Fluid Level: When the jack is fully retracted, the fluid level in the reservoir should be approximately one inch below the fill cap. If fluid needs to be added, use Dexron III automatic transmission fluid (the same as used for a GM automobile).

Operation

To raise and lower the jack, simply push and hold the rocker switch in the up or down position. The jack will "hold" a position by releasing the switch at any time. There is no need to take the "weight" off the jack when storing the trailer, as the positive check valves in the system will not allow the jack to "bleed down".

Warning: With <u>any</u> hydraulic application, holding any position on a cylinder must be done with safety in mind. Failure in the system may cause the leg to retract or extend on its own. When working under or near the trailer, always use jack stands of appropriate rating to support the weight of the trailer.

The keyed switch <u>must</u> be in the off position and the key must be removed when the jack is not in use and/or when the trailer is in transit. If unit is <u>not</u> supplied with a keyed switch, the +12v battery lead must be fed through a power disconnect switch to fully isolate the system during inactivity or travel. The minimum switch rating must be 60 ampere DC.

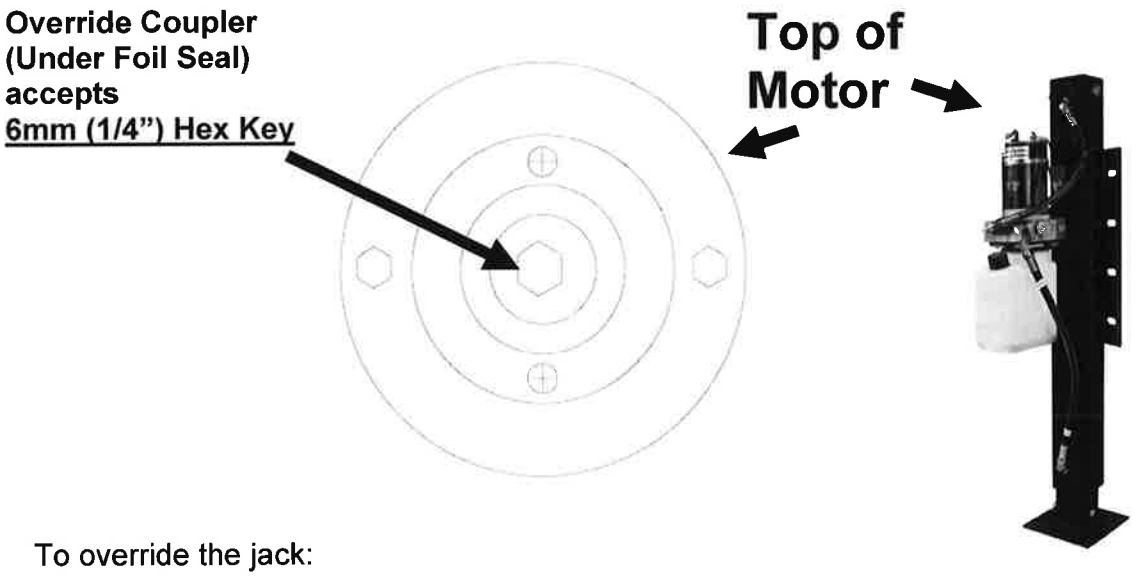
Manual Override

The AJ series single jack has provisions for manual override. The following procedure gives systematic instructions on how to operate the manual override to either retract or extend the jacks.

Required Items

- Reversible Drill- capable of producing a minimum of 2000 r.p.m. for the pump to develop appropriate pressure output.
- 6mm (¼") Hex Key (Allen wrench) bit or driver

The hydraulic power unit is equipped with a manual override output shaft. The override shaft is located on the top of the motor in the center under a foil seal.



To override the jack:

Remove the foil seal from the top of the motor. Use a small flat head screwdriver if necessary.

Insert the 6mm (¼") hex key (Allen) bit on the manual override shaft located at the top of the motor.

To Retract:
Run drill in the **clockwise** direction at 2000 r.p.m. (minimum). The jack will retract.

To Extend:
Run drill in the **counterclockwise** direction at 2000 r.p.m. (minimum). The jack will extend.

Problem Solving

"The jack runs for a few seconds, then stops" The battery is weak or battery capacity is diminished. Charge the battery fully. It may be necessary to "load test" the battery.

"The jack only runs in one direction" Verify appropriate battery voltage. Verify that all wires are attached appropriately. Verify proper function of the switchgear.

"I push the switch and nothing happens" Ensure both positive and negative 12V have adequate connection. Ensure full charge on your battery. The vast majority of calls to the Equalizer Systems Help Desk are related to low battery voltage. Ensure that the unit is properly grounded to the battery with a minimum #10-gauge wire. Check all associated wiring.

"The jack is jerky when retracting" This may be caused by air in the system, low fluid level, or incorrect hose installation. To purge leg, add fluid as necessary and run the leg to full extension and retraction at least twice. If problem persists, call Equalizer Systems for assistance.

Fluid Level: When the jack is <u>fully</u> retracted, the fluid level in the reservoir should be approximately one inch below the fill cap. If fluid needs to be added, use Dexron III automatic transmission fluid (the same as used for a GM automobile).

Avoid the most common issue! Always ensure & verify proper charge on the batteries!

Proper and adequate grounding of the pump is essential!

90% of the phone calls to the Equalizer Systems Help Desk are found to be a weak / discharged battery or poor ground!

If your problem is not listed or persists, call Equalizer Systems at

(800) 846-9659

Please gain prior authorization for warranty service or repair.

Limited Warranty Policy

RV or Vehicle Manufacturer Installed Systems or Components:
1. Only warranty claims with prior written or verbal authorization from Equalizer Systems will be recognized, all other claims will be denied.

2. Equalizer Systems warrants slide out and leveling system components for a period of three years from the date of original sale of the vehicle. This warranty covers defects in material and workmanship only. Equalizer Systems is not liable for any damage due to abuse, neglect, misuse, negligence, misapplication, error of operation, accidental or purposeful damage or damage due to an "act of God" such as, wind or rain damage, flood, lightning or other natural occurrence of the like. Equalizer Systems limited warranty is applicable to the Equalizer Systems components only and does not apply to the vehicle, apparatus or property to which it is attached. Warranty parts will be shipped at no charge if the repair is authorized by an Equalizer Systems representative. Purchased components used in authorized warranty repairs will be reimbursed at the original purchase price.

3. Labor and freight expenses due to warrantable parts defects or workmanship will be reimbursed for a period of one year from the date of original sale of the vehicle. Freight expenses will either be prepaid by Equalizer Systems or reimbursed at the UPS Ground rate only. Any additional shipping charges or requirements are the obligation of the vehicle owner or service center performing the warranty repair. The owner or service center's obligation may include overseas shipping charges, border fees, brokerage fees and any other additional fee of the like.

4. Warranty labor will be reimbursed only for claims that have prior written or verbal authorization from an Equalizer Systems representative. Warranty labor compensation is required to correspond with the "Warranty Parts Replacement Time Guideline" published by Equalizer Systems. Any warranty repair not listed on this guideline will require prior authorization from an Equalizer Systems representative. A reasonable time allowance will be determined by the Equalizer Systems representative. Any warranty repair that is not listed on this guideline that is performed without prior authorization will be denied without exception. Time associated with learning about the repair or excessive diagnostic and installation time will not be reimbursed. Warranty labor will be reimbursed at the authorized service center's published shop rate if the rate is reasonable for that region. Overtime labor will not be reimbursed without exception.

5. Labor, parts and freight credit (if applicable) will be sent after the parts are tested and the warranty claim is validated. Returned parts that are found to be in normal operating condition are not warrantable and will be charged to the owner or service center. Equalizer Systems reserves the right to charge back the service center for labor claim payments previously submitted if the installation of the warranted part is found to be inadequate at a later date.

6. Claims will be denied if the date submitted is greater than 30 days from the repair date.

7. Prior authorization is required before parts may be sent back to Equalizer Systems. A Return Authorization Number is required for items to be accepted.

8. Complete systems are not warranted unless authorized by an Equalizer Systems representative. There are absolutely no exceptions to this clause.

9. This warranty begins upon the original sale date of the vehicle and is transferable, with limitation, to subsequent owners upon furnishing the original sale date of the vehicle and proof of purchase. Only the remainder of the three year parts warranty is applicable. Warranty labor and freight are only applicable to original owner of the vehicle.

10. Equalizer Systems is not liable for loss of time, manufacturing costs, labor material, loss of profits, direct or indirect damages incurred by the vehicle manufacturer.

11. Excessive warranty labor resulting from inadequate access to the Equalizer Systems product will not be reimbursed.

12. Equalizer Systems will not pay a markup on warranty parts unless required by law.

13. Travel expenses, hotel, telephone, fuel or any other expenses of the like are not covered under warranty.

Replacement Parts:
1. Replacement parts are warranted under the same guidelines listed above for the remainder of the original warranty or 90 days, whichever is longer. Proof of warranty repair date and original vehicle purchase date are required.

No additional warranties, expressed or implied, are authorized by Equalizer Systems

This warranty voids all previous issues. Questions concerning this warranty should be directed to:

Equalizer Systems
P.O. Box 668
Elkhart, IN 46515
(800) 846-9659
(574) 266-6083 fax

01/26/06

SUNFORCE®

15 WATT SOLAR 12 VOLT BATTERY CHARGER KIT

User's Manual

15 Watt Solar 12 Volt Battery Charger and 7 Amp Charge Controller Kit
Solar Power Panel
15 Watt/ 1 Amp
Read this manual BEFORE and DURING the use of this product

Specifications and Included Items:
Power Rating: Up to 15Watts
Current: 1Amp @ 15V
Includes 7 Amp Charge Controller, 12 Volt DC Plug, Alligator Clamps, 11.5ft Wire, Solar Panel, 4 Stainless Steel Mounting Screws,

Installation
Solar Panel
Securely place the solar panel in the sun. If permanent installation is desired, use the included four stainless steel mounting screws. To allow for heat dissipation, use a heat-resistant spacer between the panel and the surface, such as the rubber washers included.

7 Amp Charge Controller
Connect to Battery:
Attach the Battery Charge Controller positive (+) wire to the positive (+) battery terminal. Then connect the Battery Charge Controller negative (-) wire to the negative (-) battery terminal. WARNING: The alligator clips MUST be placed on to the battery in the aforementioned order—wrong connections may cause sparking or explosion.

Connect to Solar Panel:
Strip the solar panel wires. Connect positive (+) wire from the solar panel to the positive (+) of the charge controller and the negative (-) wire from the solar panel to the negative (-) of the charge controller. Ensure the connections are secure.

Operation:
The Green LED indicates a full-charged battery. When the battery reaches 14.2 V, the Solar Charge Controller will cut out voltage thereby ensuring no overcharging. Yellow light indicates a battery that is being charged. When the battery reaches 13 V, the Battery Charge Controller will cut-in and begin charging.

NOTE:
This solar panel has a built-in blocking diode that protects the battery from reverse discharge.

FAQ
What type of batteries does this panel charge?
All rechargeable 12V batteries such as those used in Cars, Boats, RV's, Motorcycles and PWC's.

Can I use this solar panel outdoors?
Yes, this solar panel is weatherproof. The weatherproofing includes UV protection and protects from weather effects of -35°F-175°F (-37°C - 79°C).

How long does it take to charge a battery?
This solar panel can generate 1 Amp per hour under ideal conditions. For example, under ideal conditions, the panel will generate 15 Watts x 7 hours x 7 days per week for a total of 735 Watts of power. You can connect multiple panels together to decrease overall charging time. By adding a second panel of the same output you will double your output and half the length of the charge time.

How can I run AC appliances with my Solar Power Panel?

You may connect the battery to a power inverter (sold separately) that converts DC power to AC power and run your appliances from the inverter.

Can I overcharge my battery?

Yes, a Sunforce 7 Amp Solar Charge Controller must be used to prevent battery from overcharging.

Can I extend the wire from my solar panel to the charge controller?

Without loss of power or voltage the 11.5' wire may be extended to a maximum length of 25' with 16 gauge wire. Ensure proper connections.

How do I know if the panel is working?

To measure the voltage of the panel use a voltmeter. The voltage reading should be between 16 and 25 Volts in full sun.

Can I start/drive my vehicle while the unit is connected to the battery?

No, for safety reasons please make sure the panel is not in use while you are using your vehicle.

How many panels can I connect to my 7 Amp Solar Charge Controller?

You can connect up to 105 Watts of Solar Power to the 7 Amp Solar Charge Controller. Panels should be connected in parallel – positive to positive, negative to negative.

When will the Charged Indication light (green) light up?

The charged indication green light will light up when the battery voltage reaches 14.2 Volts and the SCC will prevent the solar panels from overcharging the battery. It is normal for the SCC LED to light on and off as the battery voltage cuts in and out.

NOTES:

• It is normal for both lights to flicker on and off during normal operation. The Battery Charge Controller should be placed within 5' of the battery in a dry, well-ventilated area. This Battery Charge Controller can support up to 105 W and up to 7 A of array current. All connections should be parallel to ensure 12 V system (positive to positive and negative to negative).

TROUBLESHOOTING

Problem:

My panel is not functioning.

- Ensure battery is operational
- Ensure that the distance of the wires is not longer than 25' and all connections are secure.

GENERAL TESTING PROTOCOL

Always test outdoors under optimal sunlight

Test Solar Panel for Voltage

Connect Voltmeter to the panel separately from the charge controller and observe Open Voltage. Open Voltage can range from 16 Volts to 24 Volts.

Test Connection to Charge Controller for Voltage.

Reconnect solar panel, and connect to charge controller as per instructions. Measure the Open Circuit Voltage at the battery side of the charge controller. Open circuit voltage should read 5-10% lower than without charge controller. Open circuit measurement will read between 15 and 23.5.

Connect charge controller to battery
First, disconnect solar panels and connect charge controller to battery. Always connect charge controller to battery first and remove last. Observe polarity – positive to positive and negative to negative.

Reconnect Solar Panels to Charge Controller
If battery voltage is 14.2 or higher, the GREEN light should be on. If battery voltage is between 13 and 14.2, the YELLOW LED should be on. If battery voltage is 13 or lower, the YELLOW LED should be on.

If all testing results within the above indicated ranges, solar system is in acceptable range. If Voltage readings indicate lower ranges, repeat above connections and retest. Finally, it is common to have 12V Battery issues such as dead cells or non-rechargeable battery problems.

Warranty
The 12 Volt Solar Battery Charger is covered by a five year limited warranty. Sunforce Products Inc warrants to the original purchaser that this product is free from defects in materials and workmanship for a period of one year from date of purchase. Power output is warranted for five years from date of purchase.
The 7 Amp Solar Charge Controller is covered by a five year limited warranty. Sunforce Products Inc warrants to the original purchaser that this product is free from defects in materials and workmanship for a period of one year from date of purchase.
To obtain warranty service please contact Sunforce Products for further instruction, at 1-888-478-6435 or email info@sunforceproducts.com. Proof of purchase including date, and an explanation of complaint is required for warranty service.

For more information or technical support
Pour plus d'information ou support technique

1-888-478-6435
www.sunforceproducts.com
info@sunforceproducts.com

Made in China
Fabriqué en Chine

Hose Line FireTrainer® O-100

Operation & Maintenance Manual

KFT™

HOSE LINE FIRETRAINER® O-100

OPERATION AND MAINTENANCE MANUAL

Document No. TM4A1-006-506
Revision: N 05-29-2026

KFT Fire Trainer, LLC
(Formerly known as Kidde Fire Trainers, LLC)
5 Pearl Court, Unit D
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Tel: 1-800-899-1701 EXT. 502

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Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Section 1 SAFETY

IN THE INTEREST OF SAFETY

NOTE

IF THE HOSE LINE FIRETRAINER® O-100 is to be controlled by any of the following optional Kidde control systems, refer to the appropriate HOSE LINE FIRETRAINER® O-100 Control System Users Manual, Section 1 for Safety precautions.

- **MULTI CONTROL**
- **WIRELESS CONTROL**
- **BATTERY POWERED CONTROL**

WARNING • AVERTISSEMENT

1. Read this Manual completely ***BEFORE*** use. If you have any questions please call Kidde Fire Trainers, Inc. at 800 288-3973 EXT 502.
2. Refer to the appropriate Appendix section of this manual for specific Mock-Up installation instructions.
3. **DO NOT USE IN AN ENCLOSED AREA**. The fires generated by this simulator are of relative high intensity and could ignite surrounding materials.
4. Use only **PROPANE** in ***approved*** storage containers for the fire generation of this appliance.
5. Become familiar with the connections and controls.
6. Use only **110-VOLT AC** power source.
7. Ensure that the Propane Supply is kept **"Up-Wind"** a minimum of 40 feet away from the Fire Pan and Pilot Box Assembly.
8. Be careful to keep appliance cords away from Fire Pan Surfaces.
9. Always position the Fire Pan and Pilot Box Assembly such that the electrical, control and gas connections face the propane supply.
10. Ensure that only water is used as the extinguishing agent for this Kidde Fire Trainer.

WARNING • AVERTISSEMENT

USING EXTINGUISHING AGENTS OTHER THAN WATER CAN RESULT IN DIMINISHED TRAINER LIFE AND PERFORMANCE AND ADVERSELY IMPACT SAFE TRAINER OPERATION.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

11. Be sure all parts of the Fire Pan and Pilot Box Assembly are cool before touching them with bare hands, cleaning cloth or cleaning materials. **(Only Use Water For Cleaning)**

12. **DO NOT** leave small children near the Fire Pan and Pilot Box Assembly **AT ANYTIME**.

13. **DO NOT** stand near the Fire Pan and Pilot Box Assembly when initiating flame ignition.

14. **DO NOT** repair or service any part of the Fire Pan and Pilot Box Assembly, unless it is specifically recommended in this guide to do so.

15. **DO NOT** use any props with the Fire Pan that has not been supplied for this purpose by Kidde Fire Trainers, Inc.

15. **DO NOT** complete repair or service as recommended by this guide unless:
    a. Vapor & liquid propane supply valves are shut off and the vapor and liquid propane has been burnt off (Refer to Section 6 of this manual).
    b. All components have had sufficient time to cool.
    c. Electrical power is shut off.

16. **DO NOT** operate the Fire Pan and Pilot Box Assembly unless completely assembled with all component access plates and covers in the secured position.

COLD WEATHER OPERATION GUIDELINES

Kidde Fire Trainers, Inc. strongly recommends that all **OUTDOOR-INDUSTRIAL FIRETRAINER® O-100** systems not be operated for training purposes when the ambient temperature is below 32°F. Operation below this temperature may impact the effectiveness of student training.

As water is used as the extinguishing agent in the training scenarios, it should be apparent that freezing water and ice may become hazardous to trainees and other site personnel. It is with this safety consideration in mind that Kidde Fire Trainers issues this cold weather operation recommendation.

NOTE

THIS RECOMMENDATION DOES NOT AND SHOULD NOT BE CONSTRUED TO IMPLY THAT THE FIRE TRAINER WILL NOT OPERATE BELOW THE KIDDE FIRE TRAINERS RECOMMENDED OPERATING TEMPERATURE.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

⚠ CAUTION

WHEN NOT IN USE THE FIRE PAN/MOCK-UP WATER BATH AND PILOT BOX ASSEMBLY SHOULD BE DRAINED OF WATER AND DURING COLD WEATHER PERIODS, WHEN THE ANTICIPATED AMBIENT TEMPERATURE IS GOING TO BE BELOW 32°F

WHEN TRAINING DURING THE ABOVE MENTIONED CONDITION, FILL WATER BATH AND TOP OF PILOT BOX ASSEMBLY JUST PRIOR TO OPERATION AND DRAIN IMMEDIATELY UPON THE COMPLETION OF TRAINING.

FAILURE TO FOLLOW THESE CAUTIONS MAY RESULT IN SERIOUS DAMAGE TO THE ASSOCIATED EQUIPMENT.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Section 2 OVERVIEW

INTRODUCTION

Kidde's portable **HOSE LINE FIRETRAINER® O-100** can be used as a standalone burner pan to replicate a fuel spill or ground fire in addition to using it with several types of mock-ups to create various fire scenarios.

NOTE

REFER TO THE APPENDIX SECTION OF THIS MANUAL FOR SPECIFIC MOCK-UP INSTALLATION INSTRUCTIONS.

The **FIRETRAINER® O-100** provides a hot, challenging fire to train both new recruits and seasoned firefighters. The **FIRETRAINER® O-100** is a completely portable unit that can be setup and operated by one person. All the fires are controlled by the instructor utilizing a Control Pendant and are easily ignited and turned off. This means greater fire fighter safety and higher numbers of firefighters that can be trained per day.

The **FIRETRAINER® O-100** is constructed of stainless steel, making it robust and easy to maintain, and it burns environmentally acceptable propane. Safety is paramount, with the operator/instructor controlling a "deadman" button on the control pendant that, when released, stops the fire.

The standard **FIRETRAINER® O-100** is composed of the following main assemblies:

- Pilot Box
- Control Pendant
- Fire Pan

NOTE

IF THE HOSE LINE FIRETRAINER® O-100 is to be controlled by any of the following optional Kidde control systems, refer to the appropriate HOSE LINE FIRETRAINER® O-100 Control System Users Manual for an overview.

- **MULTI CONTROL**
- **WIRELESS CONTROL**
- **BATTERY POWERED CONTROL**

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PILOT BOX ASSEMBLY

The Pilot Box Assembly produces the pilot flame and controls the distribution of the liquid and vapor propane, for the ignition of pilot and main flames. See Figures 2-1 and 2-2

CONTROL PENDANT

The Control Pendant is a three-button hand held control device used to activate the pilot flame and the propane vapor and liquid flames from a safe distance. See Figures 2-3.

FIRE PAN ASSEMBLY

The Fire Pan is designed to recreate a flammable liquid spill fire over its 4' x 6' surface. The Fire Pan is constructed of stainless steel and has removable wheels for ease of transport. There are two Acme connectors at one end of the Fire Pan for the hookup of the vapor and liquid propane gas supply hoses from the Pilot box Assembly.

The liquid propane supply gas is distributed through a straight burner element in the Fire Pan. The vapor propane supply gas is distributed through a "U" shaped burner element that has two, quarter turn shut off valves in line. These quarter turn valves allow the burner element to be used for a variety of different props. The operation of the standard spill fire requires that both valves be in the open position. The main gas bubbles up through the water and is ignited by the standing flame from the pilot igniter located on the Pilot Box Assembly. See Figures 2-4 and 2-5.

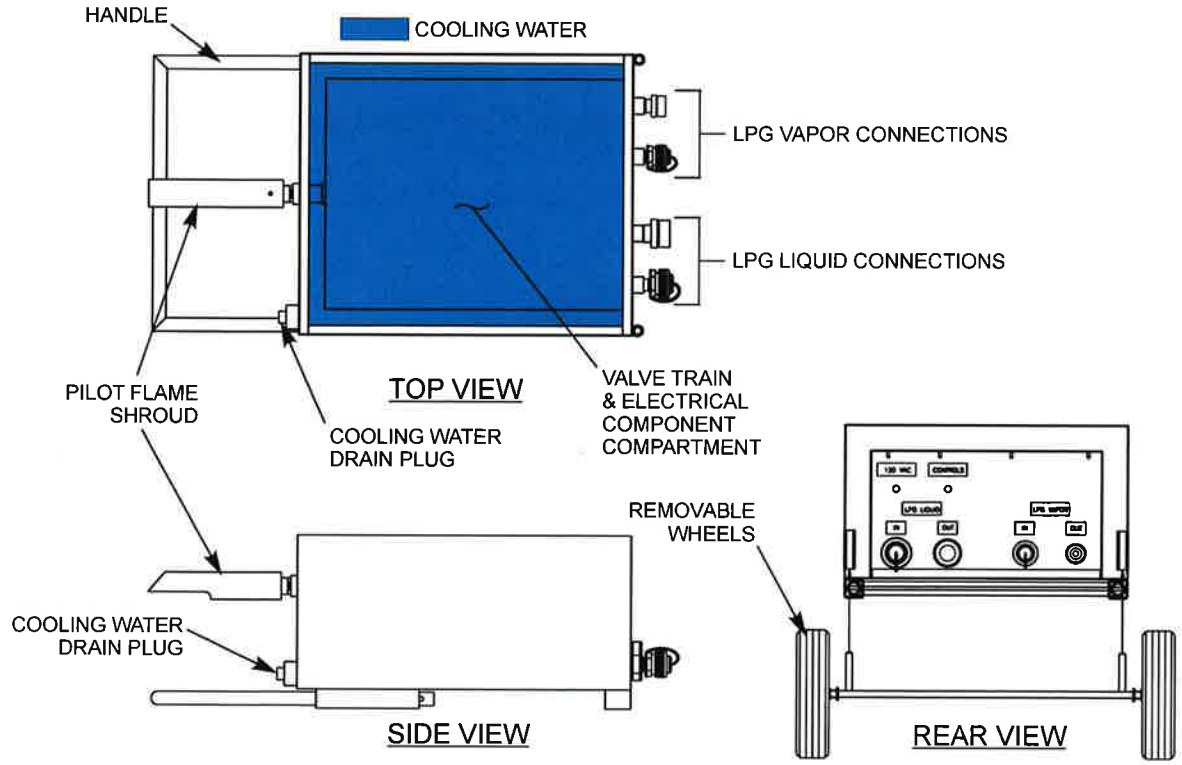

Figure 2-1 - Pilot Box Assembly

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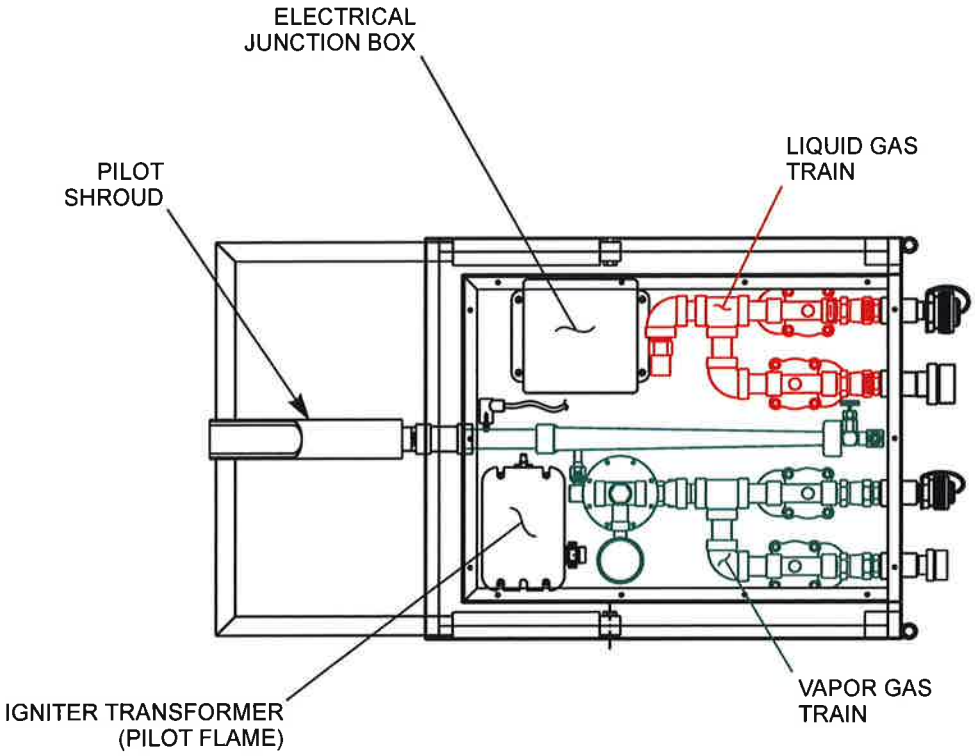

BOTTOM VIEW

Figure 2-2 - Pilot Box Assembly (Bottom View Without Cover)

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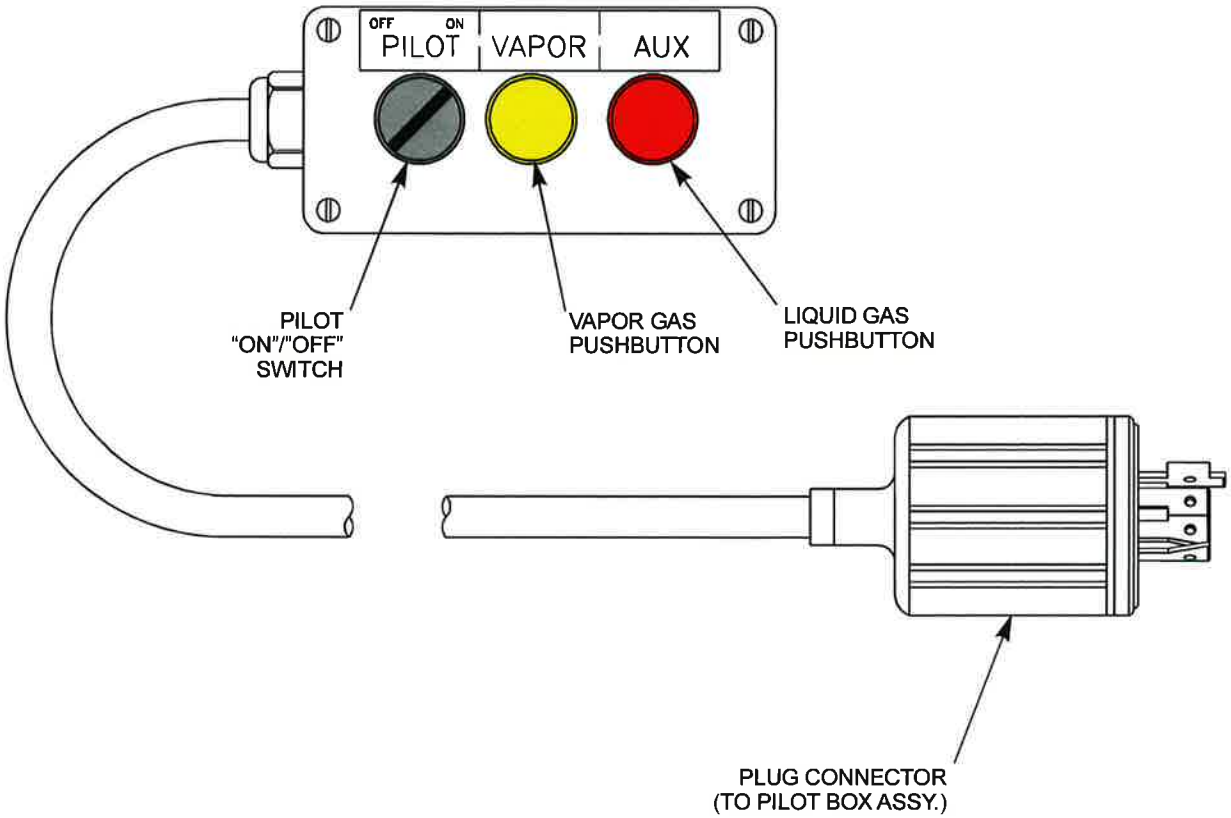


Figure 2-3 – Control Pendant Assembly

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

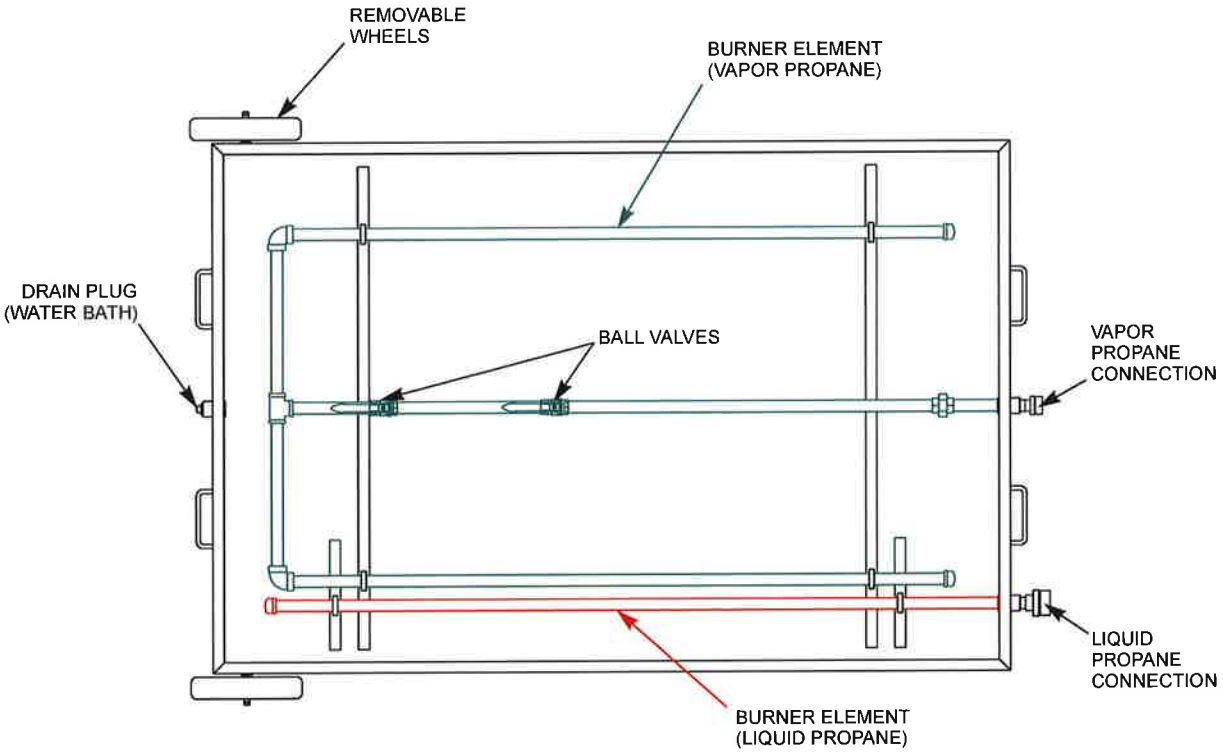


Figure 2-4 – Fire Pan Assembly

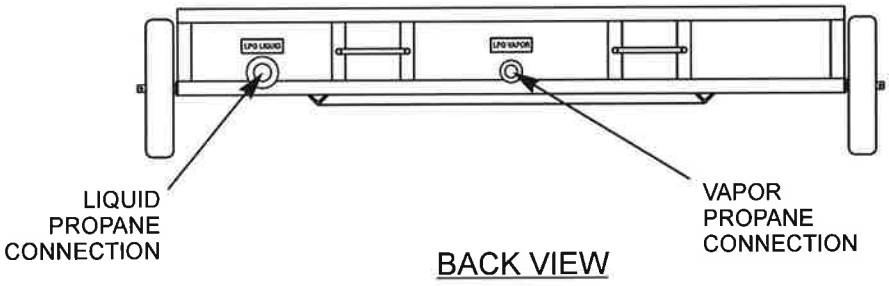


Figure 2-5 – Fire Pan Assembly

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Section 3 SETUP

NOTE

IN ADDITION TO THE SETUP STEPS LISTED BELOW, REFER TO THE APPENDIX SECTION OF THIS MANUAL FOR APPLICABLE MOCK-UP INSTALLATION INSTRUCTIONS.

NOTE

IN ADDITION TO THE SETUP STEPS LISTED BELOW, IF THE HOSE LINE FIRETRAINER® O-100 is to be controlled by any of the following optional Kidde control systems, refer to the appropriate HOSE LINE FIRETRAINER® O-100 Control System Users Manual for setup instructions.

- **MULTI CONTROL**
- **WIRELESS CONTROL**
- **BATTERY POWERED CONTROL**

1. Locate the Pilot Box Assembly outdoors; clear of all ground/aerial obstructions on level ground, 40 feet downwind of the Propane Supply with the gas, electrical and control connections facing the Propane Supply. Place the Fire Pan so that the gas connections face the pilot shroud end of the Pilot Box Assembly as shown in Figure 3-1.

NOTE

IF THE Kidde Multi Control System is used, the Multi Control Stand is considered the propane supply.

2. Remove the wheels from the Pilot Box Assembly and Fire Pan. Store wheels safely away from training area.
3. If a portable electrical power source, ie Generator is used, place it upwind of the Propane Supply and a safe distance away from it.
4. Extend all lines (gas, control, electric) and check for any damage.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF ANY OF THE CONTROL OR ELECTRICAL LINES ARE DAMAGED. DO NOT PROCEED ANY FURTHER IF ANY OF THE GAS HOSES ARE DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

⚠ WARNING • AVERTISSEMENT

WHEN UTILIZING A KIDDE HOSE LINE FIRETRAINER® O-100 Mock-Up be sure TO REFER TO THE APPENDIX SECTION OF THIS MANUAL FOR APPLICABLE MOCK-UP INSTALLATION INSTRUCTIONS AND LPG VAPOR/LIQUID CONNECTION REQUIREMENTS.

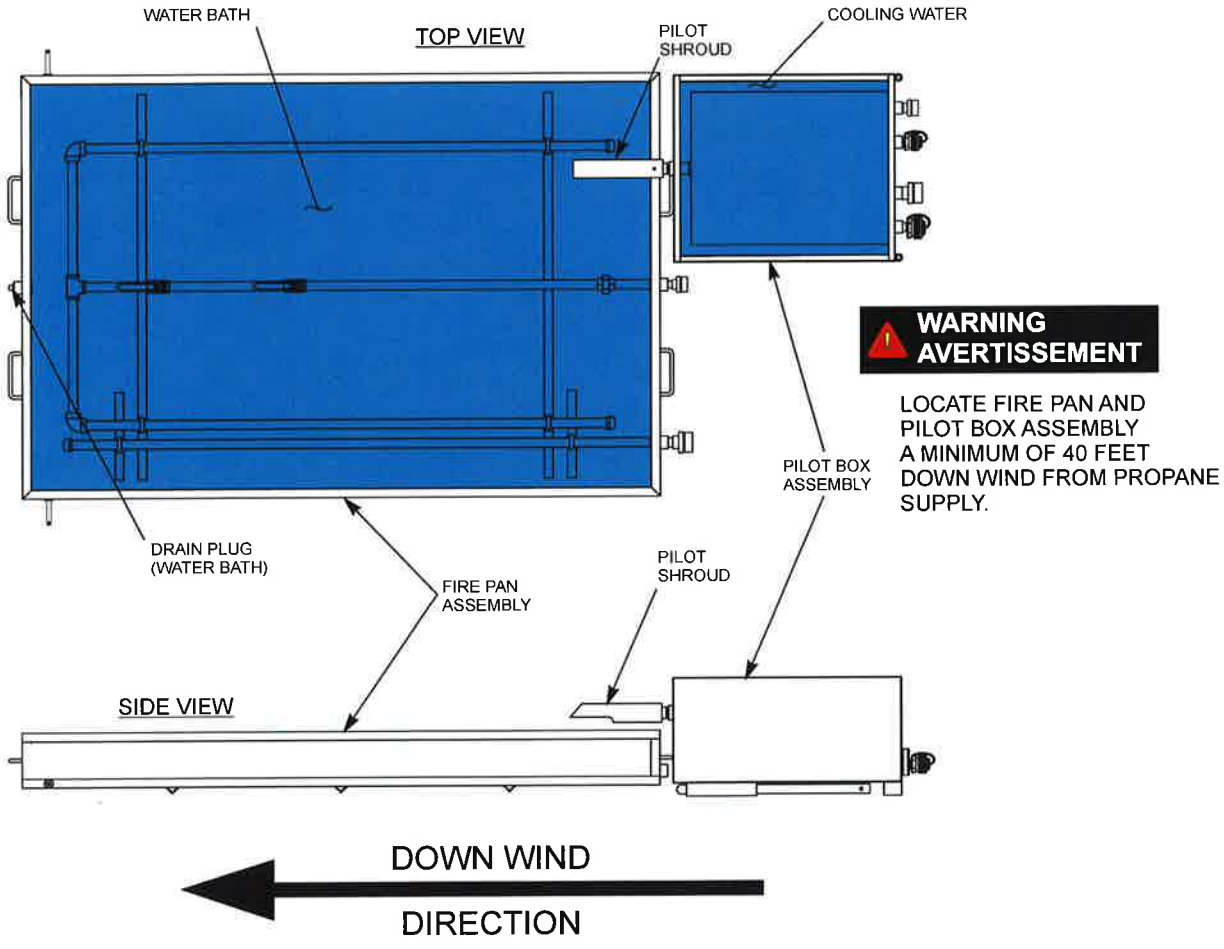


Figure 3-1. Pilot Box & Fire Pan Assembly Setup

5. At the Pilot Box Assembly remove protective thread cap from the 1-1/4" male **LPG VAPOR** connection marked **"IN".** Make sure the 1-1/4" male **LPG VAPOR** connection is free of debris, damage and that the rubber washer is present (see Figure 3-2).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE LPG VAPOR "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

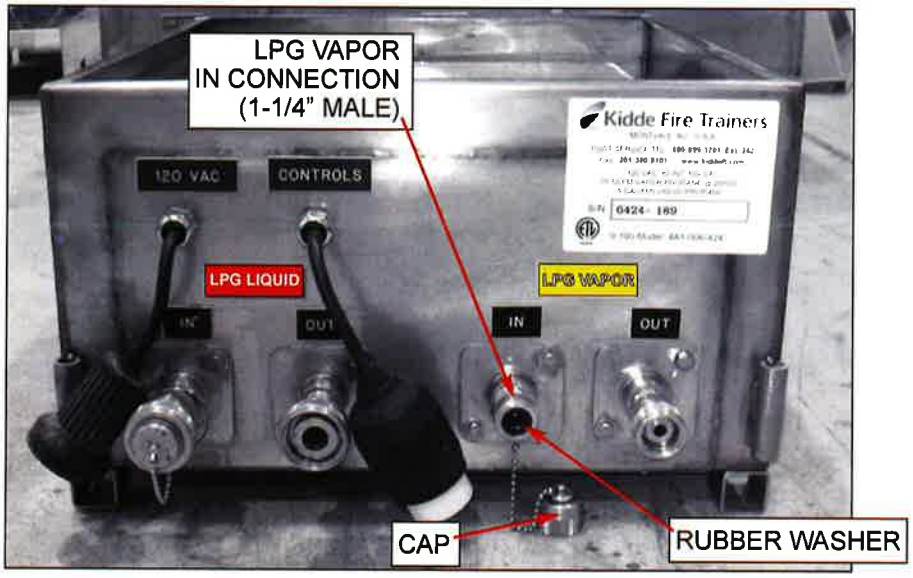


Figure 3-2. Pilot Box Assembly LPG VAPOR "IN" Connection

6. Make sure the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

7. Connect the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose to the 1-1/4" male **LPG VAPOR** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 3-3).

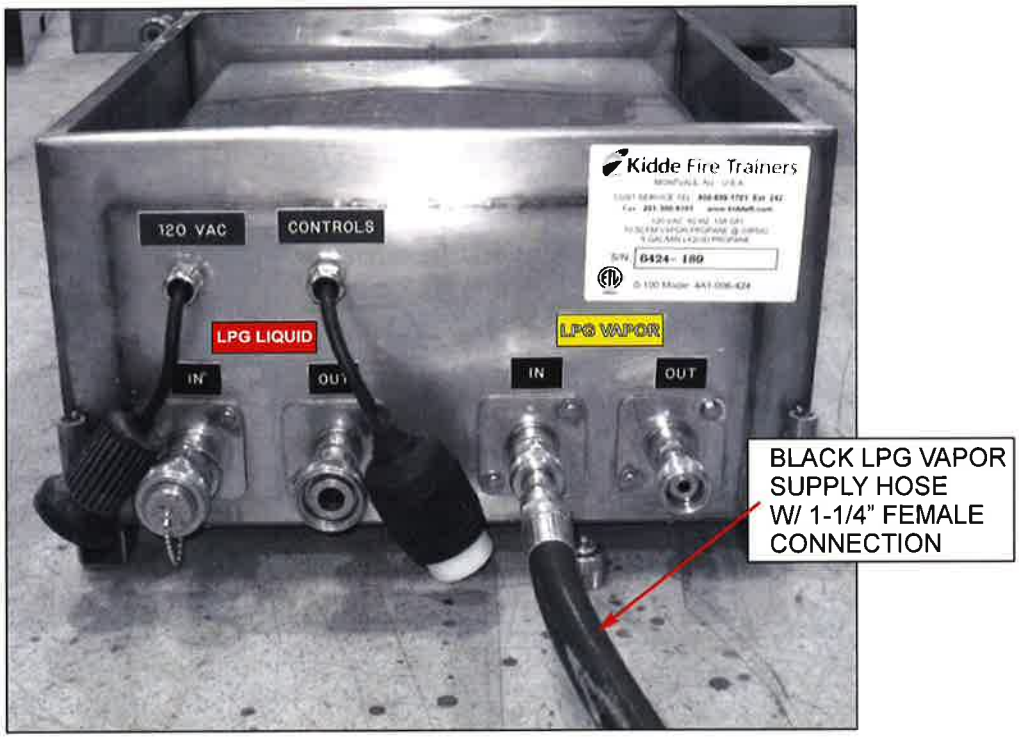


Figure 3-3. Pilot Box Assembly LPG VAPOR "IN" Supply Hose Connection

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

8. At the Pilot Box Assembly remove protective thread cap from the 1-3/4" male **LPG LIQUID** connection marked **"IN".** Make sure the 1-3/4" male **LPG LIQUID** connection is free of debris, damage and that the rubber washer is present (see Figure 3-4).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE LPG LIQUID "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

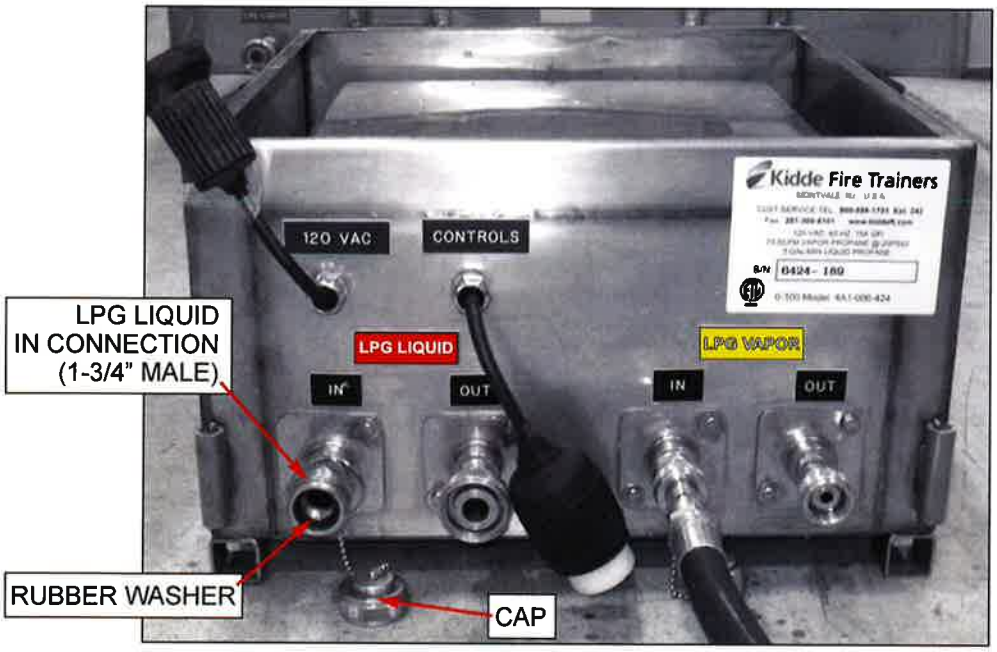


Figure 3-4. Pilot Box Assembly LPG LIQUID "IN" Connection

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

9. Make sure the 1-3/4" female connection on the 50' black rubber LPG Liquid supply hose is free of debris and damage.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

10. Connect the 1-3/4" female connection on the 50' black rubber LPG Liquid supply hose to the 1-3/4" male **LPG LIQUID** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 3-5).

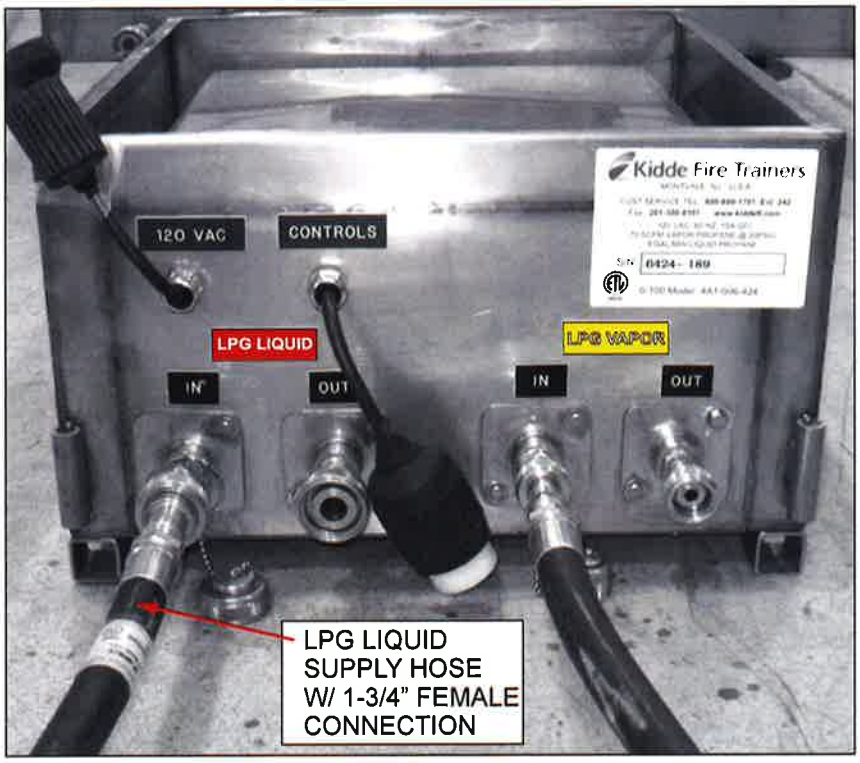


Figure 3-5. Pilot Box Assembly LPG LIQUID "IN" Supply Hose Connection

11. At the Pilot Box Assembly make sure the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE LPG VAPOR "OUT" CONNECTION HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

12. Remove protective thread cap from the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

13. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 3-6).

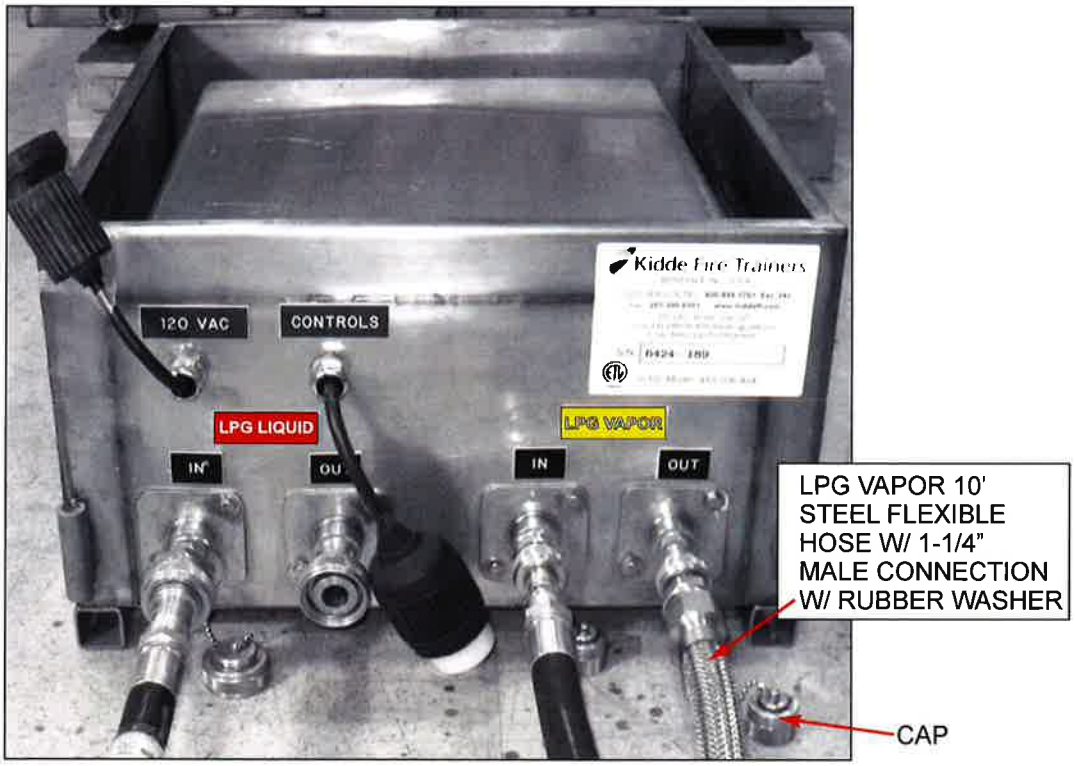


Figure 3-6. Pilot Box Assembly LPG VAPOR "OUT" Hose Connection

14. Remove protective thread cap form the 1-1/4" male connection on the remaining end of the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

15. At the Fire Pan Assembly make sure the 1-1/4" female connection marked **LPG VAPOR** is free of debris and damage.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE CONNECTION MARKED LPG VAPOR HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

16. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the 1-1/4" female connection marked **LPG VAPOR** on the Fire Pan Assembly. Attach fittings hand tight (see Figure 3-7).

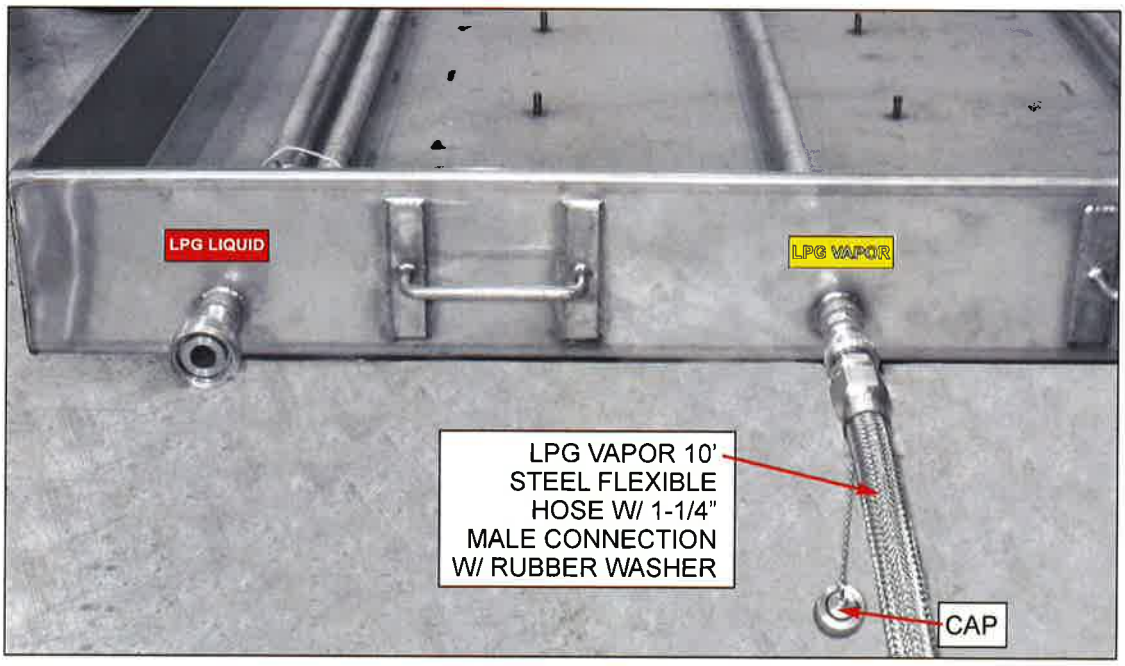


Figure 3-7. Fire Pan Assembly LPG VAPOR Hose Connection

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

17. Remove protective thread cap from the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose. Make sure the 1-3/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

18. At the Pilot Box Assembly make sure the 1-3/4" female **LPG LIQUID** connection marked **OUT** is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION MARKED LPG LIQUID "OUT" HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

19. Connect the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose to the 1-3/4" female **LPG LIQUID** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 3-8).

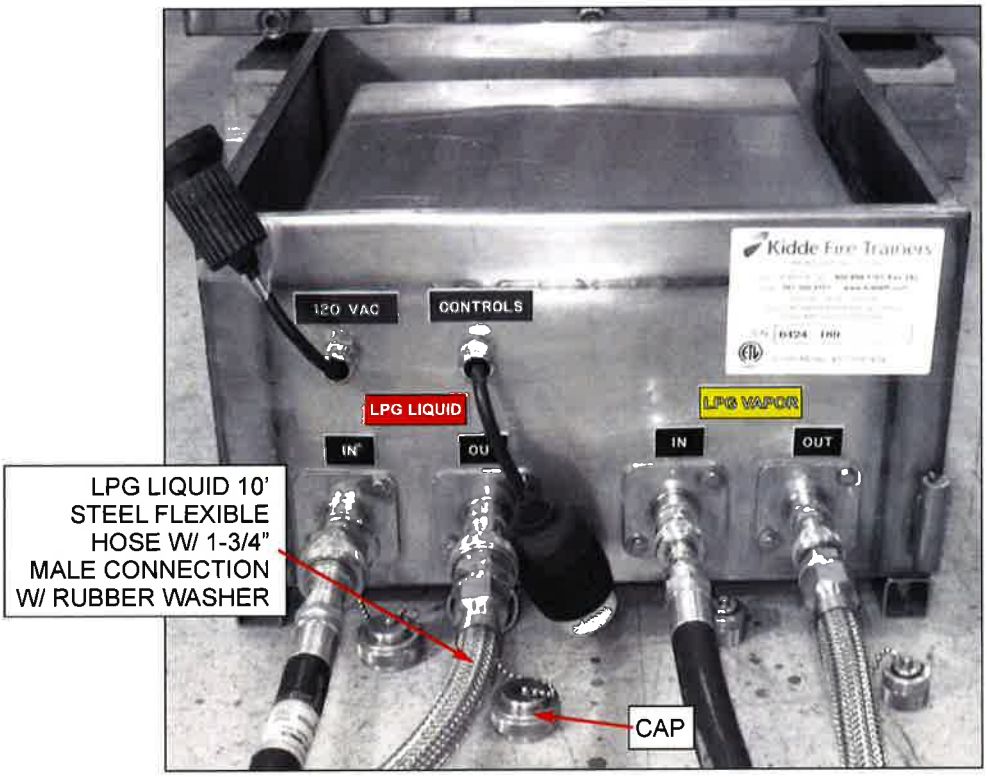


Figure 3-8. Pilot Box Assembly LPG LIQUID "OUT" Hose Connection

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

20. Remove protective thread cap form the 1-3/4" male connection on the remaining end of the 10' flexible stainless steel LPG Liquid supply hose. Make sure the 1-3/4" male connection is free of debris, damage and that the rubber washer is present.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

21. For standard Fire Pan operation, be sure the Standard Liquid Propane Burner Element is installed (see figure 2-4).

22. At the Fire Pan Assembly make sure the 1-3/4" female connection marked **LPG LIQUID** is free of debris and damage.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION MARKED LPG LIQUID HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

23. Connect the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose to the 1-3/4" female connection marked **LPG LIQUID** on the Fire Pan Assembly. Attach fittings hand tight (see Figure 3-9).

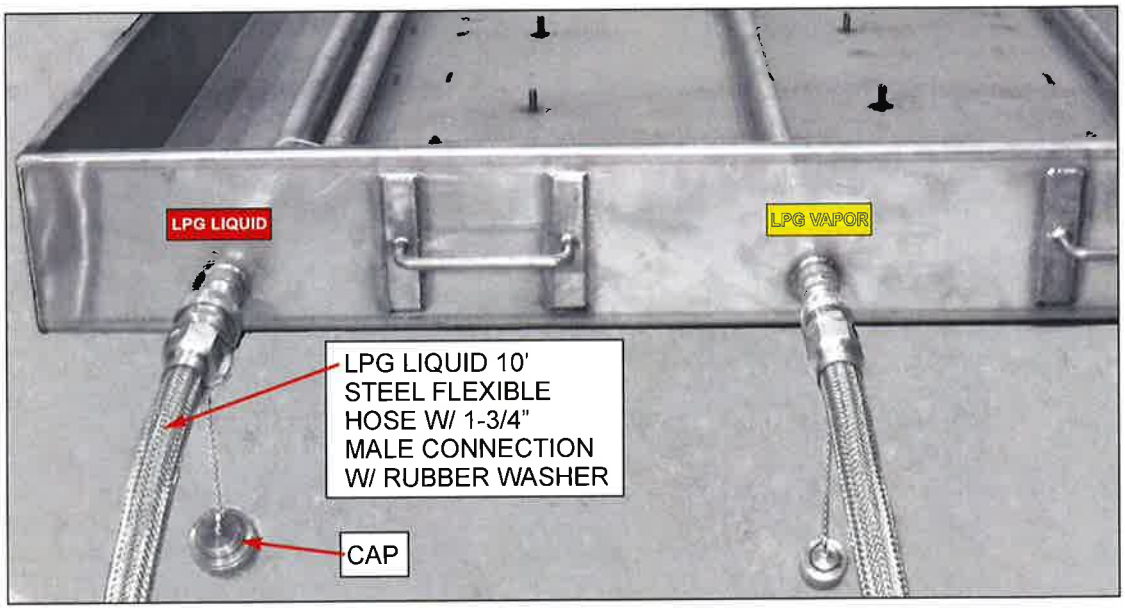


Figure 3-9. Fire Pan Assembly LPG LIQUID Hose Connection

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

⚠ WARNING • AVERTISSEMENT

ALL UNUSED LPG MANIFOLD CONNECTIONS MUST HAVE A CAP SECURELY INSTALLED PRIOR TO CONNECTING LPG SUPPLY HOSES. WHEN UTILIZING A KIDDE HOSE LINE FIRETRAINER® O-100 Mock-Up be sure TO REFER TO THE APPENDIX SECTION OF THIS MANUAL FOR APPLICABLE MOCK-UP INSTALLATION INSTRUCTIONS. FAILURE TO DO THIS WILL CAUSE DEATH OR SERIOUS INJURY.

24. Attach the Electric Supply cord to the connection marked **"120 VAC"** on Pilot Box Assembly (see Figure 3-10).
25. Attach the Control Pendant cable to the connection marked **"CONTROLS"** on Pilot Box Assembly (see Figure 3-10).
26. Open or close the Fire Pan quarter turn valves on the main burner element in accordance with the prop being utilized (see Figure 2-4).
27. Ensure that drain plug is installed in the Fire Pan (see Figure 2-4).
28. Fill Fire Pan with water until water overflows (see Figure 3-1).
29. Ensure that drain plug is installed in the Pilot Box Assembly (see Figure 2-1).
30. Fill Pilot Box Assembly top with water until it overflows (see Figure 3-1).

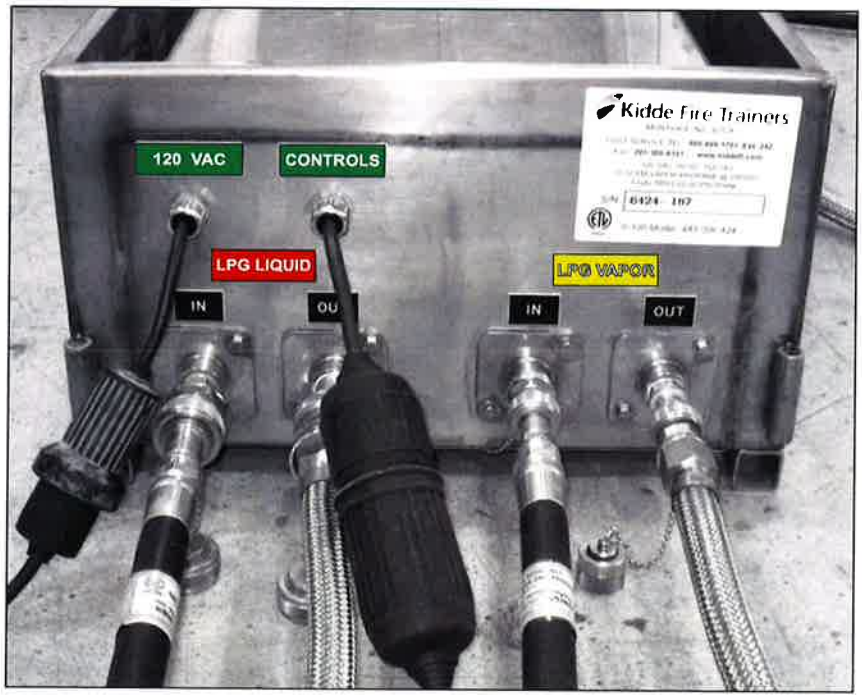


Figure 3-10. Electrical Supply & Control Pendant Connections

NOTE

IF THE HOSE LINE FIRETRAINER® O-100 is to be controlled by the Kidde Multi Control System, refer to the HOSE LINE FIRETRAINER® O-100 Multi Control System Users Manual, Sections 3.1 & 3.2.

31. Using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088), connect the remaining end of the 50' black rubber LPG vapor supply hose from the Pilot Box Assembly **LPG VAPOR** connection marked **"IN"**, to **LPG Vapor Outlet** on the Propane Supply. Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male adaptor connection. Attach fittings hand tight.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER LPG VAPOR SUPPLY HOSE IS CONNECTED TO THE LPG VAPOR OUTLET ON THE PROPANE SUPPLY TANK. FAILURE TO DO THIS WILL CAUSE DEATH OR SERIOUS INJURY.

32. Using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088), connect the remaining end of the 50' black rubber LPG liquid supply hose from the Pilot Box Assembly **LPG LIQUID** connection marked **"IN"**, to the **LPG LIQUID Output** on the Propane Supply. Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male adaptor connection. Attach fittings hand tight.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER LPG LIQUID SUPPLY HOSE IS CONNECTED TO THE LPG LIQUID OUTLET ON THE PROPANE SUPPLY TANK. FAILURE TO DO THIS WILL CAUSE DEATH OR SERIOUS INJURY.

NOTE

IF THE HOSE LINE FIRETRAINER® O-100 is to be controlled by the Kidde Battery Powered Control System, refer to the HOSE LINE FIRETRAINER® O-100 Battery Powered Control System Users Manual, Sections 3.1 & 3.2.

IF THE HOSE LINE FIRETRAINER® O-100 is to be controlled by the Kidde Wireless Control System, refer to the HOSE LINE FIRETRAINER® O-100 Wireless Control System Users Manual, Sections 3.1 & 3.2.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Section 4 STARTUP

PILOT BOX VALVES OPERATIONAL TEST

Prior to operating the trainer the Pilot Box Assembly Pilot, Vapor and Liquid solenoid valves must be operationally tested.

NOTE

WHEN PERFORMING THIS TEST WITH THE KIDDE WIRELESS CONTROL SYSTEM, REFER TO THE WIRELESS CONTROL SYSTEM USERS MANUAL, SECTIONS 3.1, 3.2 & 3.3.

WHEN PERFORMING THIS TEST WITH THE KIDDE BATTERY POWERED CONTROL SYSTEM, REFER TO THE BATTERY POWERED CONTROL SYSTEM USERS MANUAL, SECTIONS 3.1, 3.2 & 3.3.

WHEN PERFORMING THIS TEST WITH THE KIDDE MULTI CONTROL SYSTEM, REFER TO THE O-100 MULTI CONTROL SYSTEM USERS MANUAL, SECTIONS 3.1, 3.2 & 3.3.

1. Be sure "PILOT" ON/OFF Button on the Instructor Control Pendant is in the OFF position. Attach the Instructor Control Pendant to the appropriate plug on the Pilot Box Assembly.
2. Attach the electric supply cord from the Pilot Box assembly to an outdoor 120V power GFIC receptacle.
3. Turn the "PILOT" Button on the Control Pendant to the ON position and listen for the solenoid valve to open.
4. Once it has been confirmed that the Pilot solenoid valve is operating, push and hold the "VAPOR" button on the Control Pendant and listen for the solenoid valve to open.
5. Once it has been confirmed that the Vapor solenoid valve is operating, while holding the "VAPOR" button in, push and hold the "AUX" button and listen for the solenoid valve to open.
6. Once it has been confirmed that the Liquid solenoid valve is operating, release the "VAPOR" & "AUX" buttons and turn the "PILOT" Button on the Control Pendant to the OFF position
7. Disconnect the Pilot Box assembly electric supply cord from the outdoor 120V power GFIC receptacle.

⚠ WARNING • AVERTISSEMENT

IF ANY OF THE SOLENOID VALVES DID NOT OPERATE, DO NOT USE PILOT BOX ASSEMBLY UNTIL ALL SOLENOID VALVES ARE OPERATIONAL.

8. Once it has been confirmed that the Pilot, Vapor and Liquid solenoid valves are operational continue to **PILOT FLAME TEST**.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

PILOT FLAME TEST

Prior to operating the trainer the Pilot Box Assembly Pilot, Vapor and Liquid solenoid valves must be operationally tested.

NOTE

WHEN PERFORMING THIS TEST WITH THE KIDDE WIRELESS CONTROL SYSTEM, REFER TO THE WIRELESS CONTROL SYSTEM USERS MANUAL, SECTIONS 3.1,3.2 & 3.4.

WHEN PERFORMING THIS TEST WITH THE KIDDE BATTERY POWERED CONTROL SYSTEM, REFER TO THE BATTERY POWERED CONTROL SYSTEM USERS MANUAL, SECTIONS 3.1, 3.2 & 3.4.

WHEN PERFORMING THIS TEST WITH THE KIDDE MULTI CONTROL SYSTEM, REFER TO THE O-100 MULTI CONTROL SYSTEM USERS MANUAL, SECTION 3.1, 3.2 & 3.4.

1. Open the vapor propane supply valve.
2. Assure there are no leaks in hose and at all connections.

⚠ WARNING • AVERTISSEMENT

IF A LEAK IS DETECTED TURN PROPANE SUPPLY VALVE OFF IMMEDIATELY. MAKE THE APPROPRIATE REPAIRS. DO NOT USE TRAINER UNTIL ALL LEAKS ARE ELIMINATED.

3. Attach the electric supply cord from the Pilot Box assembly to an outdoor 120V power GFIC receptacle.
4. Turn the "PILOT" Button on the Control Pendant to the ON position. The Pilot may take a few seconds to ignite initially because it takes a while for the gas to get through the lines.
5. Once Pilot Flame has ignited Turn the "PILOT" Button on the Control Pendant to the OFF position.
6. Close vapor propane supply valve SLOWLY.
7. Disconnect the Pilot Box assembly electric supply cord from the outdoor 120V power GFIC receptacle.

⚠ WARNING • AVERTISSEMENT

IF THE PILOT FLAME DID NOT OPERATE, DO NOT USE PILOT BOX ASSEMBLY UNTIL PILOT FLAME IS OPERATIONAL.

8. Once it has been confirmed that the Pilot flame is operational continue to **LIQUID PROPANE LEAK CHECK**.

LIQUID PROPANE LEAK CHECK

WARNING • AVERTISSEMENT

THE PILOT BOX VALVES OPERATIONAL TEST AND THE PILOT FLAME TEST PROCEDURES MUST BE SUCCESSFULLY COMPLETED PRIOR TO PERFORMING THE LIQUID PROPANE LEAK CHECK!

DO NOT PERFORM NEXT STEP UNTIL YOU VERIFY THAT THERE IS A HOSE CONNECTED FROM THE APPLICABLE LIQUID PROPANE SUPPLY OUTLET CONNECTION TO THE PILOT BOX ASSEMBLY LPG LIQUID PROPANE CONNECTION MARKED "IN".

NOTE

WHEN PERFORMING THIS TEST WITH THE KIDDE WIRELESS CONTROL SYSTEM, REFER TO THE WIRELESS CONTROL SYSTEM USERS MANUAL, SECTIONS 3.1,3.2 & 3.5.

WHEN PERFORMING THIS TEST WITH THE KIDDE BATTERY POWERED CONTROL SYSTEM, REFER TO THE BATTERY POWERED CONTROL SYSTEM USERS MANUAL, SECTIONS 3.1, 3.2 & 3.5.

WHEN PERFORMING THIS TEST WITH THE KIDDE MULTI CONTROL SYSTEM, REFER TO THE O-100 MULTI CONTROL SYSTEM USERS MANUAL, SECTION 3.1, 3.2 & 3.5.

1. **SEE ABOVE WARNING PRIOR TO PERFORMING THIS STEP!** At the propane supply tank, open the main liquid propane supply valve **SLOWLY** until line pressure balances, then open main Liquid Propane Supply valve completely.
2. Assure there are no leaks in hose and at all connections.
3. Close liquid propane supply valve.

WARNING • AVERTISSEMENT

IF A LEAK IS DETECTED TURN PROPANE SUPPLY VALVE OFF IMMEDIATELY. MAKE THE APPROPRIATE REPAIRS. DO NOT USE TRAINER UNTIL ALL LEAKS ARE ELIMINATED.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

STARTUP

1. Run (gas, control and electric) lines straight back from Pilot Box Assembly.
2. Install cable/hose covers. Refer to figure 4-1.

⚠ CAUTION

DO NOT OPERATE O-100 TRAINER WITHOUT BOTH CABLE/HOSE COVERS IN PLACE! REFER TO FIGURE 4-1.

⚠ WARNING • AVERTISSEMENT

ONLY WHEN ALL THE ABOVE OPERATIONAL AND LEAK CHECKS HAVE BEEN SUCCESSFULLY PERFORMED YOU ARE READY TO TRAIN.

3. At start of training scenario, ensure that no one is within 25 feet of the Fire Pan.
4. Attach the electric supply cord from the Pilot Box assembly to an outdoor 120V power GFIC receptacle.
5. Decide if you are going to need Vapor or Vapor and Liquid for the training scenario to be conducted. Open the appropriate Propane Supply valve(s) SLOWLY.

NOTE

WHEN USING THE WIRELESS CONTROL SYSTEM, REFER TO THE WIRELESS CONTROL SYSTEM USERS MANUAL, SECTIONS 3.1, 3.2 & 3.6.

WHEN USING THE KIDDE BATTERY POWERED CONTROL SYSTEM, REFER TO THE BATTERY POWERED CONTROL SYSTEM USERS MANUAL, SECTIONS 3.1, 3.2 & 3.6.

WHEN USING THE KIDDE MULTI CONTROL SYSTEM, REFER TO THE O-100 MULTI CONTROL SYSTEM USERS MANUAL, SECTION 3.1, 3.2 & 3.6.

6. Only when above conditions are satisfied, turn the "PILOT" Button on the Control Pendant to the ON position. The Pilot may take a few seconds to ignite initially because it takes a while for the gas to get through the lines.

⚠ WARNING • AVERTISSEMENT

PRIOR TO PERFORMING THE NEXT STEP BE SURE THAT PILOT HAS IGNITED.

7. **SEE ABOVE WARNING PRIOR TO PERFORMING THIS STEP!** Once Pilot Flame has ignited, the vapor main flame for the vapor burner element can be ignited by pushing and holding the "VAPOR" button on the Control Pendant.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

WARNING • AVERTISSEMENT

PRIOR TO PERFORMING THE NEXT STEP BE SURE THAT FLAME FROM VAPOR BURNER ELEMENT IS VISIBLE.

8. **SEE ABOVE WARNING PRIOR TO PERFORMING THIS STEP!** Once the vapor flame is visible the AUX button can be pressed to ignite the liquid propane that supplies the liquid propane burner element

NOTE

WHEN THE VAPOR/AUX. BUTTON IS RELEASED THE CORRESPONDING SOLENOID VALVE, IN THE PILOT BOX ASSY., CLOSES AND GAS FLOW TO THE FIRE PAN STOPS. THE FLAMES WILL CONTINUE TO BURN FOR A SHORT PERIOD DUE TO THE RESIDUAL GAS IN THE SUPPLY LINES.

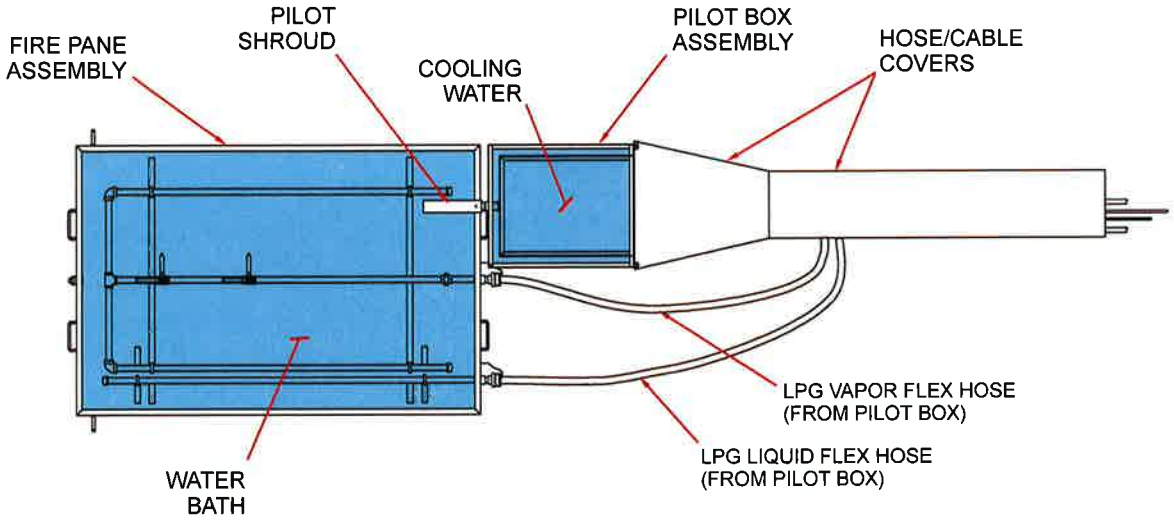


Figure 4-1 – Hose/Cable Covers

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Section 5 OPERATION

WARNING • AVERTISSEMENT

ONLY WHEN ALL THE ABOVE OPERATIONAL AND LEAK CHECKS HAVE BEEN SUCCESSFULLY PERFORMED YOU ARE READY TO TRAIN.

NOTE

WHEN USING THE WIRELESS CONTROL SYSTEM, REFER TO THE WIRELESS CONTROL SYSTEM USERS MANUAL, SECTIONS 3.1, 3.2 & 3.6.

WHEN USING THE KIDDE BATTERY POWERED CONTROL SYSTEM, REFER TO THE BATTERY POWERED CONTROL SYSTEM USERS MANUAL, SECTIONS 3.1, 3.2 & 3.6.

WHEN USING THE KIDDE MULTI CONTROL SYSTEM, REFER TO THE O-100 MULTI CONTROL SYSTEM USERS MANUAL, SECTION 3.1, 3.2 & 3.6.

1. The control of the fire rests with the instructor. The instructor can shut down the fires if an emergency arises by releasing the VAPOR/AUX. button(s) on the Control Pendant.

NOTE

WHEN THE VAPOR/AUX. BUTTON IS RELEASED THE CORRESPONDING SOLENOID VALVE, IN THE PILOT BOX ASSY., CLOSES AND GAS FLOW TO THE FIRE PAN STOPS. THE FLAMES WILL CONTINUE TO BURN FOR A SHORT PERIOD DUE TO THE RESIDUAL GAS IN THE SUPPLY LINES.

2. All training scenarios should be explained fully to the students and the explanation should include a description of the Fire Pan and its operation.

3. The Constant Pilot cannot be extinguished by the application of agent so there will always be a source of ignition for the fuel being released. On the Control Pendant there is a pilot switch. Turn this switch to the ON position. Once the switch is turned on take a look at the pilot at the control box to visually confirm that the pilot is running.

4. Main flame is controlled by the push buttons on the Control Pendant that is marked either " VAPOR" or " AUX. " = Liquid. The Vapor Flame must be started first.

NOTE

WHEN THE VAPOR/AUX. BUTTON IS RELEASED THE CORRESPONDING SOLENOID VALVE, IN THE PILOT BOX ASSY., CLOSES AND GAS FLOW TO THE FIRE PAN STOPS. THE FLAMES WILL CONTINUE TO BURN FOR A SHORT PERIOD DUE TO THE RESIDUAL GAS IN THE SUPPLY LINES.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

5. When the students have successfully performed the proper fire fighting technique the instructor should turn off the main flame by releasing the appropriate push button on the Control Pendant, but leave the pilot running. When the main flame is fully extinguished turn the Control Pendant "PILOT" switch to the OFF position.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Section 6 SHUTDOWN & DISASSEMBLY

1. At the end of a training session there will still be Fuel in the hoses from the Propane Supply Tank to the Fire Pan. This will have to be burnt off before the equipment is taken apart. It is important that this procedure be followed even if you are not taking equipment apart but are just shutting down at the end of a training session.

NOTE

WHEN PERFORMING THIS GAS BURN OFF PROCEDURE WITH THE WIRELESS CONTROL SYSTEM, REFER TO THE WIRELESS CONTROL SYSTEM USERS MANUAL, SECTIONS 3.1 & 3.7.

WHEN PERFORMING THIS GAS BURN OFF PROCEDURE WITH THE KIDDE BATTERY POWERED CONTROL SYSTEM, REFER TO THE BATTERY POWERED CONTROL SYSTEM USERS MANUAL, SECTIONS 3.1 & 3.7.

WHEN PERFORMING THIS GAS BURN OFF PROCEDURE WITH THE O-100 Multi Control System, REFER TO THE O-100 Multi Control System USERS MANUAL, SECTIONS 3.1 & 3.7.

2. Close Liquid Propane valve at that Propane Storage Tank.

3. Start the Pilot by turning the "PILOT" switch to the **ON** position. Verify that Pilot ignited.

⚠ WARNING • AVERTISSEMENT

PRIOR TO PERFORMING THE NEXT STEP BE SURE THAT PILOTS HAVE IGNITED.

4. **SEE ABOVE WARNING PRIOR TO PERFORMING THIS STEP!** Once Pilot Flame has ignited, push and hold the "VAPOR" button on the Control Pendant.

⚠ WARNING • AVERTISSEMENT

PRIOR TO PERFORMING THE NEXT STEP BE SURE THAT FLAME FROM VAPOR BURNER ELEMENT IS VISIBLE.

5. **SEE ABOVE WARNING PRIOR TO PERFORMING THIS STEP!** Once the vapor flame is visible press the "AUX" button to ignite the liquid propane that supplies the liquid propane burner element.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

6. Leave Pilot and vapor/liquid propane flames on for 2 minuets, at which time you should see the flame height reduce, indicating that the residual Liquid gas has burned off.

7. When all the residual Liquid propane has burned off, release "VAPOR"/"AUX" button, turn "PILOT" switch to the **OFF** position.

8. Close Vapor Propane valve at that Propane Storage Tank.

9. Start the Pilot by turning the "PILOT" switch to the **ON** position. Verify that Pilot ignited.

WARNING • AVERTISSEMENT

PRIOR TO PERFORMING THE NEXT STEP BE SURE THAT PILOTS HAVE IGNITED.

10. **SEE ABOVE WARNING PRIOR TO PERFORMING THIS STEP!** Once Pilot Flame has ignited, push and hold the "VAPOR" button on the Control Pendant.

11. Leave Pilot and Vapor main flame on until the residual Vapor gas is burned off and Pilot and Vapor main flames self-extinguish.

12. When ALL flames have self-extinguished, release "VAPOR" button, turn "PILOT" switch to the **OFF** position.

13. Allow Pilot Box Assembly, Fire Pan/Burn Pan and any mock-ups to cool down before working on them.

14. Unhook ALL Liquid/Vapor propane gas hoses from Propane Supply connection, Pilot Box Assembly, Fire Pan and Multi Control System (if used). Inspect hoses for damage. Safely store all hoses.

15. Install protective thread caps on the Liquid/Vapor Propane IN connections on the Pilot Box Assembly and Multi Control System (if used).

16. Unhook the control pendant and Inspect for damage. Safely store.

17. Unhook the electrical and control cables and Inspect for damage. Safely store cables. Safely store Multi Control System (If used). Safely store Battery Powered Control System (if used). Safely store Wireless Control System (if used)

18. Drain water from the Pilot Box assembly and Fire/Burner Pans by removing drain plugs, wash down as required. **ONLY USE WATER**. Drain any residual water.

19. Install wheels on Fire Pan and Pilot Box and store all Equipment in a safe location.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

⚠ CAUTION

WHEN NOT IN USE THE FIRE PAN WATER BATH AND PILOT BOX ASSEMBLY SHOULD BE DRAINED OF WATER AND DURING COLD WEATHER PERIODS, WHEN THE ANTICIPATED AMBIENT TEMPERATURE IS GOING TO BE BELOW 32°F

WHEN TRAINING DURING THE ABOVE MENTIONED CONDITION, FILL FIRE PAN WATER BATH AND TOP OF THE PILOT BOX ASSEMBLY JUST PRIOR TO OPERATION AND DRAIN IMMEDIATELY UPON THE COMPLETION OF TRAINING.

FAILURE TO FOLLOW THESE CAUTIONS MAY RESULT IN SERIOUS DAMAGE TO THE ASSOCIATED EQUIPMENT.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Section 7 CARE AND MAINTENANCE

1. This device will be a very functional and operationally effective unit if cleaned and maintained properly.

2. Replace caps on all Acme fittings on Fire Pan and Control Box before cleaning or storage. Clean surfaces with water only. *Never* use flammable liquids to clean surfaces.

WARNING • AVERTISSEMENT

CLEAN SURFACES WITH WATER ONLY. NEVER USE FLAMMABLE LIQUIDS TO CLEAN SURFACES. FAILURE TO FOLLOW THESE PROCEDURES MAY RESULT IN SERIOUS INJURY OR DEATH.

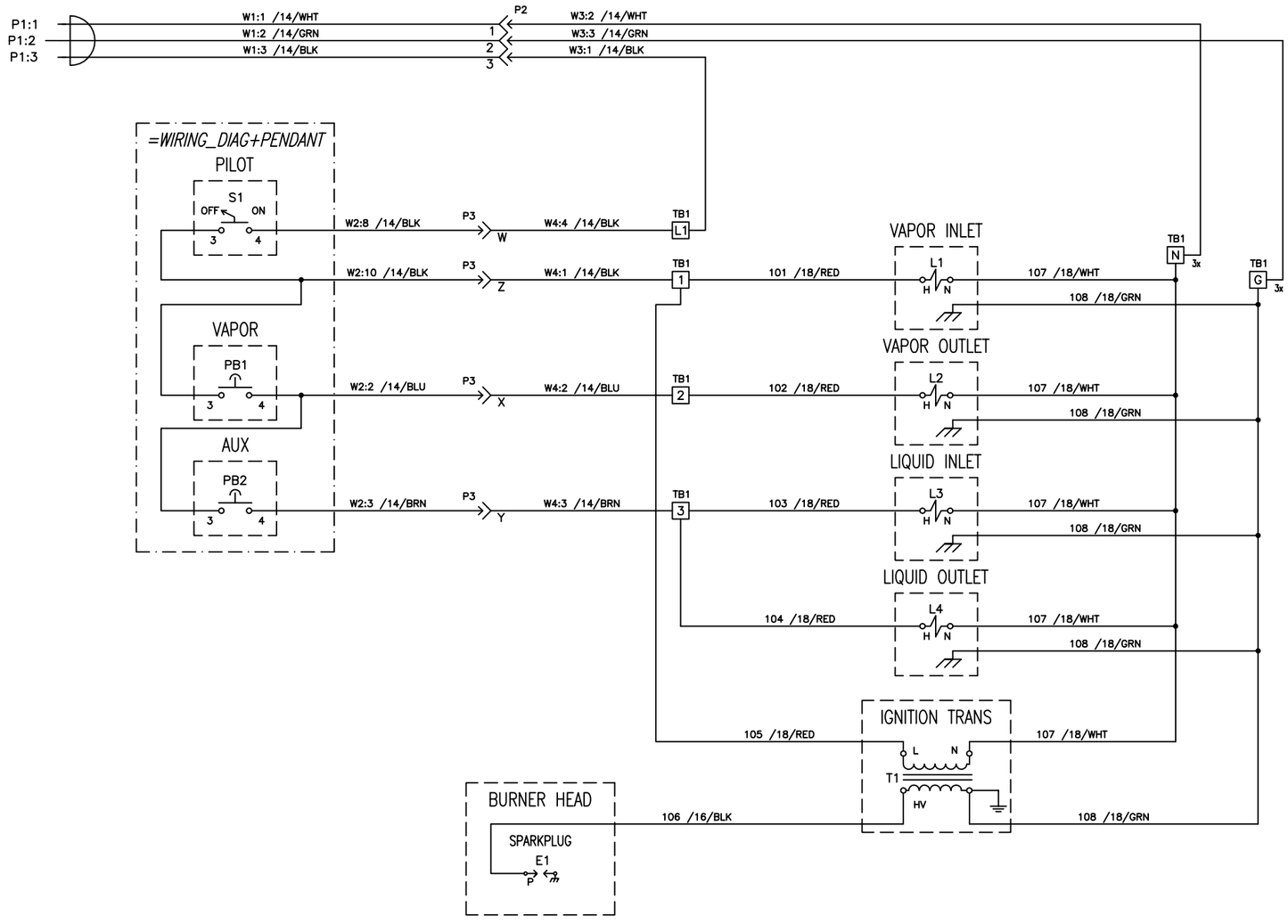
Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

WIRING DIAGRAM

DWG NO	WD4A1-006-460	SH 1

REVISIONS

REV	DRAWING CHANGE RECORD NO	DATE	APVD
A	ECO 30444	6/23/05	BA/DB



PROPRIETARY INFORMATION OF Kidde Fire Trainers
The information contained herein is proprietary to Kidde Fire Trainers, it is not to be copied, disclosed, or disseminated without written permission of Kidde Fire Trainers and then only to the extent authorized.

PROJECT		INSTALLATION			
ICS_0-100		WIRING_DIAG			
		LOCATION/S			
4A1-006-460	0-100	DWN	S HUYNH	08/06/04	
NEXT ASSY	USED ON	CHKD	S HUYNH	08/20/04	
APPLICATION		APVD	D BRADY	08/24/04	

Kidde Fire Trainers

TITLE
ELECTRICAL JUNCTION BOX 0-100
WIRING DIAGRAM
ELECTRICAL ENCL ASSY

SIZE	CAGE CODE	DWG NO
D	06003	WD4A1-006-460

SCALE	NA	CAD FILE: WD4A6460A	SHEET 1 OF 1

APPENDIX A

Set-Up Instructions

Introduction

Appendix A provides the information necessary to properly set-up the **O-100 Pressure Vessel Fire Trainer** for use with the **HOSE-LINE FIRETRAINER® O-100.**

WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 PRESSURE VESSEL FIRE TRAINER A THOROUGH UNDERSTANDING AND FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

IN ADDITION TO THE SETUP STEPS LISTED BELOW, IF THE HOSE LINE FIRETRAINER® O-100 IS TO BE CONTROLLED BY ANY OF THE FOLLOWING OPTIONAL KIDDE CONTROL SYSTEMS, REFER TO THE APPROPRIATE HOSE LINE FIRETRAINER® O-100 CONTROL SYSTEM USERS MANUAL FOR SETUP INSTRUCTIONS.

- **MULTI CONTROL**
- **WIRELESS CONTROL**
- **BATTERY POWERED CONTROL**

Kit Component List (1R1-002-089)

Item No.	Component	Quantity	Part #
A	Burner Element Base	1	3D1-003-388
B	Nuts , 3/8" Hex	4	MS35691-19
C	Screws, 3/8" Hex HD	4	MS35307-362
D	Washers, Lock, 3/8"	4	MS35338-141
E	Washers, Flat, 3/8"	8	MS15795-814
F	Silicone sealant	1 Tube	76435T21
G	Stainless Steel Hose (1)	1	2C1-004-046-006
H	Union, Pipe, 150#, 2" NPT	1	1601-400
I	Nipple, Pipe, SCH40, 2" NPT X 2-1/2" LG	1	1801-404
J	Nipple, Pipe, SCH40, 3/4" NPT X 2" LG	1	1801-153
K	Coupling, RDC, 150# BLK, 2"x3/4" NPT	1	1040-396
L	Propane Tank Mockup	1	4D1-006-900

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WARNING • AVERTISSEMENT

UNDERSTANDING AND FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

IN ADDITION TO THE SETUP STEPS LISTED BELOW, IF THE HOSE LINE FIRETRAINER® O-100 IS TO BE CONTROLLED BY ANY OF THE FOLLOWING OPTIONAL KIDDE CONTROL SYSTEMS, REFER TO THE APPROPRIATE HOSE LINE FIRETRAINER® O-100 CONTROL SYSTEM USERS MANUAL FOR SETUP INSTRUCTIONS.

- **MULTI CONTROL**
- **WIRELESS CONTROL**
- **BATTERY POWERED CONTROL**

Pressure Vessel Mockup Set-Up instructions

Refer to Figures 1 & 2 for details and the Kit Component List for component identification.

1. Locate the Pressure Vessel mockup (L) outdoors; clear of all ground/aerial obstructions on level ground, a minimum of 40 feet downwind of the Propane Supply Tank.
2. Position the 4' x 6' Fire Pan Assembly under the Pressure Vessel mockup with the gas connections facing the Propane Supply Tank. Be sure to align the Fire Pan Assembly so the burner element base (A) is inline with the center of the 6" diameter pipe that passes through the Pressure Vessel mockup.
3. Position the Pilot Box Assembly with the gas connections facing the Propane Supply Tank.
4. Thread the union (H) onto burner element base (A).
5. Connect the reducer assembly to the burner element base (A).
6. Close the LPG Liquid supply valve for the liquid burner element.
7. Open the LPG Liquid supply valve for burner element base (A).
8. Be sure the cooling water tank plug is in place.
9. Open hinged dome cover and fill Pressure Vessel mockup with water at least half way.
10. To complete the Pressure Vessel fire trainer setup you must refer to the **HOSE LINE FIRETRAINER® O-100** Operation and Maintenance Manual, Section 3.

WARNING • AVERTISSEMENT

TO COMPLETE THE PRESSURE VESSEL TRAINER SETUP YOU MUST REFER SEC 3 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

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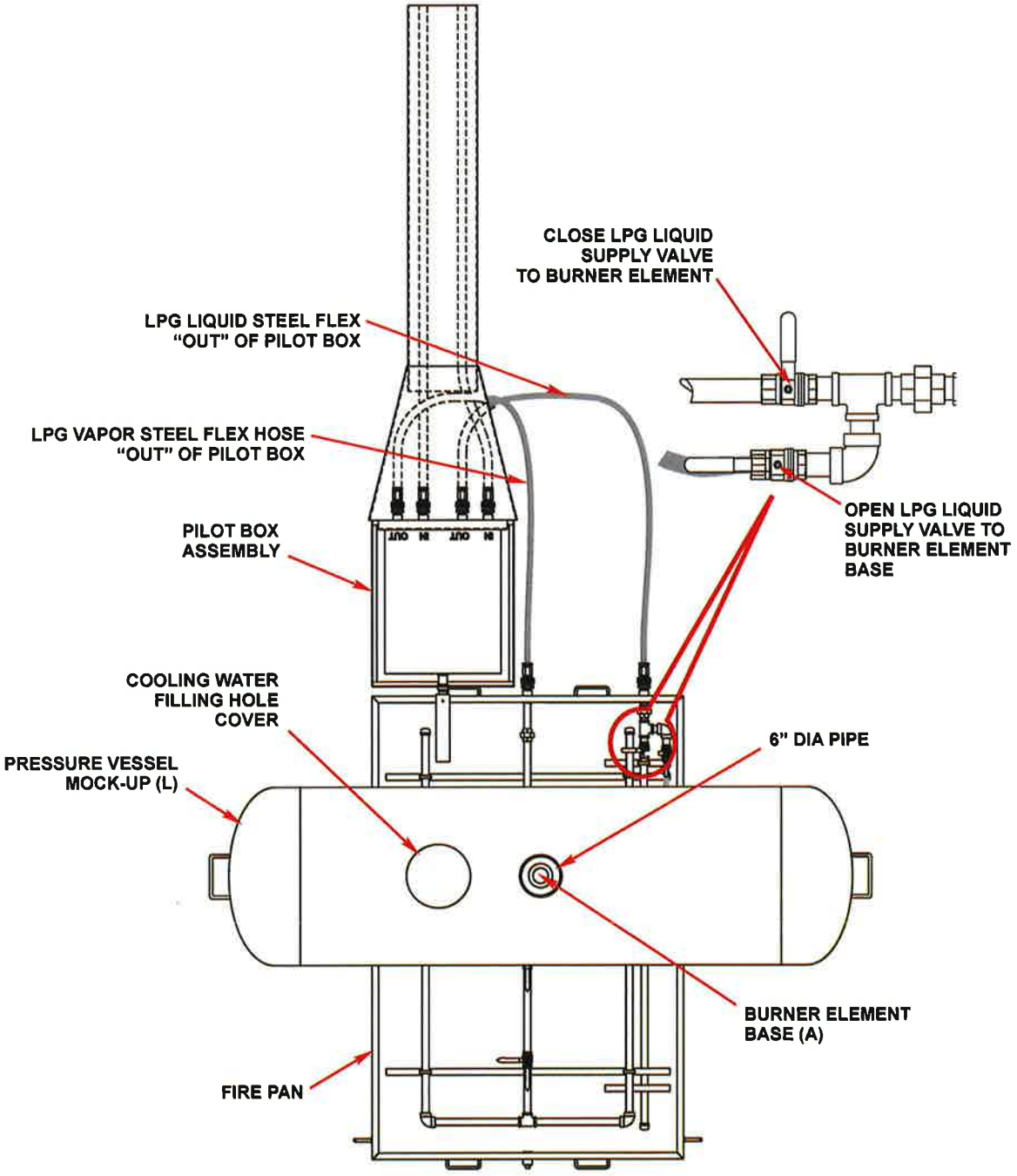

Figure 1. Propane Tank Mock-Up Setup

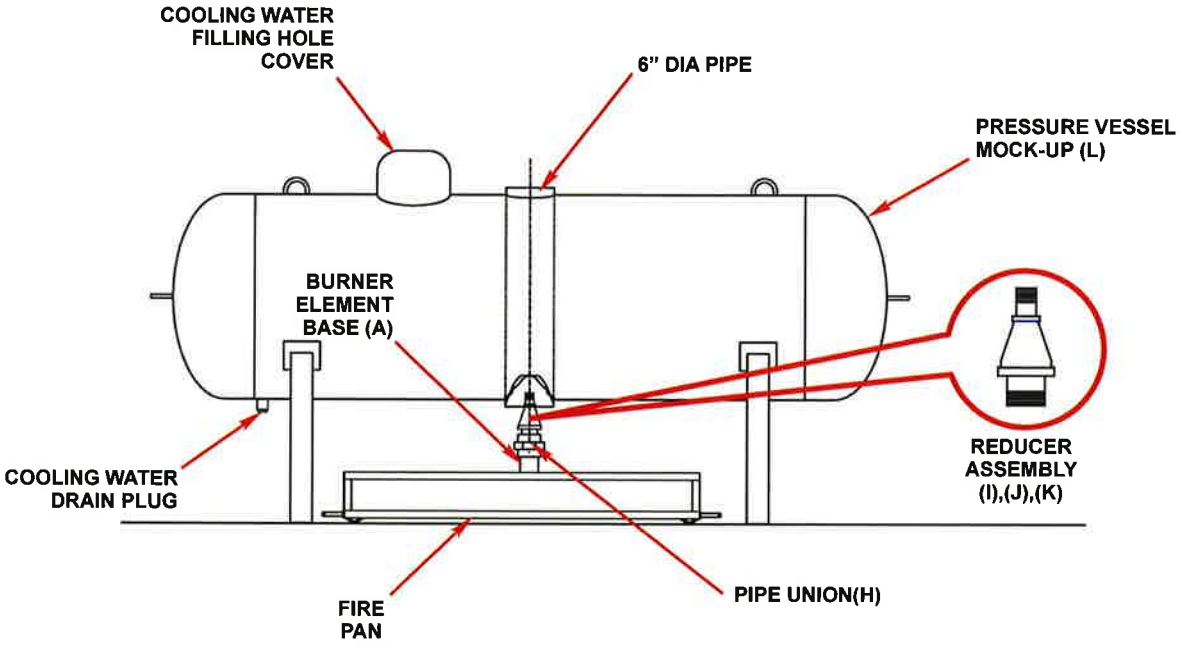

Figure 2. Pressure Vessel Mockup Installation and Setup

Pressure Vessel Fire Trainer Startup

1. Refer to the **HOSE LINE FIRETRAINER® O-100** Operation and Maintenance Manual, Section 4, for the Startup procedures.

WARNING • AVERTISSEMENT

YOU MUST FOLLOW THE STARTUP PROCEDURES IN SECTION 4 OF THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

Pressure Vessel Fire Trainer Operation

1. Refer to the **HOSE LINE FIRETRAINER® O-100** Operation and Maintenance Manual, Section 5, for the Operation procedures.

WARNING • AVERTISSEMENT

YOU MUST FOLLOW THE OPERATION PROCEDURES IN SECTION 5 OF THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

Pressure Vessel Fire Trainer Shutdown

1. Refer to the **HOSE LINE FIRETRAINER® O-100** Operation and Maintenance Manual, Section 6, for the Shutdown procedures.

WARNING • AVERTISSEMENT

YOU MUST FOLLOW THE SHUTDOWN PROCEDURES IN SECTION 6 OF THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

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APPENDIX B

Set-Up Instructions

Introduction

Appendix B provides the information necessary to properly set-up the **O-100 Christmas Tree Fire Trainer** for use with the **HOSE-LINE FIRETRAINER® O-100.**

WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 CHRISTMAS TREE FIRE TRAINER A THOROUGH UNDERSTANDING AND FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

IN ADDITION TO THE SETUP STEPS LISTED BELOW, IF THE HOSE LINE FIRETRAINER® O-100 IS TO BE CONTROLLED BY ANY OF THE FOLLOWING OPTIONAL KIDDE CONTROL SYSTEMS, REFER TO THE APPROPRIATE HOSE LINE FIRETRAINER® O-100 CONTROL SYSTEM USERS MANUAL FOR SETUP INSTRUCTIONS.

- **MULTI CONTROL**
- **WIRELESS CONTROL**
- **BATTERY POWERED CONTROL**

Christmas Tree Mockup Set-Up instructions

Refer to Figure 1 for details and component identification

1. Locate the 4' x 6' Fire Pan Assembly outdoors; clear of all ground/aerial obstructions on level ground, a minimum of 40 feet downwind of the Propane Supply Tank.
2. Thread the bottom of the Christmas Tree mock-up into the pipe union located on the burner element base.
3. Position the Pilot Box Assembly as shown in figure 1, with the gas connections facing the Propane Supply Tank.
4. Close the shutoff valve for the liquid burner element (see figure 1).
5. Open the shutoff valve for the Christmas tree mockup (see figure 1).
6. To complete the Christmas Tree fire trainer setup you must refer to the **HOSE LINE FIRETRAINER® O-100** Operation and Maintenance Manual, Section 3.

WARNING • AVERTISSEMENT

TO COMPLETE THE CHRISTMAS TREE FIRE TRAINER SETUP YOU MUST REFER SEC 3 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

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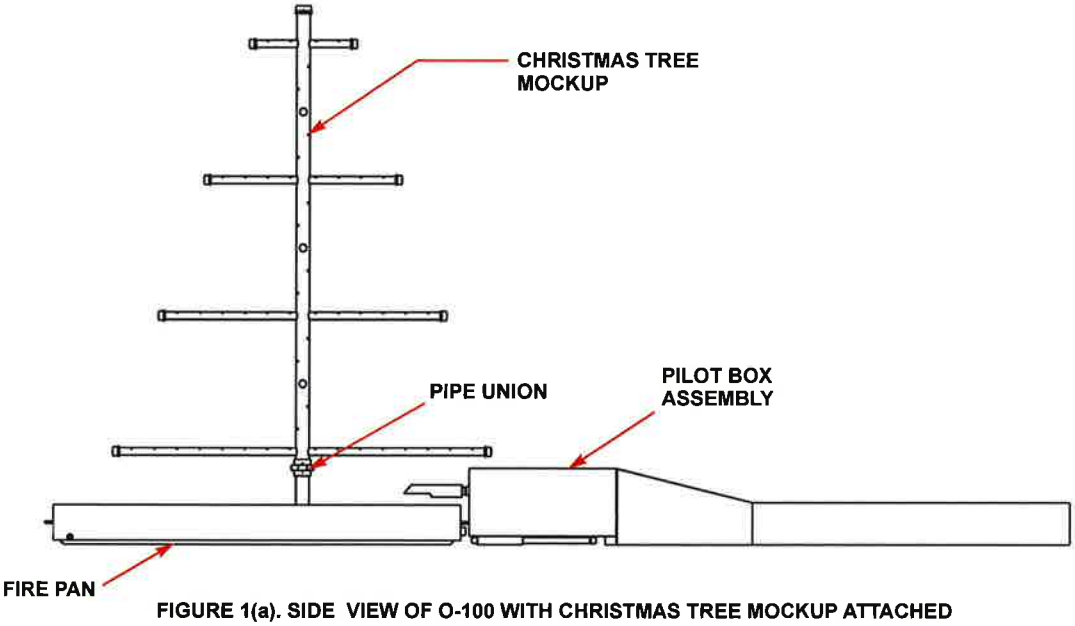

FIGURE 1(a). SIDE VIEW OF O-100 WITH CHRISTMAS TREE MOCKUP ATTACHED

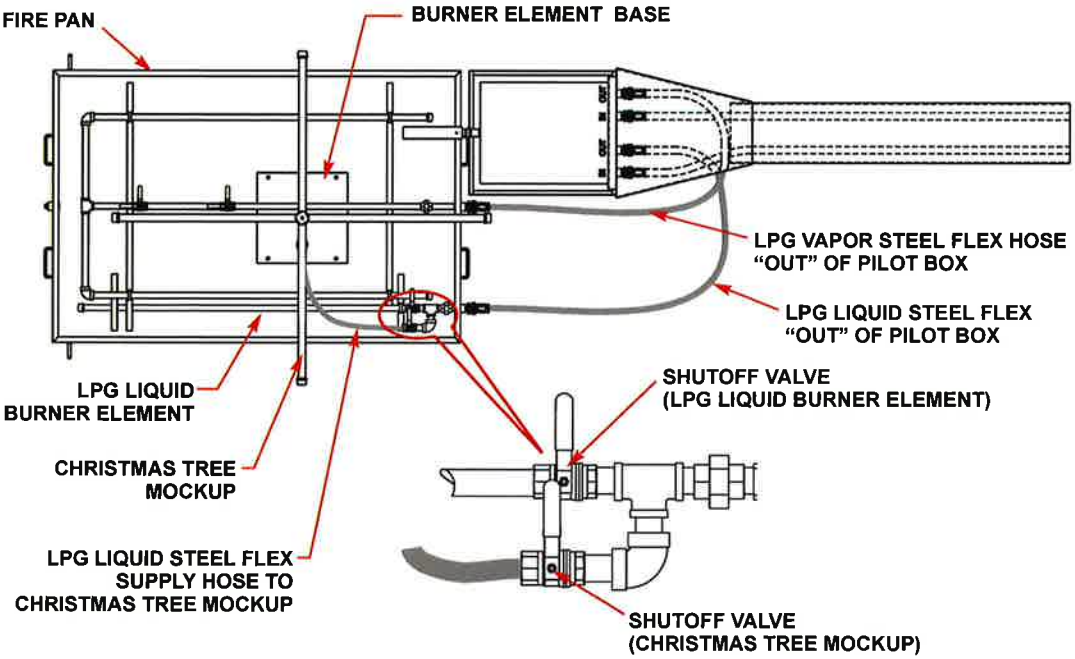

FIGURE 1(b). TOP VIEW OF O-100 WITH CHRISTMAS TREE MOCKUP ATTACHED

Figure 1. O-100 Christmas Tree Fire Trainer Setup and Overview

Christmas Tree Fire Trainer Startup

1. Refer to the **HOSE LINE FIRETRAINER® O-100** Operation and Maintenance Manual, Section 4, for the Startup procedures.

WARNING • AVERTISSEMENT

YOU MUST FOLLOW THE STARTUP PROCEDURES IN SECTION 4 OF THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

Christmas Tree Fire Trainer Operation

1. Refer to the **HOSE LINE FIRETRAINER® O-100** Operation and Maintenance Manual, Section 5, for the Operation procedures.

WARNING • AVERTISSEMENT

YOU MUST FOLLOW THE OPERATION PROCEDURES IN SECTION 5 OF THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

Christmas Tree Fire Trainer Shutdown

1. Refer to the **HOSE LINE FIRETRAINER® O-100** Operation and Maintenance Manual, Section 6, for the Shutdown procedures.

WARNING • AVERTISSEMENT

YOU MUST FOLLOW THE SHUTDOWN PROCEDURES IN SECTION 6 OF THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

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APPENDIX C

Set-Up Instructions

Introduction

Appendix C provides the information necessary to properly set-up the **O-100 Barbeque Fire Trainer** for use with the **HOSE-LINE FIRETRAINER® O-100.**

WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 BARBEQUE FIRE TRAINER A THOROUGH UNDERSTANDING AND FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

IN ADDITION TO THE SETUP STEPS LISTED BELOW, IF THE HOSE LINE FIRETRAINER® O-100 IS TO BE CONTROLLED BY ANY OF THE FOLLOWING OPTIONAL KIDDE CONTROL SYSTEMS, REFER TO THE APPROPRIATE HOSE LINE FIRETRAINER® O-100 CONTROL SYSTEM USERS MANUAL FOR SETUP INSTRUCTIONS.

- **MULTI CONTROL**
- **WIRELESS CONTROL**
- **BATTERY POWERED CONTROL**

Kit Component List

Component	Quantity	Part #
Barbeque Fire Assembly	1	1R1-002-091

Installation Instructions

1. Locate the Barbeque Mock-Up outdoors; clear of all ground/aerial obstructions on level ground, a minimum of 40 feet downwind of the Propane Supply.

NOTE

IF THE Kidde Multi Control System is used, the Multi Control Stand is considered the propane supply.

2. Position the Barbeque Mock-Up and Pilot Box Assembly as shown in figure 1.
3. Remove the wheels from the Pilot Box Assembly. Store wheels safely away from training area.
4. Extend all lines (Vapor Propane gas, control, electric) and check for any damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF ANY OF THE CONTROL OR ELECTRICAL LINES ARE DAMAGED. DO NOT PROCEED ANY FURTHER IF ANY OF THE GAS HOSES ARE DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

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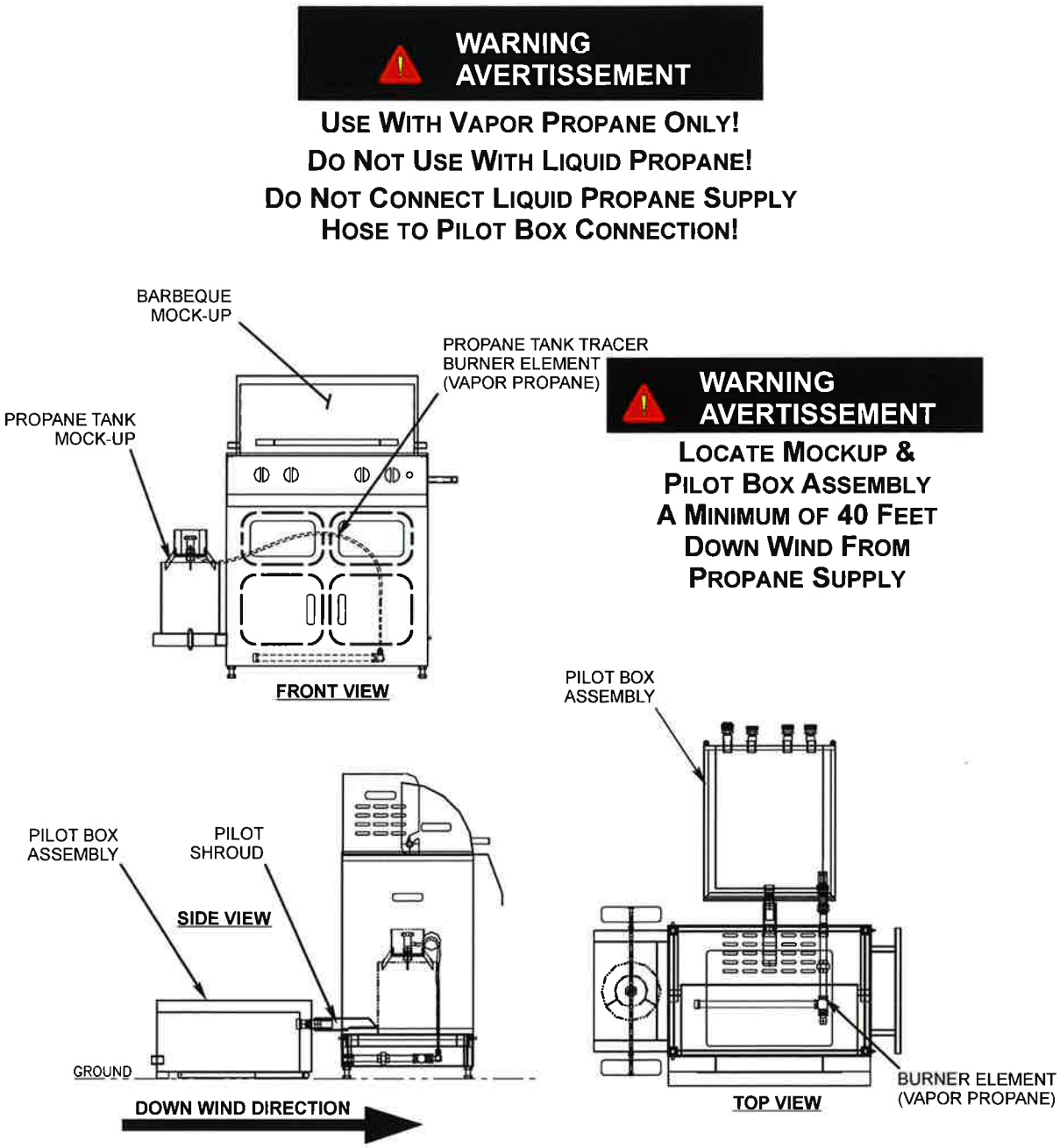

Figure 1. O-100 Barbeque Fire Trainer Setup and Overview

5. At the Pilot Box Assembly remove protective thread cap from the 1-1/4" male **LPG VAPOR** connection marked **"IN".** Make sure the 1-1/4" male **LPG VAPOR** connection is free of debris, damage and that the rubber washer is present (see Figure 2).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE LPG VAPOR "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

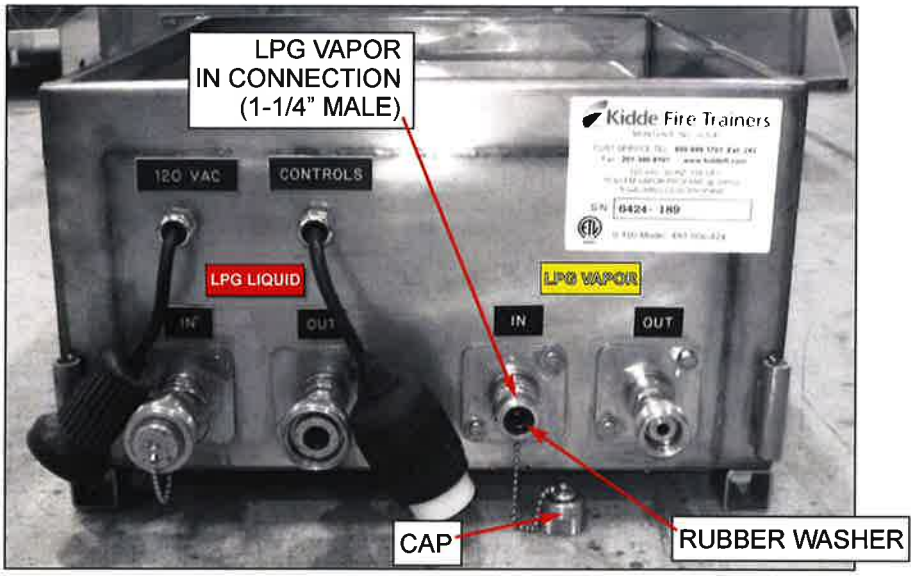

Figure 2. Pilot Box Assembly LPG VAPOR "IN" Connection

6. Make sure the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

7. Connect the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose to the 1-1/4" male **LPG VAPOR** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 3).

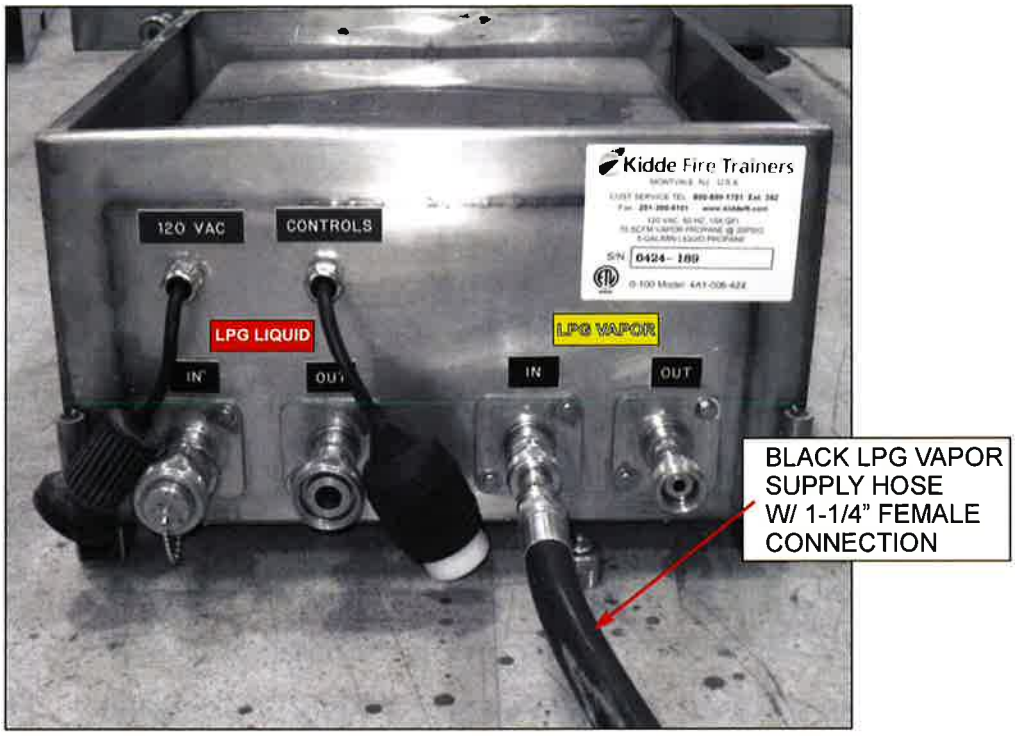

Figure 3. Pilot Box Assembly LPG VAPOR "IN" Supply Hose Connection

8. At the Pilot Box Assembly make sure the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE LPG VAPOR "OUT" CONNECTION HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

9. Remove protective thread cap from the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

10. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 4).

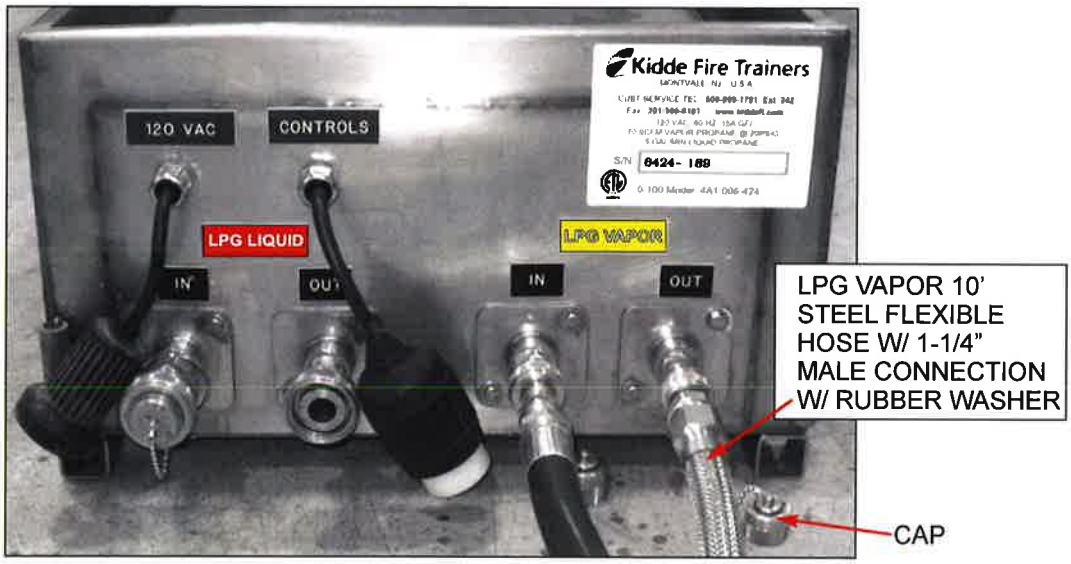

Figure 4. Pilot Box Assembly LPG VAPOR "OUT" Hose Connection

11. Remove protective thread cap form the 1-1/4" male connection on the remaining end of the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

12. At the Barbeque Mockup, make sure the LPG Vapor IN 1-1/4" female connection is free of debris and damage (see Figure 5 for location).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE LPG VAPOR 1-1/4" FEMALE CONNECTION ON THE BARBEQUE MOCKUP HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

13. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the LPG Vapor 1-1/4" female connection on the Barbeque Mockup. Attach fittings hand tight (see Figure 5).

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

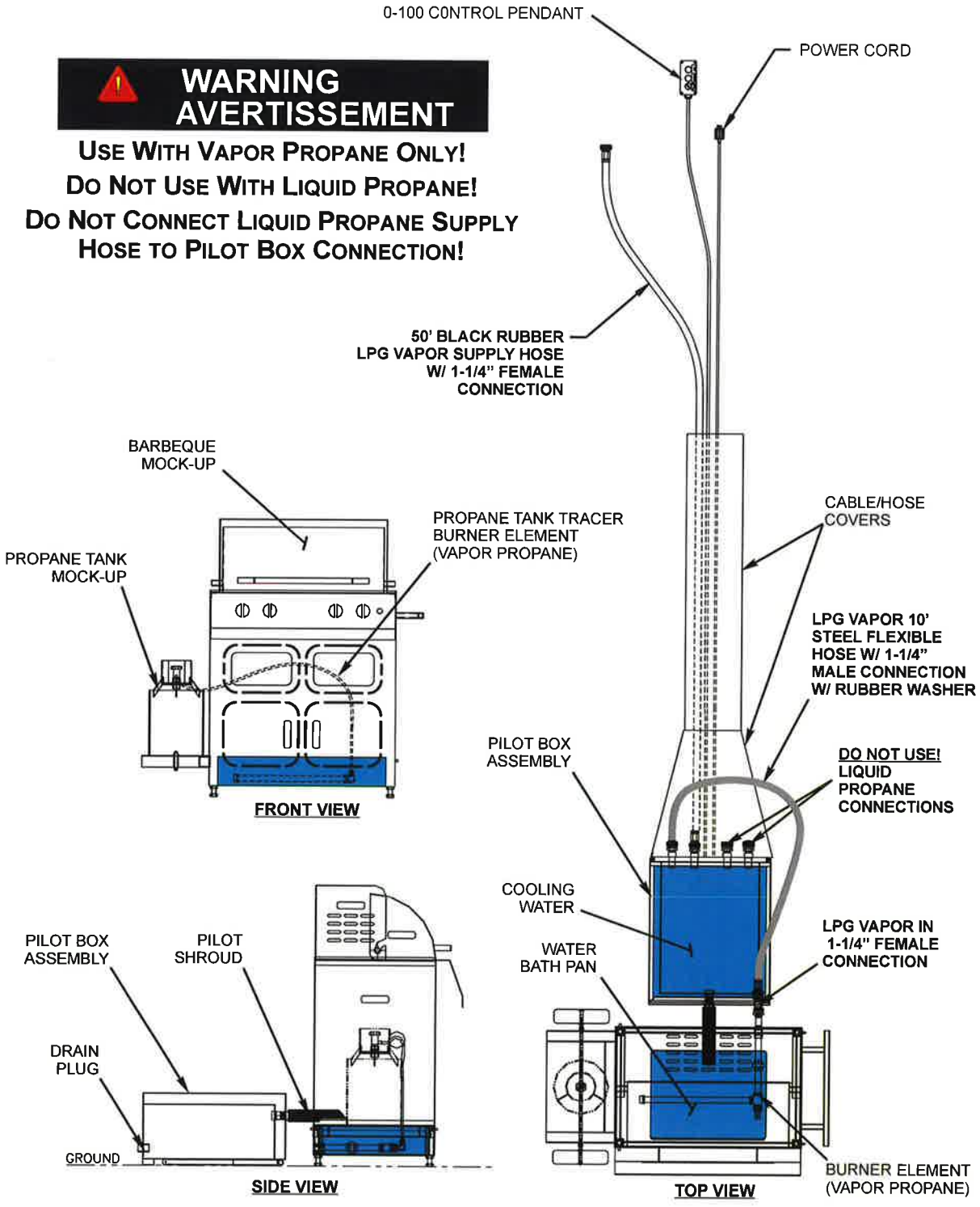

Figure 5. O-100 Barbeque Fire Trainer Setup

14. Attach the Control Pendant cable to the connection marked **"CONTROLS"** on Pilot Box Assembly (see Figure 6).

15. Attach the Electric Supply cord to the connection marked **"120 VAC"** on Pilot Box Assembly (see Figure 6).

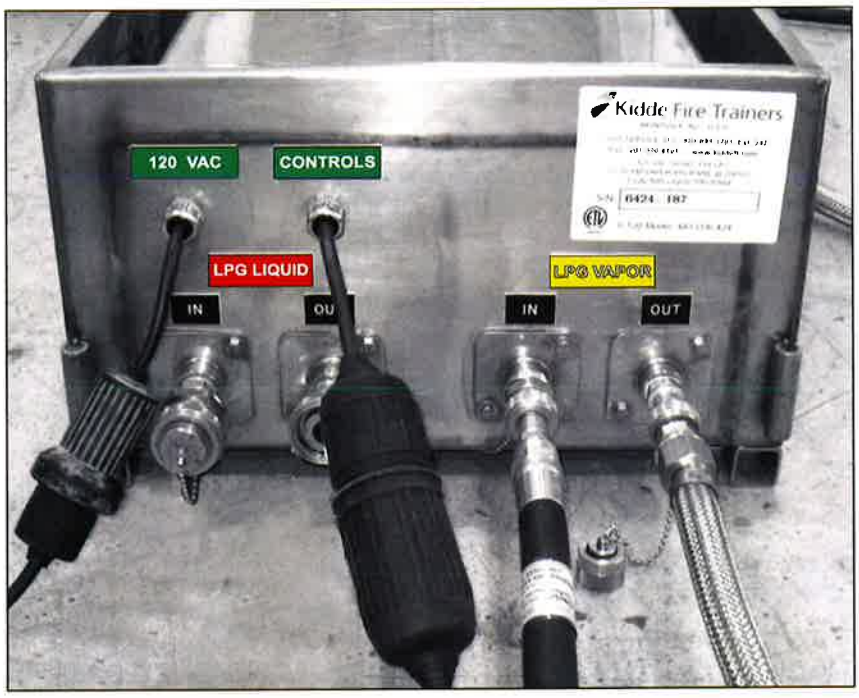

Figure 6. Electrical Supply & Control Pendant Connections

16. Ensure that drain plug is installed in the Pilot Box Assembly (see Figure 5).
17. Fill Pilot Box Assembly top with water until it overflows (see Figure 5).
18. Ensure that drain plug is installed in the Barbeque Fire Assembly water bath pan.
19. Fill the bottom of the Barbeque Fire Assembly with water. Access is from the back of the assembly (see Figure 5).

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

NOTE

IF THE HOSE LINE FIRETRAINER® O-100 is to be controlled by the Kidde Multi Control System, refer to the HOSE LINE FIRETRAINER® O-100 Multi Control System Users Manual, Sections 3.1 & 3.2.

20. Using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088), connect the remaining end of the 50' black rubber LPG vapor supply hose from the Pilot Box Assembly **LPG VAPOR** connection marked **"IN"**, to **LPG Vapor Outlet** on the Propane Supply. Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male adaptor connection. Attach fittings hand tight.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER LPG VAPOR SUPPLY HOSE IS CONNECTED TO THE LPG VAPOR OUTLET ON THE PROPANE SUPPLY TANK. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Prior to Operation

Pilot Box Valves Operational Test

1. Perform the Pilot Box LPG Vapor Valve Operational Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Box Valves Operational Test.

WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT BOX VALVES OPERATIONAL TEST HAS BEEN SUCCESSFULLY COMPLETED.

Pilot Flame Test

1. Perform the Pilot Flame Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Flame Test.

WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT FLAME TEST HAS BEEN SUCCESSFULLY COMPLETED.

Operation

1. Run (gas, control and electric) lines straight back from Pilot Box Assembly.
2. Install cable/hose covers. Refer to Figure 5.

CAUTION

DO NOT OPERATE O-100 TRAINER WITHOUT BOTH CABLE/HOSE COVERS IN PLACE! REFER TO FIGURE 5.

WARNING • AVERTISSEMENT

ONLY WHEN ALL THE ABOVE OPERATIONAL AND LEAK CHECKS HAVE BEEN SUCCESSFULLY PERFORMED YOU ARE READY TO TRAIN.

3. Attach the electric supply cord from the Pilot Box assembly to an outdoor 120V power GFIC receptacle.

NOTE

FOR OPERATION INSTRUCTIONS REFER TO THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL. FOLLOW ALL PROCEDURES FOR VAPOR PROPANE ONLY. ANY REFERENCES TO LIQUID PROPANE DOES NOT APPLY.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

⚠ WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 BARBEQUE FIRE TRAINER A THOROUGH UNDERSTANDING AND FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FOLLOW ALL PROCEDURES FOR VAPOR PROPANE ONLY! ANY REFERENCES TO LIQUID PROPANE DOES NOT APPLY! FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

Shutdown & Disassembly

⚠ WARNING • AVERTISSEMENT

AT THE END OF A TRAINING SESSION THERE WILL STILL BE FUEL IN THE HOSES FROM THE PROPANE SUPPLY TANK TO THE MOCKUP/FIRE PAN. THIS WILL HAVE TO BE BURNT OFF BEFORE THE EQUIPMENT IS TAKEN APART. IT IS IMPORTANT THAT THIS PROCEDURE BE FOLLOWED EVEN IF YOU ARE NOT TAKING EQUIPMENT APART BUT ARE JUST SHUTTING DOWN AT THE END OF A TRAINING SESSION. FOR THIS PROCEDURE REFER TO SECTION 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

NOTE

FOR SHUTDOWN & DISASSEMBLY INSTRUCTIONS REFER TO SEC 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

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Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

APPENDIX D

Set-Up Instructions

Introduction

Appendix D provides the information necessary to properly set-up the **O-100 Dumpster Fire Trainer** for use with the **HOSE-LINE FIRETRAINER® O-100.**

WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 DUMPSTER FIRE TRAINER A THOROUGH UNDERSTANDING AND FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

IN ADDITION TO THE SETUP STEPS LISTED BELOW, IF THE HOSE LINE FIRETRAINER® O-100 IS TO BE CONTROLLED BY ANY OF THE FOLLOWING OPTIONAL KIDDE CONTROL SYSTEMS, REFER TO THE APPROPRIATE HOSE LINE FIRETRAINER® O-100 CONTROL SYSTEM USERS MANUAL FOR SETUP INSTRUCTIONS.

- **MULTI CONTROL**
- **WIRELESS CONTROL**
- **BATTERY POWERED CONTROL**

Components List

Component	Quantity	Part #
Dumpster Fire Assembly	1	4B1-006-927

Installation Instructions

1. Locate the Dumpster Mock-Up outdoors; clear of all ground/aerial obstructions on level ground, a minimum of 40 feet downwind of the Propane Supply.

NOTE

IF THE Kidde Multi Control System is used, the Multi Control Stand is considered the propane supply.

2. Install leveling pads by placing a hex nut on the threaded portion of each leveling pad. Run each hex nut down to the pad, as shown in Figure 1.
3. Insert a leveling pad into each of the receivers located on the rear of the Pilot Box assembly support angles, as shown in Figure 1. Place a second hex nut on the threaded portion of each leveling pad.
4. Mount both of the Pilot Box assembly support angles to the Dumpster Mock-Up, as shown in Figure 1.
5. Locate the Pilot Box Assembly on the support angles as shown in figure 1. Slide Pilot Box Assembly against the back of the Dumpster Mock-Up.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

6. At the back end of each support angle adjust leveling pads so they are both against the ground. Lock pads in position with the hex nuts as shown in figure 1.
7. Extend all lines (gas, control, electric) and check for any damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF ANY OF THE CONTROL OR ELECTRICAL LINES ARE DAMAGED. DO NOT PROCEED ANY FURTHER IF ANY OF THE GAS HOSES ARE DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

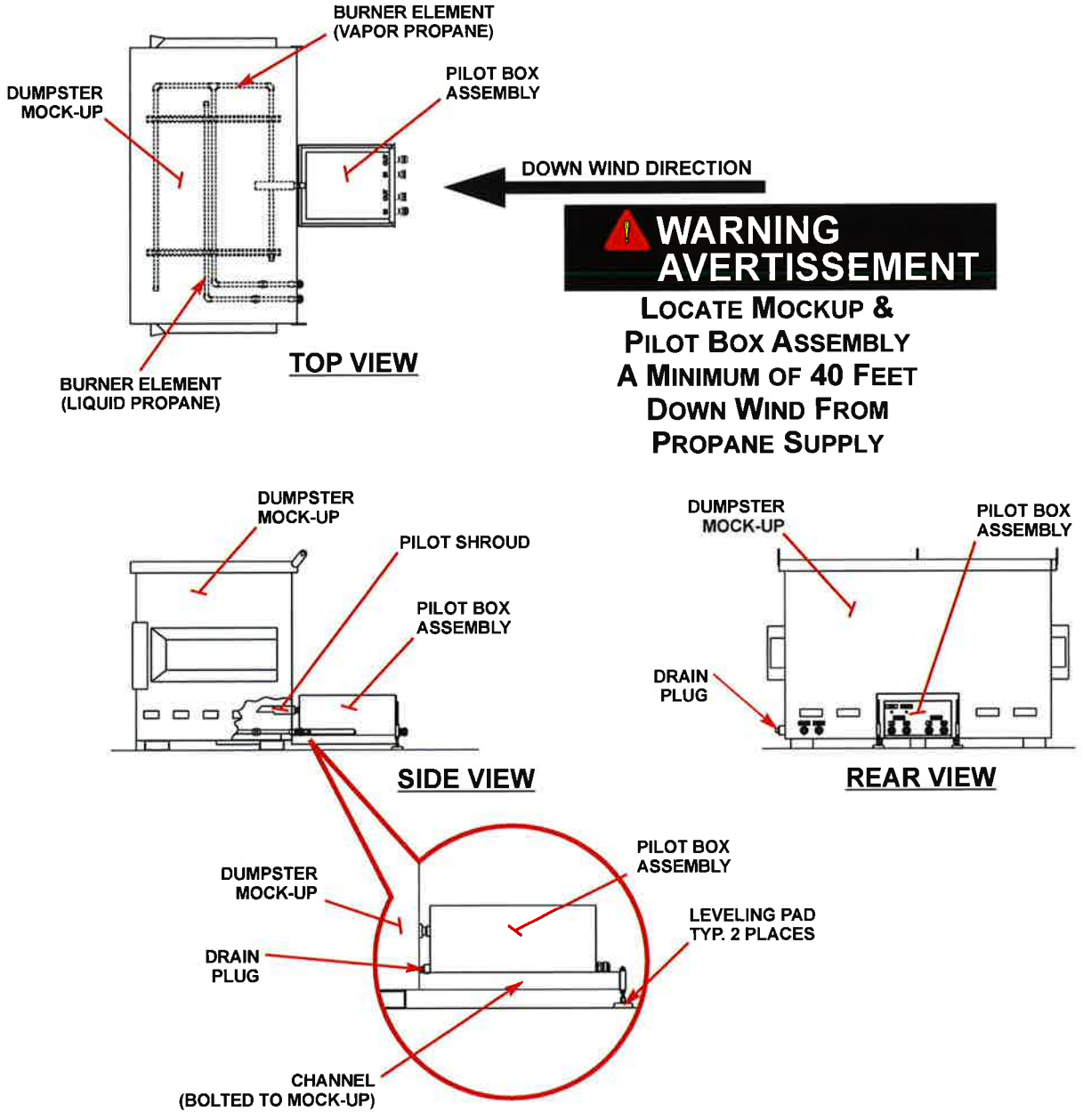

Figure 1. O-100 Dumpster Fire Trainer Setup and Overview

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

8. At the Pilot Box Assembly remove protective thread cap from the 1-1/4" male **LPG VAPOR** connection marked **"IN"**. Make sure the 1-1/4" male **LPG VAPOR** connection is free of debris, damage and that the rubber washer is present (see Figure 2).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE LPG VAPOR "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

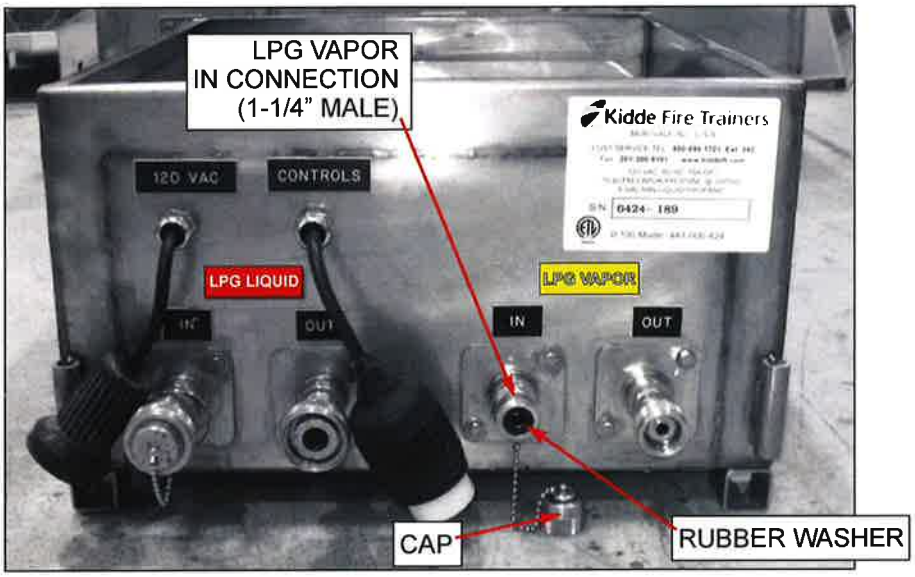

Figure 2. Pilot Box Assembly LPG VAPOR "IN" Connection

9. Make sure the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

10. Connect the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose to the 1-1/4" male **LPG VAPOR** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 3).

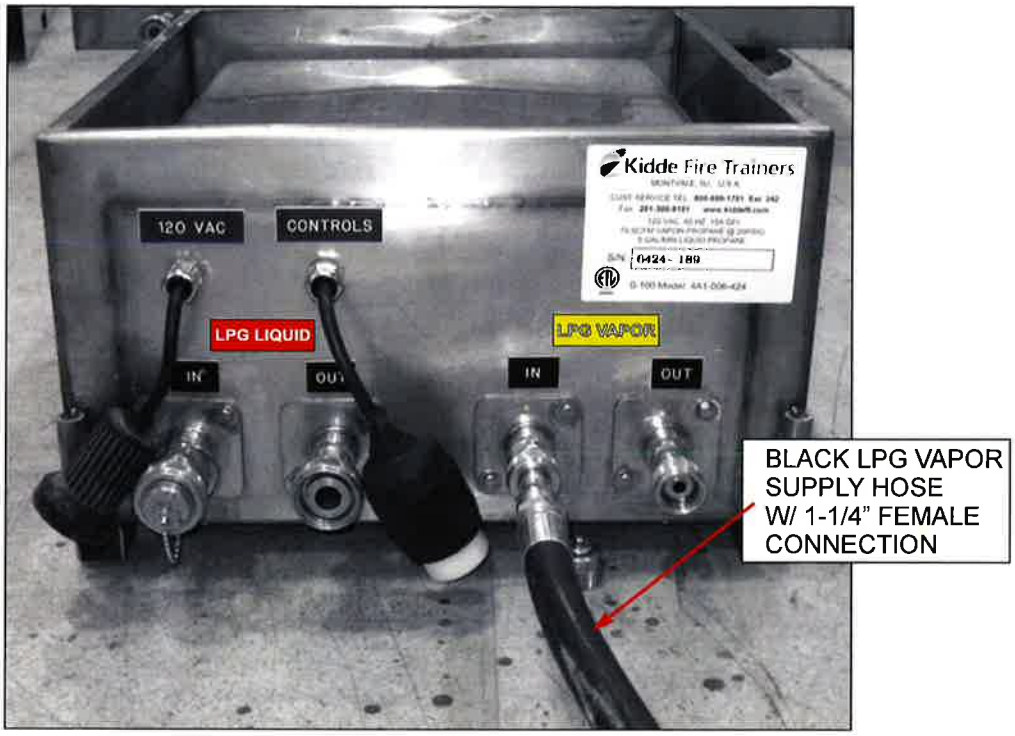

Figure 3. Pilot Box Assembly LPG VAPOR "IN" Supply Hose Connection

11. At the Pilot Box Assembly remove protective thread cap from the 1-3/4" male **LPG LIQUID** connection marked **"IN".** Make sure the 1-3/4" male **LPG LIQUID** connection is free of debris, damage and that the rubber washer is present (see Figure 4).

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE LPG LIQUID "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

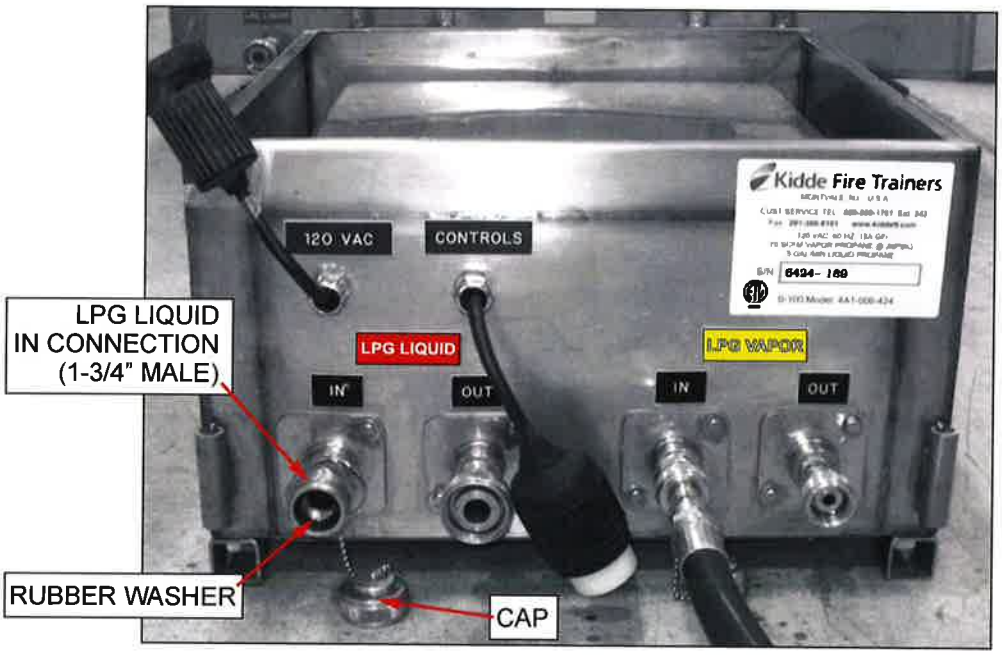

Figure 4. Pilot Box Assembly LPG LIQUID "IN" Connection

12. Make sure the 1-3/4" female connection on the 50' black rubber LPG Liquid supply hose is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

13. Connect the 1-3/4" female connection on the 50' black rubber LPG Liquid supply hose to the 1-3/4" male **LPG LIQUID** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 5).

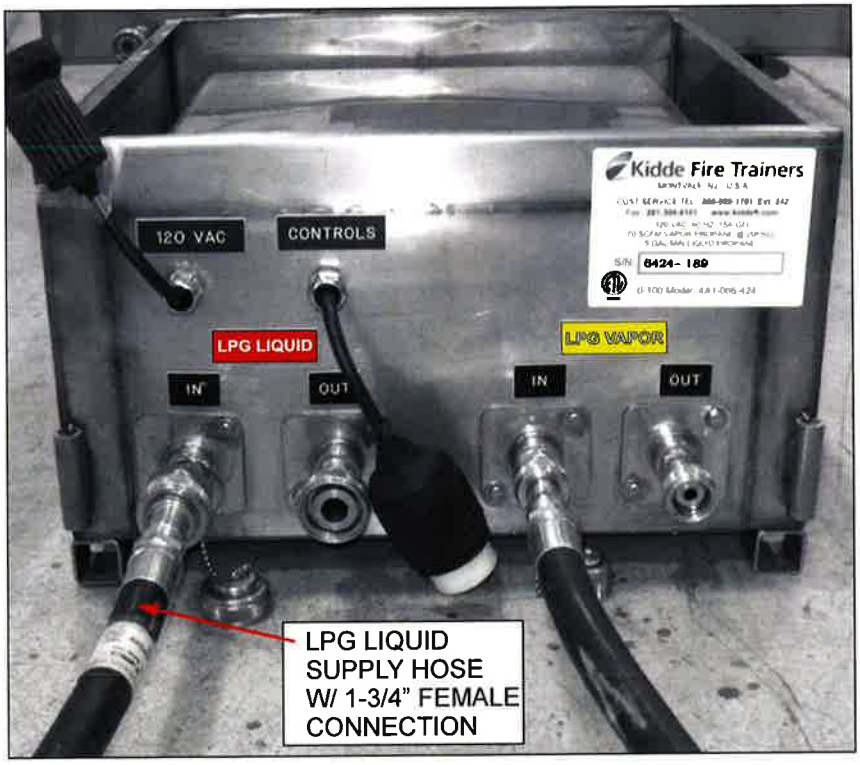

Figure 5. Pilot Box Assembly LPG LIQUID "IN" Supply Hose Connection

14. At the Pilot Box Assembly make sure the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** is free of debris and damage.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE LPG VAPOR "OUT" CONNECTION HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

15. Remove protective thread cap from the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

16. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 6).

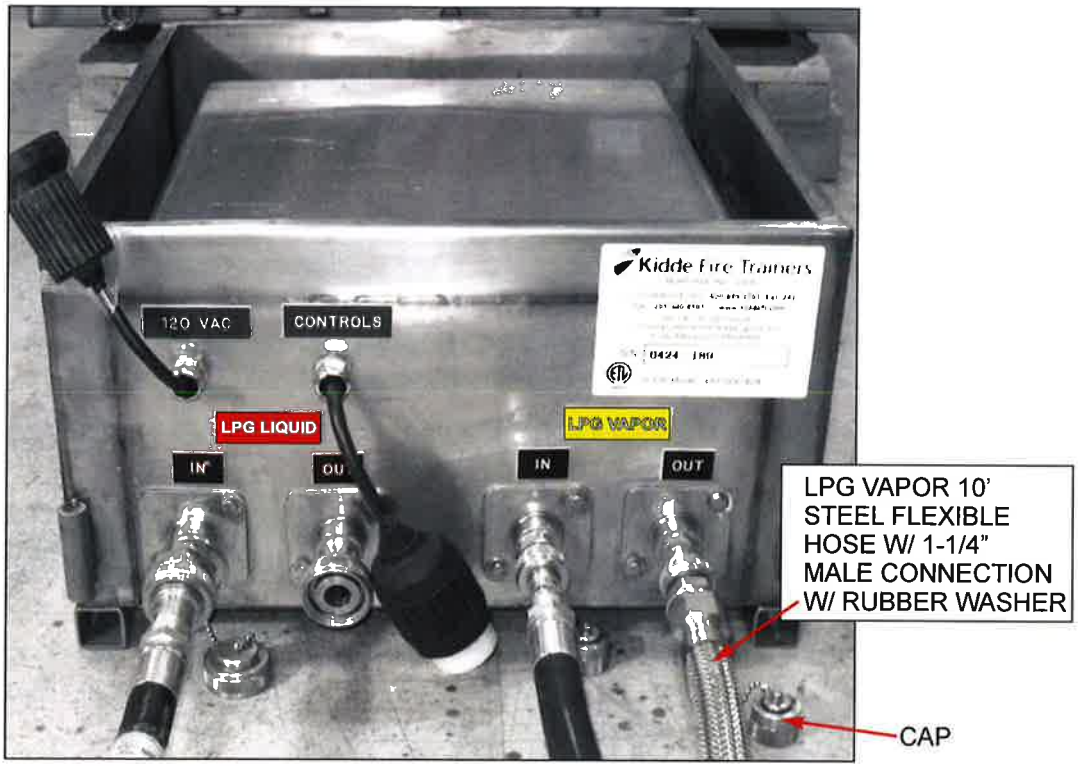

Figure 6. Pilot Box Assembly LPG VAPOR "OUT" Hose Connection

17. Remove protective thread cap form the 1-1/4" male connection on the remaining end of the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

18. At the Dumpster Mockup, make sure the 1-1/4" female connection marked **LPG VAPOR** is free of debris and damage. (see Figure 8 for location).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE CONNECTION MARKED LPG VAPOR, ON THE DUMPSTER MOCKUP HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

19. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the 1-1/4" female connection marked **LPG VAPOR** on the Dumpster Mockup. Attach fittings hand tight (see Figure 8).

20. Remove protective thread cap from the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose. Make sure the 1-3/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

21. At the Pilot Box Assembly make sure the 1-3/4" female **LPG LIQUID** connection marked **OUT** is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION MARKED LPG LIQUID "OUT" HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

22. Connect the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose to the 1-3/4" female **LPG LIQUID** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 7).

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

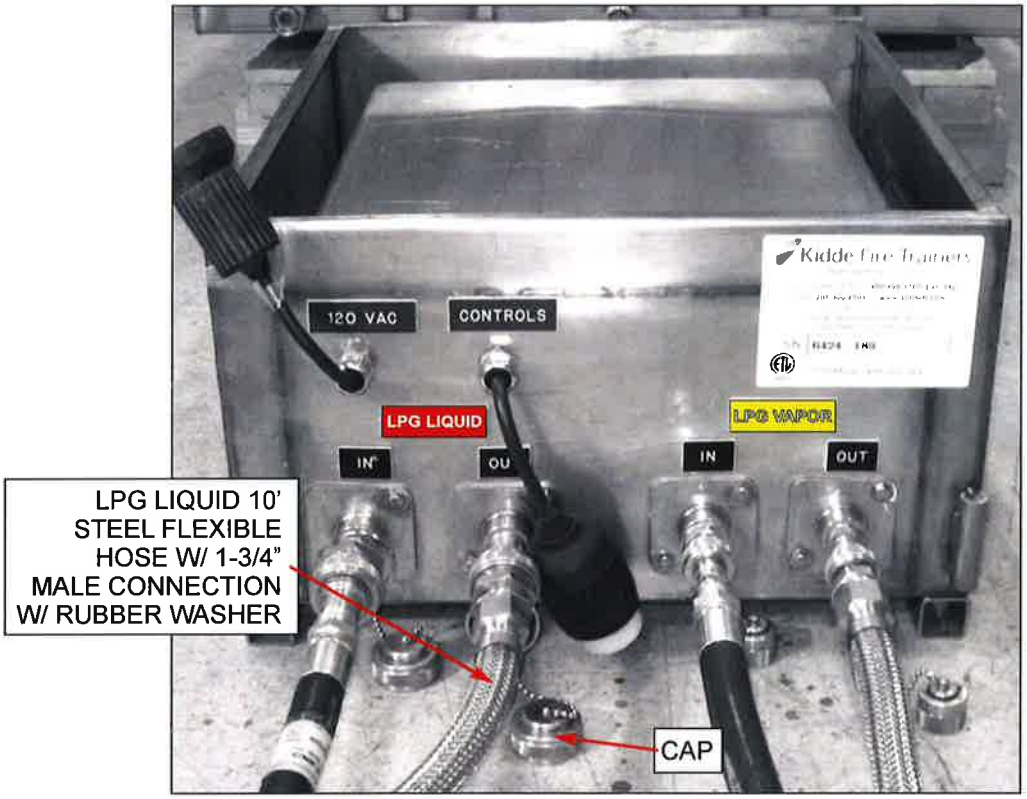

Figure 7. Pilot Box Assembly LPG LIQUID "OUT" Hose Connection

23. Remove protective thread cap form the 1-3/4" male connection on the remaining end of the 10' flexible stainless steel LPG Liquid supply hose. Make sure the 1-3/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

24. At the Dumpster Mockup make sure the 1-3/4" female connection marked **LPG LIQUID** is free of debris and damage (See Figure 8 for location).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION MARKED LPG LIQUID, ON THE DUMPSTER MOCKUP HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

25. Connect the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose to the 1-3/4" female connection marked **LPG LIQUID** on the Dumpster Mockup. Attach fittings hand tight (see Figure 8).

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

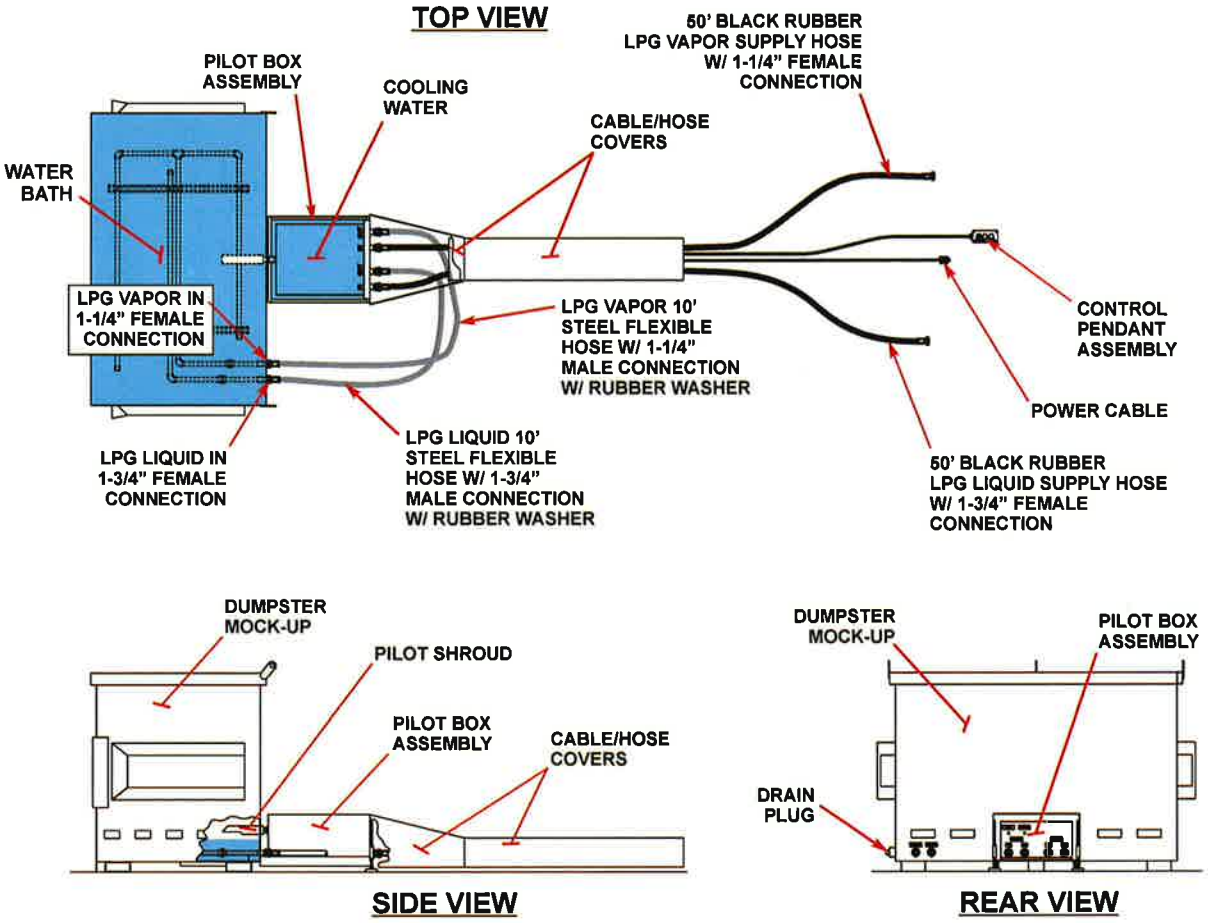

Figure 8. O-100 Dumpster Fire Trainer Setup

26. Attach the Control Pendant cable to the connection marked **"CONTROLS"** on Pilot Box Assembly (see Figure 9).
27. Attach the Electric Supply cord to the connection marked **"120 VAC"** on Pilot Box Assembly (see Figure 9).

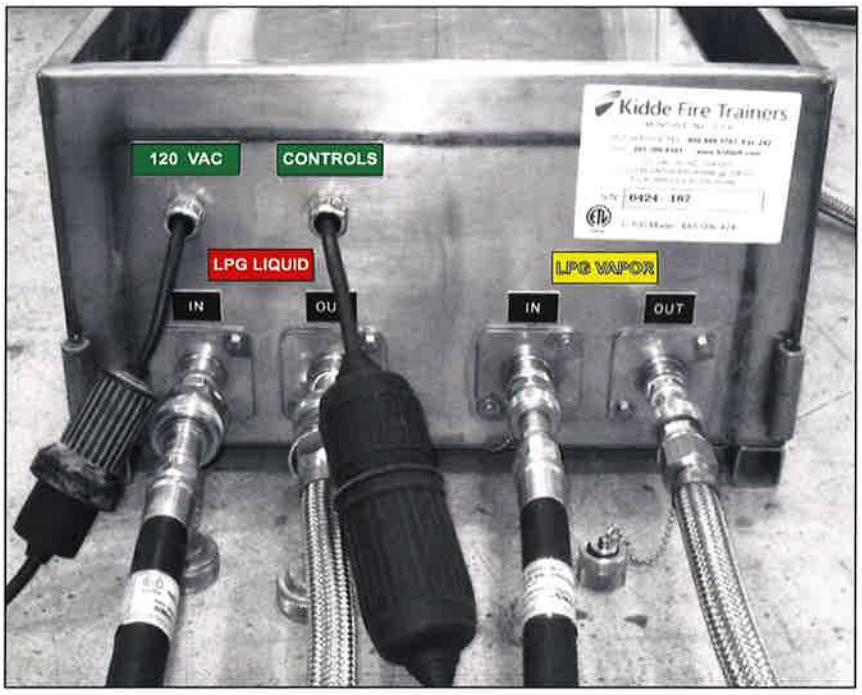

Figure 9. Electrical Supply & Control Pendant Connections

28. Ensure that drain plug is installed in the Dumpster Mock-Up. Fill Dumpster with water, until burn pan overflows. Refer to Figure 8 for location.
29. Ensure that water drain plug is installed in the Pilot Box Assembly.
30. Fill Pilot Box Assembly top with water until it overflows (see Figure 8).

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

NOTE

IF THE HOSE LINE FIRETRAINER® O-100 is to be controlled by the Kidde Multi Control System, refer to the HOSE LINE FIRETRAINER® O-100 Multi Control System Users Manual, Sections 3.1 & 3.2.

31. Using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088), connect the remaining end of the 50' black rubber LPG vapor supply hose from the Pilot Box Assembly **LPG VAPOR** connection marked **"IN"**, to **LPG Vapor Outlet** on the Propane Supply. Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male adaptor connection. Attach fittings hand tight.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER LPG VAPOR supply hose is connected to the LPG VAPOR outlet on the propane supply tank.

32. Using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088), connect the remaining end of the 50' black rubber LPG liquid supply hose from the Pilot Box Assembly **LPG LIQUID** connection marked **"IN"**, to the **LPG LIQUID Output** on the Propane Supply. Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male adaptor connection. Attach fittings hand tight.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER LPG LIQUID supply hose is connected to the LPG LIQUID outlet on the propane supply tank.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Prior to Operation

Pilot Box Valves Operational Test

1. Perform the Pilot Box Valves Operational Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Box Valves Operational Test.

WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT BOX VALVES OPERATIONAL TEST HAS BEEN SUCCESSFULLY COMPLETED.

Pilot Flame Test

1. Perform the Pilot Flame Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Flame Test.

WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT FLAME TEST HAS BEEN SUCCESSFULLY COMPLETED.

Liquid Propane Leak check

1. Perform the Liquid Propane Leak Check. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Liquid Propane Leak Check.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER UNTIL THE LIQUID PROPANE LEAK CHECK HAS BEEN SUCCESSFULLY COMPLETED.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Operation

1. Run (gas, control and electric) lines straight back from Pilot Box Assembly.
2. Install cable/hose covers. Refer to Figure 8.

⚠ CAUTION

DO NOT OPERATE O-100 TRAINER WITHOUT BOTH CABLE/HOSE COVERS IN PLACE! REFER TO FIGURE 8.

⚠ WARNING • AVERTISSEMENT

ONLY WHEN ALL THE ABOVE OPERATIONAL AND LEAK CHECKS HAVE BEEN SUCCESSFULLY PERFORMED YOU ARE READY TO TRAIN.

3. Attach the electric supply cord from the Pilot Box assembly to an outdoor 120V power GFIC receptacle.

NOTE

FOR OPERATION INSTRUCTIONS REFER TO THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

⚠ WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 HORIZONTAL PROPANE TANK FIRE TRAINER A THOROUGH FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

Shutdown & Disassembly

⚠ WARNING • AVERTISSEMENT

AT THE END OF A TRAINING SESSION THERE WILL STILL BE FUEL IN THE HOSES FROM THE PROPANE SUPPLY TANK TO THE MOCKUP/FIRE PAN. THIS WILL HAVE TO BE BURNT OFF BEFORE THE EQUIPMENT IS TAKEN APART. IT IS IMPORTANT THAT THIS PROCEDURE BE FOLLOWED EVEN IF YOU ARE NOT TAKING EQUIPMENT APART BUT ARE JUST SHUTTING DOWN AT THE END OF A TRAINING SESSION. FOR THIS PROCEDURE REFER TO SECTION 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

NOTE

FOR SHUTDOWN & DISASSEMBLY INSTRUCTIONS REFER TO SEC 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

APPENDIX E

Set-Up Instructions

Introduction

Appendix E provides the information necessary to properly set-up the **O-100 Horizontal Propane Tank Fire Trainer** for use with the **HOSE-LINE FIRETRAINER® O-100.**

WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 HORIZONTAL PROPANE TANK FIRE TRAINER A THOROUGH UNDERSTANDING AND FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

IN ADDITION TO THE SETUP STEPS LISTED BELOW, IF THE HOSE LINE FIRETRAINER® O-100 IS TO BE CONTROLLED BY ANY OF THE FOLLOWING OPTIONAL KIDDE CONTROL SYSTEMS, REFER TO THE APPROPRIATE HOSE LINE FIRETRAINER® O-100 CONTROL SYSTEM USERS MANUAL FOR SETUP INSTRUCTIONS.

- **MULTI CONTROL**
- **WIRELESS CONTROL**
- **BATTERY POWERED CONTROL**

Installation Instructions

1. Locate the Propane Tank Mock-Up outdoors; clear of all ground/aerial obstructions on level ground, a minimum of 40 feet downwind of the Propane Supply.

NOTE

IF THE Kidde Multi Control System is used, the Multi Control Stand is considered the propane supply.

2. Position the Pilot Box Assembly so the pilot shroud is over the LPG Vapor Burner Element as shown in figure 1.
3. Remove the wheels from the Pilot Box Assembly (see figure 1). Store wheels safely away from training area.
4. Extend all lines (gas, control, electric) and check for any damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF ANY OF THE CONTROL OR ELECTRICAL LINES ARE DAMAGED. DO NOT PROCEED ANY FURTHER IF ANY OF THE GAS HOSES ARE DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

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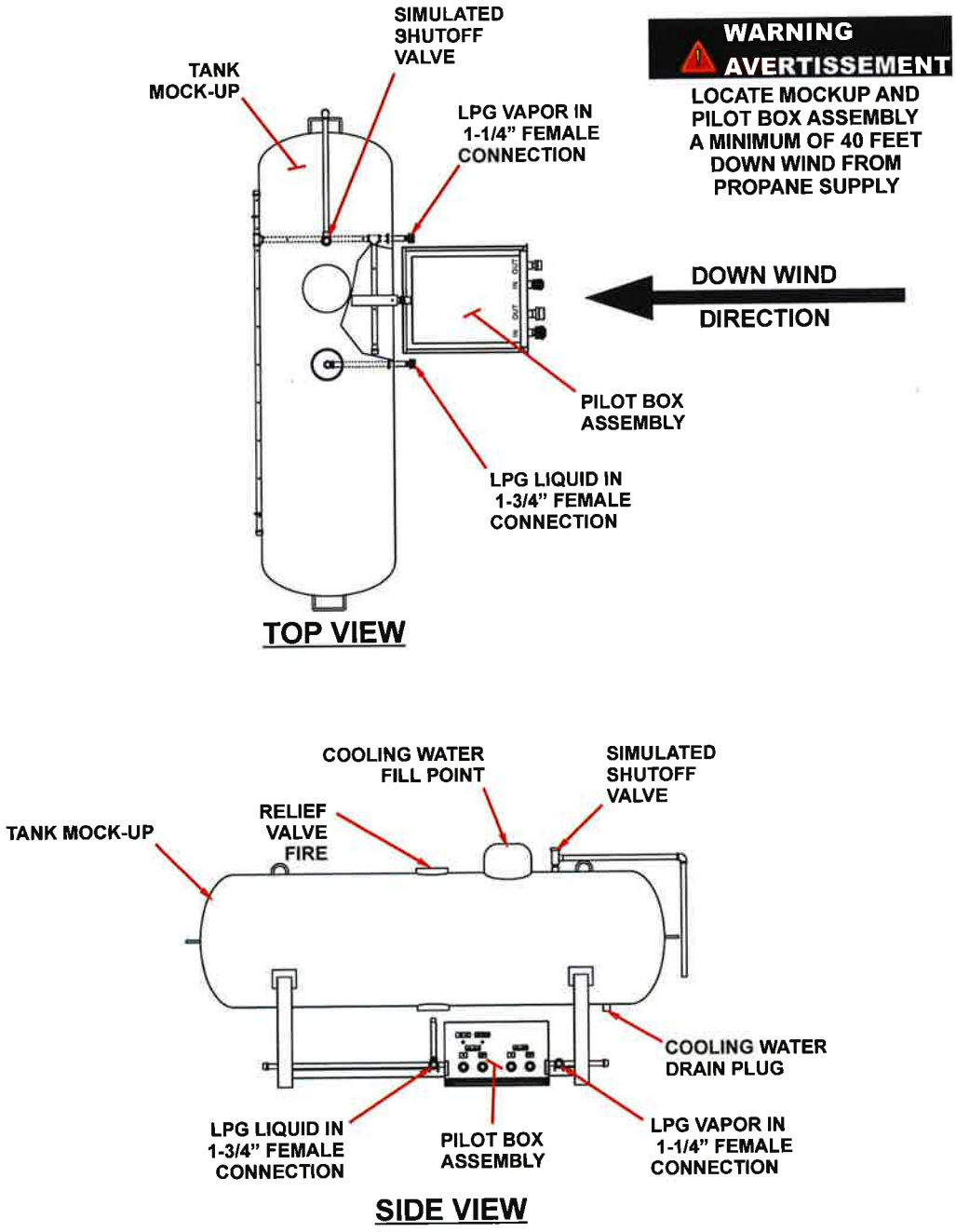

Figure 1. O-100 Horizontal Propane Tank Fie Trainer Setup and Overview

5. At the Pilot Box Assembly remove protective thread cap from the 1-1/4" male **LPG VAPOR** connection marked **"IN".** Make sure the 1-1/4" male **LPG VAPOR** connection is free of debris, damage and that the rubber washer is present (see Figure 2).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE LPG VAPOR "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

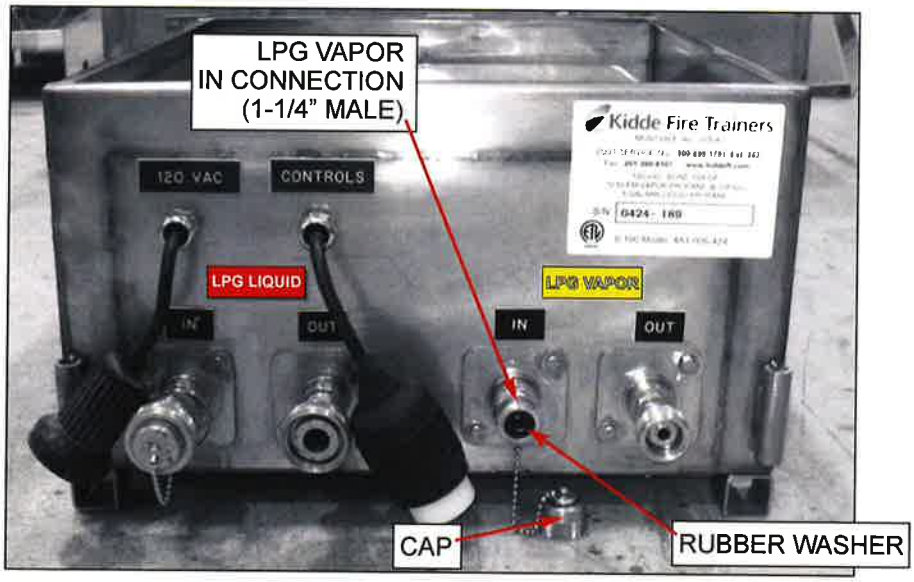

Figure 2. Pilot Box Assembly LPG VAPOR "IN" Connection

6. Make sure the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

7. Connect the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose to the 1-1/4" male **LPG VAPOR** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 3).

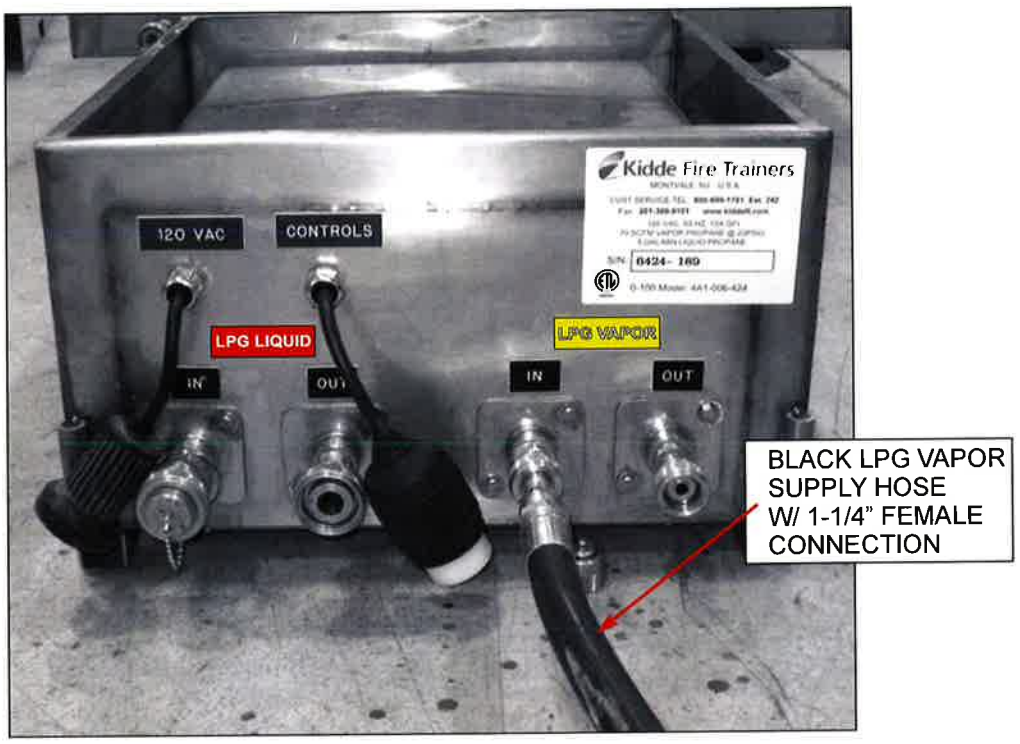

Figure 3. Pilot Box Assembly LPG VAPOR "IN" Supply Hose Connection

8. At the Pilot Box Assembly remove protective thread cap from the 1-3/4" male **LPG LIQUID** connection marked **"IN".** Make sure the 1-3/4" male **LPG LIQUID** connection is free of debris, damage and that the rubber washer is present (see Figure 4).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE LPG LIQUID "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

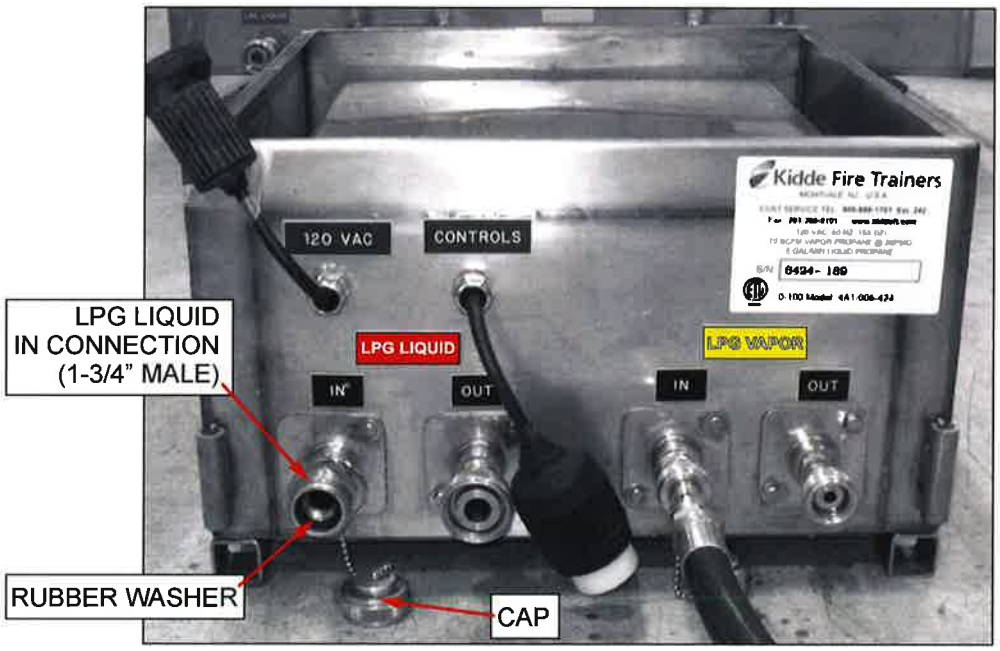

Figure 4. Pilot Box Assembly LPG LIQUID "IN" Connection

9. Make sure the 1-3/4" female connection on the 50' black rubber LPG Liquid supply hose is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

10. Connect the 1-3/4" female connection on the 50' black rubber LPG Liquid supply hose to the 1-3/4" male **LPG LIQUID** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 5).

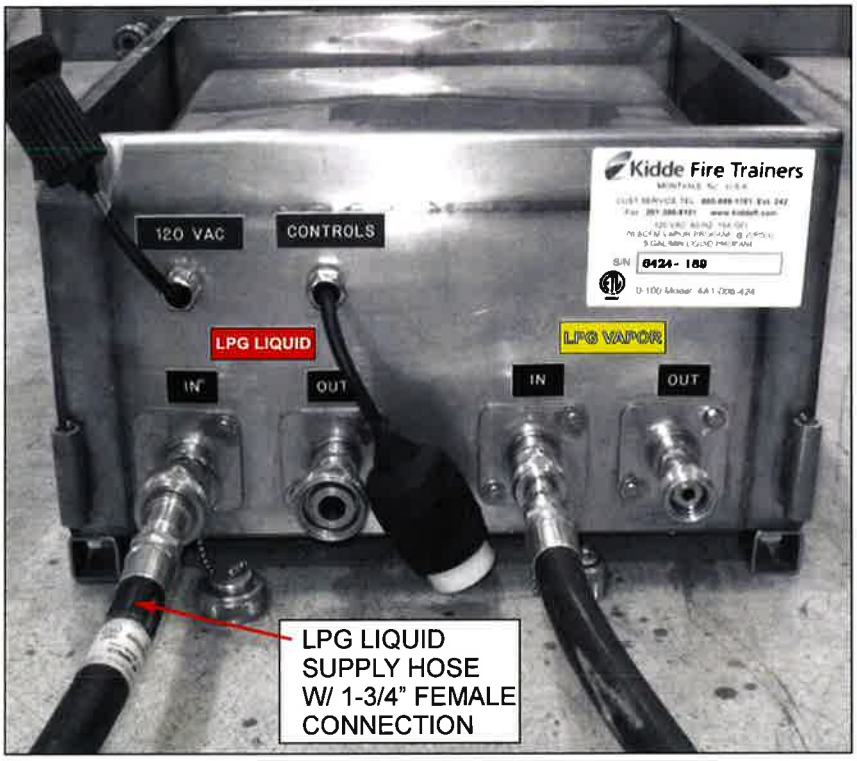

Figure 5. Pilot Box Assembly LPG LIQUID "IN" Supply Hose Connection

11. At the Pilot Box Assembly make sure the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE LPG VAPOR "OUT" CONNECTION HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

12. Remove protective thread cap from the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

13. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 6).

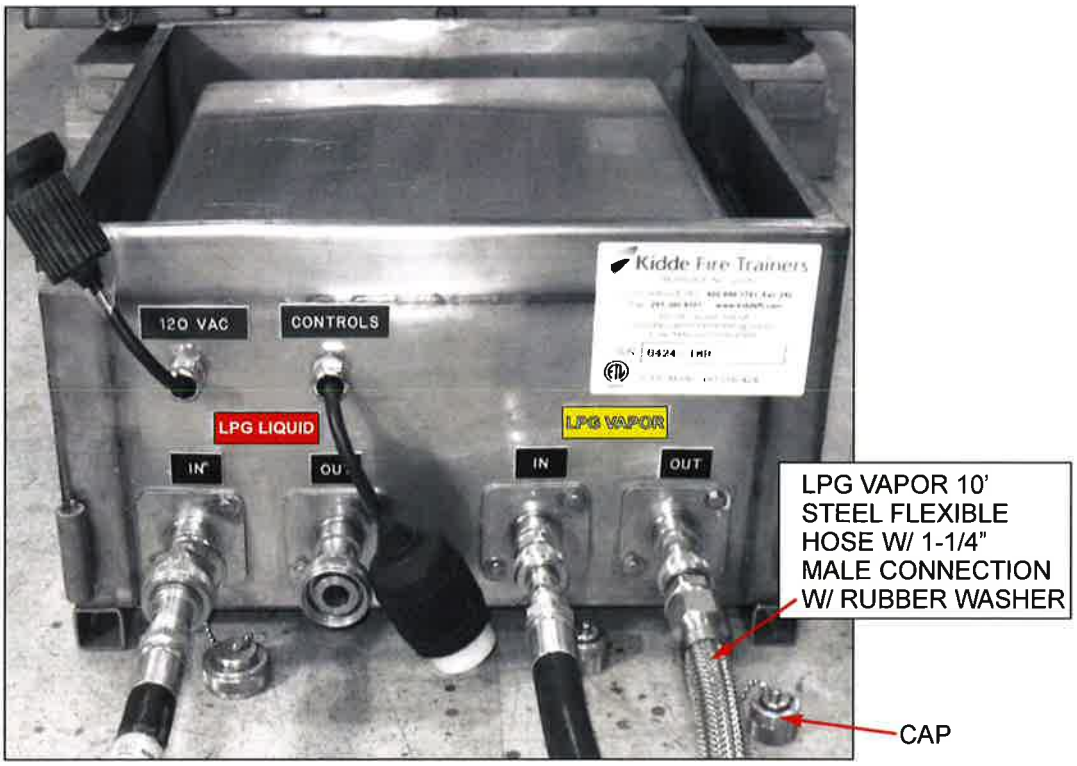

Figure 6. Pilot Box Assembly LPG VAPOR "OUT" Hose Connection

14. Remove protective thread cap form the 1-1/4" male connection on the remaining end of the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

15. At the Horizontal Tank Mockup, make sure the LPG Vapor IN 1-1/4" female connection is free of debris and damage (see Figure 8 for location).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE LPG VAPOR 1-1/4" FEMALE CONNECTION ON THE HORIZONTAL TANK MOCKUP HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

16. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the LPG Vapor 1-1/4" female connection on the Horizontal Tank Mockup. Attach fittings hand tight (see Figure 8).

17. Remove protective thread cap from the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose. Make sure the 1-3/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

18. At the Pilot Box Assembly make sure the 1-3/4" female **LPG LIQUID** connection marked **OUT** is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION MARKED LPG LIQUID "OUT" HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

19. Connect the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose to the 1-3/4" female **LPG LIQUID** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 7).

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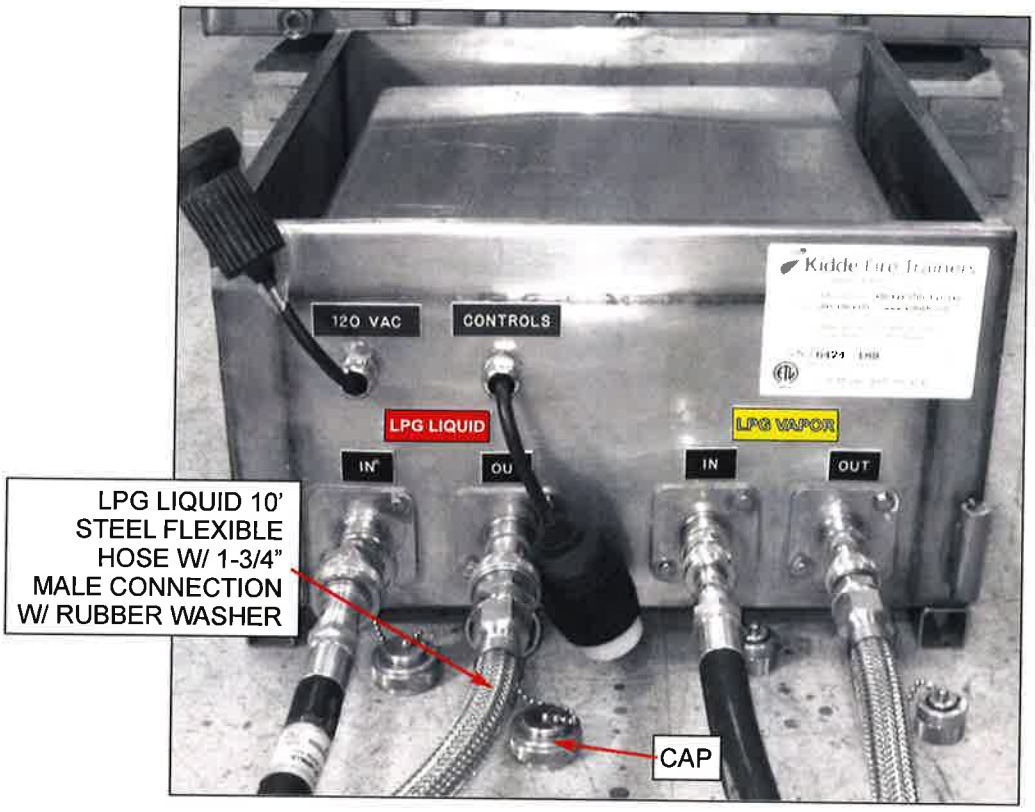

Figure 7. Pilot Box Assembly LPG LIQUID "OUT" Hose Connection

20. Remove protective thread cap form the 1-3/4" male connection on the remaining end of the 10' flexible stainless steel LPG Liquid supply hose. Make sure the 1-3/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

21. At the Horizontal Tank Mockup make sure the LPG Liquid IN 1-3/4" female connection is free of debris and damage (See Figure 8 for location).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE LPG LIQUID 1-3/4" FEMALE CONNECTION ON THE HORIZONTAL TANK MOCKUP HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

22. Connect the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose to the LPG Liquid 1-3/4" female connection on the Horizontal Tank Mockup. Attach fittings hand tight (see Figure 8).

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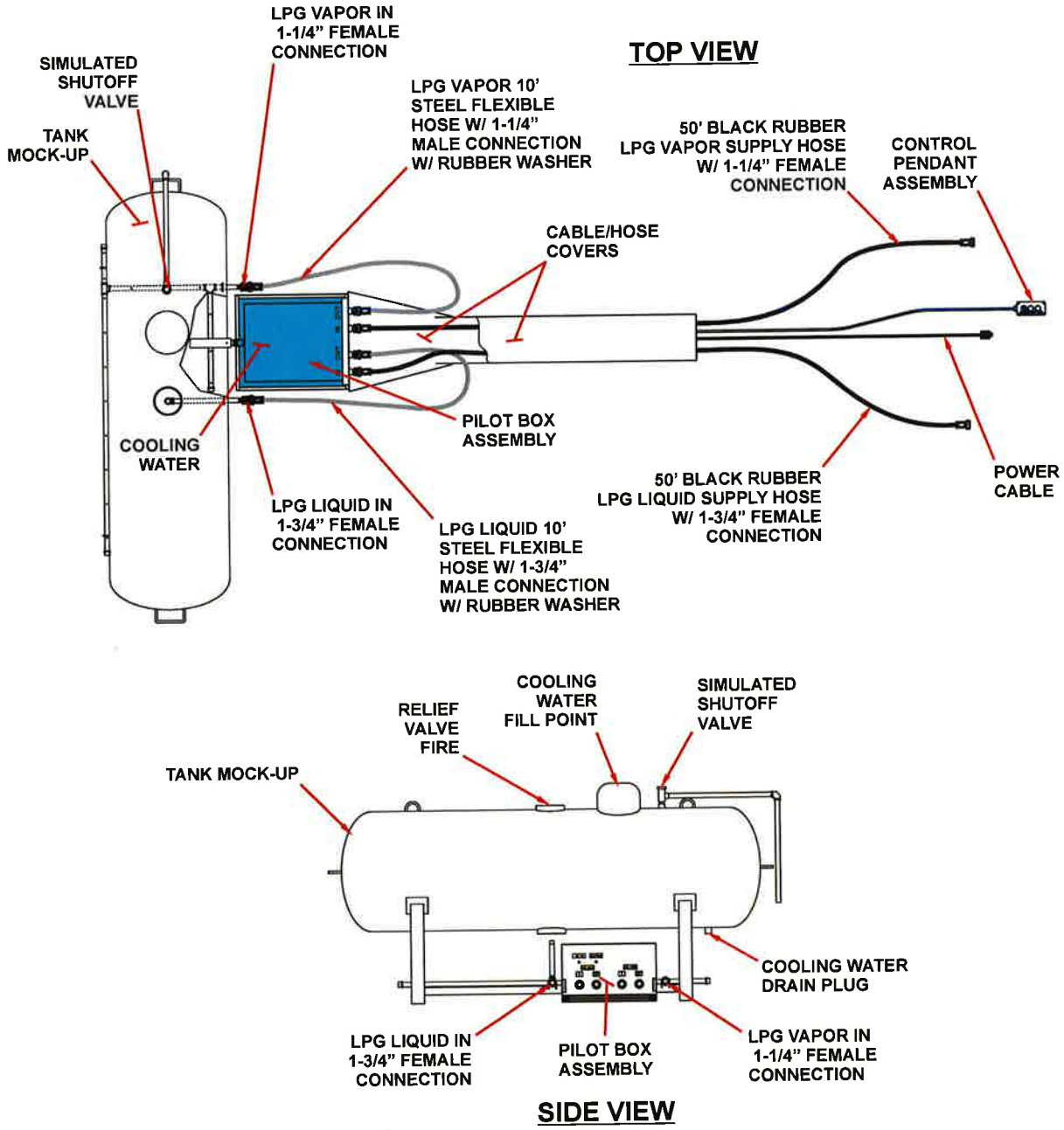

Figure 8. O-100 Horizontal Propane Tank Fie Trainer Setup

23. Attach the Control Pendant cable to the connection marked **"CONTROLS"** on Pilot Box Assembly (see Figure 9).

24. Attach the Electric Supply cord to the connection marked **"120 VAC"** on Pilot Box Assembly (see Figure 9).

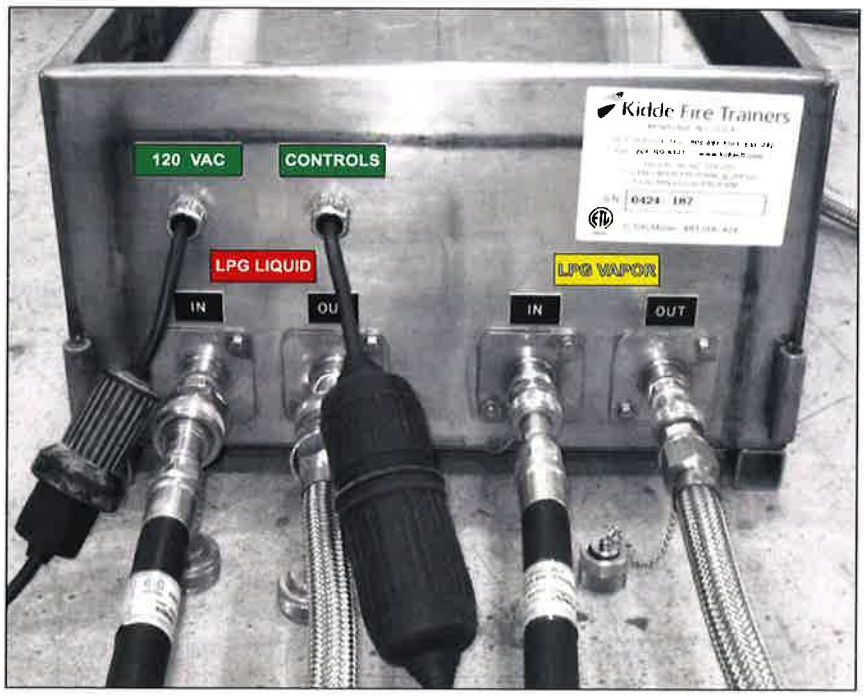

Figure 9. Electrical Supply & Control Pendant Connections

25. Ensure that drain plug is installed in the Propane Tank Mock-Up. Lift cover and fill Tank prop with water, until it overflows. Close cover. Refer to Figure 8 for location.

26. Ensure that water drain plug is installed in the Pilot Box Assembly. Drain plug located at front of Pilot Box Assembly.

27. Fill Pilot Box Assembly top with water until it overflows (see Figure 8).

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

NOTE

IF THE HOSE LINE FIRETRAINER® O-100 is to be controlled by the Kidde Multi Control System, refer to the HOSE LINE FIRETRAINER® O-100 Multi Control System Users Manual, Sections 3.1 & 3.2.

28. Using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088), connect the remaining end of the 50' black rubber LPG vapor supply hose from the Pilot Box Assembly **LPG VAPOR** connection marked **"IN"**, to **LPG Vapor Outlet** on the Propane Supply. Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male adaptor connection. Attach fittings hand tight.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER LPG VAPOR supply hose is connected to the LPG VAPOR outlet on the propane supply tank.

29. Using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088), connect the remaining end of the 50' black rubber LPG liquid supply hose from the Pilot Box Assembly **LPG LIQUID** connection marked **"IN"**, to the **LPG LIQUID Output** on the Propane Supply. Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male adaptor connection. Attach fittings hand tight.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER LPG LIQUID supply hose is connected to the LPG LIQUID outlet on the propane supply tank.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Prior to Operation

Pilot Box Valves Operational Test

1. Perform the Pilot Box Valves Operational Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Box Valves Operational Test.

WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT BOX VALVES OPERATIONAL TEST HAS BEEN SUCCESSFULLY COMPLETED.

Pilot Flame Test

1. Perform the Pilot Flame Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Flame Test.

WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT FLAME TEST HAS BEEN SUCCESSFULLY COMPLETED.

Liquid Propane Leak check

1. Perform the Liquid Propane Leak Check. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Liquid Propane Leak Check.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER UNTIL THE LIQUID PROPANE LEAK CHECK HAS BEEN SUCCESSFULLY COMPLETED.

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Operation

1. Run (gas, control and electric) lines straight back from Pilot Box Assembly.
2. Install cable/hose covers. Refer to Figure 8.

⚠ CAUTION

DO NOT OPERATE O-100 TRAINER WITHOUT BOTH CABLE/HOSE COVERS IN PLACE! REFER TO FIGURE 8.

⚠ WARNING • AVERTISSEMENT

ONLY WHEN ALL THE ABOVE OPERATIONAL AND LEAK CHECKS HAVE BEEN SUCCESSFULLY PERFORMED YOU ARE READY TO TRAIN.

3. Attach the electric supply cord from the Pilot Box assembly to an outdoor 120V power GFIC receptacle.

NOTE

FOR OPERATION INSTRUCTIONS REFER TO THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

⚠ WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 HORIZONTAL PROPANE TANK FIRE TRAINER A THOROUGH FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

Shutdown & Disassembly

⚠ WARNING • AVERTISSEMENT

AT THE END OF A TRAINING SESSION THERE WILL STILL BE FUEL IN THE HOSES FROM THE PROPANE SUPPLY TANK TO THE MOCKUP/FIRE PAN. THIS WILL HAVE TO BE BURNT OFF BEFORE THE EQUIPMENT IS TAKEN APART. IT IS IMPORTANT THAT THIS PROCEDURE BE FOLLOWED EVEN IF YOU ARE NOT TAKING EQUIPMENT APART BUT ARE JUST SHUTTING DOWN AT THE END OF A TRAINING SESSION. FOR THIS PROCEDURE REFER TO SECTION 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

NOTE

FOR SHUTDOWN & DISASSEMBLY INSTRUCTIONS REFER TO SEC 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

APPENDIX F

Set-Up Instructions

Introduction

Appendix F provides the information necessary to properly set-up the **O-100 Split Flange Fire Trainer** for use with the **HOSE-LINE FIRETRAINER® O-100.**

WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 SPLIT FLANGE FIRE TRAINER A THOROUGH UNDERSTANDING AND FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

IN ADDITION TO THE SETUP STEPS LISTED BELOW, IF THE HOSE LINE FIRETRAINER® O-100 IS TO BE CONTROLLED BY ANY OF THE FOLLOWING OPTIONAL KIDDE CONTROL SYSTEMS, REFER TO THE APPROPRIATE HOSE LINE FIRETRAINER® O-100 CONTROL SYSTEM USERS MANUAL FOR SETUP INSTRUCTIONS.

- **MULTI CONTROL**
- **WIRELESS CONTROL**
- **BATTERY POWERED CONTROL**

Components List

Component	Quantity	Part #
Split Flange Fire Assembly	1	4A1-007-022

Installation Instructions

1. Locate the Split Flange Mock-Up outdoors; clear of all ground/aerial obstructions on level ground, a minimum of 40 feet downwind of the Propane Supply.

NOTE

IF THE Kidde Multi Control System is used, the Multi Control Stand is considered the propane supply.

2. Position the Pilot Box Assembly as shown in figure 1.
3. Remove the wheels from the Pilot Box Assembly (see figure 2). Store wheels safely away from training area.
4. Extend all lines (gas, control, electric) and check for any damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF ANY OF THE CONTROL OR ELECTRICAL LINES ARE DAMAGED. DO NOT PROCEED ANY FURTHER IF ANY OF THE GAS HOSES ARE DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

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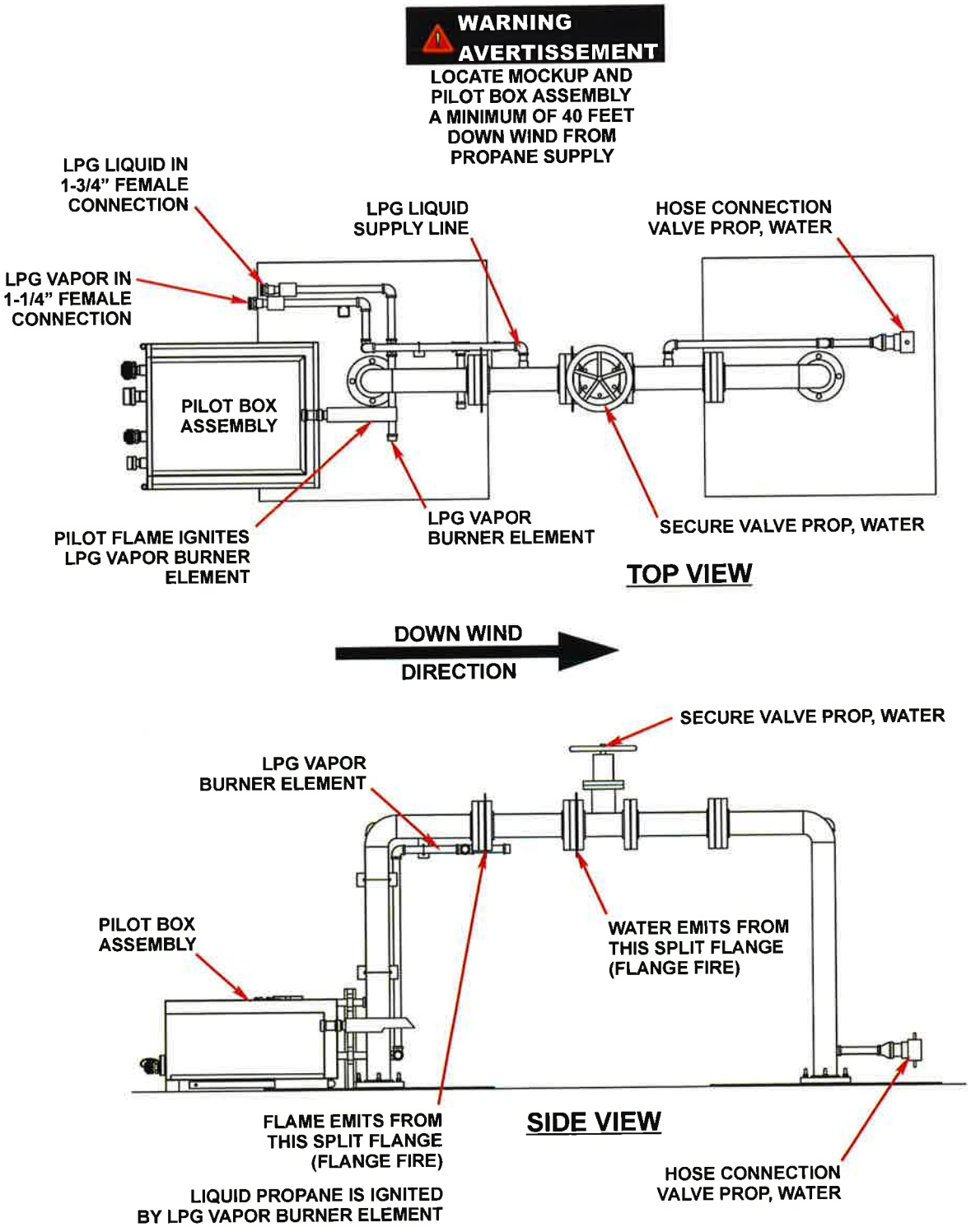

Figure 1. O-100 Split Flange Fire Trainer Setup and Overview

5. At the Pilot Box Assembly remove protective thread cap from the 1-1/4" male **LPG VAPOR** connection marked **"IN".** Make sure the 1-1/4" male **LPG VAPOR** connection is free of debris, damage and that the rubber washer is present (see Figure 2).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE LPG VAPOR "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

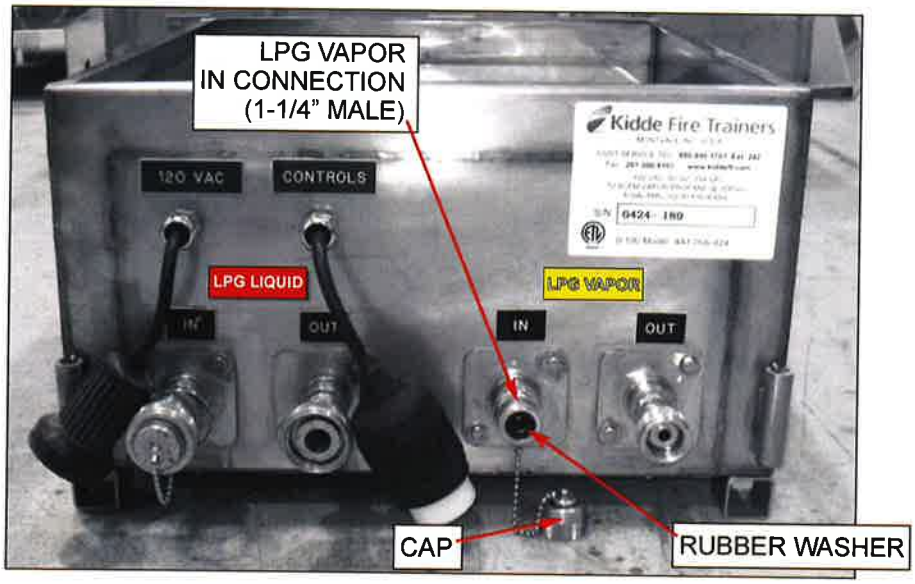

Figure 2. Pilot Box Assembly LPG VAPOR "IN" Connection

6. Make sure the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

7. Connect the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose to the 1-1/4" male **LPG VAPOR** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 3).

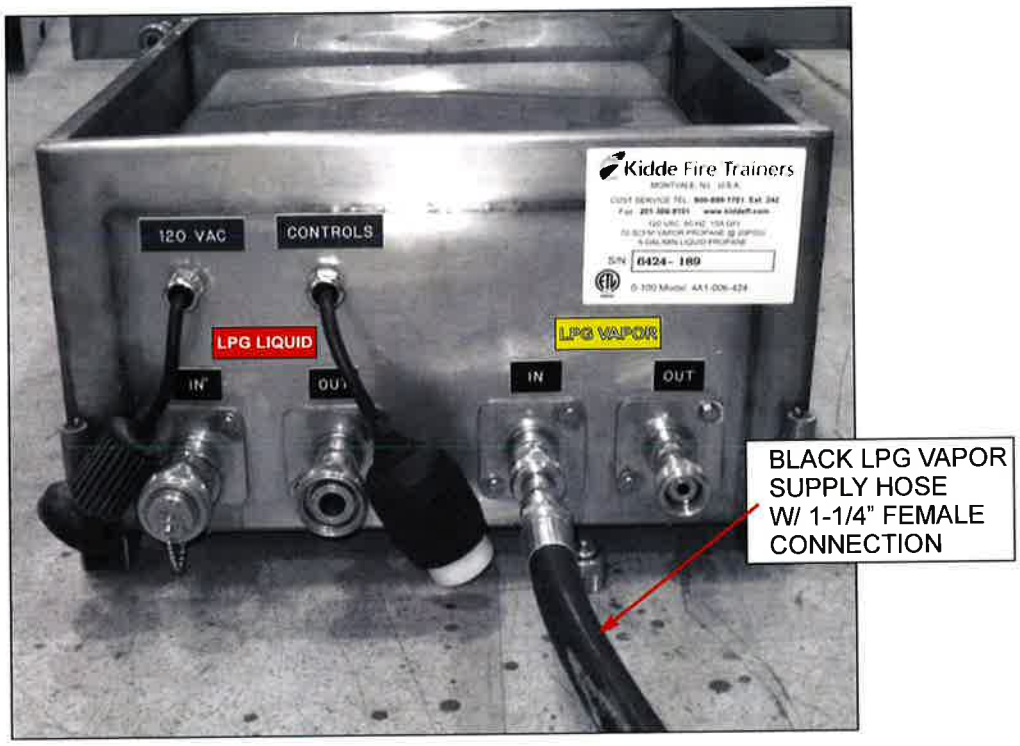

Figure 3. Pilot Box Assembly LPG VAPOR "IN" Supply Hose Connection

8. At the Pilot Box Assembly remove protective thread cap from the 1-3/4" male **LPG LIQUID** connection marked **"IN"**. Make sure the 1-3/4" male **LPG LIQUID** connection is free of debris, damage and that the rubber washer is present (see Figure 4).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE LPG LIQUID "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

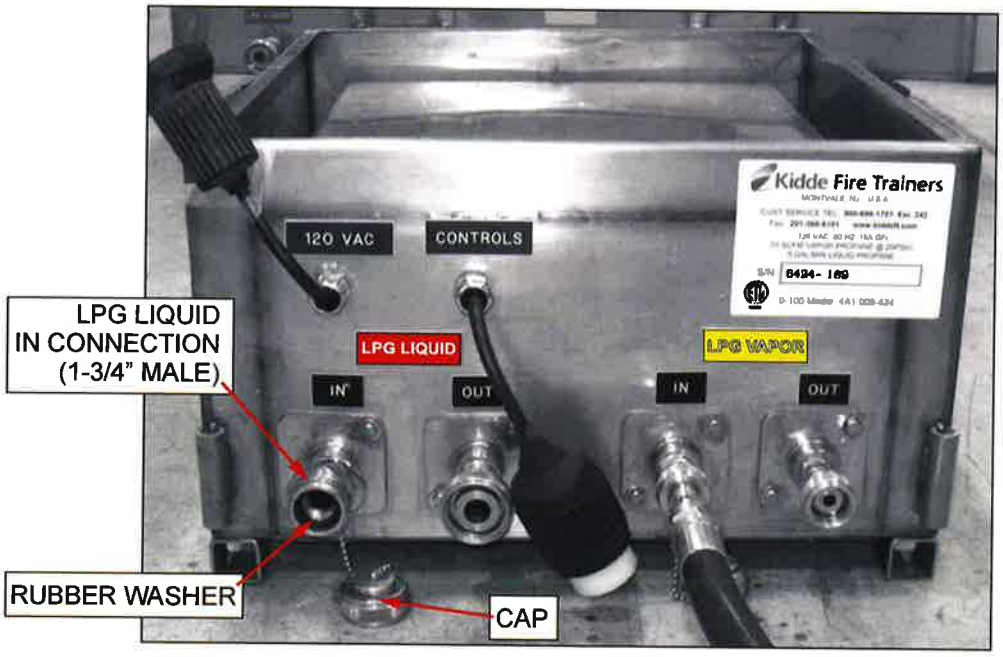

Figure 4. Pilot Box Assembly LPG LIQUID "IN" Connection

9. Make sure the 1-3/4" female connection on the 50' black rubber LPG Liquid supply hose is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

10. Connect the 1-3/4" female connection on the 50' black rubber LPG Liquid supply hose to the 1-3/4" male **LPG LIQUID** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 5).

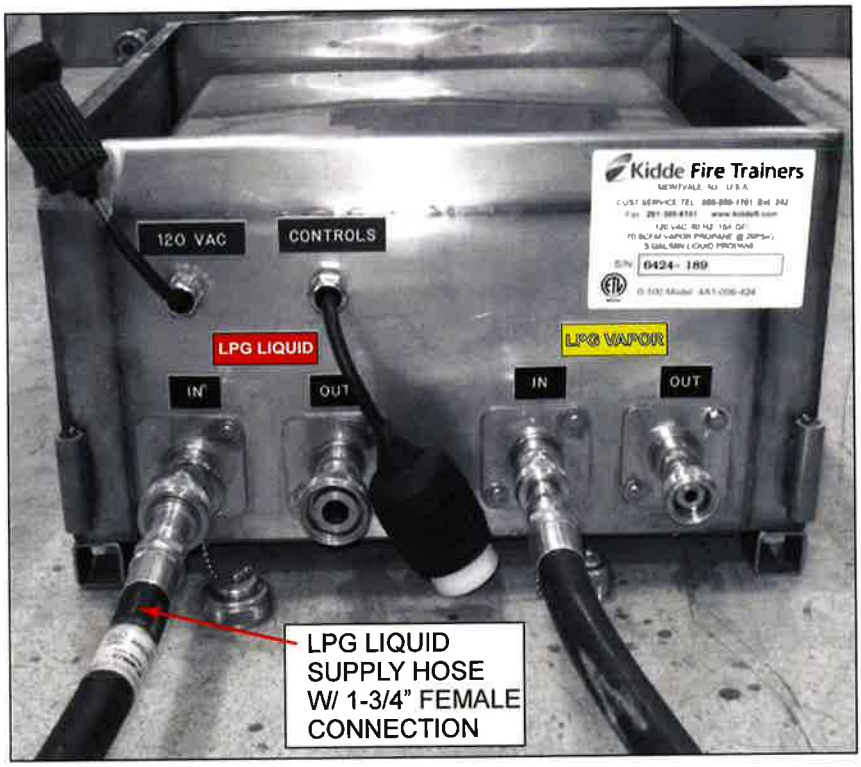

Figure 5. Pilot Box Assembly LPG LIQUID "IN" Supply Hose Connection

11. At the Pilot Box Assembly make sure the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE LPG VAPOR "OUT" CONNECTION HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

12. Remove protective thread cap from the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

13. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 6).

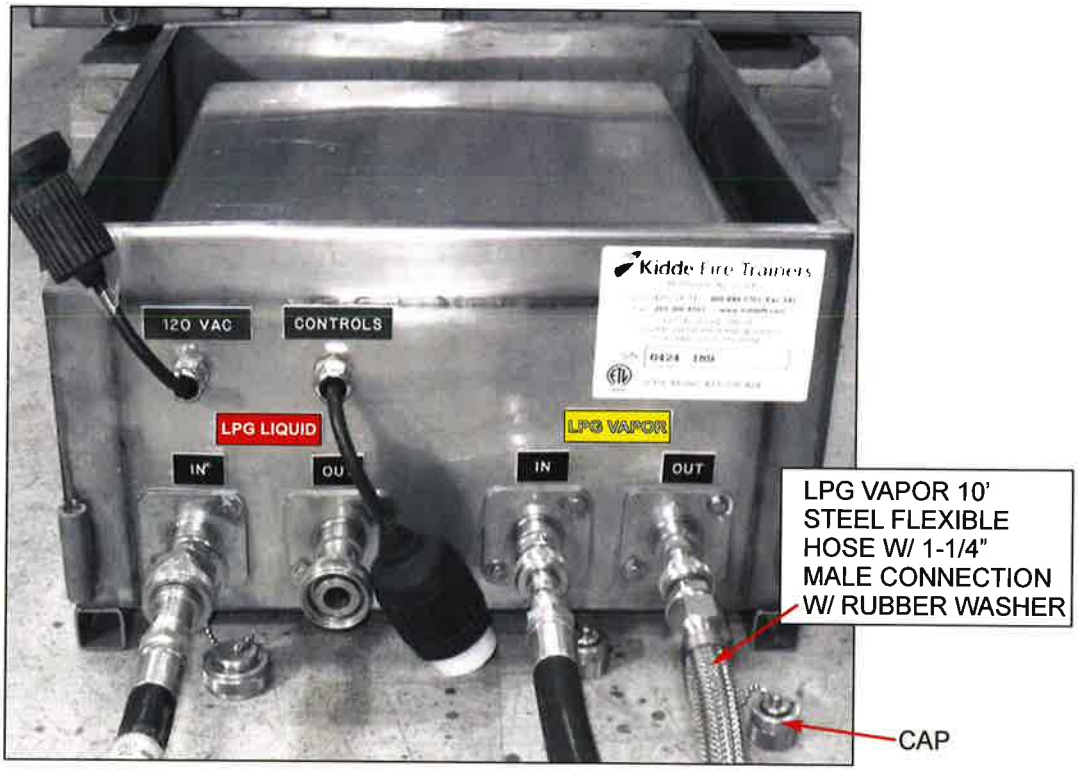

Figure 6. Pilot Box Assembly LPG VAPOR "OUT" Hose Connection

14. Remove protective thread cap form the 1-1/4" male connection on the remaining end of the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

15. At the Split Flange Mockup, make sure the LPG Vapor IN 1-1/4" female connection is free of debris and damage (see Figure 8 for location).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE LPG VAPOR 1-1/4" FEMALE CONNECTION ON THE SPLIT FLANE MOCKUP HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

16. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the LPG Vapor 1-1/4" female connection on the Split Flange Mockup. Attach fittings hand tight (see Figure 8).

17. Remove protective thread cap from the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose. Make sure the 1-3/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

18. At the Pilot Box Assembly make sure the 1-3/4" female **LPG LIQUID** connection marked **OUT** is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION MARKED LPG LIQUID "OUT" HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

19. Connect the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose to the 1-3/4" female **LPG LIQUID** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 7).

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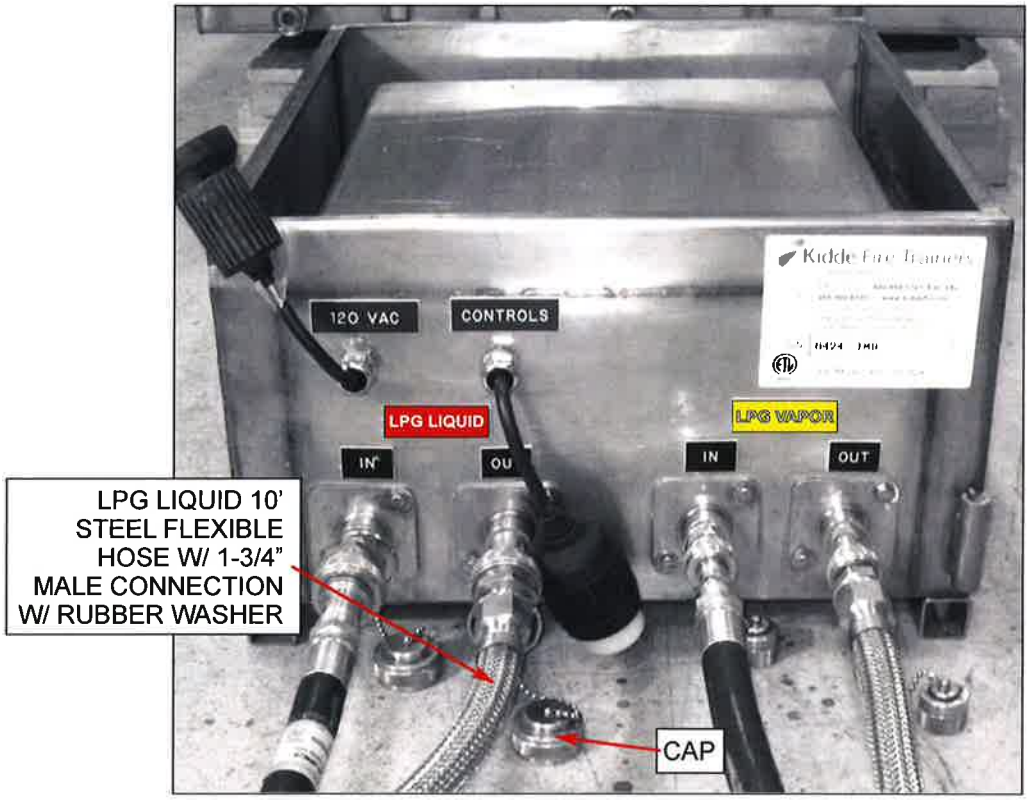

Figure 7. Pilot Box Assembly LPG LIQUID "OUT" Hose Connection

20. Remove protective thread cap form the 1-3/4" male connection on the remaining end of the 10' flexible stainless steel LPG Liquid supply hose. Make sure the 1-3/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

21. At the Split Flange Mockup make sure the LPG Liquid IN 1-3/4" female connection is free of debris and damage (See Figure 8 for location).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE LPG LIQUID 1-3/4" FEMALE CONNECTION ON THE SPLIT FLANGE MOCKUP HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

22. Connect the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose to the LPG Liquid 1-3/4" female connection on the Split Flange Mockup. Attach fittings hand tight (see Figure 8).

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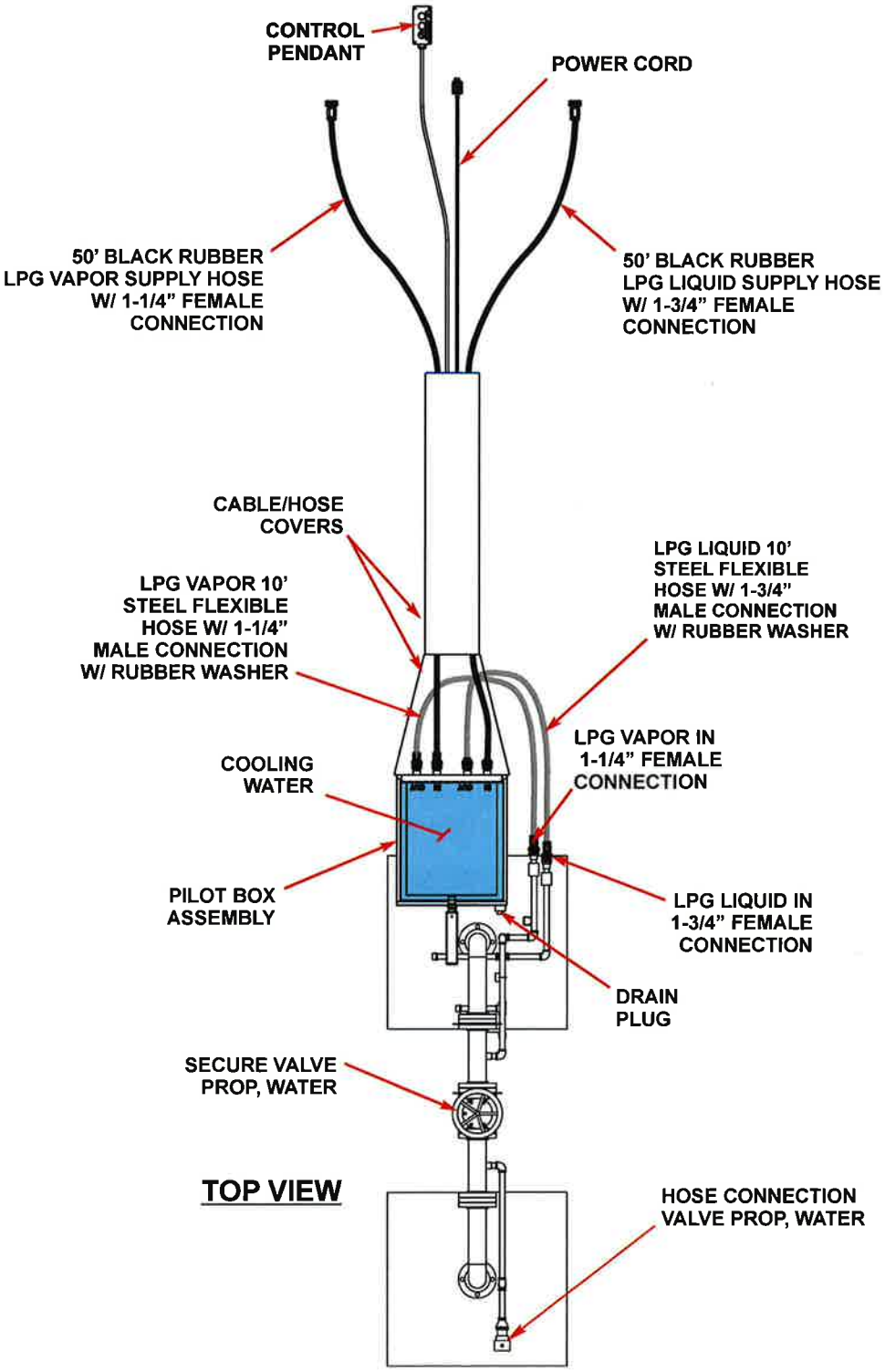

Figure 8. O-100 Split Flange Fire Trainer Setup

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

23. Attach the Control Pendant cable to the connection marked **"CONTROLS"** on Pilot Box Assembly (see Figure 9).
24. Attach the Electric Supply cord to the connection marked **"120 VAC"** on Pilot Box Assembly (see Figure 9).

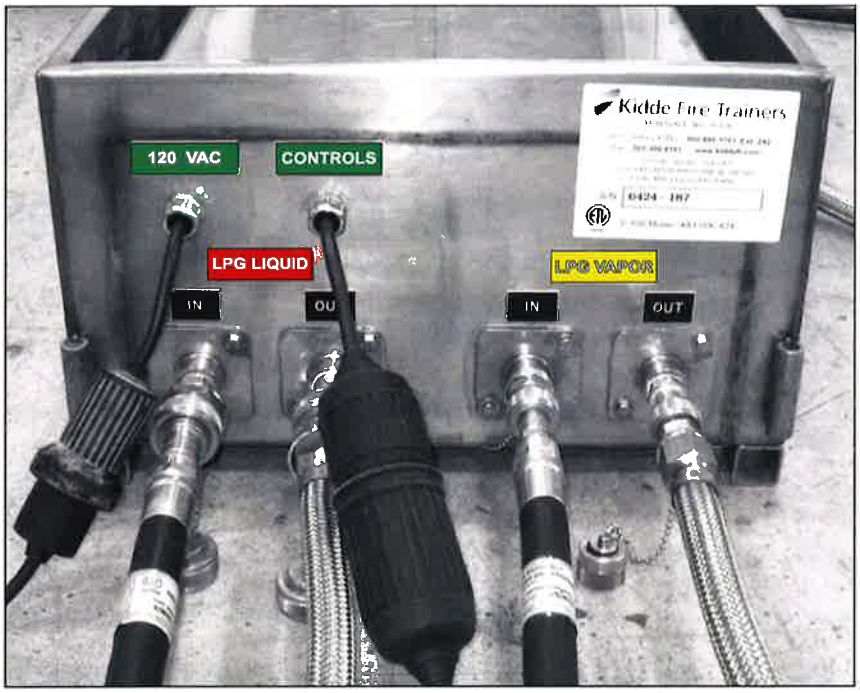

Figure 9. Electrical Supply & Control Pendant Connections

25. Ensure that drain plug is installed in the Pilot Box Assembly (see figure 8).
26. Fill Pilot Box Assembly top with water until it overflows (see figure 8).
27. Close the secure valve prop on the Split Flange Mock-Up (See figure 8).
28. Attach water supply hose to the secure valve prop water connection on the Split Flange Mock-Up (See figure 8).

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

NOTE

IF THE HOSE LINE FIRETRAINER® O-100 is to be controlled by the Kidde Multi Control System, refer to the HOSE LINE FIRETRAINER® O-100 Multi Control System Users Manual, Sections 3.1 & 3.2.

29. Using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088), connect the remaining end of the 50' black rubber LPG vapor supply hose from the Pilot Box Assembly **LPG VAPOR** connection marked **"IN"**, to **LPG Vapor Outlet** on the Propane Supply. Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male adaptor connection. Attach fittings hand tight.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER LPG VAPOR supply hose is connected to the LPG VAPOR outlet on the propane supply tank.

30. Using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088), connect the remaining end of the 50' black rubber LPG liquid supply hose from the Pilot Box Assembly **LPG LIQUID** connection marked **"IN"**, to the **LPG LIQUID Output** on the Propane Supply. Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male adaptor connection. Attach fittings hand tight.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER LPG LIQUID supply hose is connected to the LPG LIQUID outlet on the propane supply tank.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Prior to Operation

Pilot Box Valves Operational Test

1. Perform the Pilot Box Valves Operational Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Box Valves Operational Test.

WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT BOX VALVES OPERATIONAL TEST HAS BEEN SUCCESSFULLY COMPLETED.

Pilot Flame Test

1. Perform the Pilot Flame Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Flame Test.

WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT FLAME TEST HAS BEEN SUCCESSFULLY COMPLETED.

Liquid Propane Leak check

1. Perform the Liquid Propane Leak Check. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Liquid Propane Leak Check.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER UNTIL THE LIQUID PROPANE LEAK CHECK HAS BEEN SUCCESSFULLY COMPLETED.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Operation

1. Run (gas, control and electric) lines straight back from Pilot Box Assembly.
2. Install cable/hose covers. Refer to Figure 8.

⚠ CAUTION

DO NOT OPERATE O-100 TRAINER WITHOUT BOTH CABLE/HOSE COVERS IN PLACE! REFER TO FIGURE 8.

⚠ WARNING • AVERTISSEMENT

ONLY WHEN ALL THE ABOVE OPERATIONAL AND LEAK CHECKS HAVE BEEN SUCCESSFULLY PERFORMED YOU ARE READY TO TRAIN.

3. Attach the electric supply cord from the Pilot Box assembly to an outdoor 120V power GFIC receptacle.

NOTE

FOR OPERATION INSTRUCTIONS REFER TO THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

⚠ WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 SPLIT FLANGE FIRE TRAINER A THOROUGH FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

NOTE

THE SPLIT FLANGE MOCK-UP LPG LIQUID AND VAPOR LINES ARE EACH EQUIPPED WITH A BUTTERFLY VALVE AT THE HOSE CONNECTION (SEE FIGURE 8). THESE VALVES ARE FACTORY SET AND SHOULD NOT REQUIRE ADJUSTMENT. IN THE EVENT THAT AN ADJUSTMENT IS REQUIRED, LOOSEN THE SOCKET HEAD CAP SCREW AND ROTATE TOP OF VALVE (CLOCKWISE TO INCREASE FLOW) (COUNTERCLOCKWISE TO REDUCE FLOW). MAKE ADJUSTMENTS IN SMALL INCREMENTS, UNTIL DESIRED PERFORMANCE IS OBTAINED. LOCK VALVE SETTING BY RE-TIGHTENING THE SOCKET HEAD CAP SCREW.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Shutdown & Disassembly

WARNING • AVERTISSEMENT

AT THE END OF A TRAINING SESSION THERE WILL STILL BE FUEL IN THE HOSES FROM THE PROPANE SUPPLY TANK TO THE MOCKUP/FIRE PAN. THIS WILL HAVE TO BE BURNT OFF BEFORE THE EQUIPMENT IS TAKEN APART. IT IS IMPORTANT THAT THIS PROCEDURE BE FOLLOWED EVEN IF YOU ARE NOT TAKING EQUIPMENT APART BUT ARE JUST SHUTTING DOWN AT THE END OF A TRAINING SESSION. FOR THIS PROCEDURE REFER TO SECTION 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

NOTE

FOR SHUTDOWN & DISASSEMBLY INSTRUCTIONS REFER TO SEC 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

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Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

APPENDIX G

Set-Up Instructions

Introduction

Appendix G provides the information necessary to properly set-up the **O-100 Motor Generator Mock-up** for use with the **HOSE-LINE FIRETRAINER® O-100.**

WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 MOTOR GENERATOR FIRE TRAINER A THOROUGH UNDERSTANDING AND FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

IN ADDITION TO THE SETUP STEPS LISTED BELOW, IF THE HOSE LINE FIRETRAINER® O-100 IS TO BE CONTROLLED BY ANY OF THE FOLLOWING OPTIONAL KIDDE CONTROL SYSTEMS, REFER TO THE APPROPRIATE HOSE LINE FIRETRAINER® O-100 CONTROL SYSTEM USERS MANUAL FOR SETUP INSTRUCTIONS.

- **MULTI CONTROL**
- **WIRELESS CONTROL**
- **BATTERY POWERED CONTROL**

Installation instructions

1. Locate the 4' x 6' Fire Pan Assembly outdoors; clear of all ground/aerial obstructions on level ground, a minimum of 40 feet downwind of the Propane Supply Tank with the connections facing the Propane Supply Tank (see Figure 1). Remove wheels from the Fire Pan Assembly and store wheels safely away from trainer.

NOTE

IF THE Kidde Multi Control System is used, the Multi Control Stand is considered the propane supply.

2. On the vapor propane center burner element, open the first ball valve and close the second one (see Figure 1).
3. Remove the wheels from the Pilot Box Assembly and store wheels safely away from training area. Be sure the Pilot Box Assembly drain plug is installed (see Figure 1). Position the Pilot Box Assembly as shown in Figure 1, with the gas connections facing the Propane Supply Tank.
4. Extend all lines (Vapor Propane gas, control, electric) and check for any damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF ANY OF THE CONTROL OR ELECTRICAL LINES ARE DAMAGED. DO NOT PROCEED ANY FURTHER IF ANY OF THE GAS HOSES ARE DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

5. Locate the Motor Generator Mock-up assembly in the Fire Pan as shown in Figure 2. Refer to Figure 3 for assembly details.

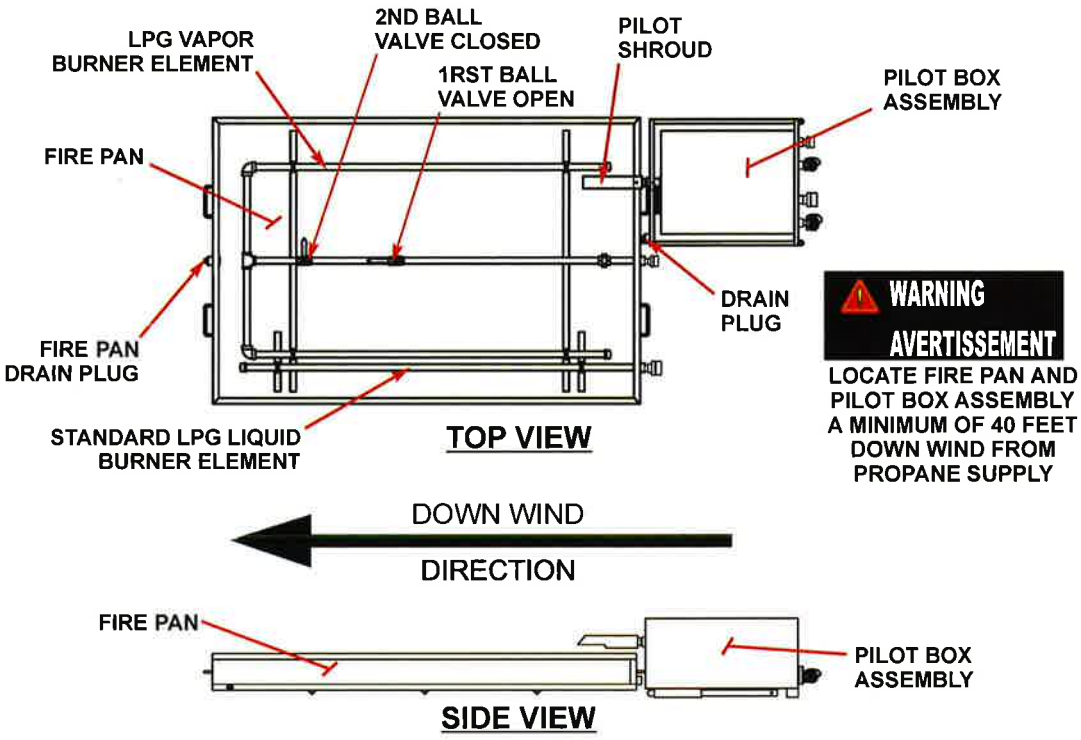

Figure 1. O-100 Fire Pan & Pilot Box Assembly Setup and Overview

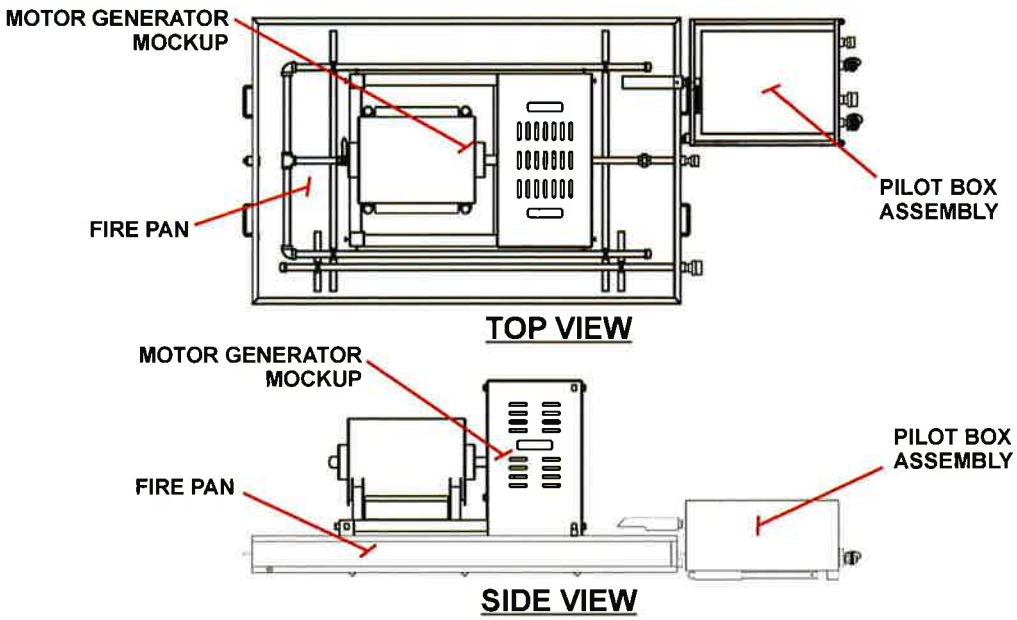

Figure 2. O-100 Motor Generator Fire Trainer Setup and Overview

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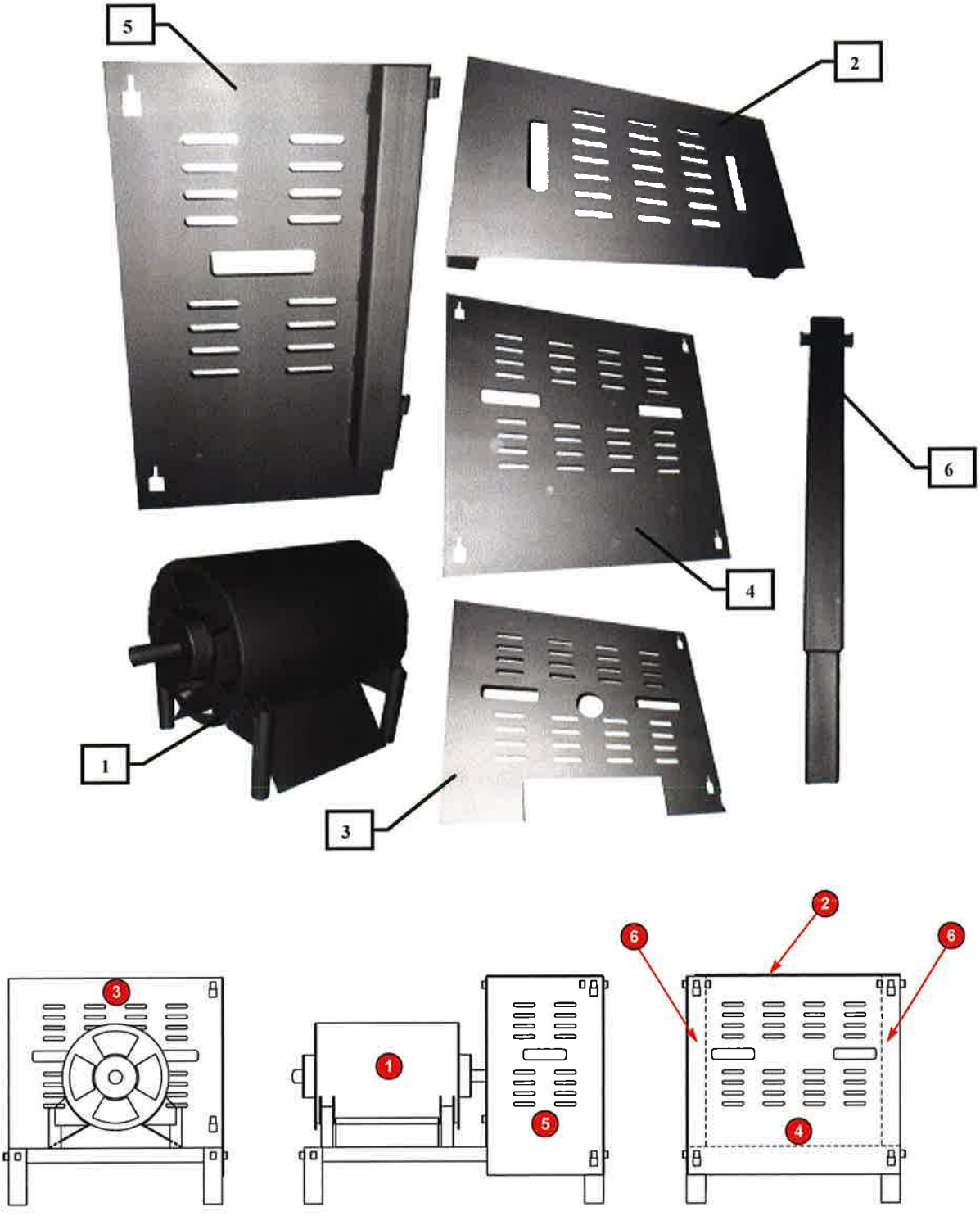

Figure 3 Motor Generator Mockup Installation and Setup

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

3 Installation Instructions

6. At the Pilot Box Assembly remove protective thread cap from the 1-1/4" male **LPG VAPOR** connection marked **"IN".** Make sure the 1-1/4" male **LPG VAPOR** connection is free of debris, damage and that the rubber washer is present (see Figure 4).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE LPG VAPOR "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

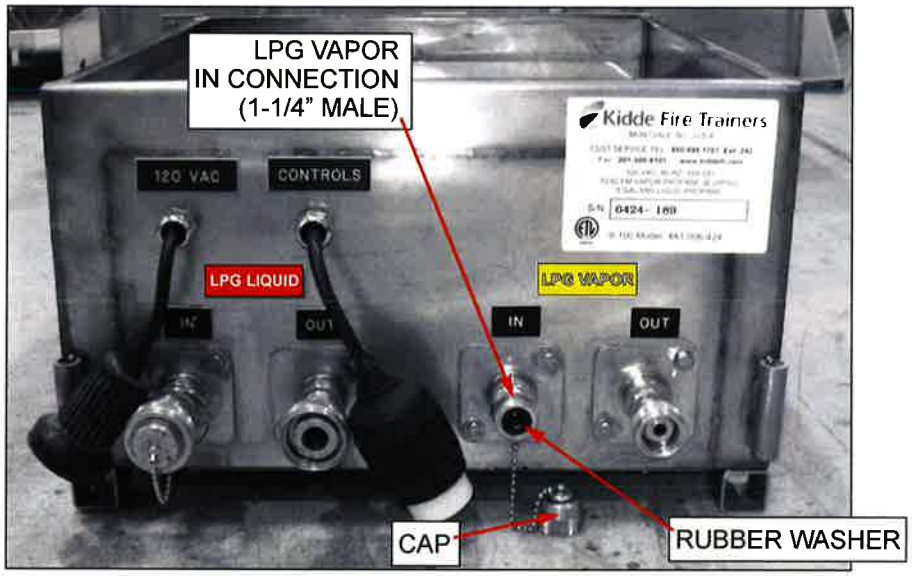

Figure 4. Pilot Box Assembly LPG VAPOR "IN" Connection

7. Make sure the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

8. Connect the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose to the 1-1/4" male **LPG VAPOR** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 5).

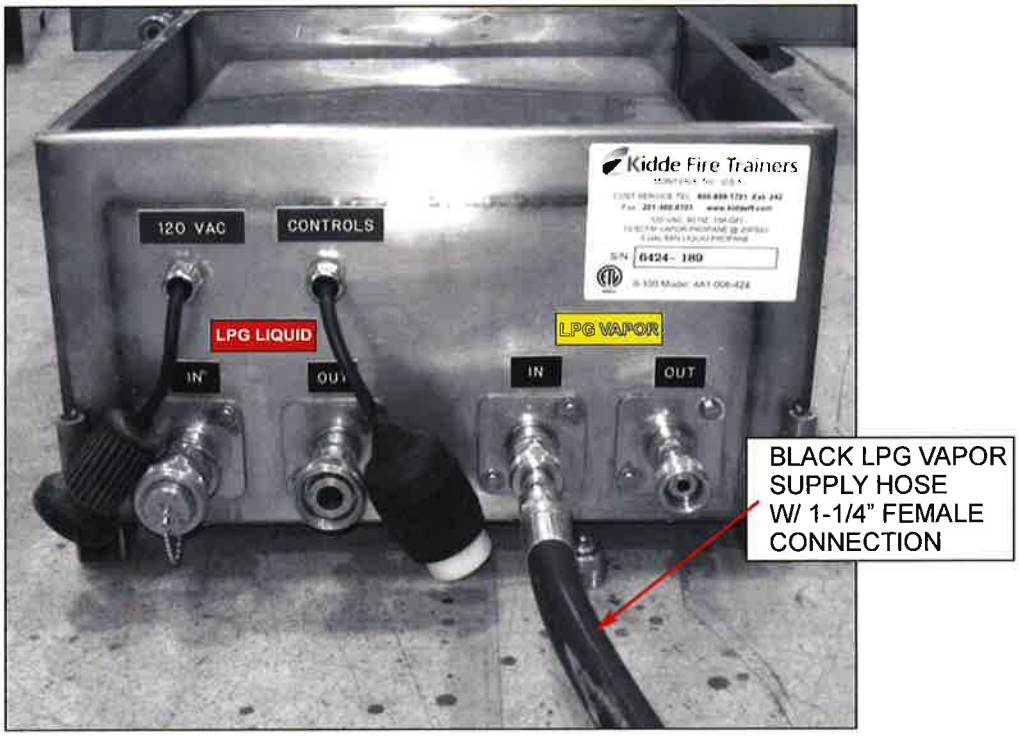

Figure 5. Pilot Box Assembly LPG VAPOR "IN" Supply Hose Connection

9. At the Pilot Box Assembly make sure the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE LPG VAPOR "OUT" CONNECTION HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

10. Remove protective thread cap from the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

11. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 6).

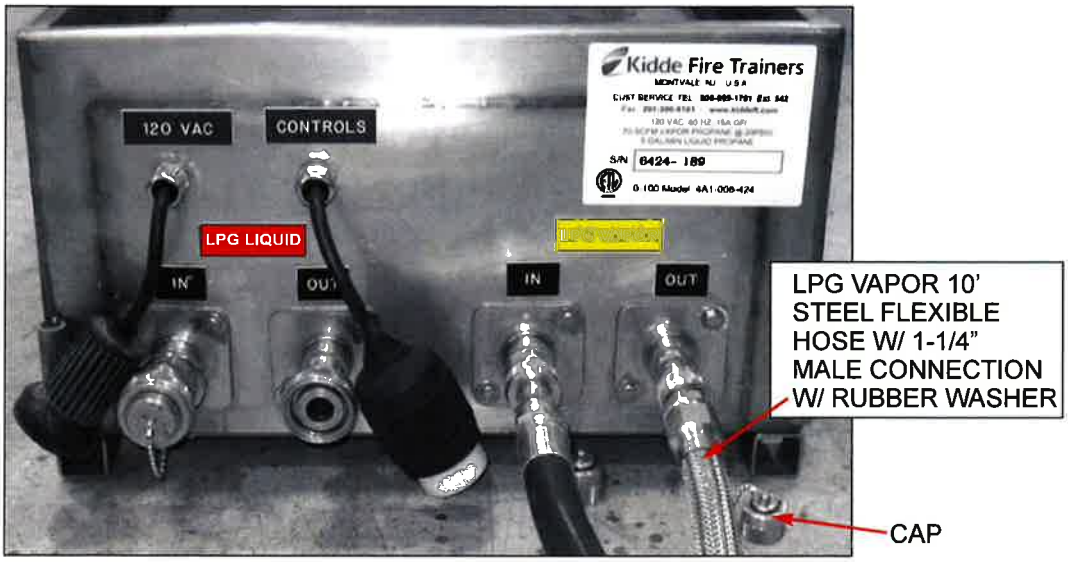

Figure 6. Pilot Box Assembly LPG VAPOR "OUT" Hose Connection

12. Remove protective thread cap form the 1-1/4" male connection on the remaining end of the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

13. At the Fire Pan Assembly make sure the 1-1/4" female connection marked **LPG VAPOR** is free of debris and damage. (see Figure 7 for location).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE LPG VAPOR 1-1/4" FEMALE CONNECTION MARKED LPG VAPOR ON THE FIRE PAN HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

14. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the 1-1/4" female connection marked **LPG VAPOR** on the Fire Pan Assembly. Attach fittings hand tight (see Figure 7).

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

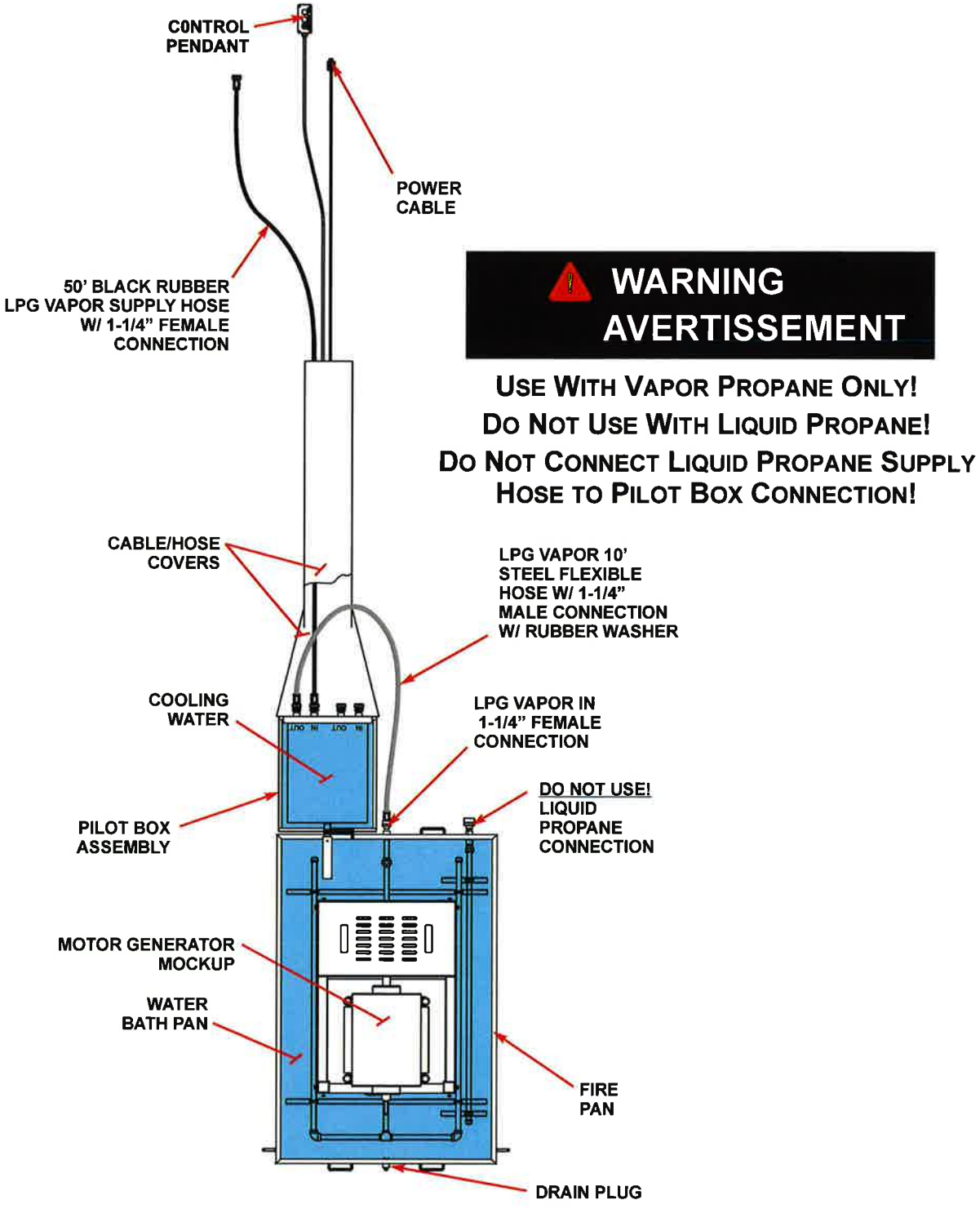

Figure 7. O-100 Motor Generator Fire Trainer Setup

15. Attach the Control Pendant cable to the connection marked **"CONTROLS"** on Pilot Box Assembly (see Figure 8).

16. Attach the Electric Supply cord to the connection marked **"120 VAC"** on Pilot Box Assembly (see Figure 8).

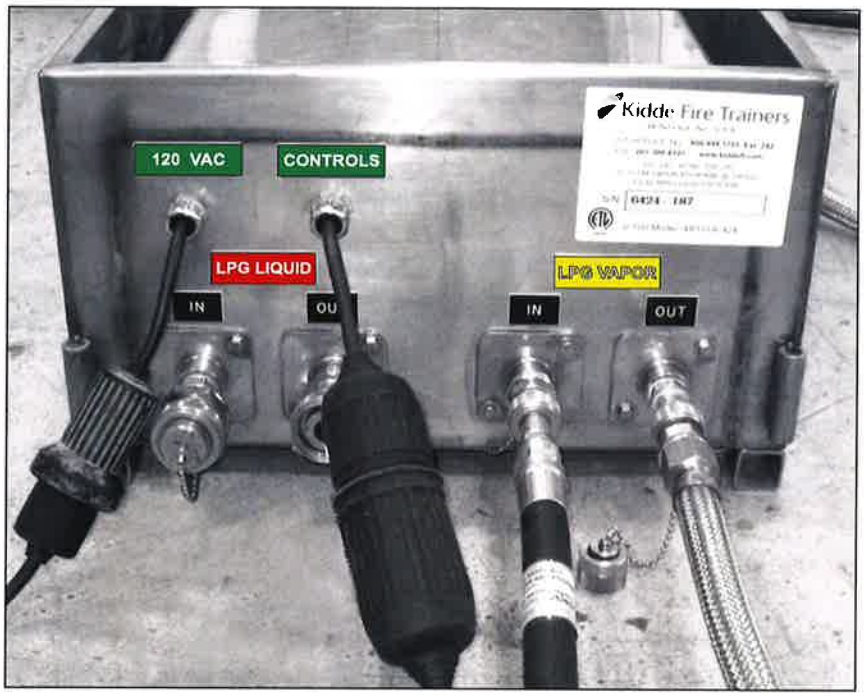

Figure 8. Electrical Supply & Control Pendant Connections

17. Fill Pilot Box Assembly top with water until it overflows (see Figure 7).

18. Ensure that drain plug is installed in the Fire Pan (see Figure 7).

19. Fill Fire Pan with water until it overflows (see Figure 7).

NOTE

IF THE HOSE LINE FIRETRAINER® O-100 is to be controlled by the Kidde Multi Control System, refer to the HOSE LINE FIRETRAINER® O-100 Multi Control System Users Manual, Sections 3.1 & 3.2.

20. Using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088), connect the remaining end of the 50' black rubber LPG vapor supply hose from the Pilot Box Assembly **LPG VAPOR** connection marked **"IN"**, to **LPG Vapor Outlet** on the Propane Supply. Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male adaptor connection. Attach fittings hand tight.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER LPG VAPOR SUPPLY HOSE IS CONNECTED TO THE LPG VAPOR OUTLET ON THE PROPANE SUPPLY TANK. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Prior to Operation

Pilot Box Valves Operational Test

1. Perform the Pilot Box LPG Vapor Valve Operational Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Box Valves Operational Test.

WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT BOX VALVES OPERATIONAL TEST HAS BEEN SUCCESSFULLY COMPLETED.

Pilot Flame Test

1. Perform the Pilot Flame Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Flame Test.

WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT FLAME TEST HAS BEEN SUCCESSFULLY COMPLETED.

Operation

1. Run (gas, control and electric) lines straight back from Pilot Box Assembly.
2. Install cable/hose covers. Refer to Figure 7.

CAUTION

DO NOT OPERATE O-100 TRAINER WITHOUT BOTH CABLE/HOSE COVERS IN PLACE! REFER TO FIGURE 7.

WARNING • AVERTISSEMENT

ONLY WHEN ALL THE ABOVE OPERATIONAL AND LEAK CHECKS HAVE BEEN SUCCESSFULLY PERFORMED YOU ARE READY TO TRAIN.

3. Attach the electric supply cord from the Pilot Box assembly to an outdoor 120V power GFIC receptacle.

NOTE

FOR OPERATION INSTRUCTIONS REFER TO THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL. FOLLOW ALL PROCEDURES FOR VAPOR PROPANE ONLY. ANY REFERENCES TO LIQUID PROPANE DOES NOT APPLY.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 MOTOR GENERATOR FIRE TRAINER A THOROUGH UNDERSTANDING AND FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FOLLOW ALL PROCEDURES FOR VAPOR PROPANE ONLY! ANY REFERENCES TO LIQUID PROPANE DOES NOT APPLY! FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

Shutdown & Disassembly

WARNING • AVERTISSEMENT

AT THE END OF A TRAINING SESSION THERE WILL STILL BE FUEL IN THE HOSES FROM THE PROPANE SUPPLY TANK TO THE MOCKUP/FIRE PAN. THIS WILL HAVE TO BE BURNT OFF BEFORE THE EQUIPMENT IS TAKEN APART. IT IS IMPORTANT THAT THIS PROCEDURE BE FOLLOWED EVEN IF YOU ARE NOT TAKING EQUIPMENT APART BUT ARE JUST SHUTTING DOWN AT THE END OF A TRAINING SESSION. FOR THIS PROCEDURE REFER TO SECTION 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

NOTE

FOR SHUTDOWN & DISASSEMBLY INSTRUCTIONS REFER TO SEC 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

APPENDIX H

Set-Up Instructions

Introduction

Appendix H provides the information necessary to properly set-up the **O-100 Paint Locker Mock-up** for use with the **HOSE-LINE FIRETRAINER® O-100.**

WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 PAINT LOCKER FIRE TRAINER A THOROUGH UNDERSTANDING AND FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

IN ADDITION TO THE SETUP STEPS LISTED BELOW, IF THE HOSE LINE FIRETRAINER® O-100 IS TO BE CONTROLLED BY ANY OF THE FOLLOWING OPTIONAL KIDDE CONTROL SYSTEMS, REFER TO THE APPROPRIATE HOSE LINE FIRETRAINER® O-100 CONTROL SYSTEM USERS MANUAL FOR SETUP INSTRUCTIONS.

- **MULTI CONTROL**
- **WIRELESS CONTROL**
- **BATTERY POWERED CONTROL**

Installation instructions

1. Locate the 4' x 6' Fire Pan Assembly outdoors; clear of all ground/aerial obstructions on level ground, a minimum of 40 feet downwind of the Propane Supply Tank with the connections facing the Propane Supply Tank (see Figure 1). Remove wheels from the Fire Pan Assembly and store wheels safely away from trainer.

NOTE

IF THE Kidde Multi Control System is used, the Multi Control Stand is considered the propane supply.

2. On the vapor propane center burner element, open the first ball valve and close the second one (see Figure 1).
3. Remove the wheels from the Pilot Box Assembly and store wheels safely away from training area. Be sure the Pilot Box Assembly drain plug is installed (see Figure 1). Position the Pilot Box Assembly as shown in Figure 1, with the gas connections facing the Propane Supply Tank.
4. Extend all lines (Vapor Propane gas, control, electric) and check for any damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF ANY OF THE CONTROL OR ELECTRICAL LINES ARE DAMAGED. DO NOT PROCEED ANY FURTHER IF ANY OF THE GAS HOSES ARE DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

5. Locate the Paint Locker Mock-up assembly in the Fire Pan as shown in Figure 2.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

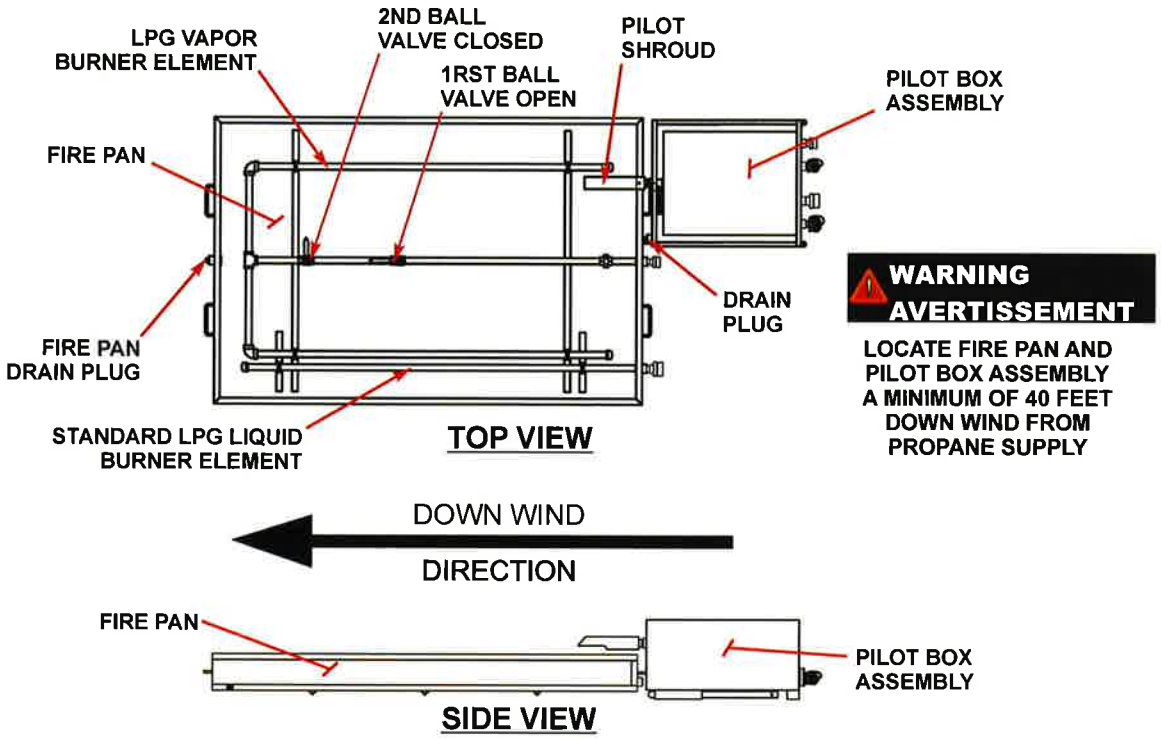

Figure 1. O-100 Fire Pan & Pilot Box Assembly Setup and Overview

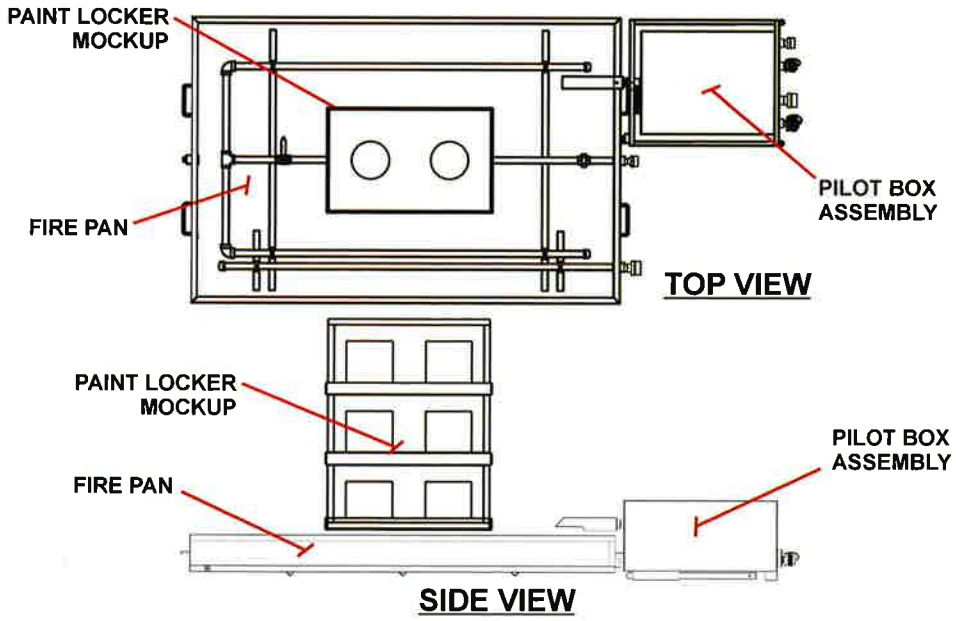

Figure 2. O-100 Paint Locker Fire Trainer Setup and Overview

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

6. At the Pilot Box Assembly remove protective thread cap from the 1-1/4" male **LPG VAPOR** connection marked **"IN"**. Make sure the 1-1/4" male **LPG VAPOR** connection is free of debris, damage and that the rubber washer is present (see Figure 3).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE LPG VAPOR "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

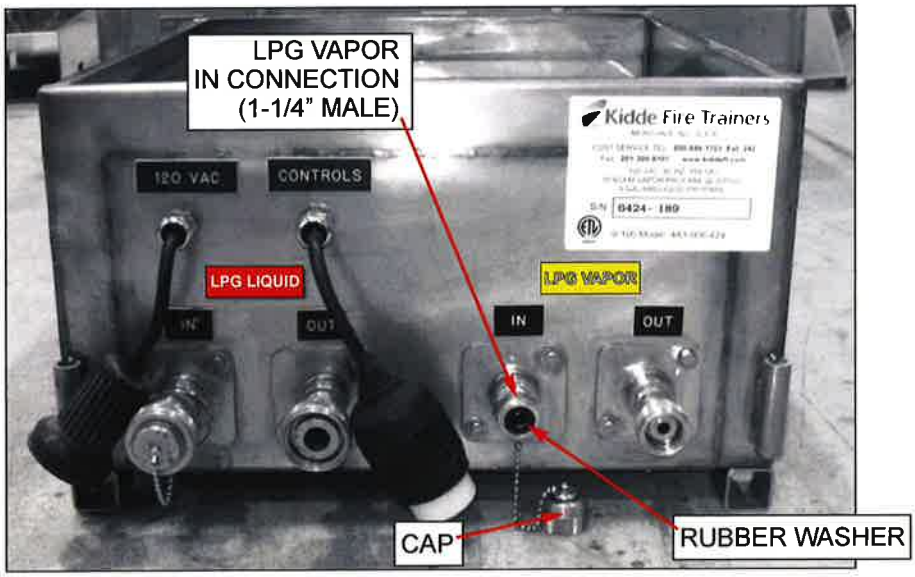

Figure 3. Pilot Box Assembly LPG VAPOR "IN" Connection

7. Make sure the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

8. Connect the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose to the 1-1/4" male **LPG VAPOR** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 4).

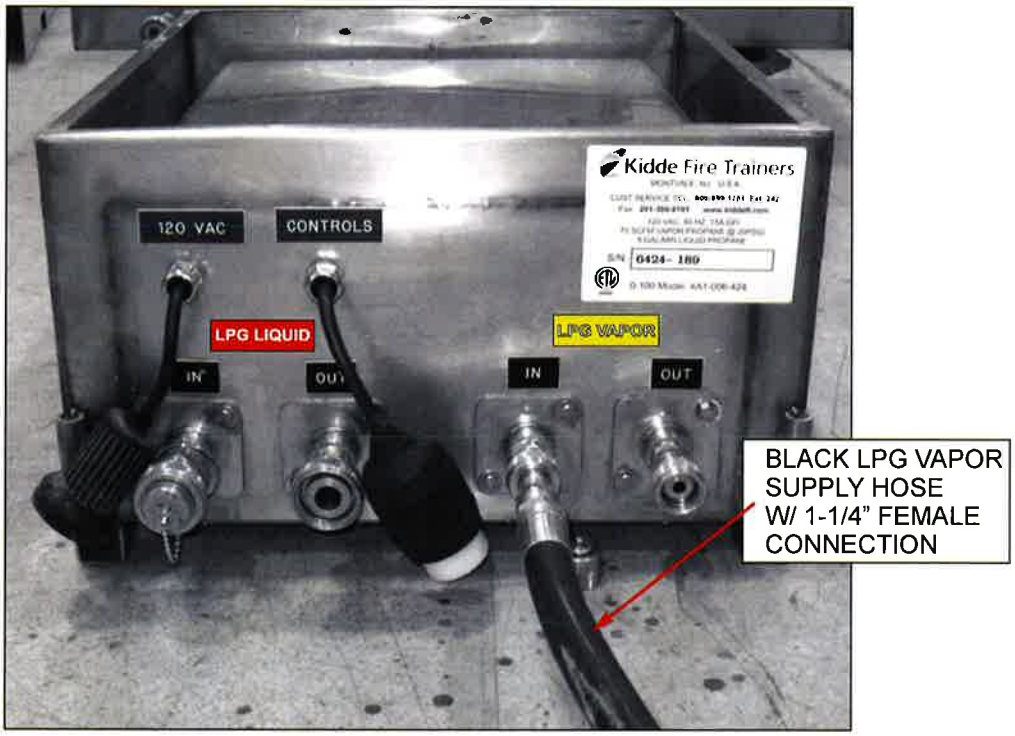

Figure 4. Pilot Box Assembly LPG VAPOR "IN" Supply Hose Connection

9. At the Pilot Box Assembly make sure the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE LPG VAPOR "OUT" CONNECTION HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

10. Remove protective thread cap from the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

11. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 5).

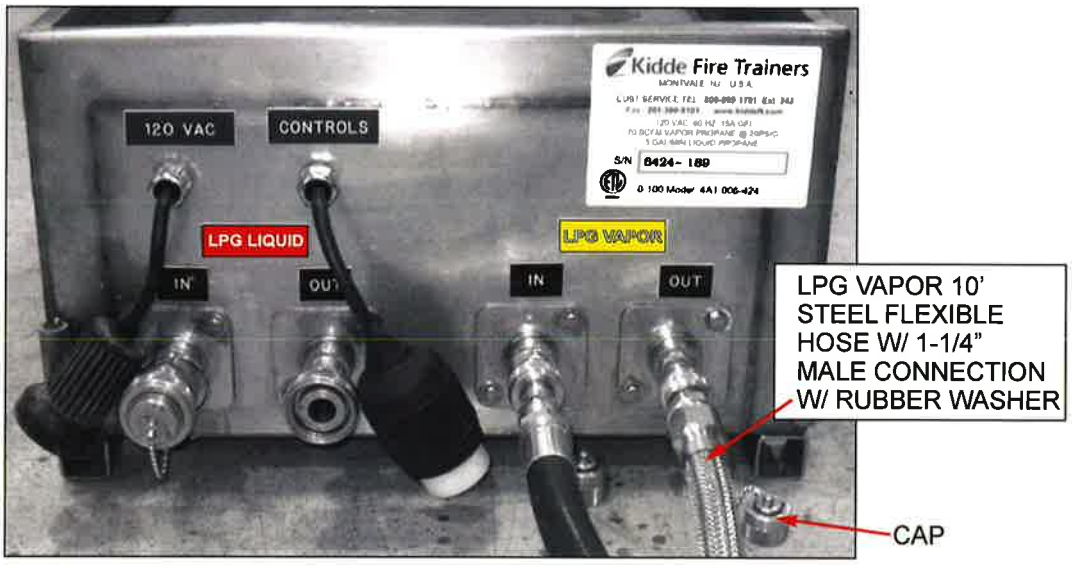

Figure 5. Pilot Box Assembly LPG VAPOR "OUT" Hose Connection

12. Remove protective thread cap form the 1-1/4" male connection on the remaining end of the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

13. At the Fire Pan Assembly make sure the 1-1/4" female connection marked **LPG VAPOR** is free of debris and damage. (see Figure 6 for location).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE LPG VAPOR 1-1/4" FEMALE CONNECTION MARKED LPG VAPOR ON THE FIRE PAN HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

14. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the 1-1/4" female connection marked **LPG VAPOR** on the Fire Pan Assembly. Attach fittings hand tight (see Figure 6).

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

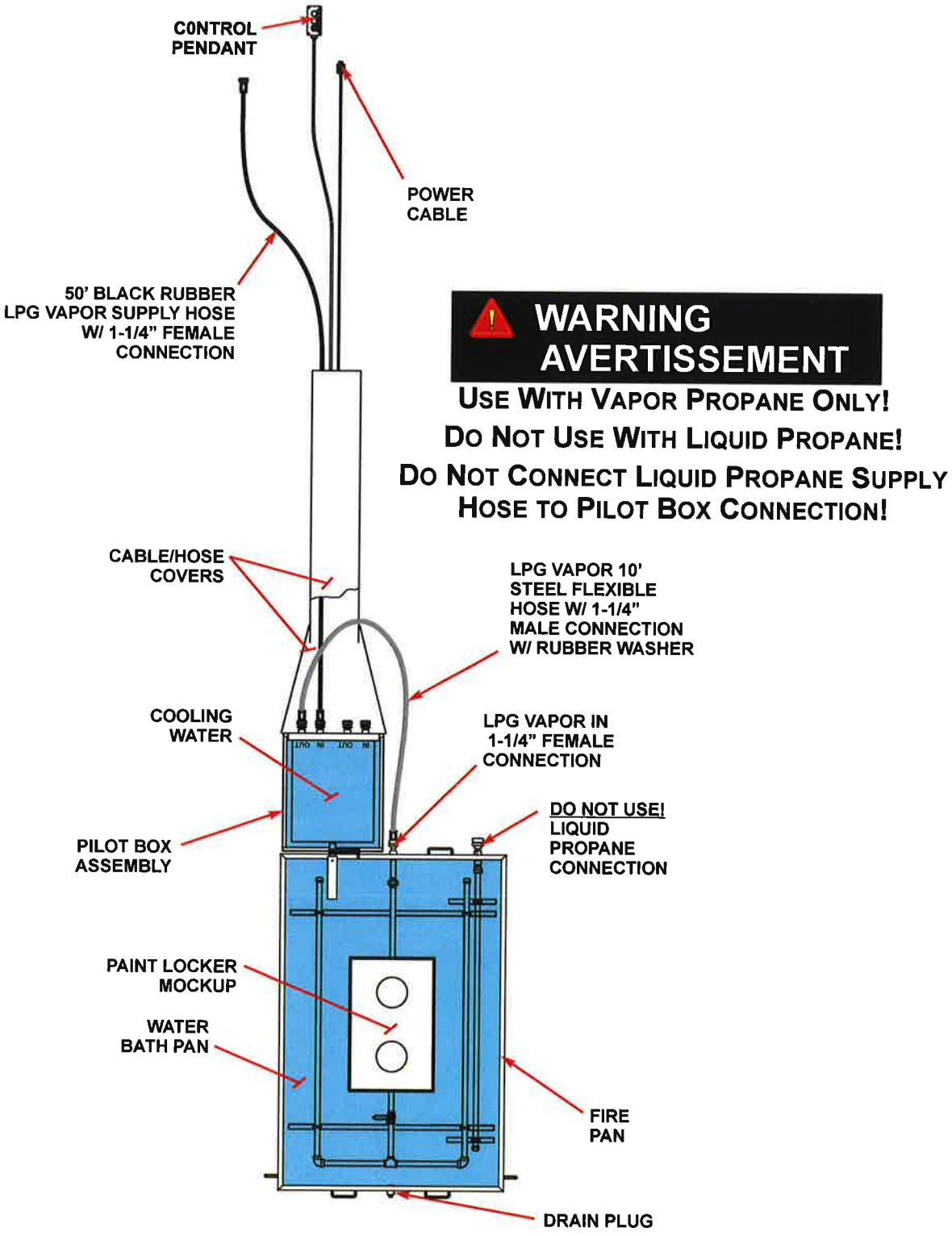

Figure 6. O-100 Paint Locker Fire Trainer Setup

15. Attach the Control Pendant cable to the connection marked **"CONTROLS"** on Pilot Box Assembly (see Figure 7).

16. Attach the Electric Supply cord to the connection marked **"120 VAC"** on Pilot Box Assembly (see Figure 7).

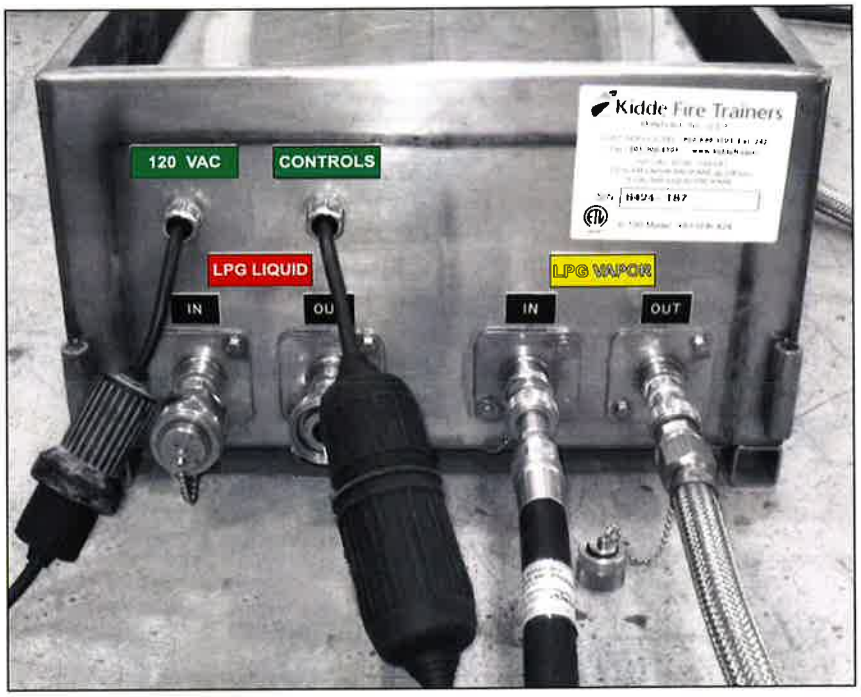

Figure 7. Electrical Supply & Control Pendant Connections

17. Fill Pilot Box Assembly top with water until it overflows (see Figure 6).

18. Ensure that drain plug is installed in the Fire Pan (see Figure 6).

19. Fill Fire Pan with water until it overflows (see Figure 6).

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

NOTE

IF THE HOSE LINE FIRETRAINER® O-100 is to be controlled by the Kidde Multi Control System, refer to the HOSE LINE FIRETRAINER® O-100 Multi Control System Users Manual, Sections 3.1 & 3.2.

20. Using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088), connect the remaining end of the 50' black rubber LPG vapor supply hose from the Pilot Box Assembly **LPG VAPOR** connection marked **"IN"**, to **LPG Vapor Outlet** on the Propane Supply. Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male adaptor connection. Attach fittings hand tight.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER LPG VAPOR SUPPLY HOSE IS CONNECTED TO THE LPG VAPOR OUTLET ON THE PROPANE SUPPLY TANK. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Prior to Operation

Pilot Box Valves Operational Test

1. Perform the Pilot Box LPG Vapor Valve Operational Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Box Valves Operational Test.

WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT BOX VALVES OPERATIONAL TEST HAS BEEN SUCCESSFULLY COMPLETED.

Pilot Flame Test

1. Perform the Pilot Flame Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Flame Test.

WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT FLAME TEST HAS BEEN SUCCESSFULLY COMPLETED.

Operation

1. Run (gas, control and electric) lines straight back from Pilot Box Assembly.
2. Install cable/hose covers. Refer to Figure 6.

CAUTION

DO NOT OPERATE O-100 TRAINER WITHOUT BOTH CABLE/HOSE COVERS IN PLACE! REFER TO FIGURE 6.

WARNING • AVERTISSEMENT

ONLY WHEN ALL THE ABOVE OPERATIONAL AND LEAK CHECKS HAVE BEEN SUCCESSFULLY PERFORMED YOU ARE READY TO TRAIN.

3. Attach the electric supply cord from the Pilot Box assembly to an outdoor 120V power GFIC receptacle.

NOTE

FOR OPERATION INSTRUCTIONS REFER TO THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL. FOLLOW ALL PROCEDURES FOR VAPOR PROPANE ONLY. ANY REFERENCES TO LIQUID PROPANE DOES NOT APPLY.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 PAINT LOCKER FIRE TRAINER A THOROUGH UNDERSTANDING AND FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FOLLOW ALL PROCEDURES FOR VAPOR PROPANE ONLY! ANY REFERENCES TO LIQUID PROPANE DOES NOT APPLY! FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

Shutdown & Disassembly

WARNING • AVERTISSEMENT

AT THE END OF A TRAINING SESSION THERE WILL STILL BE FUEL IN THE HOSES FROM THE PROPANE SUPPLY TANK TO THE MOCKUP/FIRE PAN. THIS WILL HAVE TO BE BURNT OFF BEFORE THE EQUIPMENT IS TAKEN APART. IT IS IMPORTANT THAT THIS PROCEDURE BE FOLLOWED EVEN IF YOU ARE NOT TAKING EQUIPMENT APART BUT ARE JUST SHUTTING DOWN AT THE END OF A TRAINING SESSION. FOR THIS PROCEDURE REFER TO SECTION 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

NOTE

FOR SHUTDOWN & DISASSEMBLY INSTRUCTIONS REFER TO SEC 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

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APPENDIX I

Set-Up Instructions

Introduction

Appendix I provides the information necessary to properly set-up the **O-100 Transformer Mock-up** for use with the **HOSE-LINE FIRETRAINER® O-100.**

WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 TRANSFORMER FIRE TRAINER A THOROUGH UNDERSTANDING AND FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

IN ADDITION TO THE SETUP STEPS LISTED BELOW, IF THE HOSE LINE FIRETRAINER® O-100 IS TO BE CONTROLLED BY ANY OF THE FOLLOWING OPTIONAL KIDDE CONTROL SYSTEMS, REFER TO THE APPROPRIATE HOSE LINE FIRETRAINER® O-100 CONTROL SYSTEM USERS MANUAL FOR SETUP INSTRUCTIONS.

- **MULTI CONTROL**
- **WIRELESS CONTROL**
- **BATTERY POWERED CONTROL**

Installation instructions

1. Locate the 4' x 6' Fire Pan Assembly outdoors; clear of all ground/aerial obstructions on level ground, a minimum of 40 feet downwind of the Propane Supply Tank with the connections facing the Propane Supply Tank (see Figure 1). Remove wheels from the Fire Pan Assembly and store wheels safely away from trainer.

NOTE

IF THE Kidde Multi Control System is used, the Multi Control Stand is considered the propane supply.

2. On the vapor propane center burner element, open the first ball valve and close the second one (see Figure 1).
3. Remove the wheels from the Pilot Box Assembly and store wheels safely away from training area. Be sure the Pilot Box Assembly drain plug is installed (see Figure 1). Position the Pilot Box Assembly as shown in Figure 1, with the gas connections facing the Propane Supply Tank.
4. Extend all lines (Vapor Propane gas, control, electric) and check for any damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF ANY OF THE CONTROL OR ELECTRICAL LINES ARE DAMAGED. DO NOT PROCEED ANY FURTHER IF ANY OF THE GAS HOSES ARE DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

5. Locate the Transformer Mock-up assembly in the Fire Pan as shown in Figure 2.

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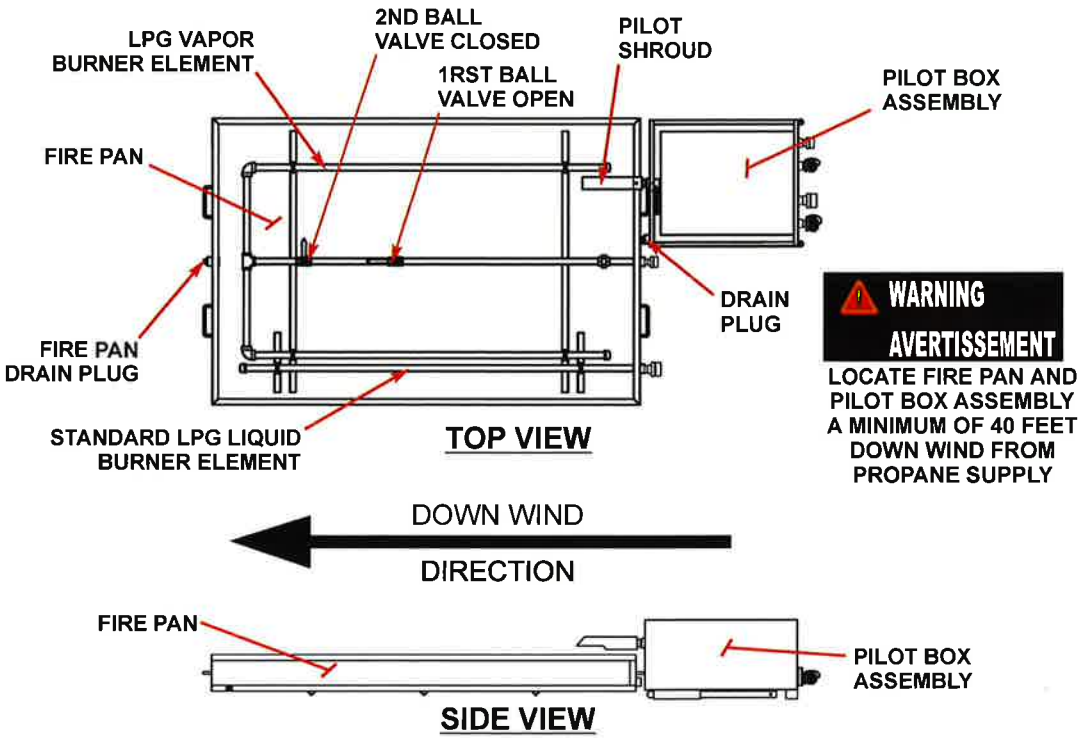

Figure 1. O-100 Fire Pan & Pilot Box Assembly Setup and Overview

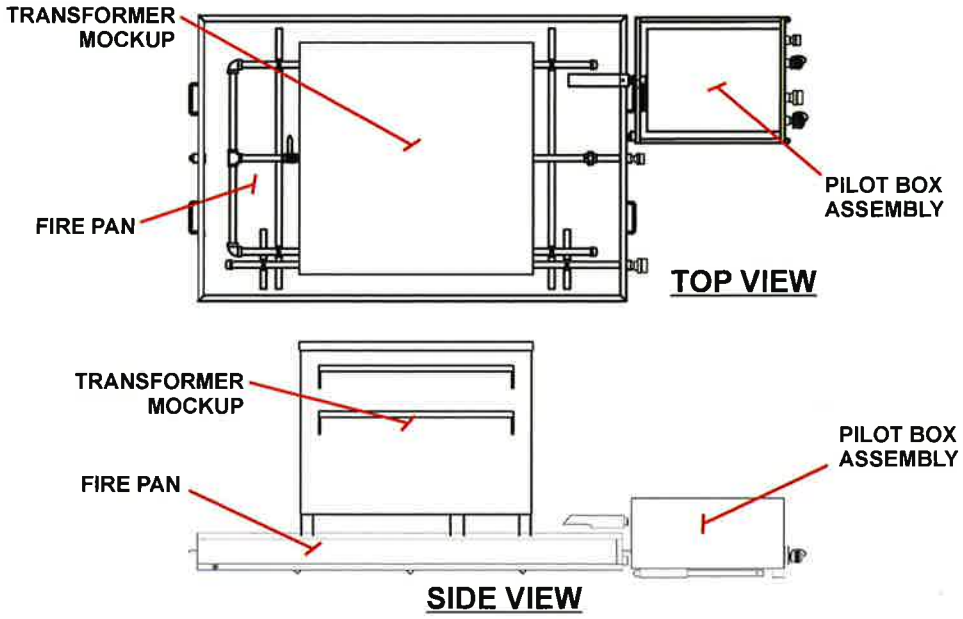

Figure 2. O-100 Transformer Fire Trainer Setup and Overview

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

6. At the Pilot Box Assembly remove protective thread cap from the 1-1/4" male **LPG VAPOR** connection marked **"IN".** Make sure the 1-1/4" male **LPG VAPOR** connection is free of debris, damage and that the rubber washer is present (see Figure 3).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE LPG VAPOR "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

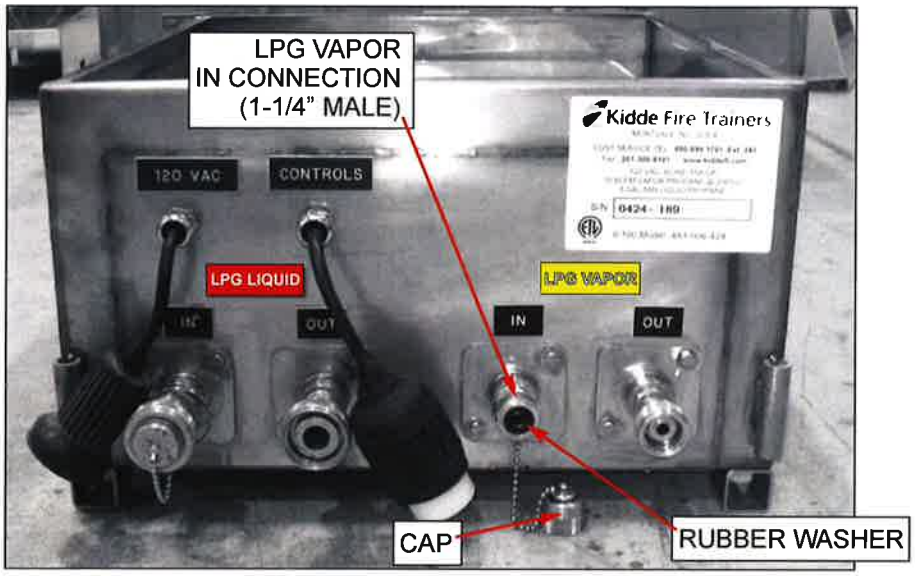

Figure 3. Pilot Box Assembly LPG VAPOR "IN" Connection

7. Make sure the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

8. Connect the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose to the 1-1/4" male **LPG VAPOR** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 4).

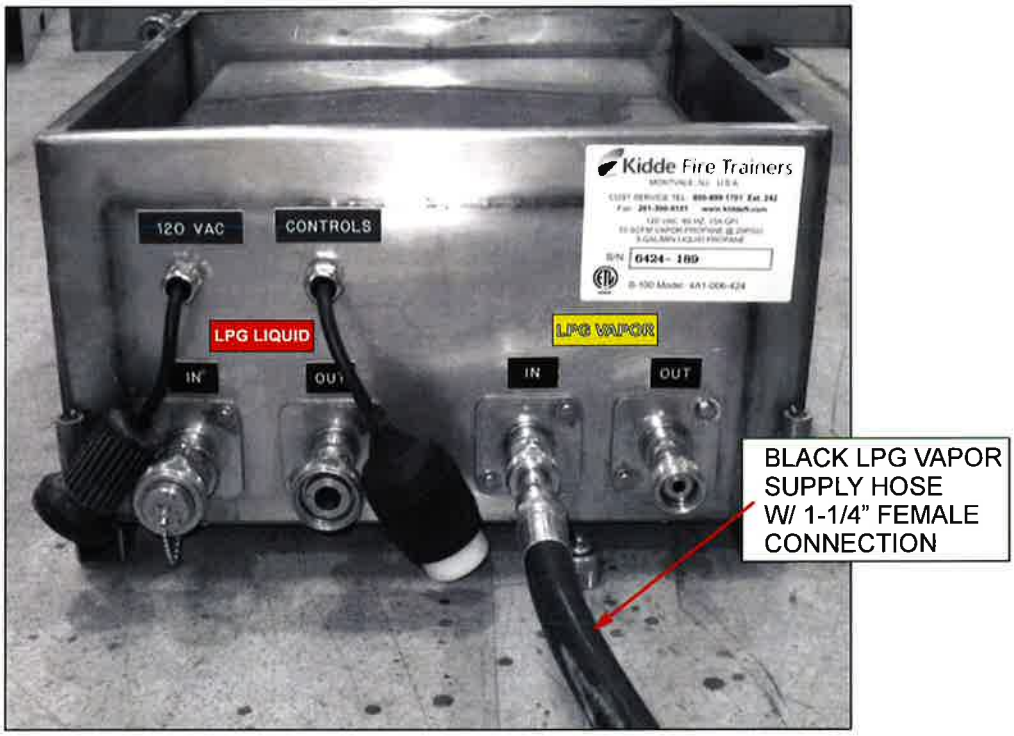

Figure 4. Pilot Box Assembly LPG VAPOR "IN" Supply Hose Connection

9. At the Pilot Box Assembly make sure the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE LPG VAPOR "OUT" CONNECTION HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

10. Remove protective thread cap from the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

11. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 5).

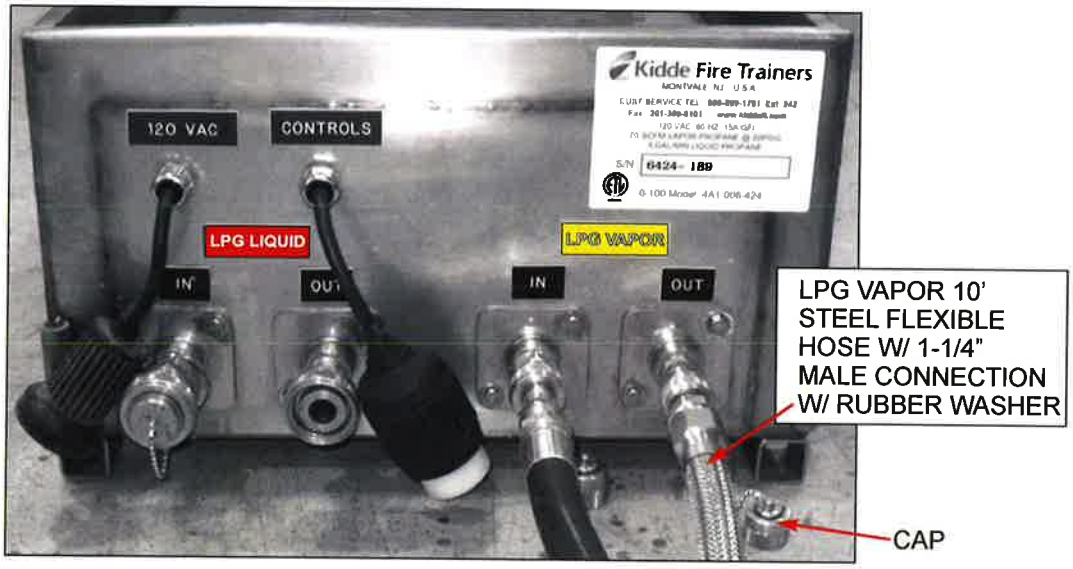

Figure 5. Pilot Box Assembly LPG VAPOR "OUT" Hose Connection

12. Remove protective thread cap form the 1-1/4" male connection on the remaining end of the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

13. At the Fire Pan Assembly make sure the 1-1/4" female connection marked **LPG VAPOR** is free of debris and damage. (see Figure 6 for location).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE LPG VAPOR 1-1/4" FEMALE CONNECTION MARKED LPG VAPOR ON THE FIRE PAN HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

14. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the 1-1/4" female connection marked **LPG VAPOR** on the Fire Pan Assembly. Attach fittings hand tight (see Figure 6).

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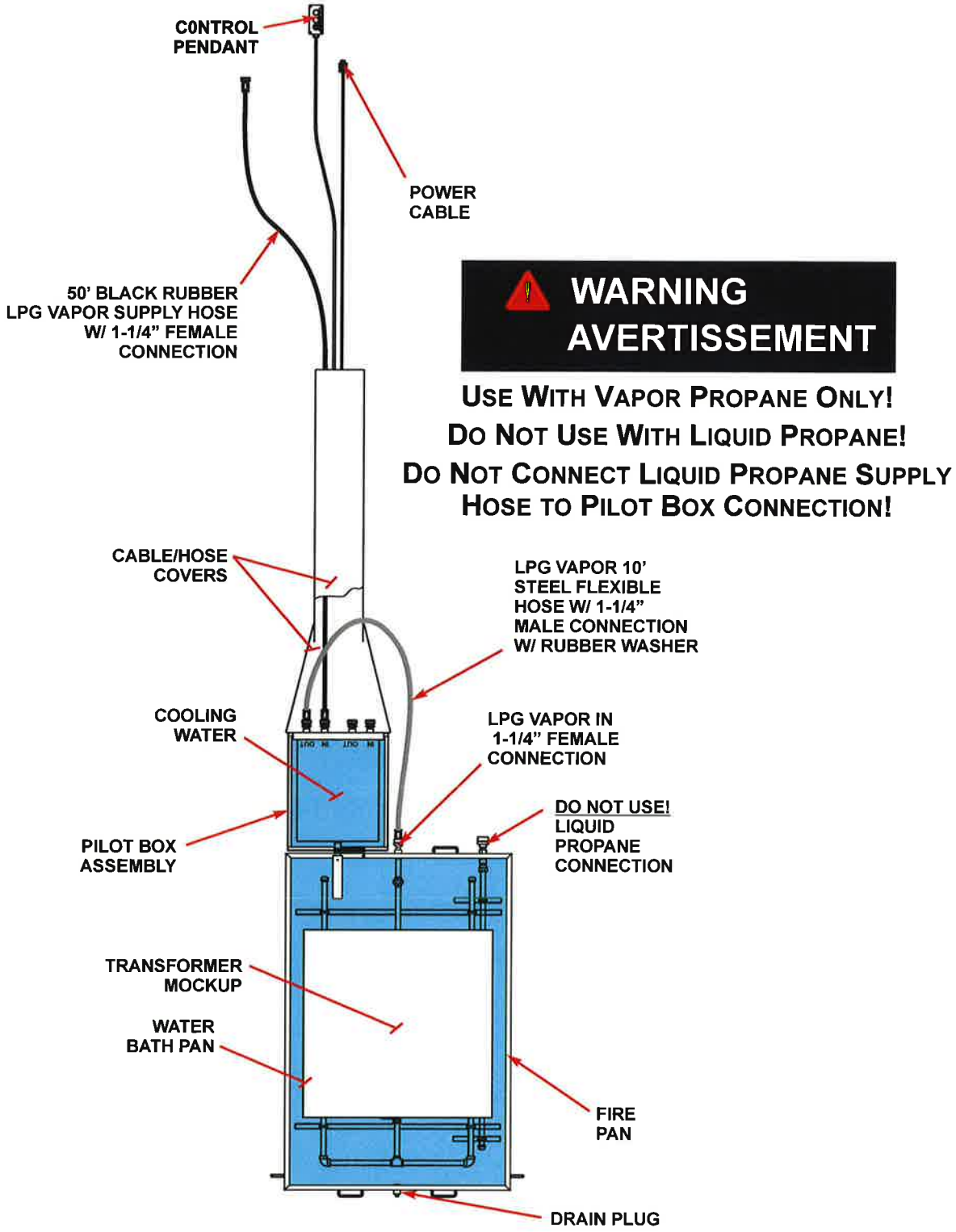

Figure 6. O-100 Transformer Fire Trainer Setup

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

15. Attach the Control Pendant cable to the connection marked **"CONTROLS"** on Pilot Box Assembly (see Figure 7).

16. Attach the Electric Supply cord to the connection marked **"120 VAC"** on Pilot Box Assembly (see Figure 7).

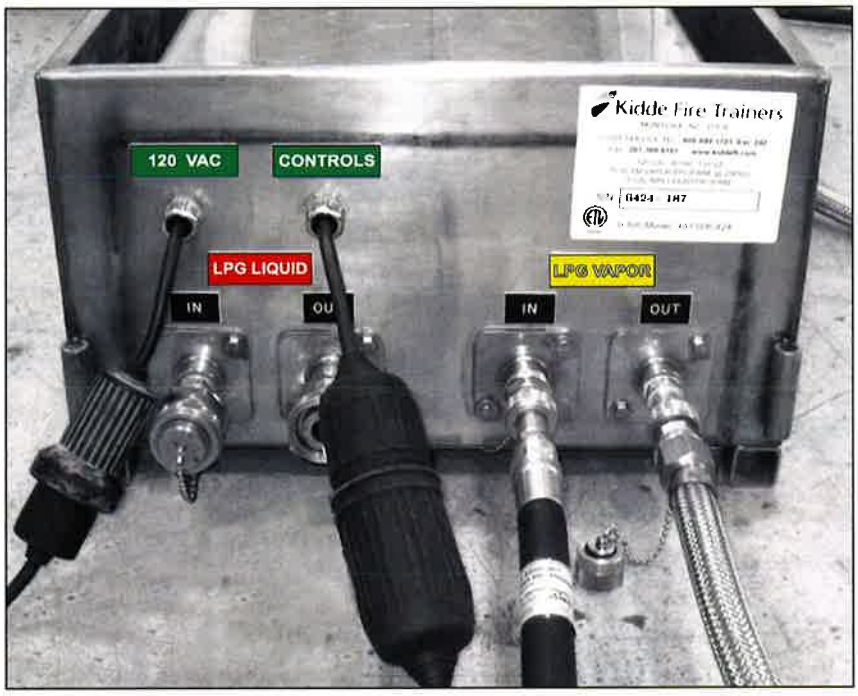

Figure 7. Electrical Supply & Control Pendant Connections

17. Fill Pilot Box Assembly top with water until it overflows (see Figure 6).

18. Ensure that drain plug is installed in the Fire Pan (see Figure 6).

19. Fill Fire Pan with water until it overflows (see Figure 6).

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

NOTE

IF THE HOSE LINE FIRETRAINER® O-100 is to be controlled by the Kidde Multi Control System, refer to the HOSE LINE FIRETRAINER® O-100 Multi Control System Users Manual, Sections 3.1 & 3.2.

20. Using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088), connect the remaining end of the 50' black rubber LPG vapor supply hose from the Pilot Box Assembly **LPG VAPOR** connection marked **"IN"**, to **LPG Vapor Outlet** on the Propane Supply. Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male adaptor connection. Attach fittings hand tight.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER LPG VAPOR SUPPLY HOSE IS CONNECTED TO THE LPG VAPOR OUTLET ON THE PROPANE SUPPLY TANK. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Prior to Operation

Pilot Box Valves Operational Test

1. Perform the Pilot Box LPG Vapor Valve Operational Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Box Valves Operational Test.

⚠ WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT BOX VALVES OPERATIONAL TEST HAS BEEN SUCCESSFULLY COMPLETED.

Pilot Flame Test

1. Perform the Pilot Flame Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Flame Test.

⚠ WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT FLAME TEST HAS BEEN SUCCESSFULLY COMPLETED.

Operation

1. Run (gas, control and electric) lines straight back from Pilot Box Assembly.
2. Install cable/hose covers. Refer to Figure 6.

⚠ CAUTION

DO NOT OPERATE O-100 TRAINER WITHOUT BOTH CABLE/HOSE COVERS IN PLACE! REFER TO FIGURE 6.

⚠ WARNING • AVERTISSEMENT

ONLY WHEN ALL THE ABOVE OPERATIONAL AND LEAK CHECKS HAVE BEEN SUCCESSFULLY PERFORMED YOU ARE READY TO TRAIN.

3. Attach the electric supply cord from the Pilot Box assembly to an outdoor 120V power GFIC receptacle.

NOTE

FOR OPERATION INSTRUCTIONS REFER TO THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL. FOLLOW ALL PROCEDURES FOR VAPOR PROPANE ONLY. ANY REFERENCES TO LIQUID PROPANE DOES NOT APPLY.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 TRANSFORMER FIRE TRAINER A THOROUGH UNDERSTANDING AND FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FOLLOW ALL PROCEDURES FOR VAPOR PROPANE ONLY! ANY REFERENCES TO LIQUID PROPANE DOES NOT APPLY! FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

Shutdown & Disassembly

WARNING • AVERTISSEMENT

AT THE END OF A TRAINING SESSION THERE WILL STILL BE FUEL IN THE HOSES FROM THE PROPANE SUPPLY TANK TO THE MOCKUP/FIRE PAN. THIS WILL HAVE TO BE BURNT OFF BEFORE THE EQUIPMENT IS TAKEN APART. IT IS IMPORTANT THAT THIS PROCEDURE BE FOLLOWED EVEN IF YOU ARE NOT TAKING EQUIPMENT APART BUT ARE JUST SHUTTING DOWN AT THE END OF A TRAINING SESSION. FOR THIS PROCEDURE REFER TO SECTION 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

NOTE

FOR SHUTDOWN & DISASSEMBLY INSTRUCTIONS REFER TO SEC 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

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Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

APPENDIX J

Set-Up Instructions

1.0 Introduction

Appendix J provides the information necessary to properly set-up the **O-100 Flame Bar Trainer** for use with the **HOSE-LINE FIRETRAINER® O-100.**

WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 FLAME BAR FIRE TRAINER A THOROUGH UNDERSTANDING AND FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

IN ADDITION TO THE SETUP STEPS LISTED BELOW, IF THE HOSE LINE FIRETRAINER® O-100 IS TO BE CONTROLLED BY ANY OF THE FOLLOWING OPTIONAL KIDDE CONTROL SYSTEMS, REFER TO THE APPROPRIATE HOSE LINE FIRETRAINER® O-100 CONTROL SYSTEM USERS MANUAL FOR SETUP INSTRUCTIONS.

- **MULTI CONTROL**
- **WIRELESS CONTROL**
- **BATTERY POWERED CONTROL**

1.1 Components List

Item No.	Component	Quantity	Part #
A	Pipe, Schedule 40, SS, ¾"	7	Various lengths
B	Coupling, Pipe, 1000#,SS, ¾" NPT	2	4464K215
C	Elbow, Pipe, 90deg, 150#, SS, ¾" NPT	2	4464K15
D	Tee, Pipe, 150#, SS, ¾" NPT	1	4464K52
E	Cap, Pipe, 1000#,SS, ¾" NPT	2	4464K88
F	Channel. Slotted, 6"	6	2D1-003-486-002
G	Clamp, Pipe, ¾"	6	P112SS
H	Gas Line, Flex, SS, ¾" FNPTx120"LG	1	2C1-004-446
I	Adaptor, BRS, 1-3/4"M ACMEx3/4"FNPT	1	ME213

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1.2 Installation Instructions

Refer to Figure 1 for details and Components List for item identifications

WARNING • AVERTISSEMENT

IF PILOT BOX ASSEMBLY IS BEING USED FROM AN EXISTING SETUP, PRIOR TO PERFORMING THE NEXT STEP, BE SURE THAT FUEL IN HOSES FROM THE PROPANE SUPPLY TANK TO THE PILOT BOX ASSEMBLY AND FIRE PAN HAVE BEEN BURNT OFF AS DIRECTED IN SECTION 6 OF THIS MANUAL. ONCE THIS IS COMPLETED BE SURE TO ALLOW PILOT BOX ASSEMBLY, FIRE PAN/BURN PAN AND ANY MOCK-UPS TO COOL DOWN BEFORE PROCEEDING.

1. Place flame bar elements on the ground in accordance with Figure 1.

CAUTION

VERIFY THAT THE HOLES IN EACH PIECE OF PIPE ARE FACING UPWARDS.

2. Assemble pipe sections (A) together with pipe fittings (B,C,D & E) shown in Figure 1, and tighten.
3. Place slotted channel (F) at the points shown in Figure 1.
4. Place pipe clamp (G) over pipe at each channel position and tighten.
5. Attach flexible stainless steel hose (H) with adaptor (I) as shown in Figure 1.
6. Place Pilot Box Assembly over the flame bar pipe as shown in Figure 1.
7. To complete the Pressure Vessel fire trainer setup you must refer to the **HOSE LINE FIRETRAINER® O-100** Operation and Maintenance Manual, Section 3.

WARNING • AVERTISSEMENT

TO COMPLETE THE PRESSURE VESSEL TRAINER SETUP YOU MUST REFER SEC 3 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

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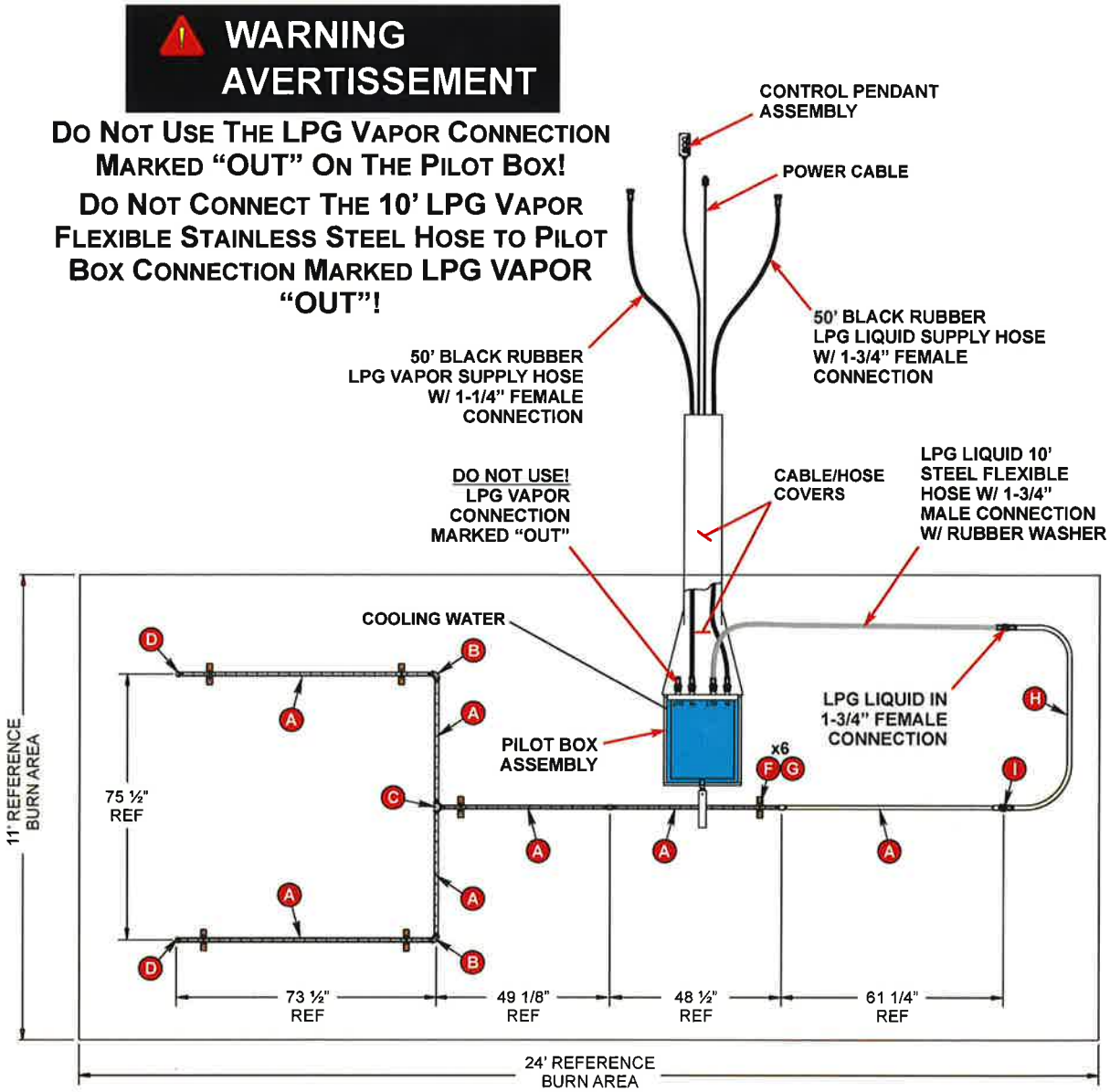

Figure 1. Flame Bar Installation Diagram

Flame Bar Fire Trainer Startup

1. Refer to the **HOSE LINE FIRETRAINER® O-100** Operation and Maintenance Manual, Section 4, for the Startup procedures.

WARNING • AVERTISSEMENT

YOU MUST FOLLOW THE STARTUP PROCEDURES IN SECTION 4 OF THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

Flame Bar Fire Trainer Operation

1. Refer to the **HOSE LINE FIRETRAINER® O-100** Operation and Maintenance Manual, Section 5, for the Operation procedures.

WARNING • AVERTISSEMENT

YOU MUST FOLLOW THE OPERATION PROCEDURES IN SECTION 5 OF THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

Flame Bar Fire Trainer Shutdown

1. Refer to the **HOSE LINE FIRETRAINER® O-100** Operation and Maintenance Manual, Section 6, for the Shutdown procedures.

WARNING • AVERTISSEMENT

YOU MUST FOLLOW THE SHUTDOWN PROCEDURES IN SECTION 6 OF THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

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APPENDIX K

Set-Up Instructions

Introduction

This Appendix provides the information necessary to properly set-up the **O-100 500 Gallon Horizontal Propane Tank Fire Trainer** for use with the **HOSE-LINE FIRETRAINER® O-100.**

WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 500 GAL. HORIZONTAL PROPANE TANK FIRE TRAINER A THOROUGH UNDERSTANDING AND FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

IN ADDITION TO THE SETUP STEPS LISTED BELOW, IF THE HOSE LINE FIRETRAINER® O-100 IS TO BE CONTROLLED BY ANY OF THE FOLLOWING OPTIONAL KIDDE CONTROL SYSTEMS, REFER TO THE APPROPRIATE HOSE LINE FIRETRAINER® O-100 CONTROL SYSTEM USERS MANUAL FOR SETUP INSTRUCTIONS.

- **MULTI CONTROL**
- **WIRELESS CONTROL**
- **BATTERY POWERED CONTROL**

Installation Instructions

1. Locate the Propane Tank Mock-Up outdoors; clear of all ground/aerial obstructions on level ground, a minimum of 40 feet downwind of the Propane Supply.

NOTE

IF THE Kidde Multi Control System is used, the Multi Control Stand is considered the propane supply.

2. Position the Pilot Box Assembly so the pilot shroud is over the LPG Vapor Burner Element as shown in figure 1.
3. Remove the wheels from the Pilot Box Assembly (see figure 1). Store wheels safely away from training area.
4. Extend all lines (gas, control, electric) and check for any damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF ANY OF THE CONTROL OR ELECTRICAL LINES ARE DAMAGED. DO NOT PROCEED ANY FURTHER IF ANY OF THE GAS HOSES ARE DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

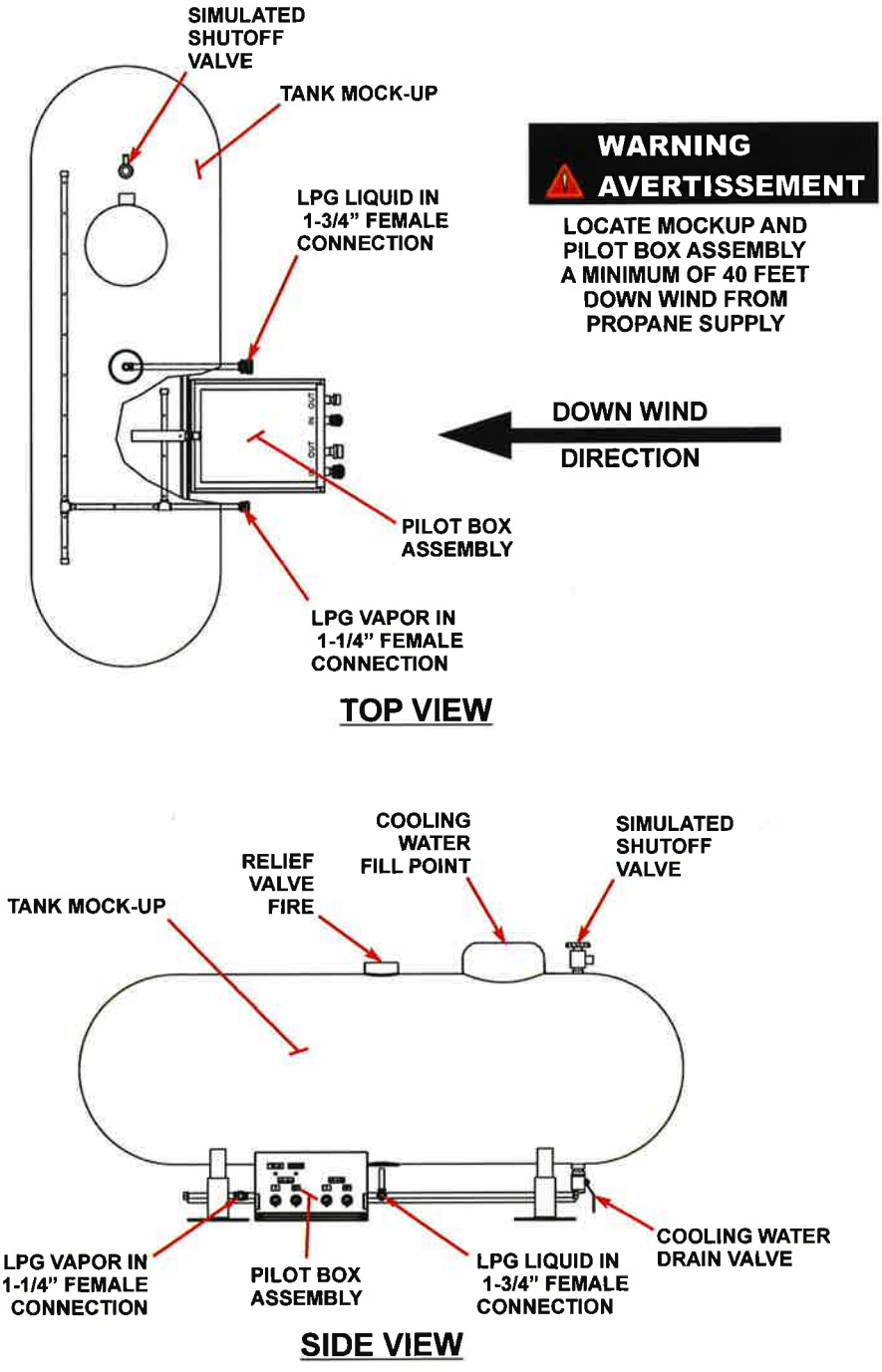

Figure 1. O-100 500 al. Horizontal Propane Tank Fie Trainer Setup and Overview

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

5. At the Pilot Box Assembly remove protective thread cap from the 1-1/4" male **LPG VAPOR** connection marked **"IN".** Make sure the 1-1/4" male **LPG VAPOR** connection is free of debris, damage and that the rubber washer is present (see Figure 2).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE LPG VAPOR "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

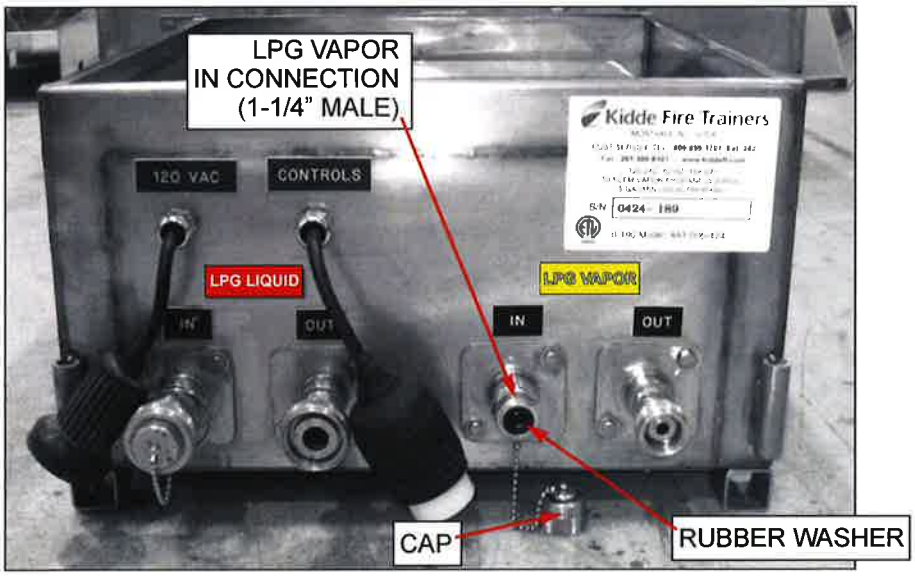

Figure 2. Pilot Box Assembly LPG VAPOR "IN" Connection

6. Make sure the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

7. Connect the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose to the 1-1/4" male **LPG VAPOR** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 3).

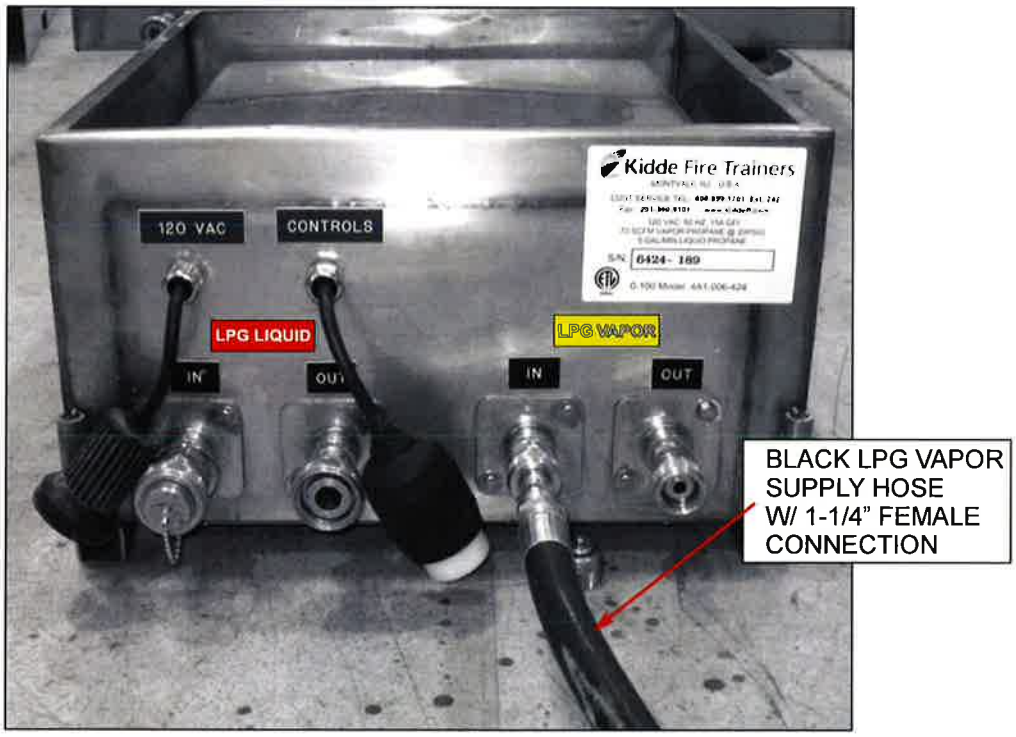

Figure 3. Pilot Box Assembly LPG VAPOR "IN" Supply Hose Connection

8. At the Pilot Box Assembly remove protective thread cap from the 1-3/4" male **LPG LIQUID** connection marked **"IN"**. Make sure the 1-3/4" male **LPG LIQUID** connection is free of debris, damage and that the rubber washer is present (see Figure 4).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE LPG LIQUID "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

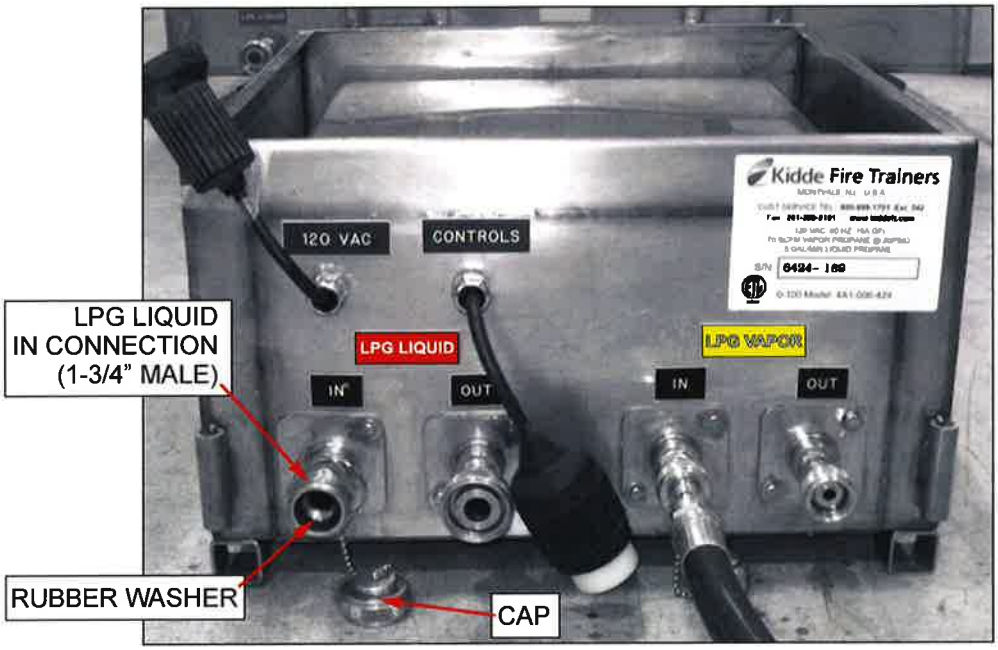

Figure 4. Pilot Box Assembly LPG LIQUID "IN" Connection

9. Make sure the 1-3/4" female connection on the 50' black rubber LPG Liquid supply hose is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

10. Connect the 1-3/4" female connection on the 50' black rubber LPG Liquid supply hose to the 1-3/4" male **LPG LIQUID** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 5).

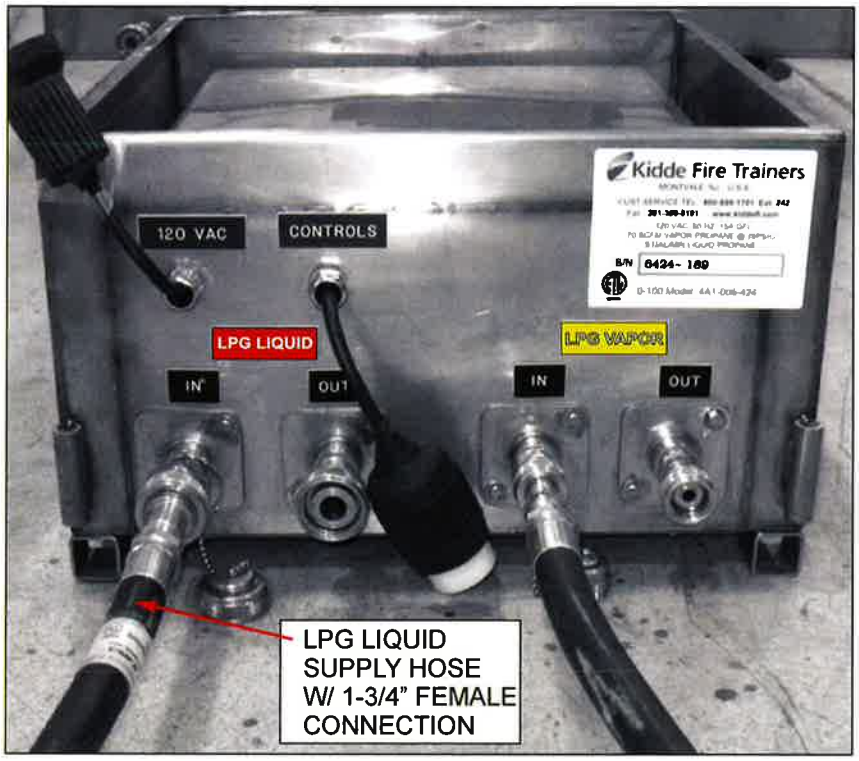

Figure 5. Pilot Box Assembly LPG LIQUID "IN" Supply Hose Connection

11. At the Pilot Box Assembly make sure the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE LPG VAPOR "OUT" CONNECTION HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

12. Remove protective thread cap from the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

13. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 6).

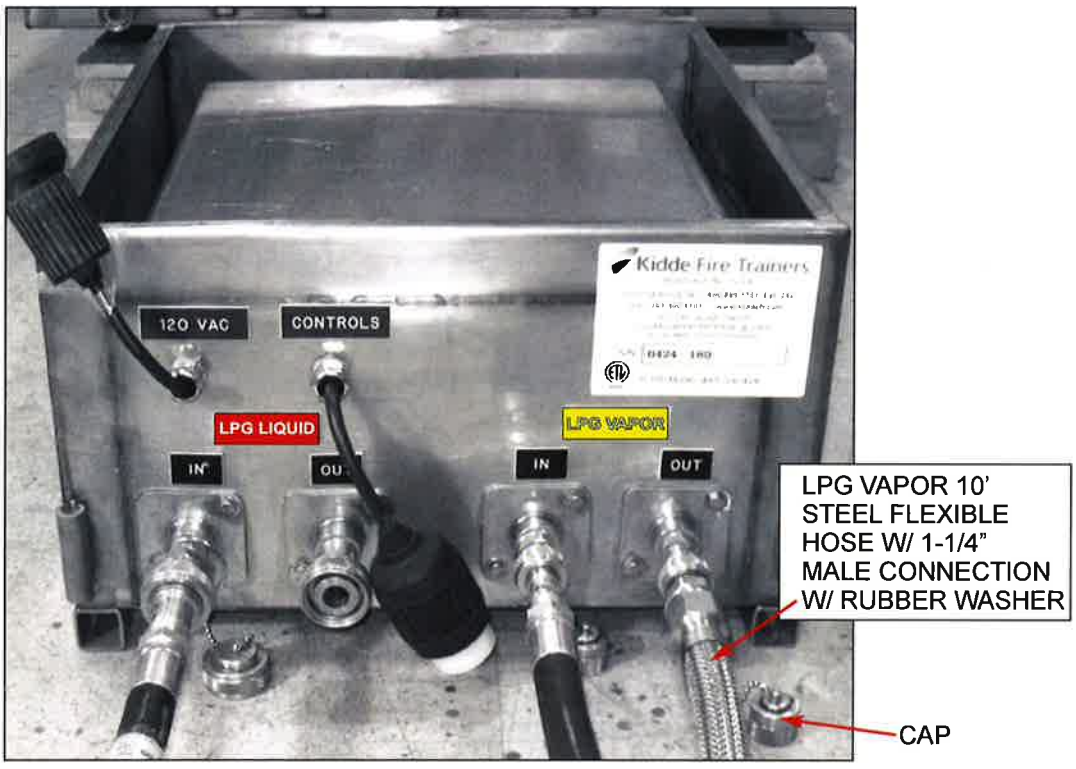

Figure 6. Pilot Box Assembly LPG VAPOR "OUT" Hose Connection

14. Remove protective thread cap form the 1-1/4" male connection on the remaining end of the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

15. At the Horizontal Tank Mockup, make sure the LPG Vapor IN 1-1/4" female connection is free of debris and damage (see Figure 8 for location).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE LPG VAPOR 1-1/4" FEMALE CONNECTION ON THE HORIZONTAL TANK MOCKUP HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

16. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the LPG Vapor 1-1/4" female connection on the Horizontal Tank Mockup. Attach fittings hand tight (see Figure 8).

17. Remove protective thread cap from the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose. Make sure the 1-3/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

18. At the Pilot Box Assembly make sure the 1-3/4" female **LPG LIQUID** connection marked **OUT** is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION MARKED LPG LIQUID "OUT" HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

19. Connect the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose to the 1-3/4" female **LPG LIQUID** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 7).

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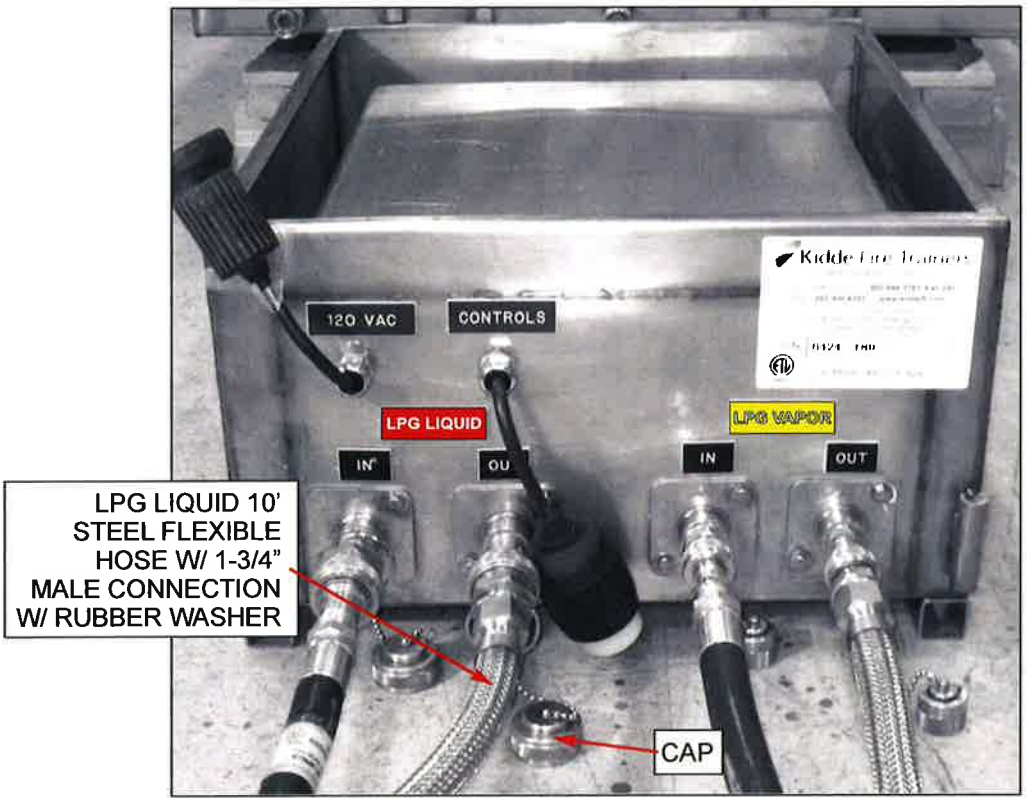

Figure 7. Pilot Box Assembly LPG LIQUID "OUT" Hose Connection

20. Remove protective thread cap form the 1-3/4" male connection on the remaining end of the 10' flexible stainless steel LPG Liquid supply hose. Make sure the 1-3/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

21. At the Horizontal Tank Mockup make sure the LPG Liquid IN 1-3/4" female connection is free of debris and damage (See Figure 8 for location).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE LPG LIQUID 1-3/4" FEMALE CONNECTION ON THE HORIZONTAL TANK MOCKUP HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

22. Connect the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose to the LPG Liquid 1-3/4" female connection on the Horizontal Tank Mockup. Attach fittings hand tight (see Figure 8).

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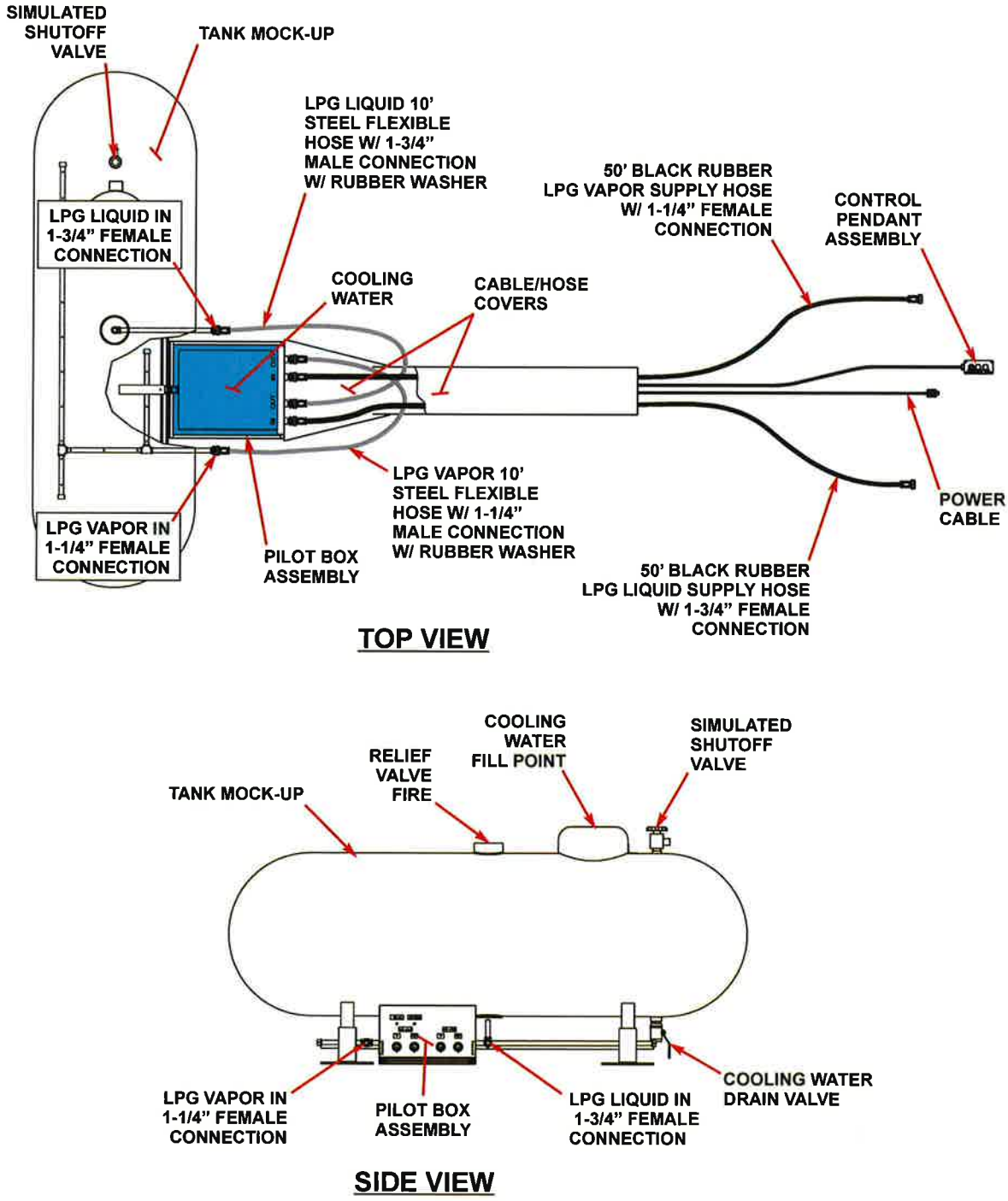

Figure 8. O-100 500 Gal. Horizontal Propane Tank Fie Trainer Setup

23. Attach the Control Pendant cable to the connection marked **"CONTROLS"** on Pilot Box Assembly (see Figure 9).
24. Attach the Electric Supply cord to the connection marked **"120 VAC"** on Pilot Box Assembly (see Figure 9).

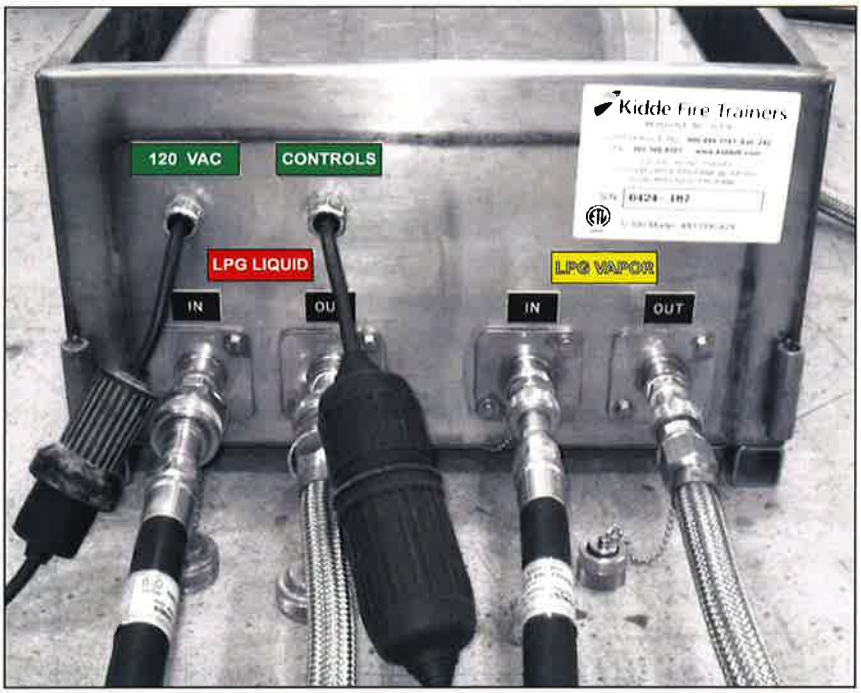

Figure 9. Electrical Supply & Control Pendant Connections

25. Ensure the tank mock-up drain valve is closed (see Figure 8). Lift cover and fill Tank prop with water, until it overflows. Close cover. Refer to Figure 8 for water fill location.
26. Ensure that water drain plug is installed in the Pilot Box Assembly. Drain plug located at front of Pilot Box Assembly.
27. Fill Pilot Box Assembly top with water until it overflows (see Figure 8).

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NOTE

IF THE HOSE LINE FIRETRAINER® O-100 is to be controlled by the Kidde Multi Control System, refer to the HOSE LINE FIRETRAINER® O-100 Multi Control System Users Manual, Sections 3.1 & 3.2.

28. Using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088), connect the remaining end of the 50' black rubber LPG vapor supply hose from the Pilot Box Assembly **LPG VAPOR** connection marked **"IN"**, to **LPG Vapor Outlet** on the Propane Supply. Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male adaptor connection. Attach fittings hand tight.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER LPG VAPOR supply hose is connected to the LPG VAPOR outlet on the propane supply tank.

29. Using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088), connect the remaining end of the 50' black rubber LPG liquid supply hose from the Pilot Box Assembly **LPG LIQUID** connection marked **"IN"**, to the **LPG LIQUID Output** on the Propane Supply. Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male adaptor connection. Attach fittings hand tight.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER LPG LIQUID supply hose is connected to the LPG LIQUID outlet on the propane supply tank.

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Prior to Operation

Pilot Box Valves Operational Test

1. Perform the Pilot Box Valves Operational Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Box Valves Operational Test.

⚠ WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT BOX VALVES OPERATIONAL TEST HAS BEEN SUCCESSFULLY COMPLETED.

Pilot Flame Test

1. Perform the Pilot Flame Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Flame Test.

⚠ WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT FLAME TEST HAS BEEN SUCCESSFULLY COMPLETED.

Liquid Propane Leak check

1. Perform the Liquid Propane Leak Check. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Liquid Propane Leak Check.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER UNTIL THE LIQUID PROPANE LEAK CHECK HAS BEEN SUCCESSFULLY COMPLETED.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Operation

1. Run (gas, control and electric) lines straight back from Pilot Box Assembly.
2. Install cable/hose covers. Refer to Figure 8.

⚠ CAUTION

DO NOT OPERATE O-100 TRAINER WITHOUT BOTH CABLE/HOSE COVERS IN PLACE! REFER TO FIGURE 8.

⚠ WARNING • AVERTISSEMENT

ONLY WHEN ALL THE ABOVE OPERATIONAL AND LEAK CHECKS HAVE BEEN SUCCESSFULLY PERFORMED YOU ARE READY TO TRAIN.

3. Attach the electric supply cord from the Pilot Box assembly to an outdoor 120V power GFIC receptacle.

NOTE

FOR OPERATION INSTRUCTIONS REFER TO THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

⚠ WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 500 GAL HORIZONTAL PROPANE TANK FIRE TRAINER A THOROUGH FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

Shutdown & Disassembly

⚠ WARNING • AVERTISSEMENT

AT THE END OF A TRAINING SESSION THERE WILL STILL BE FUEL IN THE HOSES FROM THE PROPANE SUPPLY TANK TO THE MOCKUP/FIRE PAN. THIS WILL HAVE TO BE BURNT OFF BEFORE THE EQUIPMENT IS TAKEN APART. IT IS IMPORTANT THAT THIS PROCEDURE BE FOLLOWED EVEN IF YOU ARE NOT TAKING EQUIPMENT APART BUT ARE JUST SHUTTING DOWN AT THE END OF A TRAINING SESSION. FOR THIS PROCEDURE REFER TO SECTION 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

NOTE

FOR SHUTDOWN & DISASSEMBLY INSTRUCTIONS REFER TO SEC 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

APPENDIX L

Set-Up Instructions

Introduction

Appendix L provides the information necessary to properly set-up the **O-100 4-Piece Dumpster Fire Trainer** for use with the **HOSE-LINE FIRETRAINER® O-100.**

⚠ WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 DUMPSTER FIRE TRAINER A THOROUGH UNDERSTANDING AND FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

IN ADDITION TO THE SETUP STEPS LISTED BELOW, IF THE HOSE LINE FIRETRAINER® O-100 IS TO BE CONTROLLED BY ANY OF THE FOLLOWING OPTIONAL KIDDE CONTROL SYSTEMS, REFER TO THE APPROPRIATE HOSE LINE FIRETRAINER® O-100 CONTROL SYSTEM USERS MANUAL FOR SETUP INSTRUCTIONS.

- **MULTI CONTROL**
- **WIRELESS CONTROL**
- **BATTERY POWERED CONTROL**

Components List

Component	Quantity	Part #
4-Piece Dumpster Fire Assembly	1	4D1-008-977

Installation Instructions

1. Locate the 4' x 6' Fire Pan Assembly outdoors; clear of all ground/aerial obstructions on level ground, a minimum of 40 feet downwind of the Propane Supply Tank with the gas connections facing in the direction of the propane supply. Remove the wheels from the Fire Pan and safely store.

NOTE

IF THE Kidde Multi Control System is used, the Multi Control Stand is considered the propane supply.

2. At the Fire Pan, be sure the Standard Liquid Propane Burner Element is installed (see Figure 1). Be sure both LPG vapor burner element ball valves are open and the drain plug is installed (see Figure 1).
3. If you have a Fire Pan configured for the Christmas tree fire trainer, (refer to Appendix B, Figure 1), you must remove the Christmas tree mockup, close the LPG Liquid supply valve to the Christmas tree mockup and open the LPG Liquid supply valve to the LPG Liquid burner element (refer to Appendix B, Figure 1).

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

4. Assemble the dumpster mock-up around the Fire Pan Assembly (see Figure 1).
5. Be sure the drain plug is installed in the Pilot Box Assembly. Remove the wheels from the Pilot Box Assembly and safely store. Push Pilot box handle all the way in.
6. Locate the Pilot Box Assembly on the gas connection side of the Fire Pan (see Figure 1).
7. Slide the Pilot Box Assembly in so the pilot shroud protrudes inside the hole in the side of the dumpster mock-up, with the front of the Pilot Box Assembly against the side of the dumpster mock-up (see Figure 1).

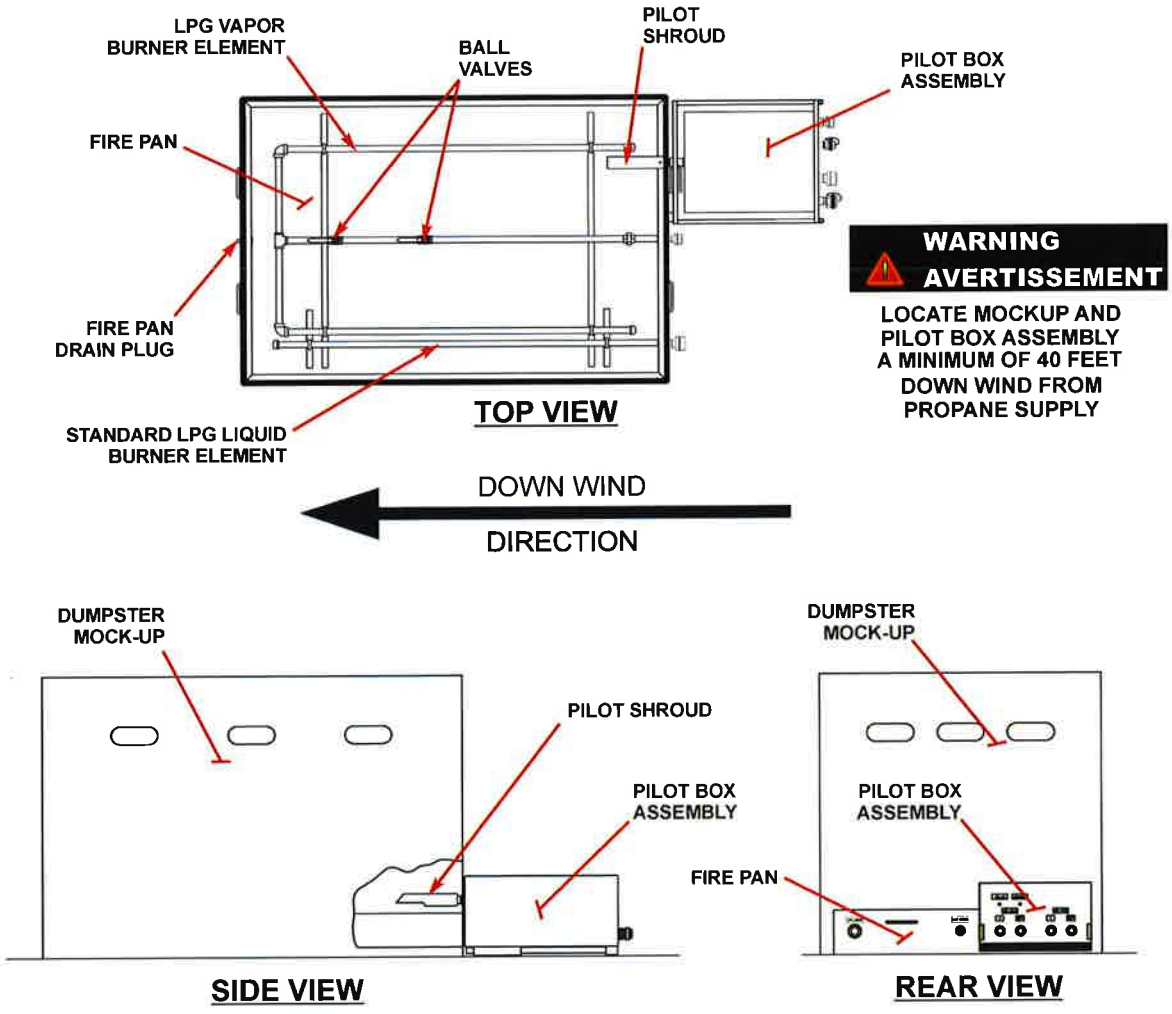

Figure 1. O-100 Dumpster Fie Trainer Setup and Overview

8. Extend all lines (gas, control, electric) and check for any damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF ANY OF THE CONTROL OR ELECTRICAL LINES ARE DAMAGED. DO NOT PROCEED ANY FURTHER IF ANY OF THE GAS HOSES ARE DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

9. At the Pilot Box Assembly remove protective thread cap from the 1-1/4" male **LPG VAPOR** connection marked **"IN"**. Make sure the 1-1/4" male **LPG VAPOR** connection is free of debris, damage and that the rubber washer is present (see Figure 2).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE LPG VAPOR "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

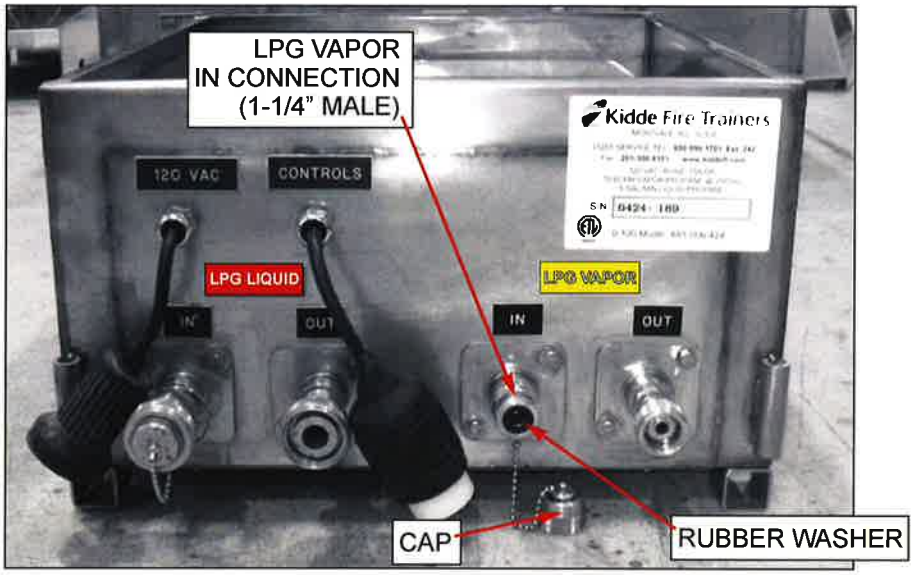

Figure 2. Pilot Box Assembly LPG VAPOR "IN" Connection

10. Make sure the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

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11. Connect the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose to the 1-1/4" male **LPG VAPOR** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 3).

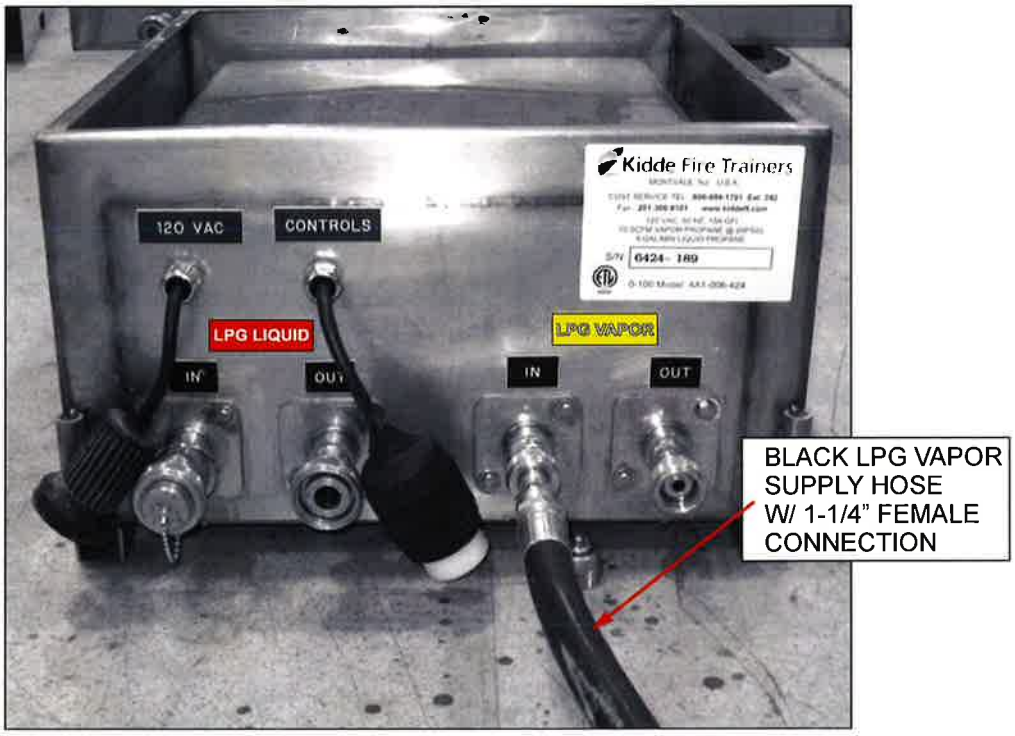

Figure 3. Pilot Box Assembly LPG VAPOR "IN" Supply Hose Connection

12. At the Pilot Box Assembly remove protective thread cap from the 1-3/4" male **LPG LIQUID** connection marked **"IN"**. Make sure the 1-3/4" male **LPG LIQUID** connection is free of debris, damage and that the rubber washer is present (see Figure 4).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE LPG LIQUID "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

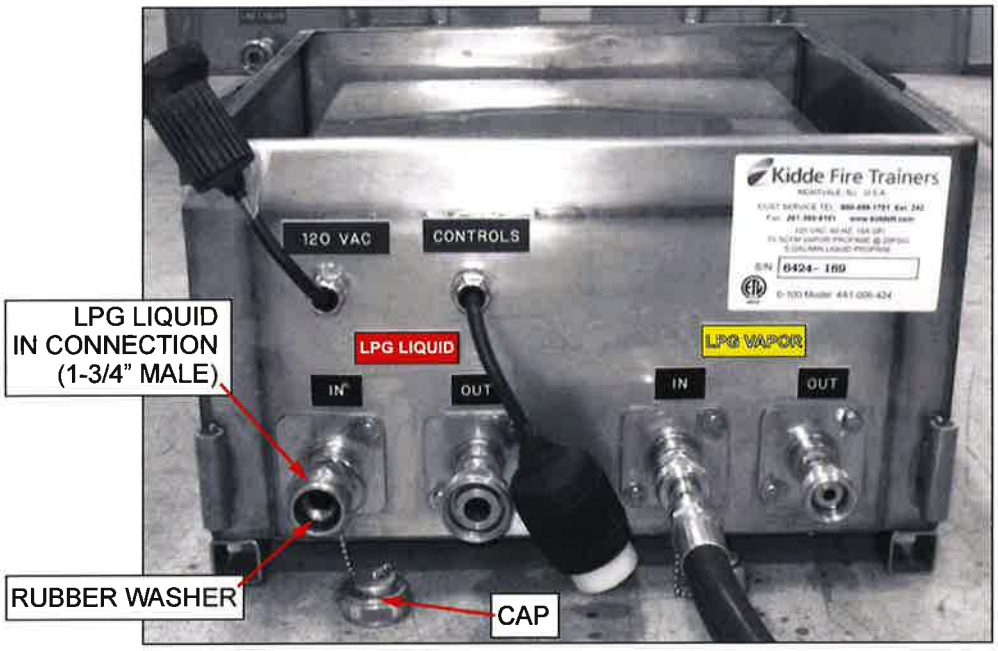

Figure 4. Pilot Box Assembly LPG LIQUID "IN" Connection

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

13. Make sure the 1-3/4" female connection on the 50' black rubber LPG Liquid supply hose is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

14. Connect the 1-3/4" female connection on the 50' black rubber LPG Liquid supply hose to the 1-3/4" male **LPG LIQUID** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 5).

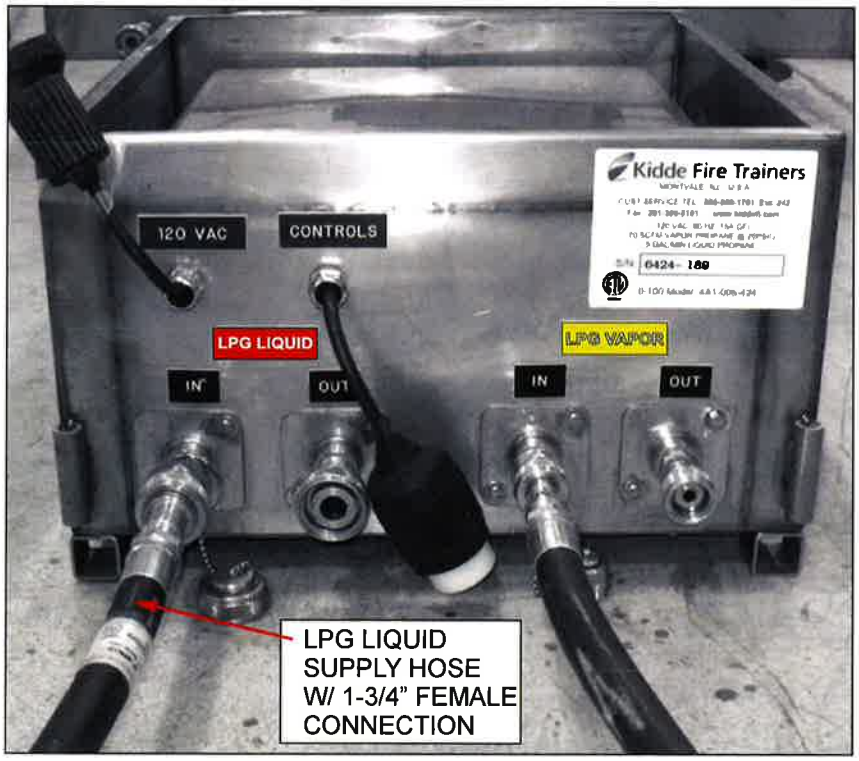

Figure 5. Pilot Box Assembly LPG LIQUID "IN" Supply Hose Connection

15. At the Pilot Box Assembly make sure the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE LPG VAPOR "OUT" CONNECTION HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

16. Remove protective thread cap from the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

17. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 6).

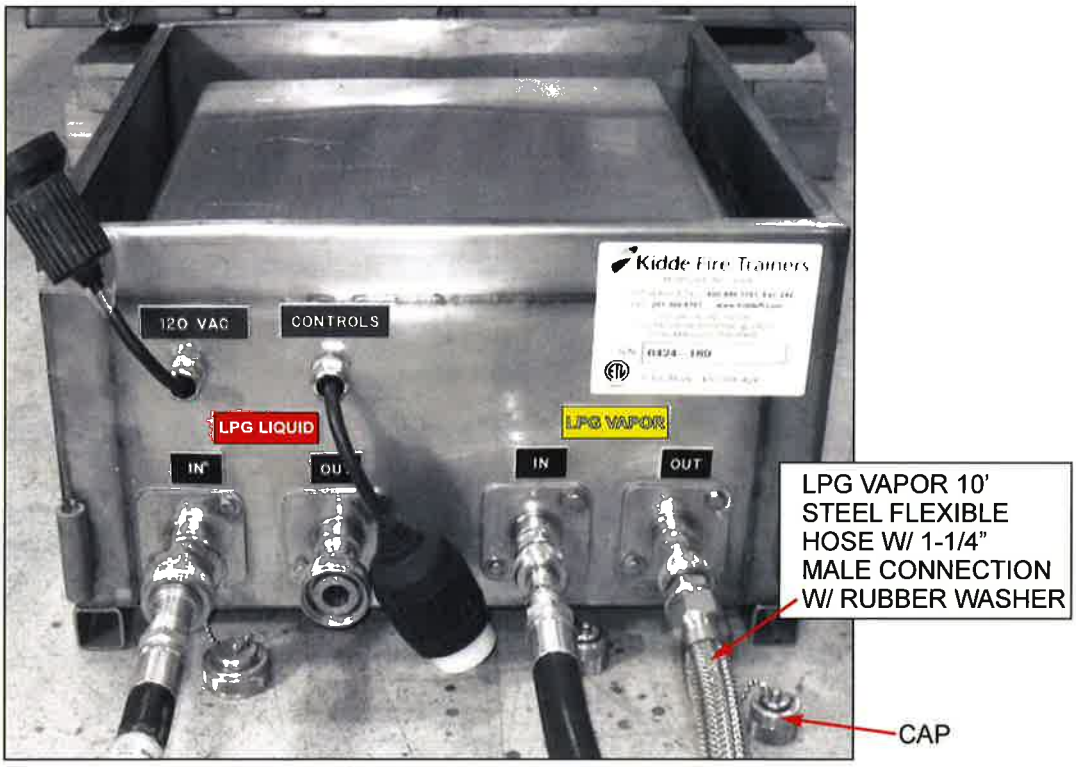

Figure 6. Pilot Box Assembly LPG VAPOR "OUT" Hose Connection

18. Remove protective thread cap form the 1-1/4" male connection on the remaining end of the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

19. At the Fire Pan Assembly make sure the 1-1/4" female connection marked **LPG VAPOR** is free of debris and damage. (see Figure 8 for location).

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE LPG VAPOR 1-1/4" FEMALE CONNECTION MARKED LPG VAPOR ON THE FIRE PAN HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

20. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the 1-1/4" female connection marked **LPG VAPOR** on the Fire Pan Assembly. (see Figure 8).

21. Remove protective thread cap from the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose. Make sure the 1-3/4" male connection is free of debris, damage and that the rubber washer is present.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

22. At the Pilot Box Assembly make sure the 1-3/4" female **LPG LIQUID** connection marked **OUT** is free of debris and damage.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION MARKED LPG LIQUID "OUT" HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

23. Connect the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose to the 1-3/4" female **LPG LIQUID** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 7).

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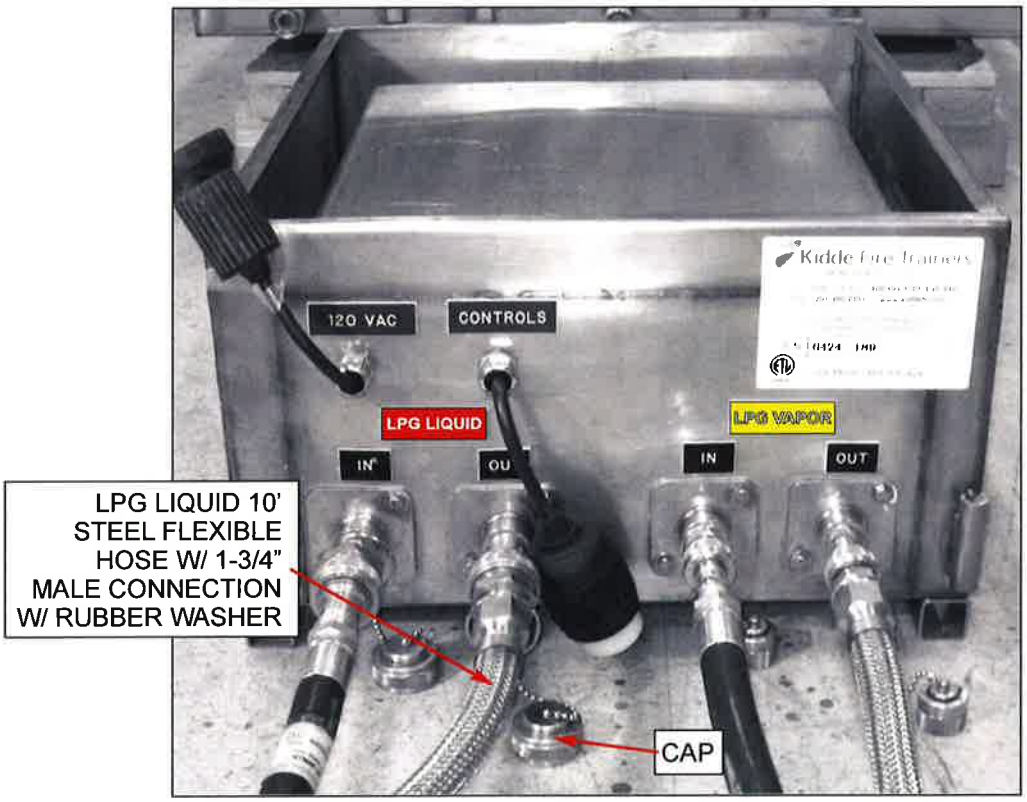

Figure 7. Pilot Box Assembly LPG LIQUID "OUT" Hose Connection

24. Remove protective thread cap form the 1-3/4" male connection on the remaining end of the 10' flexible stainless steel LPG Liquid supply hose. Make sure the 1-3/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

25. At the Fire Pan Assembly make sure the 1-3/4" female connection marked **LPG LIQUID** is free of debris and damage (See Figure 8 for location).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE LPG LIQUID 1-3/4" FEMALE CONNECTION MARKED LPG LIQUID ON THE FIE PAN HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

26. Connect the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose to the 1-3/4" female connection marked **LPG LIQUID** on the Fire Pan Assembly. Attach fittings hand tight (see Figure 8).

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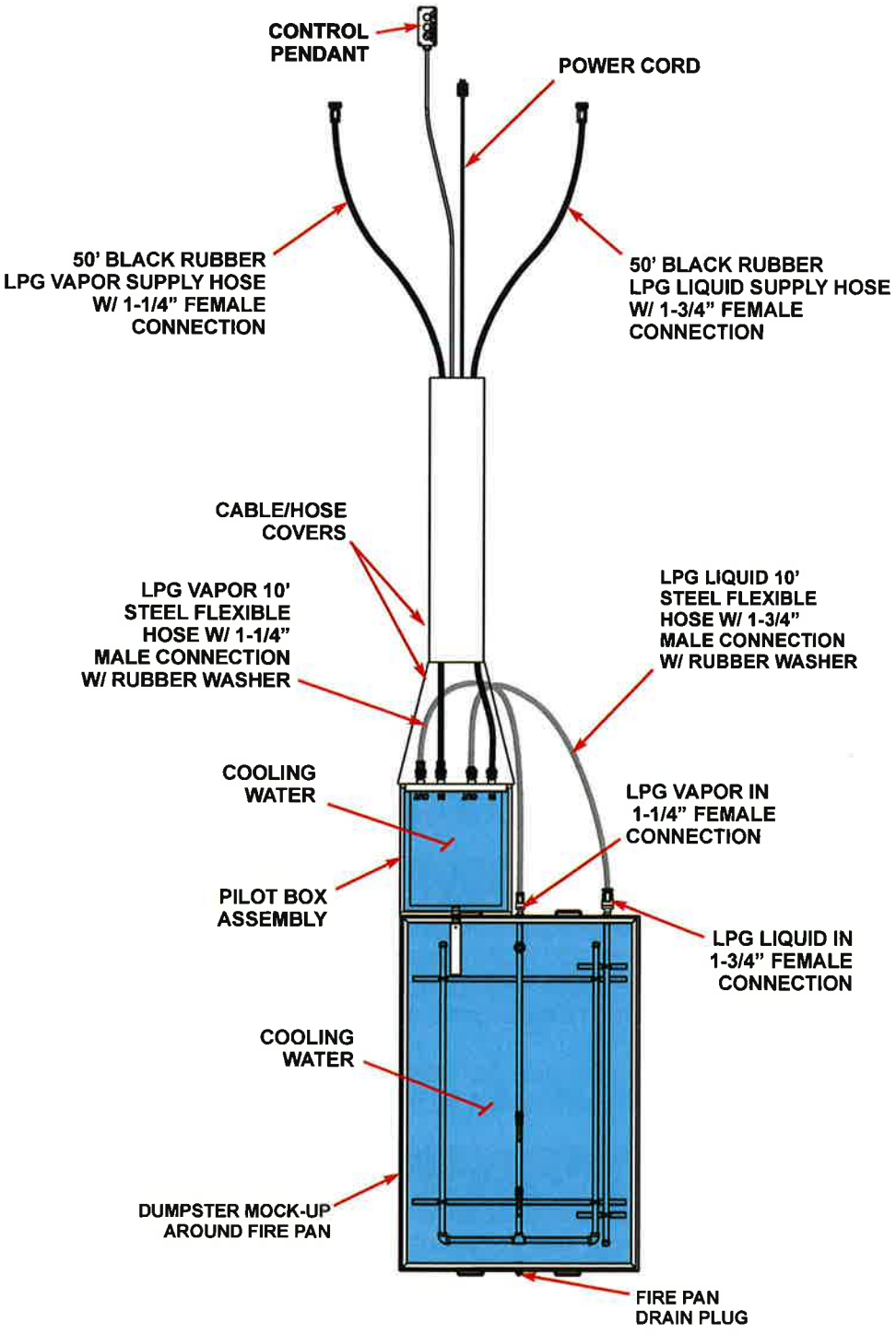

Figure 8. O-100 Dumpster Fie Trainer Setup (Top View)

27. Attach the Control Pendant cable to the connection marked **"CONTROLS"** on Pilot Box Assembly (see Figure 9).

28. Attach the Electric Supply cord to the connection marked **"120 VAC"** on Pilot Box Assembly (see Figure 9).

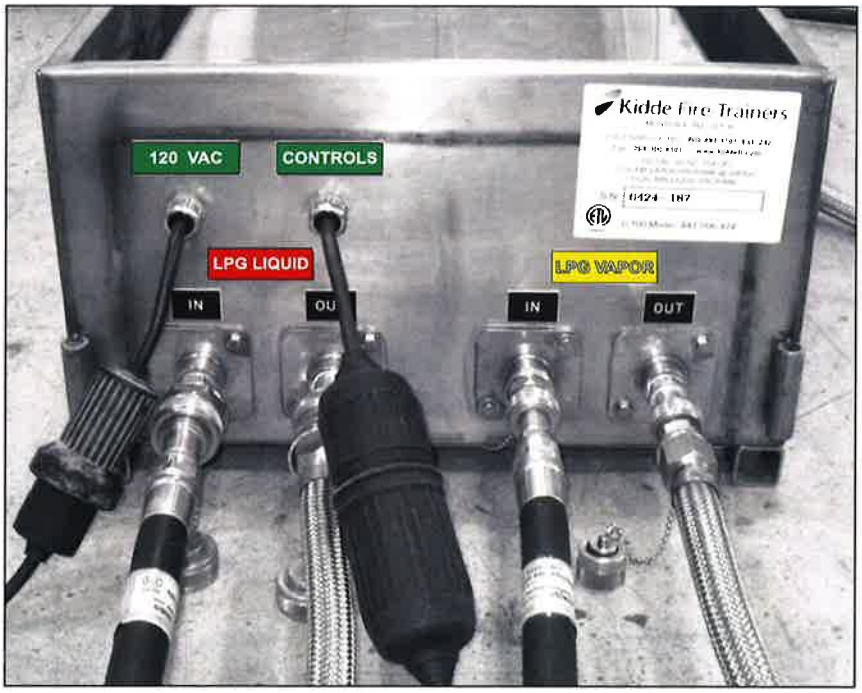

Figure 9. Electrical Supply & Control Pendant Connections

29. Ensure that water drain plug is installed in the Pilot Box Assembly. Drain plug located at front of Pilot Box Assembly.

30. Fill Pilot Box Assembly top with water until it overflows (see Figure 8).

31. Ensure that drain plug is installed in the Fire Pan (see Figure 8)

32. Fill Fire Pan with water until water overflows (see Figure 8).

NOTE

IF THE HOSE LINE FIRETRAINER® O-100 is to be controlled by the Kidde Multi Control System, refer to the HOSE LINE FIRETRAINER® O-100 Multi Control System Users Manual, Sections 3.1 & 3.2.

33. Using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088), connect the remaining end of the 50' black rubber LPG vapor supply hose from the Pilot Box Assembly **LPG VAPOR** connection marked **"IN"**, to **LPG Vapor Outlet** on the Propane Supply. Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male adaptor connection. Attach fittings hand tight.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER LPG VAPOR supply hose is connected to the LPG VAPOR outlet on the propane supply tank.

34. Using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088), connect the remaining end of the 50' black rubber LPG liquid supply hose from the Pilot Box Assembly **LPG LIQUID** connection marked **"IN"**, to the **LPG LIQUID Output** on the Propane Supply. Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male adaptor connection. Attach fittings hand tight.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER LPG LIQUID supply hose is connected to the LPG LIQUID outlet on the propane supply tank.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Prior to Operation

Pilot Box Valves Operational Test

1. Perform the Pilot Box Valves Operational Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Box Valves Operational Test.

WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT BOX VALVES OPERATIONAL TEST HAS BEEN SUCCESSFULLY COMPLETED.

Pilot Flame Test

1. Perform the Pilot Flame Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Flame Test.

WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT FLAME TEST HAS BEEN SUCCESSFULLY COMPLETED.

Liquid Propane Leak check

1. Perform the Liquid Propane Leak Check. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Liquid Propane Leak Check.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER UNTIL THE LIQUID PROPANE LEAK CHECK HAS BEEN SUCCESSFULLY COMPLETED.

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Operation

1. Run (gas, control and electric) lines straight back from Pilot Box Assembly.
2. Install cable/hose covers. Refer to Figure 8.

⚠ CAUTION

DO NOT OPERATE O-100 TRAINER WITHOUT BOTH CABLE/HOSE COVERS IN PLACE! REFER TO FIGURE 8.

⚠ WARNING • AVERTISSEMENT

ONLY WHEN ALL THE ABOVE OPERATIONAL AND LEAK CHECKS HAVE BEEN SUCCESSFULLY PERFORMED YOU ARE READY TO TRAIN.

3. Attach the electric supply cord from the Pilot Box assembly to an outdoor 120V power GFIC receptacle.

NOTE

FOR OPERATION INSTRUCTIONS REFER TO THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

⚠ WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 HORIZONTAL PROPANE TANK FIRE TRAINER A THOROUGH FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

Shutdown & Disassembly

⚠ WARNING • AVERTISSEMENT

AT THE END OF A TRAINING SESSION THERE WILL STILL BE FUEL IN THE HOSES FROM THE PROPANE SUPPLY TANK TO THE MOCKUP/FIRE PAN. THIS WILL HAVE TO BE BURNT OFF BEFORE THE EQUIPMENT IS TAKEN APART. IT IS IMPORTANT THAT THIS PROCEDURE BE FOLLOWED EVEN IF YOU ARE NOT TAKING EQUIPMENT APART BUT ARE JUST SHUTTING DOWN AT THE END OF A TRAINING SESSION. FOR THIS PROCEDURE REFER TO SECTION 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

NOTE

FOR SHUTDOWN & DISASSEMBLY INSTRUCTIONS REFER TO SEC 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

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APPENDIX M

Set-Up Instructions

Introduction

Appendix M provides the information necessary to properly set-up the **O-100 Gas Cylinders Trainer** for use with the **HOSE-LINE FIRETRAINER® O-100.**

WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 GAS CYLINDERS FIRE TRAINER A THOROUGH UNDERSTANDING AND FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

IN ADDITION TO THE SETUP STEPS LISTED BELOW, IF THE HOSE LINE FIRETRAINER® O-100 IS TO BE CONTROLLED BY ANY OF THE FOLLOWING OPTIONAL KIDDE CONTROL SYSTEMS, REFER TO THE APPROPRIATE HOSE LINE FIRETRAINER® O-100 CONTROL SYSTEM USERS MANUAL FOR SETUP INSTRUCTIONS.

- **MULTI CONTROL**
- **WIRELESS CONTROL**
- **BATTERY POWERED CONTROL**

Installation Instructions

1. Locate the Gas Cylinders Mock-Up outdoors; clear of all ground/aerial obstructions on level ground, a minimum of 40 feet downwind of the Propane Supply.

NOTE

IF THE Kidde Multi Control System is used, the Multi Control Stand is considered the propane supply.

2. Remove the wheels from the Pilot Box Assembly and store wheels safely away from training area. Place the Pilot Box Assembly, vapor propane cylinder and liquid propane cylinder as shown in figure 1. Position arrangement so the Pilot Box Assembly connections are facing the propane supply.
3. Extend all lines (gas, control, electric) and check for any damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF ANY OF THE CONTROL OR ELECTRICAL LINES ARE DAMAGED. DO NOT PROCEED ANY FURTHER IF ANY OF THE GAS HOSES ARE DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

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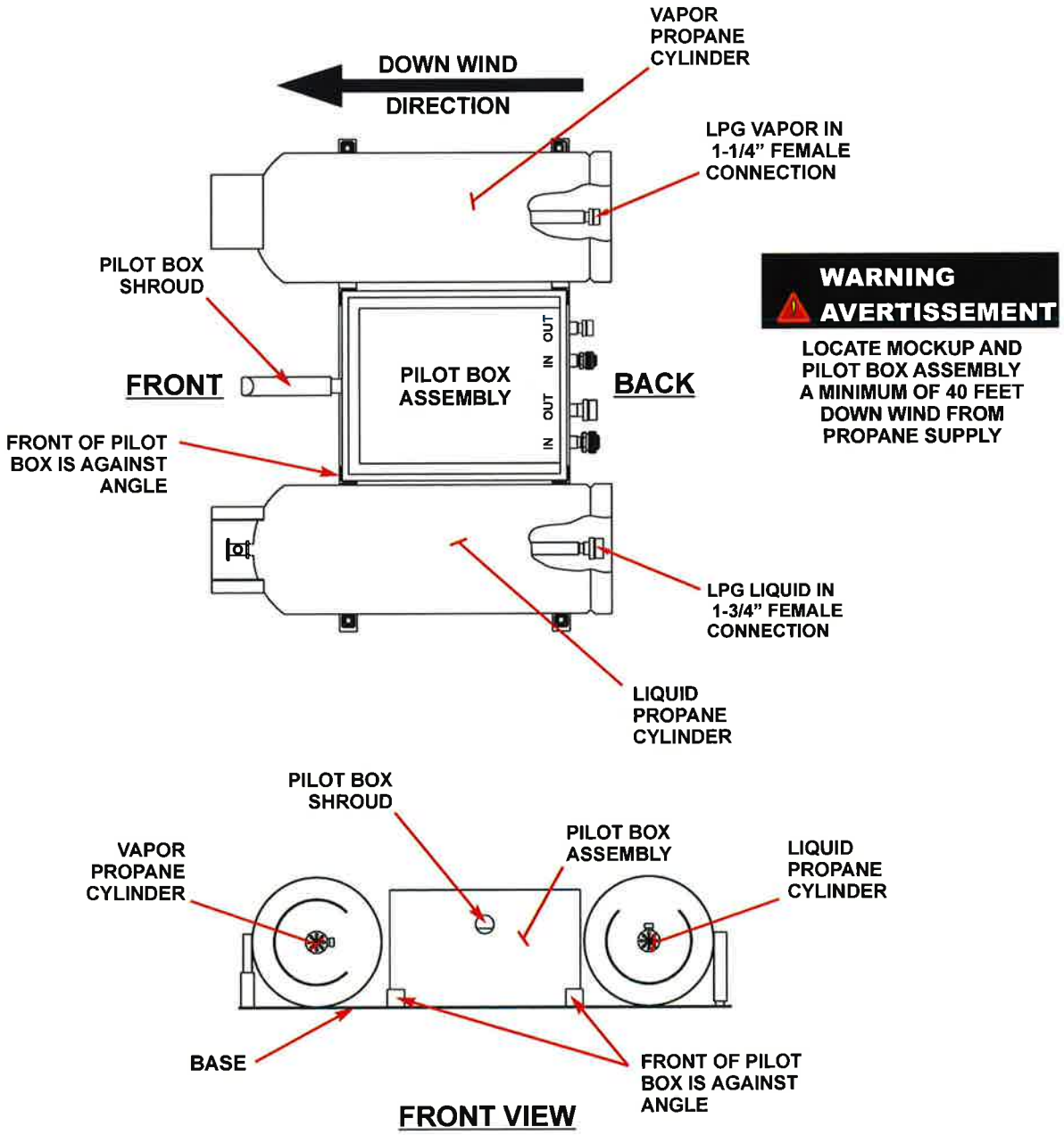

Figure 1. O-100 Gas Cylinders Fie Trainer Setup and Overview

4. At the Pilot Box Assembly remove protective thread cap from the 1-1/4" male **LPG VAPOR** connection marked **"IN"**. Make sure the 1-1/4" male **LPG VAPOR** connection is free of debris, damage and that the rubber washer is present (see Figure 2).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE LPG VAPOR "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

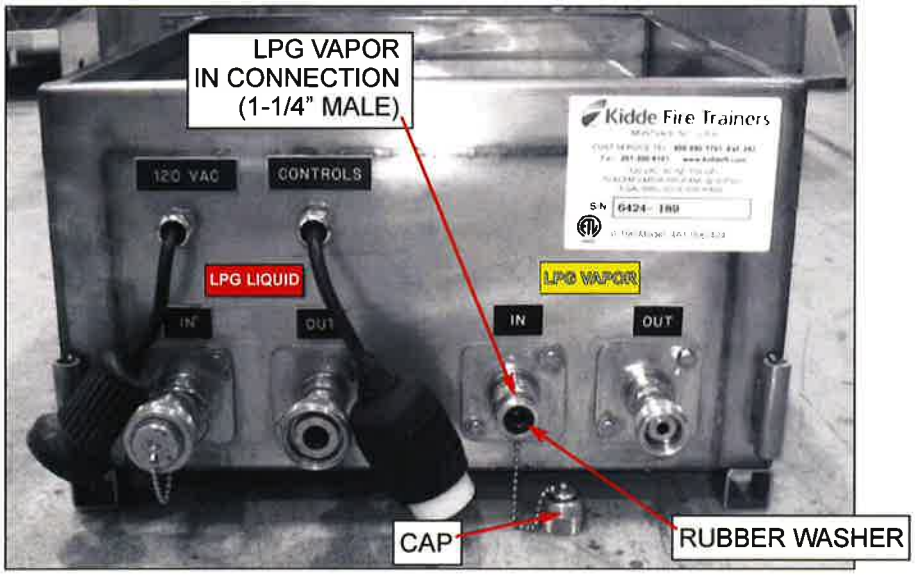

Figure 2. Pilot Box Assembly LPG VAPOR "IN" Connection

5. Make sure the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

6. Connect the 1-1/4" female connection on the 50' black rubber LPG Vapor supply hose to the 1-1/4" male **LPG VAPOR** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 3).

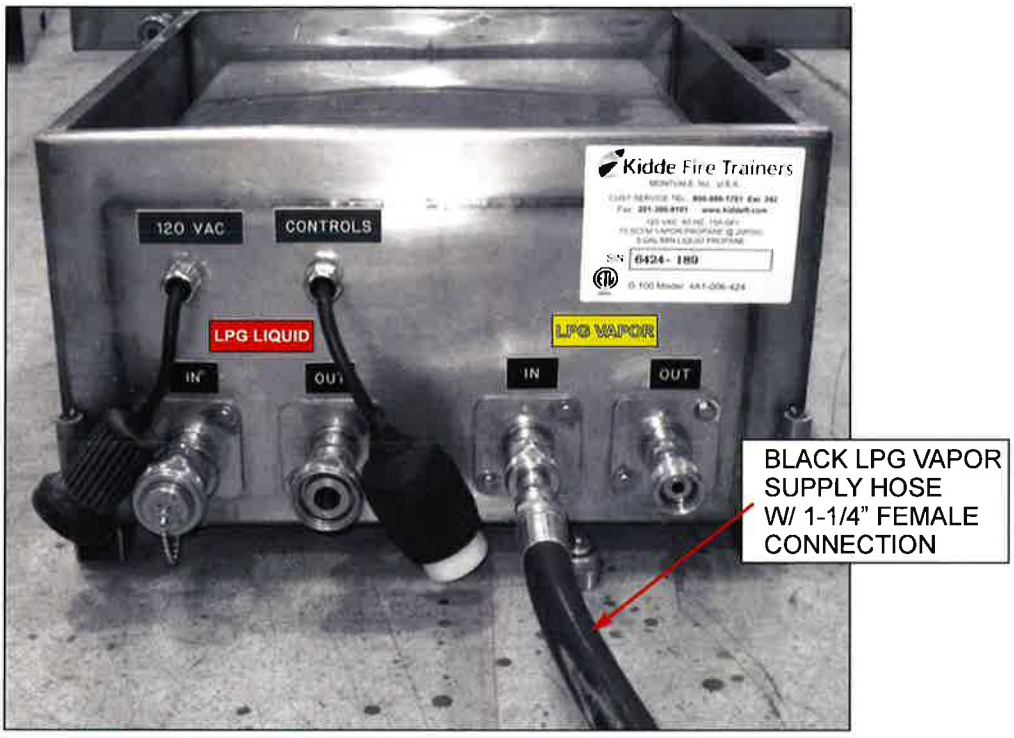

Figure 3. Pilot Box Assembly LPG VAPOR "IN" Supply Hose Connection

7. At the Pilot Box Assembly remove protective thread cap from the 1-3/4" male **LPG LIQUID** connection marked **"IN".** Make sure the 1-3/4" male **LPG LIQUID** connection is free of debris, damage and that the rubber washer is present (see Figure 4).

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE LPG LIQUID "IN" CONNECTION HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

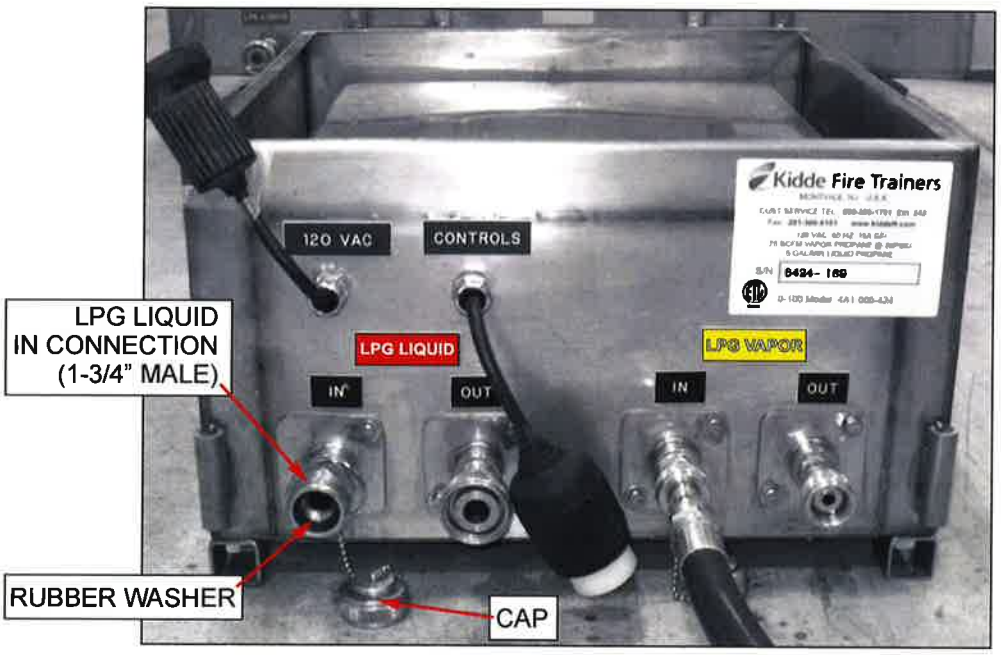

Figure 4. Pilot Box Assembly LPG LIQUID "IN" Connection

8. Make sure the 1-3/4" female connection on the 50' black rubber LPG Liquid supply hose is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION ON THE 50' BLACK RUBBER LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

9. Connect the 1-3/4" female connection on the 50' black rubber LPG Liquid supply hose to the 1-3/4" male **LPG LIQUID** connection marked **"IN"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 5).

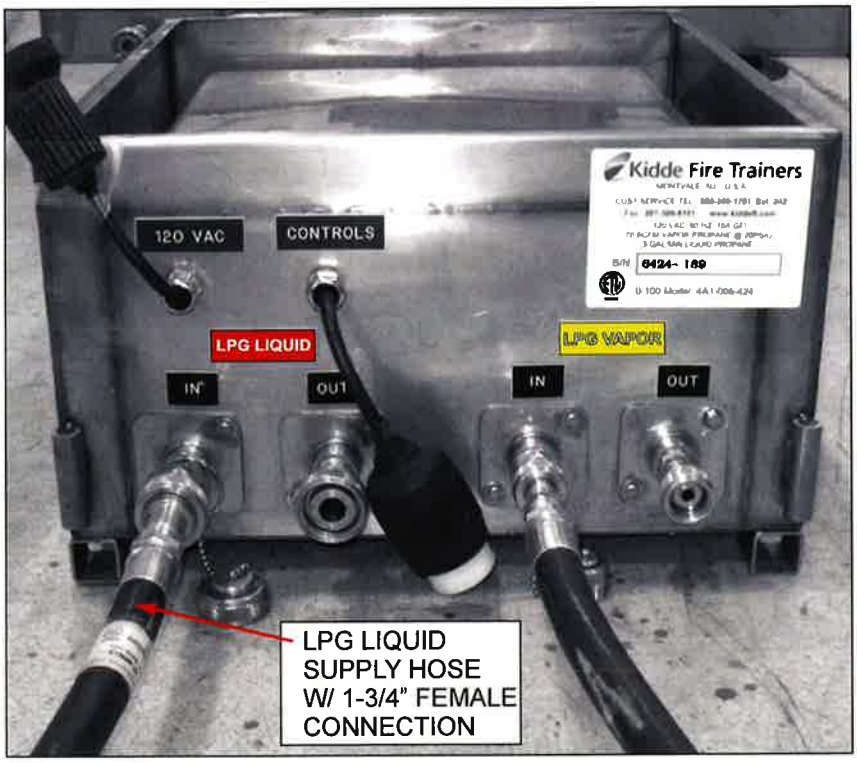

Figure 5. Pilot Box Assembly LPG LIQUID "IN" Supply Hose Connection

10. At the Pilot Box Assembly make sure the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" FEMALE LPG VAPOR "OUT" CONNECTION HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

11. Remove protective thread cap from the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

12. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the 1-1/4" female **LPG VAPOR** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 6).

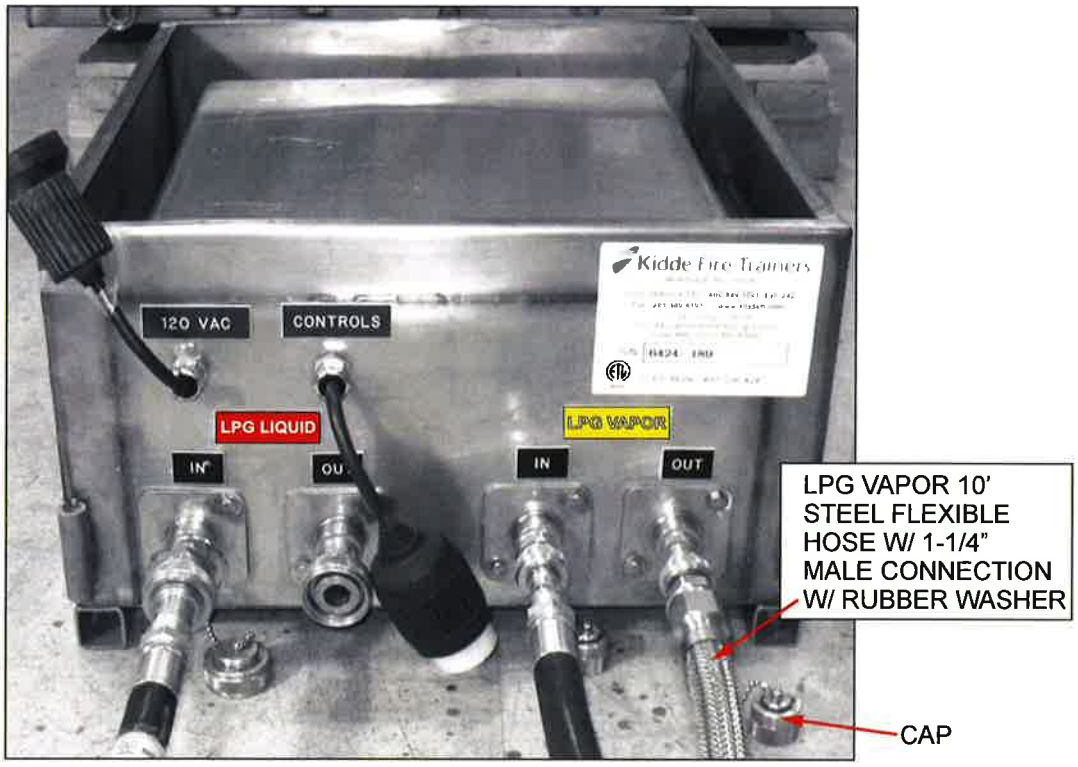

Figure 6. Pilot Box Assembly LPG VAPOR "OUT" Hose Connection

13. Remove protective thread cap form the 1-1/4" male connection on the remaining end of the 10' flexible stainless steel LPG Vapor supply hose. Make sure the 1-1/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-1/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG VAPOR SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

14. At the Gas cylinder Mockup, make sure the LPG Vapor IN 1-1/4" female connection is free of debris and damage (see Figure 8 for location).

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE LPG VAPOR 1-1/4" FEMALE CONNECTION ON THE GAS CYLINDER MOCKUP HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

15. Connect the 1-1/4" male connection on the 10' flexible stainless steel LPG Vapor supply hose to the LPG Vapor 1-1/4" female connection on the Gas Cylinder Mockup. Attach fittings hand tight (see Figure 8).

16. Remove protective thread cap from the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose. Make sure the 1-3/4" male connection is free of debris, damage and that the rubber washer is present.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

17. At the Pilot Box Assembly make sure the 1-3/4" female **LPG LIQUID** connection marked **OUT** is free of debris and damage.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" FEMALE CONNECTION MARKED LPG LIQUID "OUT" HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

18. Connect the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose to the 1-3/4" female **LPG LIQUID** connection marked **"OUT"** on the Pilot Box Assembly. Attach fittings hand tight (see Figure 7).

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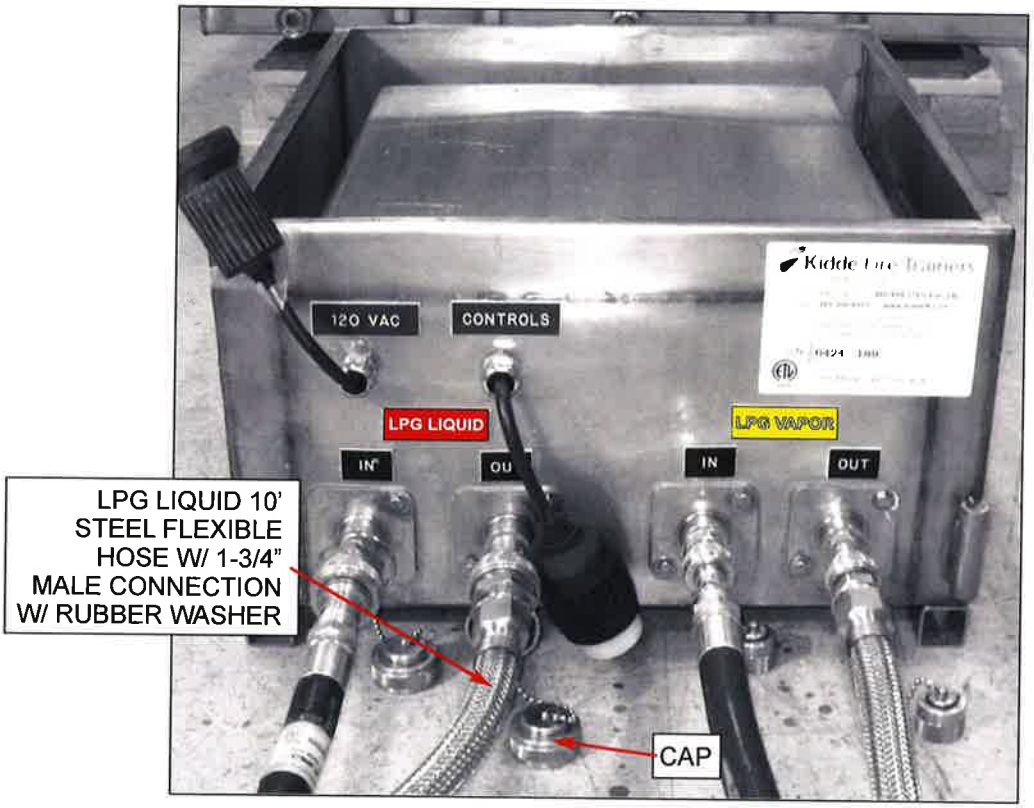

Figure 7. Pilot Box Assembly LPG LIQUID "OUT" Hose Connection

19. Remove protective thread cap form the 1-3/4" male connection on the remaining end of the 10' flexible stainless steel LPG Liquid supply hose. Make sure the 1-3/4" male connection is free of debris, damage and that the rubber washer is present.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE 1-3/4" MALE CONNECTION ON THE 10' FLEXIBLE STAINLESS STEEL LPG LIQUID SUPPLY HOSE HAS DEBRIS IN IT, IS DAMAGED OR THE RUBBER WASHER IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

20. At the Gas Cylinders Mockup make sure the LPG Liquid IN 1-3/4" female connection is free of debris and damage (See Figure 8 for location).

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE LPG LIQUID 1-3/4" FEMALE CONNECTION ON THE GAS CYLINDER MOCKUP HAS DEBRIS IN IT OR IS DAMAGED. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

21. Connect the 1-3/4" male connection on the 10' flexible stainless steel LPG Liquid supply hose to the LPG Liquid 1-3/4" female connection on the Gas Cylinder Mockup. Attach fittings hand tight (see Figure 8).

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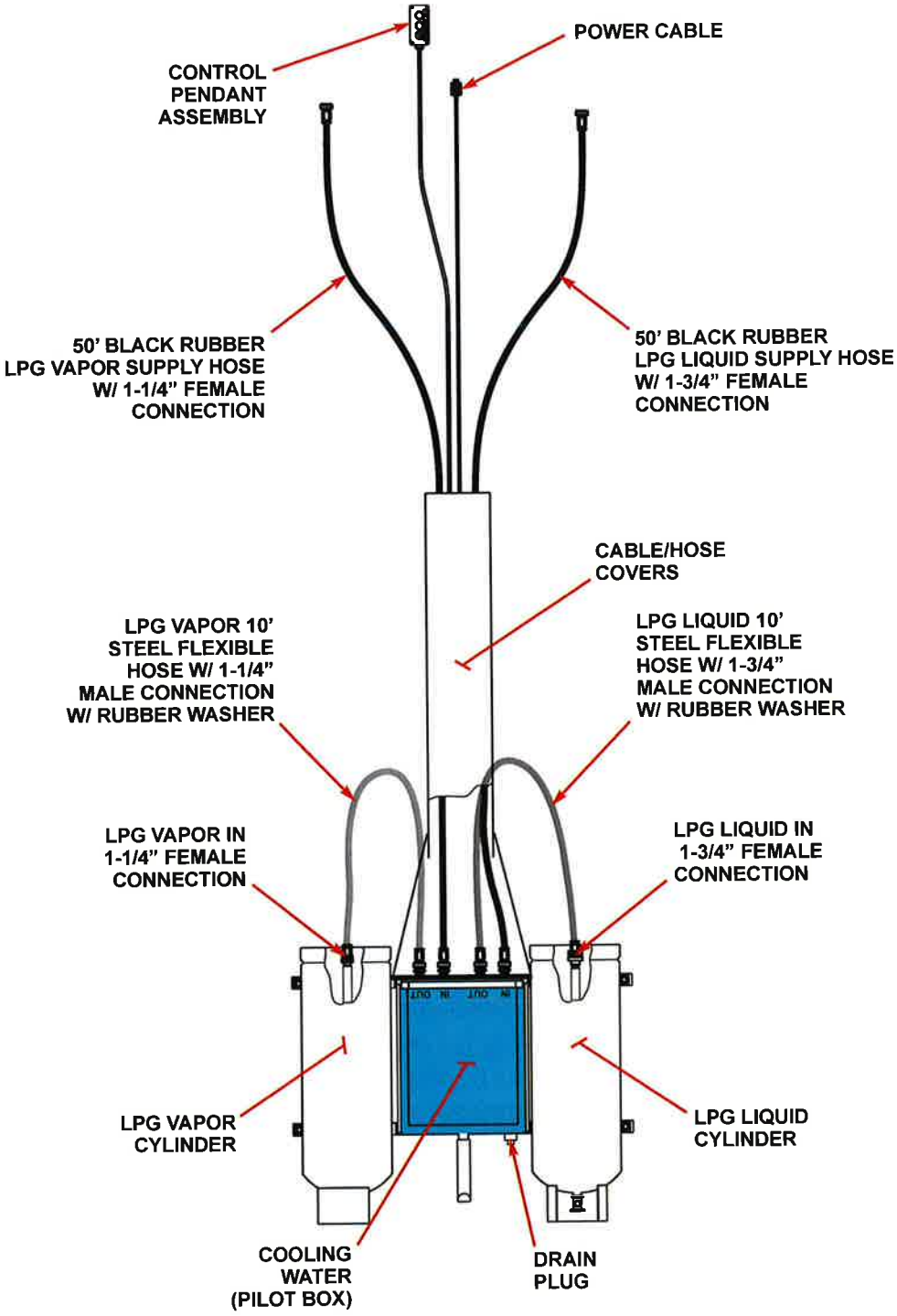

Figure 8. O-100 Gas Cylinders Fie Trainer Setup (Top View)

22. Attach the Control Pendant cable to the connection marked **"CONTROLS"** on Pilot Box Assembly (see Figure 9).
23. Attach the Electric Supply cord to the connection marked **"120 VAC"** on Pilot Box Assembly (see Figure 9).

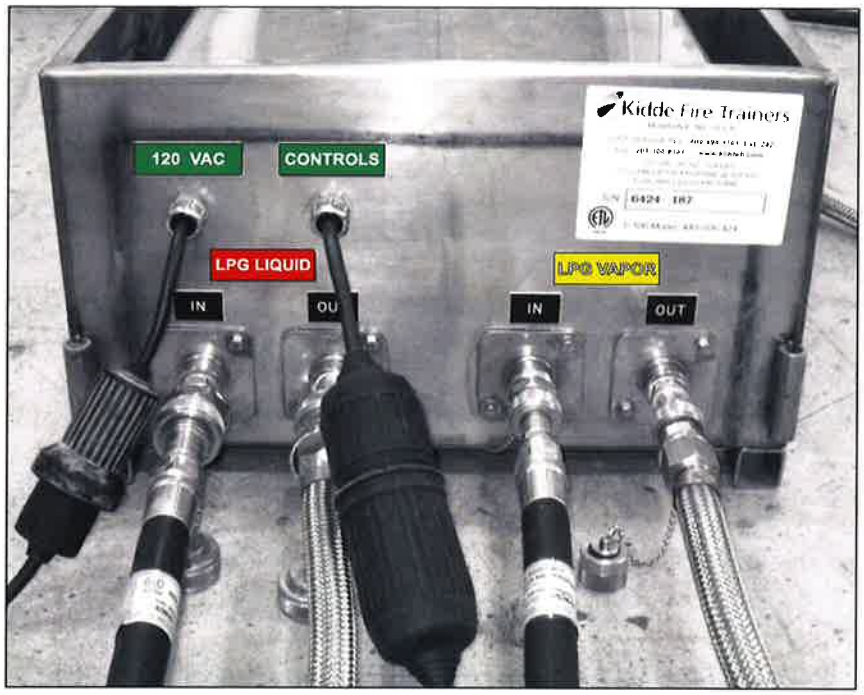

Figure 9. Electrical Supply & Control Pendant Connections

24. Ensure that water drain plug is installed in the Pilot Box Assembly. Drain plug located at front of Pilot Box Assembly. Refer to Figure 8 for location.
25. Fill Pilot Box Assembly top with water until it overflows (see Figure 8).

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

NOTE

IF THE HOSE LINE FIRETRAINER® O-100 is to be controlled by the Kidde Multi Control System, refer to the HOSE LINE FIRETRAINER® O-100 Multi Control System Users Manual, Sections 3.1 & 3.2.

26. Using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088), connect the remaining end of the 50' black rubber LPG vapor supply hose from the Pilot Box Assembly **LPG VAPOR** connection marked **"IN"**, to **LPG Vapor Outlet** on the Propane Supply. Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male adaptor connection. Attach fittings hand tight.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER LPG VAPOR supply hose is connected to the LPG VAPOR outlet on the propane supply tank.

27. Using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088), connect the remaining end of the 50' black rubber LPG liquid supply hose from the Pilot Box Assembly **LPG LIQUID** connection marked **"IN"**, to the **LPG LIQUID Output** on the Propane Supply. Prior to making the connection, check connections for dirt, damage and that the rubber washer is present in the male adaptor connection. Attach fittings hand tight.

WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER IF THE CONNECTIONS HAVE DEBRIS IN THEM, ARE DAMAGED OR THE RUBBER WASHER IN THE MALE ADAPTER CONNECTION IS DAMAGED OR MISSING. DO NOT USE TRAINER UNTIL SITUATION IS CORRECTED.

BE SURE THE BLACK RUBBER LPG LIQUID supply hose is connected to the LPG LIQUID outlet on the propane supply tank.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

Prior to Operation

Pilot Box Valves Operational Test

1. Perform the Pilot Box Valves Operational Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Box Valves Operational Test.

⚠ WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT BOX VALVES OPERATIONAL TEST HAS BEEN SUCCESSFULLY COMPLETED.

Pilot Flame Test

1. Perform the Pilot Flame Test. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Pilot Flame Test.

⚠ WARNING • AVERTISSEMENT

DO NOT PERFORM NEXT STEP UNTIL THE PILOT FLAME TEST HAS BEEN SUCCESSFULLY COMPLETED.

Liquid Propane Leak check

1. Perform the Liquid Propane Leak Check. Refer to the **HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL,** SEC 4, Liquid Propane Leak Check.

⚠ WARNING • AVERTISSEMENT

DO NOT PROCEED ANY FURTHER UNTIL THE LIQUID PROPANE LEAK CHECK HAS BEEN SUCCESSFULLY COMPLETED.

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Operation

1. Run (gas, control and electric) lines straight back from Pilot Box Assembly.
2. Install cable/hose covers. Refer to Figure 8.

⚠ CAUTION

DO NOT OPERATE O-100 TRAINER WITHOUT BOTH CABLE/HOSE COVERS IN PLACE! REFER TO FIGURE 8.

⚠ WARNING • AVERTISSEMENT

ONLY WHEN ALL THE ABOVE OPERATIONAL AND LEAK CHECKS HAVE BEEN SUCCESSFULLY PERFORMED YOU ARE READY TO TRAIN.

3. Attach the electric supply cord from the Pilot Box assembly to an outdoor 120V power GFIC receptacle.

NOTE

FOR OPERATION INSTRUCTIONS REFER TO THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

⚠ WARNING • AVERTISSEMENT

PRIOR TO OPERATION OF THE O-100 HORIZONTAL PROPANE TANK FIRE TRAINER A THOROUGH FAMILIARITY WITH THIS APPENDIX AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

Shutdown & Disassembly

⚠ WARNING • AVERTISSEMENT

AT THE END OF A TRAINING SESSION THERE WILL STILL BE FUEL IN THE HOSES FROM THE PROPANE SUPPLY TANK TO THE MOCKUP/FIRE PAN. THIS WILL HAVE TO BE BURNT OFF BEFORE THE EQUIPMENT IS TAKEN APART. IT IS IMPORTANT THAT THIS PROCEDURE BE FOLLOWED EVEN IF YOU ARE NOT TAKING EQUIPMENT APART BUT ARE JUST SHUTTING DOWN AT THE END OF A TRAINING SESSION. FOR THIS PROCEDURE REFER TO SECTION 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

NOTE

FOR SHUTDOWN & DISASSEMBLY INSTRUCTIONS REFER TO SEC 6 IN THE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL.

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APPENDIX N

Introduction

The Pilot Box Assembly and Pan Assembly located throughout the O-100 Manual and Appendices shows the labels in English. Appendix N provides those labels on the Pilot Box Assembly as shown in Figure 1-1 and the Pan Assembly as shown in Figure 1-2 are in French.

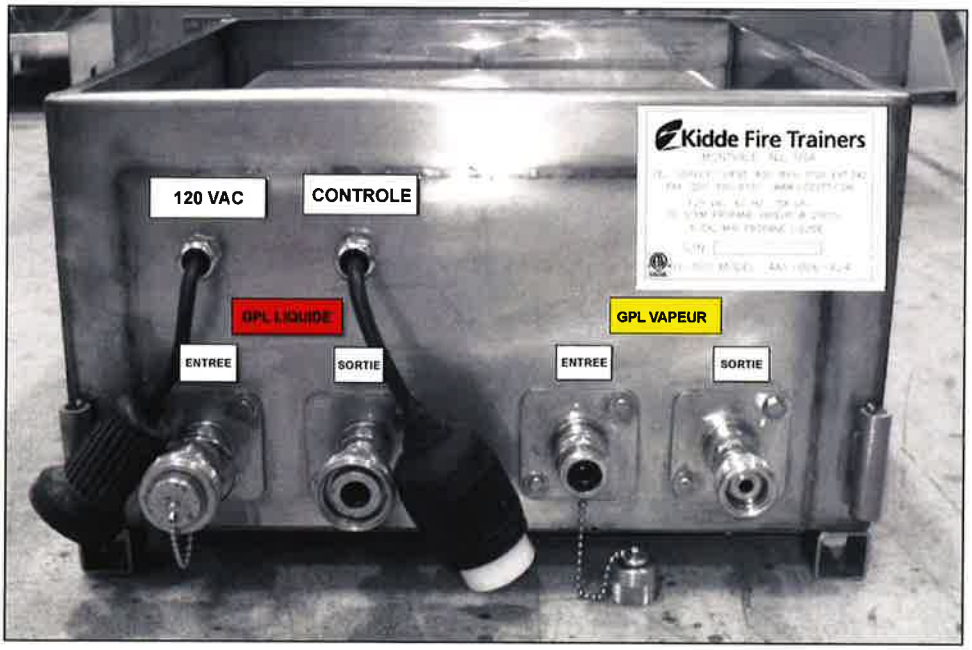

Figure 1-1. Pilot Box Assembly

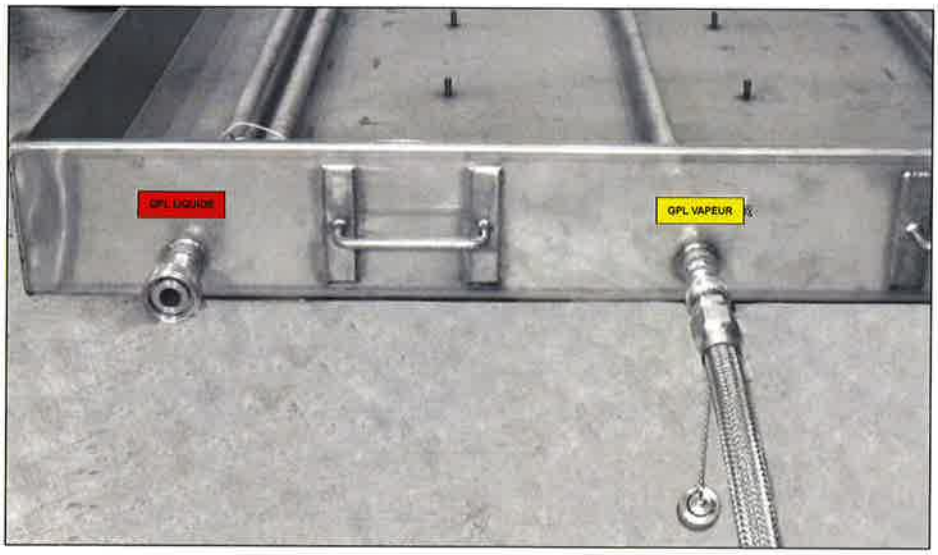

Figure 1-2. Fire Pan Assembly

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DWG NO.	2C1-004-860	SH	1

REVISIONS

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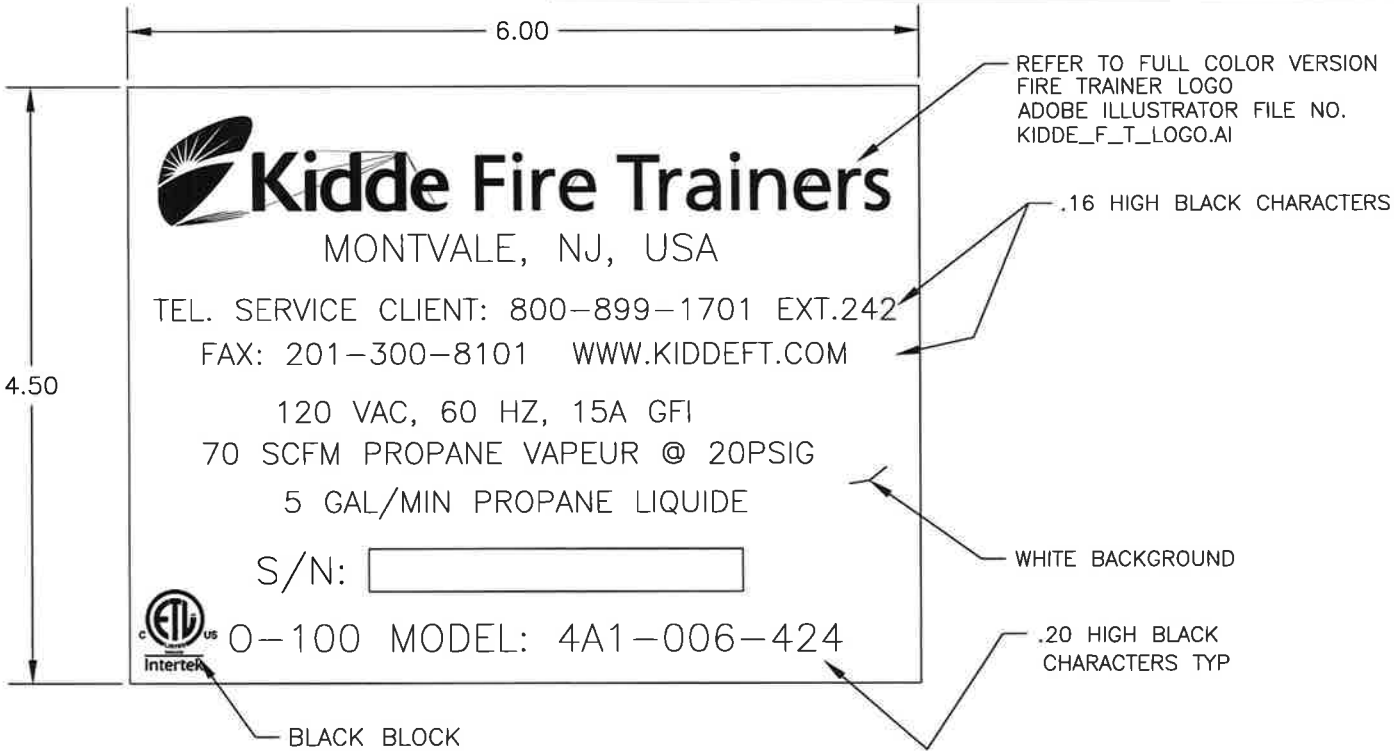

NOTES:
1-SUB-SURFACE SCREEN-PRINTED .010" TEXTURED LEXAN.

2-2 MIL ADHESIVE FOR OUTDOOR USE.

3-USE MANUFACTURER'S FILE NO. WHEN ORDERING.

PURCHASE FROM:
NAMEPLATES FOR INDUSTRY, INC.
213 THEODORE RICE BLVD.
NEW BEDFORD, MA 02745
TEL: 800-999-8900

4-UV WEATHERABLE LEXAN ONLY.

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ITEM	QTY	UOM	SPECIFICATIONS	DESCRIPTION

MATERIAL LIST

--	--
NEXT ASSY	USED ON
APPLICATION	

MATERIAL: SEE NOTE 4

FINISH:

TOLERANCES UNLESS OTHERWISE SPECIFIED
INCH
xx DEC ± .06
xxx DEC ± .015
ANGLES ± 1°
METRIC
xx MM ± -
xxx MM ± -

CONTRACT NO.

DWN	JM	11/14/13
CHKD	LM	11/14/13
APVD		

Kidde Fire Trainers

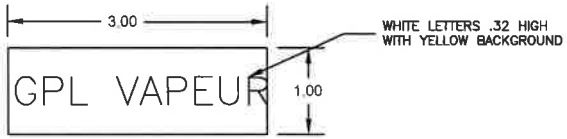
TITLE: LABEL, O-100

SIZE	CAGE CODE	DWG NO.
B	06003	2C1-004-860

SCALE -	CAD FILE NO. 2C4860A01	SHEET 1 OF 1

DWG NO	3D1-003-782	SH 1

REVISIONS			
REV	DRAWING CHANGE RECORD NO	DATE	APVD



-001 DETAIL
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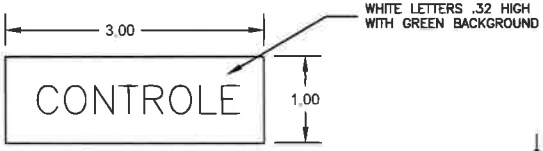

-004 DETAIL
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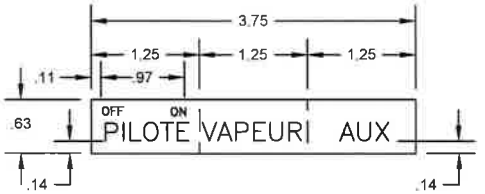
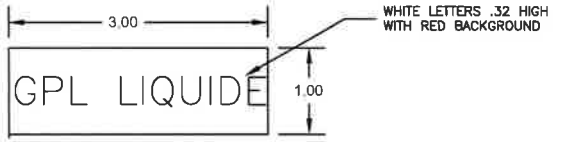
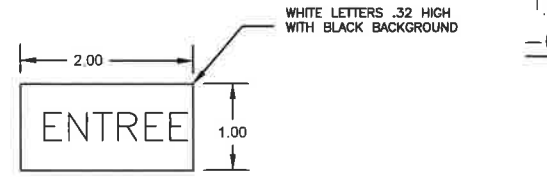


-007 DETAIL
SCALE 1:1

BLACK LAMECOID SELF ADHESIVE LABEL
WHITE LETTERS .21 HIGH ON THE LARGE
AND .09 HIGH ON THE SMALLER AS ON
A BLACK BACKGROUND LARGE AS
POSSIBLE TO FIT SIZE



-002 DETAIL
SCALE 1:1



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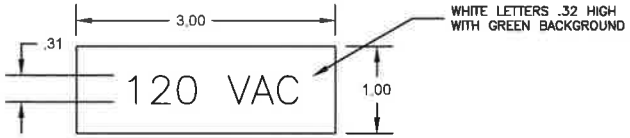

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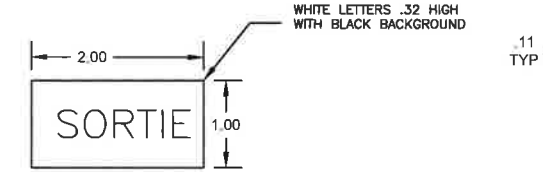

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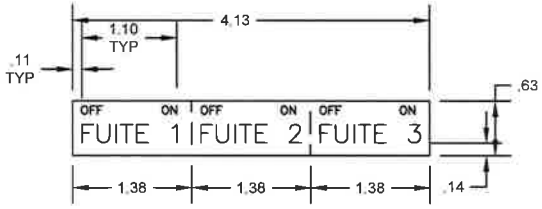

-008 DETAIL
SCALE 1:1

BLACK LAMECOID SELF ADHESIVE LABEL
WHITE LETTERS .21 HIGH ON THE LARGE
AND .09 HIGH ON THE SMALLER AS ON
A BLACK BACKGROUND LARGE AS
POSSIBLE TO FIT SIZE

NOTE: UNLESS OTHERWISE SPECIFIED,
ALL CHARACTERS CENTERED ON
LABEL.

ITEM	QTY	UOM	SPECIFICATIONS	DESCRIPTION
MATERIAL LIST				

PROJECT		TOLERANCES UNLESS OTHERWISE SPECIFIED INCH xx DEC ± .06 xxx DEC ± .015 ANGLES ± 1° METRIC xx MM ± - xxx MM ± -	MATERIAL	Kidde Fire Trainers	
			FINISH	TITLE	
				LABELS, O-100	
4A1-006-424	O-100		DWN JM 11/14/13	SIZE C	CAGE CODE 06003 / DWG NO 3D1-003-782
NEXT ASSY	USED ON		CHKD LM 11/14/13	SCALE 1:1	CAD FILE: 3D3782-01 / SHEET 1 OF 1
APPLICATION			APVD		

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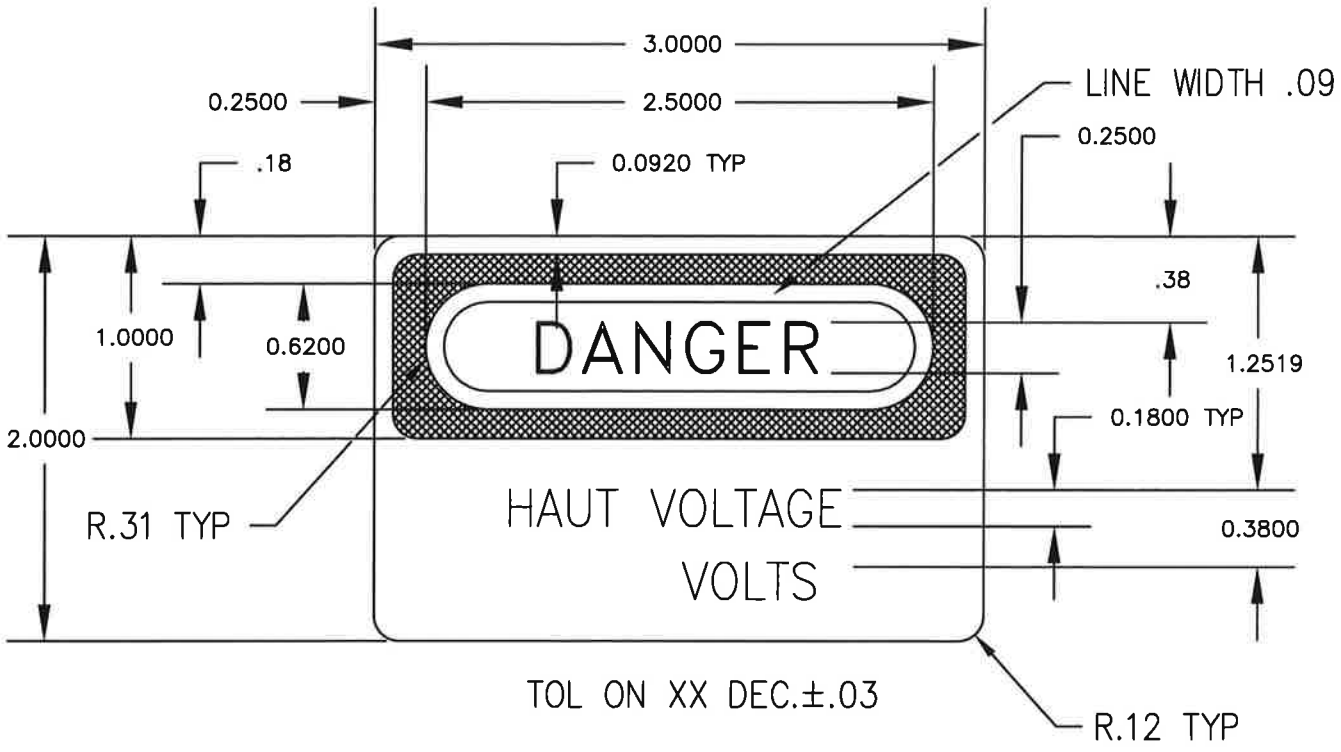

Nr.PIECE 1D1-002-184 DASH	VOLTS
001	7500
002	PAS UTILISE
003	6000
004	18000
005	PAS UTILISE
006	4000

NOTES:

1. -PLATE TO BE MADE IN ACCORDANCE WITH MIL-P-19834 B TYPE 1 (ADHESIVE-BACKED METAL FOIL .010 THICK).
2. -PLATE DESIGN AND COLOR TO BE IN ACCORDANCE WITH ANSI Z35.1-1972.
3. -THE UPPER PANEL SHALL HAVE THE WORD "DANGER" IN WHITE WITHIN A RED OVAL OUTLINED IN WHITE ON A BLACK RECTANGULAR BACKGROUND.
4. -THE LOWER PANEL SHALL HAVE THE WORDS "HAUT VOLTAGE, VOLTS" IN BLACK ON A WHITE BACKGROUND
5. -CHARACTERS IN UPPER/LOWER PANELS TO BE "GOTHIC" TYPE, UPPER CASE, CENTRALLY LOCATED AS SHOWN

TITLE	SIZE	CAGE CODE	DWG NO.	
PLATE, INSTRUCTION	A	06003	1D1-002-184	-001 -006
	REV	CAD FILE 1D2184-01	SHEET 1 OF 1	

Hose Line FireTrainer® O-100

Multi-Control System Manual

KFT™

HOSE LINE FIRETRAINER® O-100

MULTI CONTROL SYSTEM

USER'S MANUAL

KFT Fire Trainer, LLC
(Formerly known as Kidde Fire Trainers, LLC)
5 Pearl Court, Unit D
Allendale, NJ 07401 USA
Tel: 1-800-899-1701 EXT. 502

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Document No.: O-100 4A1006750
Date: 05-05-11 / 05-29-2026
Revision: B cover page update

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1.0 INTRODUCTION

1.1 GENERAL

This manual provides all of the necessary information to operate the:

Hose Line FireTrainer® O-100 Multi Control System

MANUFACTURED BY:

Kidde Fire Trainers Inc.

If you have any questions regarding the operation or maintenance of this trainer that are not answered in this manual contact the Kidde Fire Trainers Customer Support Group at:

KIDDE FIRE TRAINERS INC.

17 PHILIPS PARKWAY
MONTVALE, NJ 07645
USA
TEL: (201) 300-8100
WEB SITE: WWW.KIDDEFT.COM
E-MAIL: INFO@KIDDE.COM

⚠ WARNING

THIS MANUAL SUPPORTS THE KIDDE HOSE LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL AND IS INCOMPLETE WITHOUT THAT MANUAL

PRIOR TO OPERATION OF THE HOSE-LINE FIRETRAINER® O-100 MULTI CONTROL SYSTEM A THOROUGH UNDERSTANDING AND FAMILIARITY WITH THIS MANUAL AND THE ENTIRE HOSE-LINE FIRETRAINER® O-100 OPERATION AND MAINTENANCE MANUAL IS REQUIRED. FAILURE TO DO THIS COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

This manual provides the necessary information to operate and maintain the Kidde Multi Control System.

To maximize training effectiveness and use the full potential of the Multi Control System, a thorough understanding of the system and controls is necessary. This section describes the layout of the manual and the Safety Guidelines. Read and understand the Safety Guidelines and system controls before operating.

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1.2 HOW TO USE THIS MANUAL

This manual is presented in the following order:

Section 1	**Introduction**	This section provides a brief functional description how to use the manual and relevant safety guidelines and instructions.
Section 2	**System Description**	This section provides an overview and a functional description of the Multi Control System. Operator controls and indicators are also described in this section
Section 3	**Operation**	Detailed operating instructions.
Section 4	**Care and Maintenance**	This section provides Information to keep the Multi Control System functional and operationally effective.
Section 5	**Parts List**	This section provides figures and parts list tables for each of the Multi Control System major assemblies.
Section 6	**Drawings**	This section contains interconnection drawings to support the trainer

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1.3 SAFETY SYMBOLS

Various symbols are used throughout the manual to alert the reader to a potentially dangerous situation.

WARNING

THE WARNING SYMBOL IS USED WHENEVER A CONDITION EXISTS THAT COULD POTENTIALLY CAUSE DEATH OR SERIOUS INJURY.

CAUTION

THE CAUTION SYMBOL IS USED WHENEVER A CONDITION EXISTS THAT MAY CAUSE INJURY OR DAMAGE TO ASSOCIATED EQUIPMENT.

1.4 TRAINER SAFETY

WARNING

REFER TO THE SITE SPECIFIC HOSE LINE FIRETRAINER® O-100 OPERATION & MAINTENANCE MANUAL, SAFETY SECTION, PRIOR TO PROCEEDING.

1.5 MULTI CONTROL SYSTEM SAFETY

1. Always keep the Trainer(s) in line of site during operation.
2. Always shut FIRE TRAINER® system down before performing any Multi Control System maintenance, to prevent accidental operation of the fireplace.

WARNING

ALL UNUSED CONTROL STAND LPG MANIFOLD CONNECTIONS MUST HAVE A CAP SECURELY INSTALLED PRIOR TO CONNECTING LPG VAPOR/LIQUID INPUT SUPPLY HOSES TO THE CONTROL. FAILURE TO DO THIS WILL CAUSE DEATH OR SERIOUS INJURY

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2.0 SYSTEM DESCRIPTION

2.1 INTRODUCTION

The Multi Control System assembly allows the instructor to control up to three (3) **KIDDE HOSE LINE FIRETRAINER® O-100s** from a one central location.

2.2 DESCRIPTION

The Multi Control System assembly consists of a Control Console and a Control stand (see Figure 2-1).

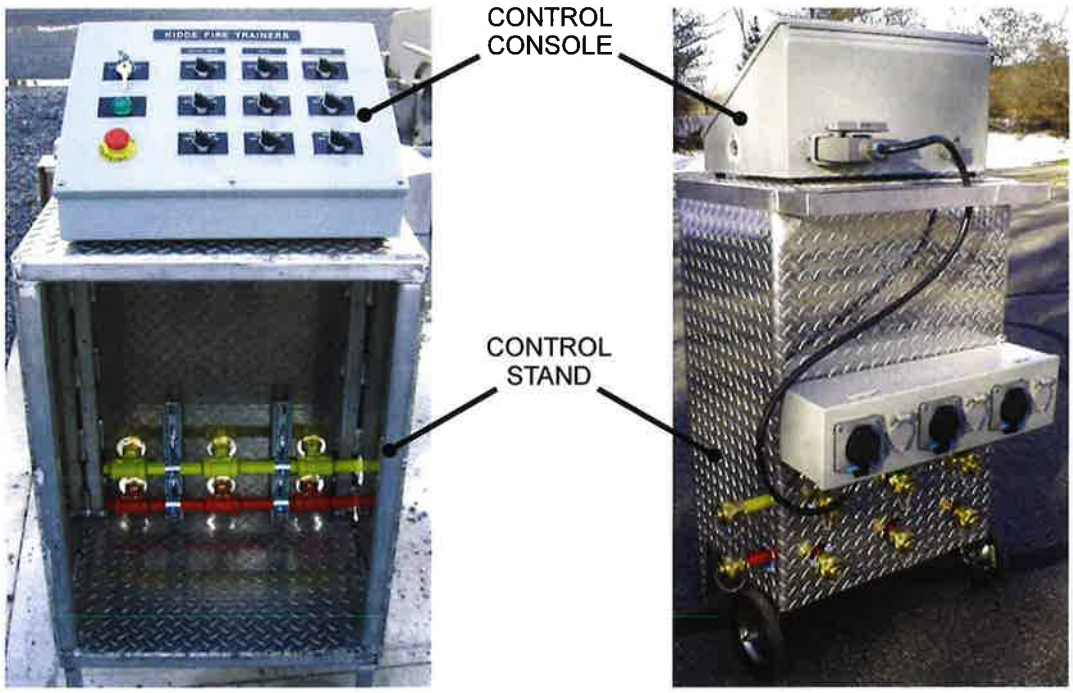

Figure 2-1 – Multi Control Assembly

2.3 CONTROL STAND ASSEMBLY

The Control Stand Assembly is used to support the Control Console. Additionally it houses the LPG Vapor/Liquid manifolds and the power/control junction box. See Figure 2-2.

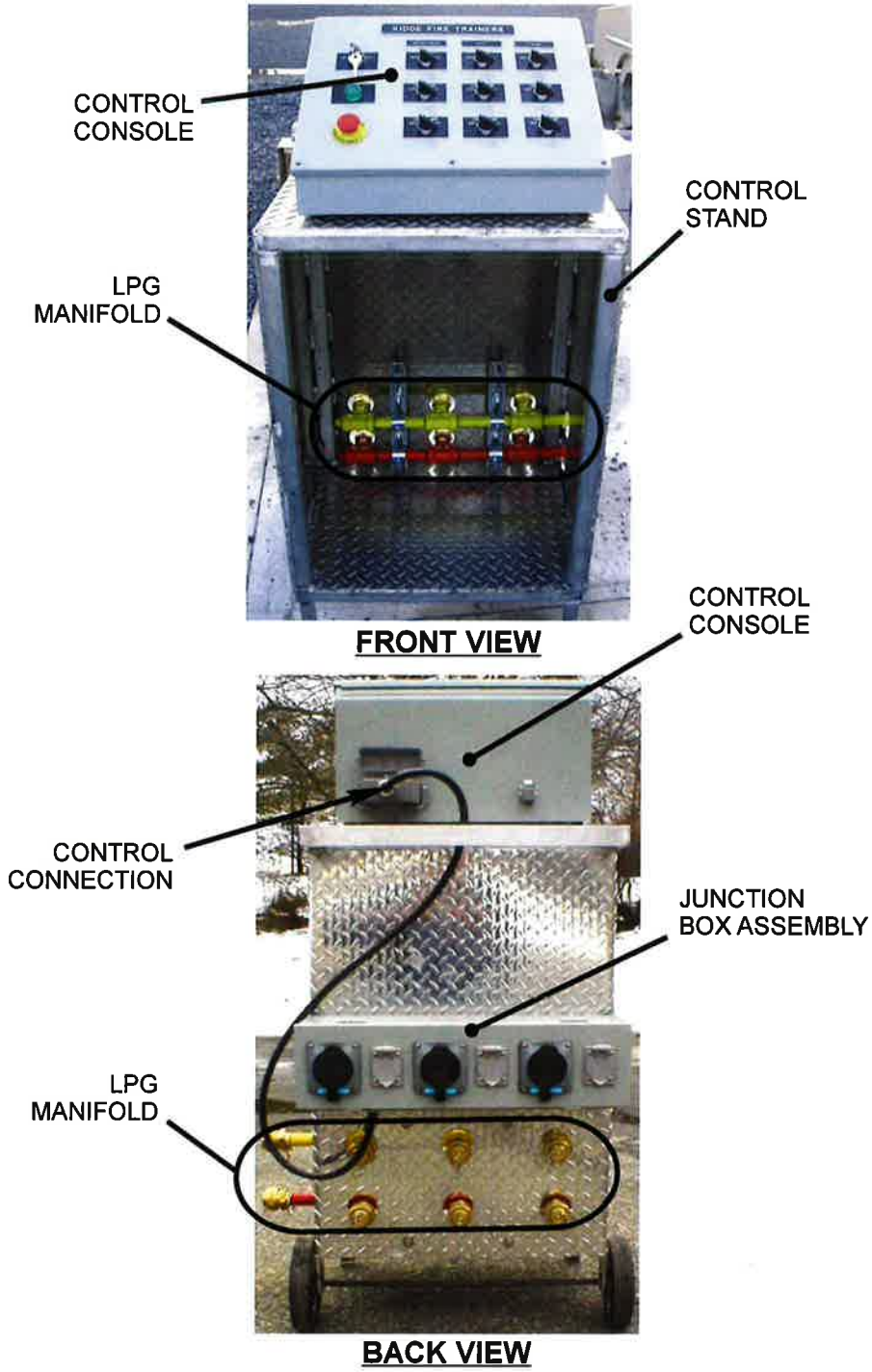

Figure 2-2 – Control Stand Assembly

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

2.3.1 GAS MANIFOLD ASSEMBLY

The Gas Manifold Assembly is located in the bottom of the control stand. The Gas Manifold Assembly is composed of the necessary piping to distribute vapor and liquid propane to all of the Pilot Box Assemblies/Burn Pans. The LPG vapor connection is painted yellow and the LPG liquid connection is painted red. See Figure 2-3 for Gas Manifold Assembly.

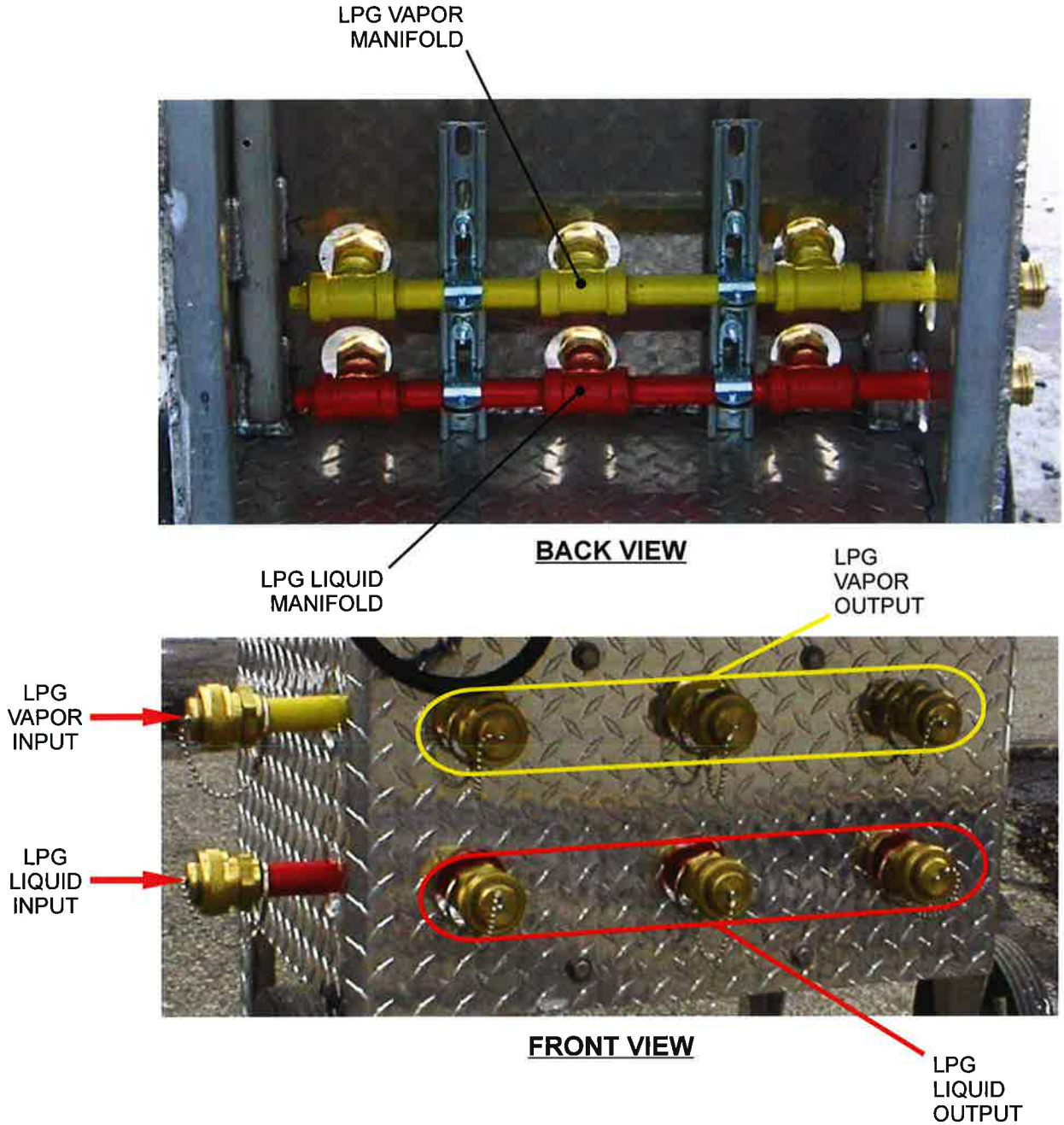

Figure 2-3 – Gas Manifold assembly

2.3.2 JUNCTION BOX ASSEMBLY

This junction box is located on the Control Stand (See Figure 2-2 for location). The control cable, connection from the Control Console to the Junction Box provides power distribution and control signals for the three Pilot Box Assemblies. See Figure 2-4.

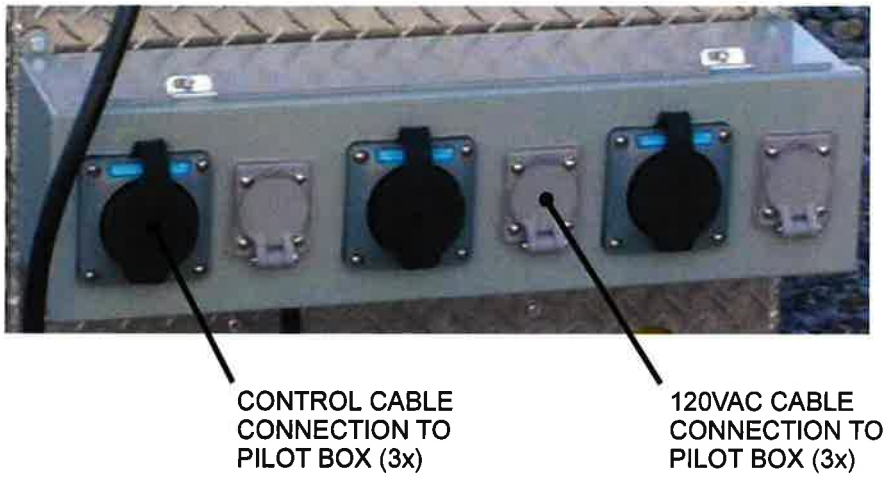

Figure 2-4 – Junction Box Assembly

2.4 CONTROL CONSOLE ASSEMBLY

This Control Console Assembly is used to control the **HOSE LINE FIRETRAINER® O-100s**. See Table 2-1 for Control/Indicator Descriptions and see Figure 2-5 for Control Console layout.

Table 2-1. Control Console-Controls and Indicators Descriptions

No.	Control	Indicator	Description
1	X		**POWER ON/OFF** - When This key switch is turned to the On position supplies power to the Control Console switches.
2		X	**POWER** - This lamp will turn on when the POWER ON/OFF Key Switch (1) has been turned to the ON position.
3	X	X	**EMERGENCY STOP** – When in the pulled out position, the lamp in this button will turn on when the POWER ON/OFF Key Switch (1) has been turned to the ON position. When this button is pushed in it shuts down power to the Burner Control Assembly, in turn closing the vapor propane and liquid s propane solenoid valves. Flames will extinguish once the residual gas in the supply lines has burned off.
4	X		**PILOT ON/OFF** – This switch controls operation of the pilot flame and has two positions: OFF - Pilot flame is off. ON - The pilot flame is enabled and ignites.
5	X		**LOW FLAME ON/OFF** – This switch controls operation of the Low flame (vapor propane) and has two positions: OFF – Low flame is off. ON - Low flame is enabled and ignites.
6	X		**HIGH FLAME ON/OFF** – This switch controls operation of the High flame (liquid propane) and has two positions: OFF – High flame is off. ON - High flame is enabled and ignites.
7	-	-	**120 VAC PLUG** – Located on side of Console, used to supply 120 VAC to Console.
8	-	-	**DEAD MAN CONNECTOR** – This is used for the dead man switch connection. Dead man switch must be pressed to start and maintain main flame

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Table 2-1. Control Console-Controls and Indicators Descriptions

No.	Control	Indicator	Description
9	-	-	**CONTROL CABLE CONNECTOR** – The control cable connects from the Control Console to the junction box located on the Control Stand.

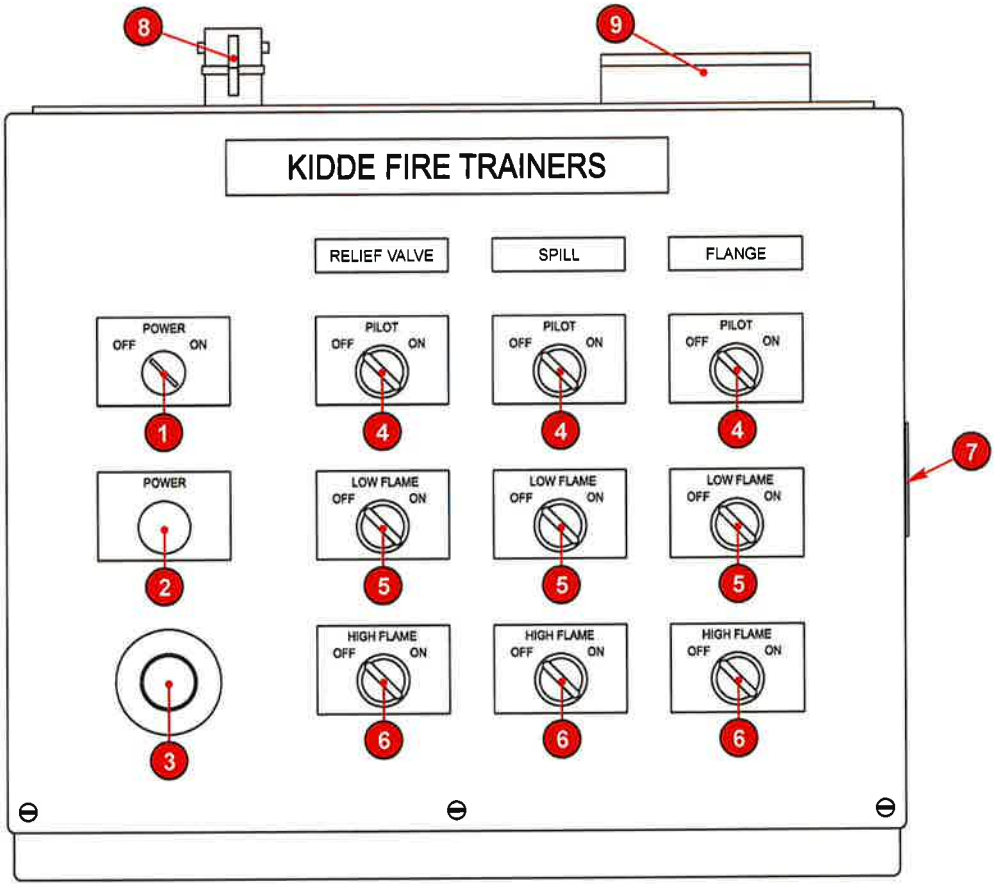

Figure 2-5 – Control Console Assembly

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2.5 DEAD MAN SWITCH

This switch is furnished with a cord that plugs into the Control Console. Once the pilot flame is ignited, the trigger on the grip of the switch must be pressed and held to operate and maintain any of the main flames. See Figure 2-6.

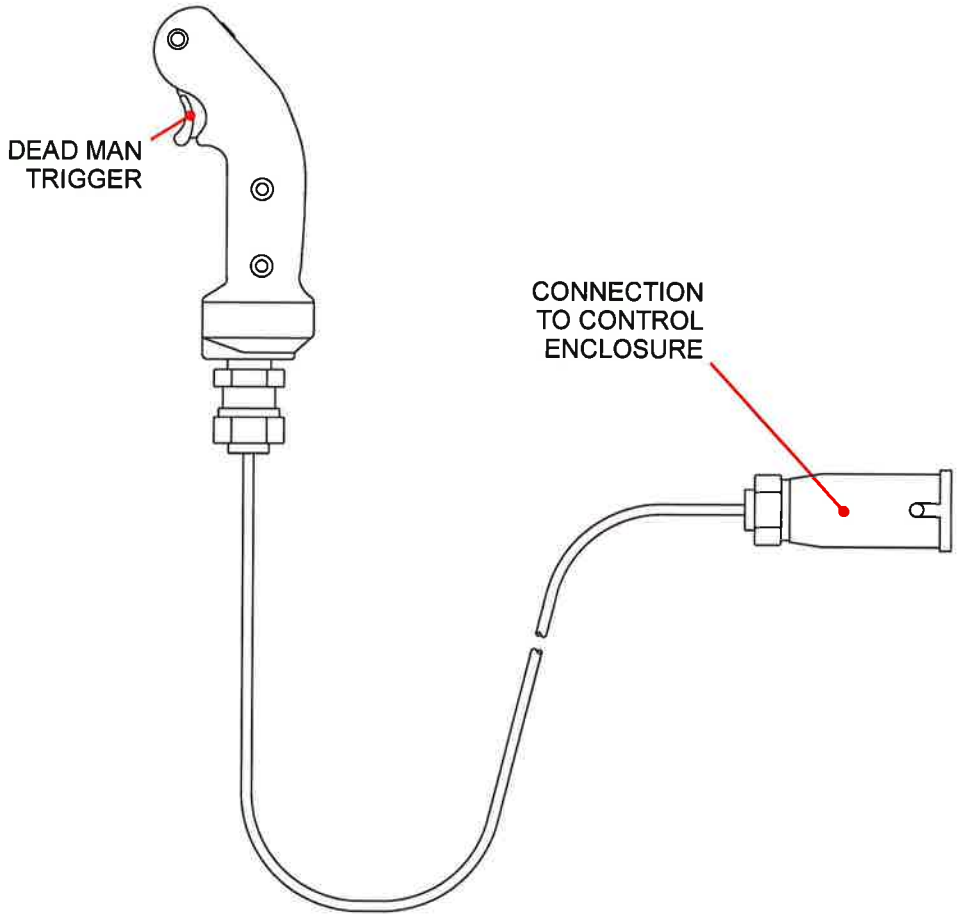

Figure 2-6 – Dead Man Switch Assembly

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3.0 OPERATION

3.1 INTRODUCTION

NOTE

THIS SECTION ONLY COVERS THE SETUP OF THE MULTI CONTROL SYSTEM. FOR COMPLETE SETUP AND STARTUP INSTRUCTIONS REFER TO THE APPLICABLE KIDDE O-100 OUTDOOR FIRE TRAINER OPERATION AND MAINTENANCE MANUAL SUPPLIED FOR YOUR SITE.

3.2 SETUP

1. Locate the Multi Control Stand 40 feet downwind of the propane supply and the O-100 Outdoor Fire Trainer mock-up 65 feet downwind of the Multi Control Stand.
2. Place the Control Console on top of the Multi Control Stand. Refer to figure 3-1.

⚠ WARNING

WHEN UTILIZING A KIDDE O-100 OUTDOOR FIRE TRAINER, BE SURE TO REFER TO THE APPLICABLE KIDDE O-100 OUTDOOR FIRE TRAINER OPERATION AND MAINTENANCE MANUAL SUPPLIED FOR YOUR SITE, FOR APPLICABLE MOCK-UP INSTALLATION INSTRUCTIONS AND LPG VAPOR/LIQUID CONNECTION REQUIREMENTS.

3. At the control stand, remove protective thread cap from the **YELLOW LPG VAPOR** connection in Zone 1. Refer to figure 3-1.

⚠ WARNING

ALL UNUSED CONTROL STAND LPG MANIFOLD CONNECTIONS MUST HAVE A CAP SECURELY INSTALLED PRIOR TO CONNECTING LPG VAPOR/LIQUID INPUT SUPPLY HOSES TO THE CONTROL STAND. REFER TO FIGURE 3-1. FAILURE TO DO THIS WILL CAUSE DEATH OR SERIOUS INJURY

4. At the Control Stand, connect the end of the LPG vapor supply hose from the Pilot Box Assembly **LPG VAPOR** connection marked **"IN"**, to the **YELLOW LPG VAPOR** Control Stand connection in Zone 1 (refer to figure 3-1). Check Acme fittings for dirt and damage before hooking up. Attach Acme fittings hand tight.
5. At the control stand, remove protective thread cap from the **RED LPG LIQUID** connection in Zone 1. Refer to figure 3-1.

⚠ WARNING

ALL UNUSED CONTROL STAND LPG MANIFOLD CONNECTIONS MUST HAVE A CAP SECURELY INSTALLED PRIOR TO CONNECTING LPG VAPOR/LIQUID INPUT SUPPLY HOSES TO THE CONTROL STAND. REFER TO FIGURE 3-1. FAILURE TO DO THIS WILL CAUSE DEATH OR SERIOUS INJURY

6. At the Control Stand, connect the end of the LPG liquid supply hose from the Pilot Box Assembly **LPG LIQUID** connection marked **"IN"**, to the **RED LPG LIQUID** Control Stand connection in Zone 1 (refer to figure 3-1). Check Acme fittings for dirt and damage before hooking up. Attach Acme fittings hand tight.

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NOTE

WHEN USING MORE THAN ONE O-100 OUTDOOR TRAINER WITH THE MULTI CONTROL SYSTEM, REPEAT STEPS 2-5 FOR CONNECTIONS TO ZONE 2 FOR A SECOND O-100 TRAINER AND ZONE 3 FOR A THIRD O-100 TRAINER. REFER TO FIGURE 3-2.

7. At the **LPG VAPOR** supply tank, using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088) connect end of the 50' gas hose to the **LPG VAPOR** tank valve connection.
8. At the **LPG LIQUID** supply tank, using the Kidde supplied propane tank connection adaptor kit (P/N 1R1-002-088) connect end of the 50' gas hose to the **LPG LIQUID** tank valve connection.
9. At the side of the control stand, remove protective thread cap from the **YELLOW LPG VAPOR** input supply connection. Refer to figure 3-1 and 3-2.
10. At the side of the control stand, connect the end of the LPG vapor supply hose to the **YELLOW LPG VAPOR** input supply connection. Check Acme fittings for dirt and damage before hooking up. Attach Acme fittings hand tight.
11. At the side of the control stand, remove protective thread cap from the **RED LPG LIQUID** input supply connection. Refer to figure 3-1 and 3-2.
12. At the side of the control stand, connect the end of the LPG liquid supply hose to the **RED LPG LIQUID** input supply connection. Check Acme fittings for dirt and damage before hooking up. Attach Acme fittings hand tight.
13. At the Pilot Box Assembly, connect the control cable.
14. At the Pilot Box Assembly, connect the power cable.
15. At the control stand, connect the Pilot Box control cable to Zone 1 Control connection on the junction box. Refer to figure 3-1.
16. At the control stand, connect the Pilot Box power cable to Zone 1 power connection on the junction box. Refer to figure 3-1.
17. Run (gas, control and electric) lines straight back from Pilot Box Assembly to the Control Stand.

NOTE

WHEN USING MORE THAN ONE O-100 OUTDOOR TRAINER WITH THE MULTI CONTROL SYSTEM, REPEAT STEPS 12 - 16 FOR CONNECTIONS TO ZONE 2 FOR A SECOND O-100 OUTDOOR TRAINER AND ZONE 3 FOR A THIRD O-100 TRAINER. REFER TO FIGURE 3-2.

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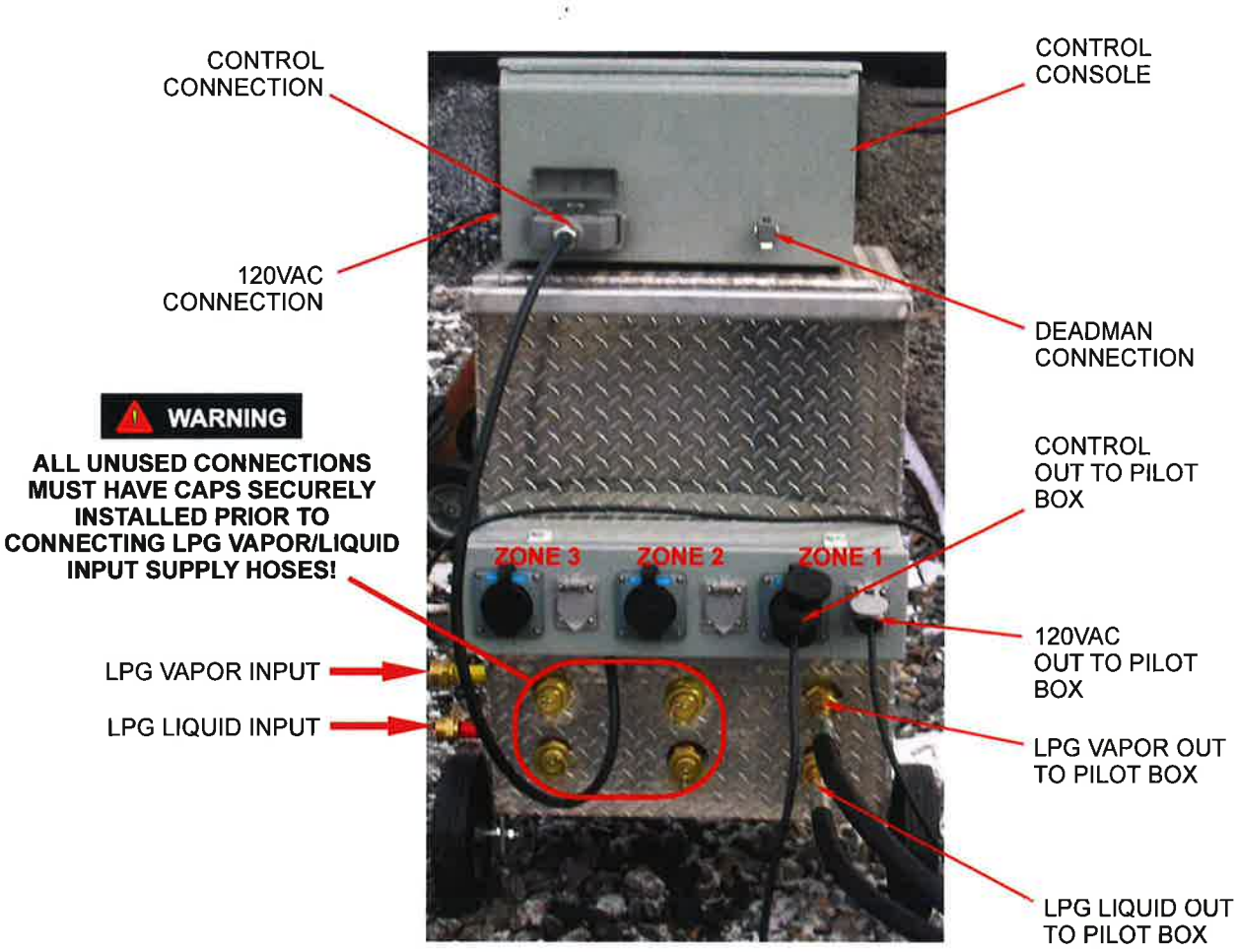

Figure 3-1 – Control Stand (One Zone Used)

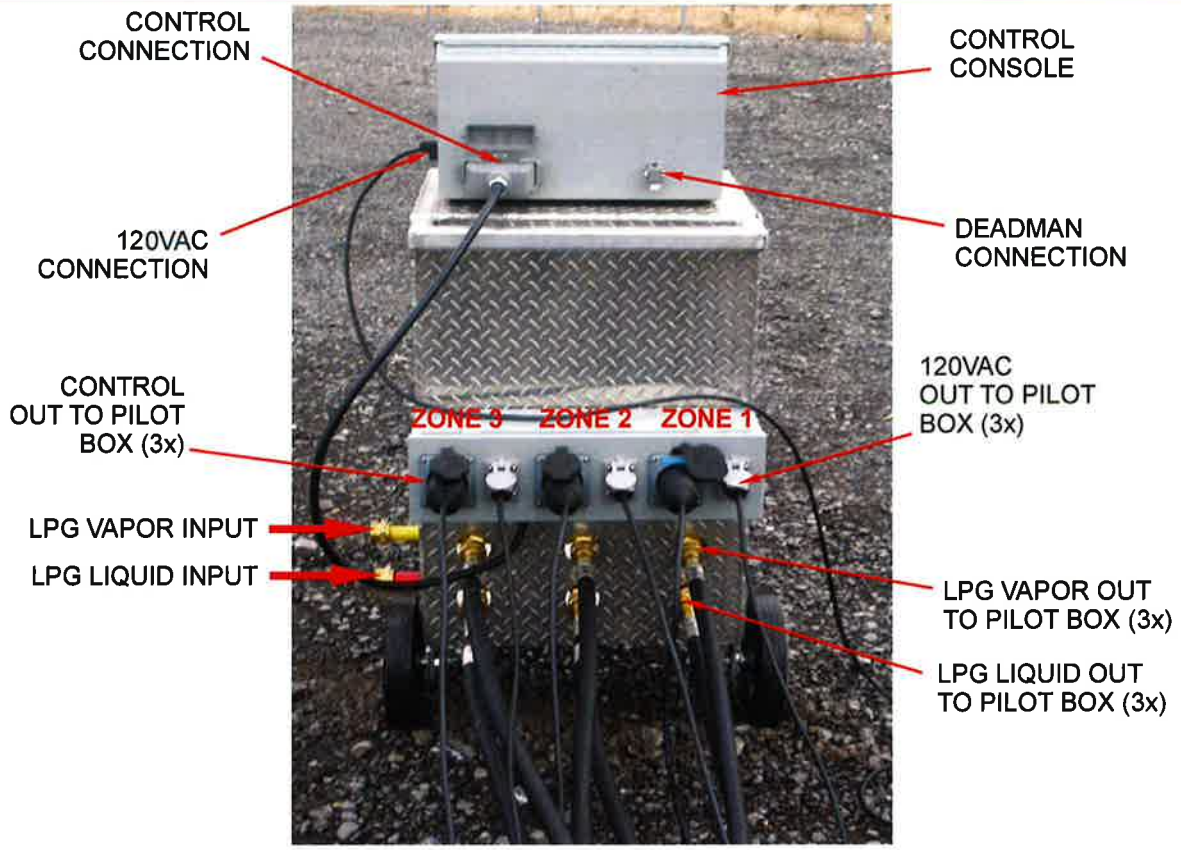

Figure 3-2 – Control Stand (Three Zones Used)

3.3 PILOT BOX VALVES OPERATIONAL TEST

Prior to operating the trainer the Pilot Box Assembly Pilot, Vapor and Liquid solenoid valves must be operationally tested.

WARNING

PERFORM THIS PROCEDURE FOR EACH PILOT BOX ASSEMBLY THAT IS CONNECTED TO THE MULTI CONTROL SYSTEM.

1. At the Control Console, be sure the Power Key is in the **"OFF"** position.
2. At the Control Console, be sure all the PILOT, LOW FLAME & HIGH FLAME switches are in the **OFF** position.
3. At the Control Console connect the control connection from the Control Stand Junction Box to the back of the Control Console. Refer to figure 3-3.
4. Connect the Deadman pendent to the connection located on the back of the Control Console. Refer to figure 3-3.
5. At the Control Console, be sure the Emergency Stop button is pulled out. Refer to Section 2, paragraphs 2.4 descriptions and layout of the Control Enclosure.

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WARNING

YOU MUST READ AND UNDERSTAND THE OWNERS MANUAL THAT CAME WITH THE POWER INVERTER BEFORE PROCEEDING.

6. If inverter is used, connect Inverter to tow vehicle and plug the power cord from the connection located on the right side of the Control Console into the inverter.
7. If shore power is used, attach the electric supply cord from the Control Console to an outdoor 120V power GFIC receptacle.
8. Turn Power Key to the **"ON"** position. Refer to Section 2, paragraphs 2.4 descriptions and layout of the Control Enclosure.
9. Green Power light should come ON and Emergency Stop Button should light up.

CONTROL CONSOLE BACK VIEW

Figure 3-3 – Control Console Connections

10. Turn the "PILOT" switch, on the Control Console to the **ON** position and listen for the solenoid valve to open.
11. Once it has been confirmed that the Pilot solenoid valve is operating, push and hold the Dead Man Button.
12. Turn the "LOW FLAME" switch, on the Control Console to the **ON** position and listen for the solenoid valve to open.
13. Once it has been confirmed that the "LOW FLAME" solenoid valve is operating, Turn the "HIGH FLAME" switch, on the Control Console to the **ON** position and listen for the solenoid valve to open.
14. Once it has been confirmed that the "HIGH FLAME" solenoid valve is operating, release the Dead Man Button and turn the "PILOT", "LOW FLAME" and "HIGH FLAME switches to the **OFF** position.
15. Disconnect the Control Console electric supply cord from the power source.

WARNING

IF ANY OF THE SOLENOID VALVES DID NOT OPERATE, DO NOT USE PILOT BOX ASSEMBLY UNTIL ALL SOLENOID VALVES ARE OPERATIONAL.

3.4 PILOT FLAME TEST

Only after procedure 3.3 has been successfully completed and prior to operating the trainer the pilot flame(s) must be operationally tested.

WARNING

PERFORM THIS PROCEDURE FOR EACH PILOT BOX ASSEMBLY THAT IS CONNECTED TO THE MULTI CONTROL SYSTEM.

1. Open the vapor propane supply valve.
2. Assure there are no leaks in all hoses and at all connections.

WARNING

IF A LEAK IS DETECTED TURN PROPANE SUPPLY VALVE OFF IMMEDIATELY. MAKE THE APPROPRIATE REPAIRS. DO NOT USE TRAINER UNTIL ALL LEAKS ARE ELIMINATED.

3. Connect the Control Console electric supply cord from the power source.
4. Turn the "PILOT" switch on the Control Console to the **ON** position. The Pilot may take a few seconds to ignite initially because it takes a while for the gas to get through the lines.

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5. Once Pilot Flame has ignited Turn the "PILOT" switch on the Control Console to the **OFF** position. The pilot flame will continue to burn for a short period due to the residual gas in the supply line.
6. Close vapor propane supply valve SLOWLY.
7. Disconnect the electric supply cord from power source.

WARNING

IF THE PILOT FLAME DID NOT OPERATE, DO NOT USE PILOT BOX ASSEMBLY UNTIL PILOT FLAME IS OPERATIONAL.

3.5 LIQUID PROPANE LEAK CHECK

WARNING

THE PROPANE LEAK CHECK MUST BE SUCCESSFULLY COMPLETED PRIOR TO OPERATING THE O-100 OUTDOOR FIRE TRAINER.

1. Ensure that all gas connections are complete and secure as previously directed.
2. Open the liquid propane supply valve SLOWLY.
3. Assure there are no leaks in all hoses and at all connections.
4. Close liquid propane supply valve.

WARNING

IF A LEAK IS DETECTED TURN PROPANE SUPPLY VALVE OFF IMMEDIATELY. MAKE THE APPROPRIATE REPAIRS. DO NOT USE TRAINER UNTIL ALL LEAKS ARE ELIMINATED.

5. Install cable/hose covers for all Pilot Box Assemblies.

CAUTION

DO NOT OPERATE O-100 TRAINER WITHOUT BOTH CABLE/HOSE COVERS IN PLACE!

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3.6 TRAINER OPERATION

WARNING

ONLY WHEN ALL THE ABOVE OPERATIONAL AND LEAK CHECKS HAVE BEEN SUCCESSFULLY PERFORMED YOU ARE READY TO TRAIN.

Refer to Section 2, paragraphs 2.4 and 2.5 descriptions and layout of the Control Console and Dead Man Switch.

1. All training scenarios should be explained fully to the students and the explanation should include a description of the Fire Pan/Mockup(s) and its operation.
2. Based on the type of mockup(s) being used, be sure all personnel are at a safe distance from the trainer.
3. Attach the electric supply cord from the connection located on the right side of the Control Console to power source.
4. Based on the type of mockup(s) being used, open the appropriate Propane Supply valve(s) SLOWLY.
5. If operating the vehicle Mock-Up, be sure that the cooling water is ON and flowing.

CAUTION

THE COOLING WATER MUST BE ON AND FLOWING DURING VEHICLE TRAINER OPERATION

6. Start the Pilots required for the scenarios by turning the "PILOT" switch to the **ON** position. The pilots will not ignite with the Dead Man Button held down. The Pilots have a built –in timer, which keeps the Operator form going to Main Flame for 5 seconds. Verify that Pilots ignite.

WARNING

PRIOR TO PERFORMING THE NEXT STEP BE SURE THAT PILOTS HAVE IGNITED.

7. **SEE ABOVE WARNING PRIOR TO PERFORMING THIS STEP!** Have the Safety Officer push and hold the Dead Man Button.

For each of the fires:

8. Turn the "LOW FIRE" switch on the Control Enclosure to the **ON** position.

WARNING

PRIOR TO PERFORMING THE NEXT STEP BE SURE THAT LOW FLAME FROM VAPOR BURNER ELEMENT IS VISIBLE.

9. Then turn the corresponding "HIGH FIRE" switch on the Control Enclosure to the **ON** position.

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NOTE

RELEASE THE DEAD MAN BUTTON TO EXTINGUISH MAIN FLAME(S), (PILOT STAYS ON). PRESS AND HOLD DEAD MAN BUTTON AGAIN TO RE-IGNITE MAIN FLAME(S)

WARNING

IN CASE OF AN EMERGENCY PRESS THE EMERGENCY STOP SWITCH ON THE CONTROL ENCLOSURE.

10. If the Control Enclosure Operator decides that the fire has been fought correctly, turn the "HIGH FIRE" to the **OFF** position. Turn the "LOW FIRE" to the **OFF** position. Or the Instructor can release the Dead Man button, which will stop ALL Low and High Fires.
11. Turn all the "HIGH FIRE" switches on the Control Enclosure to the **OFF** position.
12. Turn all the "LOW FIRE" switches on the Control Enclosure to the **OFF** position.
13. If operating the vehicle Mock-Up, once Mock-Up has cooled turn cooling water off.

3.7 TRAINER SHUTDOWN

Refer to Section 2, paragraphs 2.4 and 2.5 descriptions and layout of the Control Console and Dead Man Switch.

Once training has been completed for the day the following steps should be taken:

WARNING

PERFORM THIS PROCEDURE FOR EACH PILOT BOX ASSEMBLY THAT IS CONNECTED TO THE MULTI CONTROL SYSTEM.

At the end of a training session there will still be Fuel in the hoses from the Propane Supply Tank to the Fire Pan. This will have to be burnt off before the equipment is taken apart. It is important that this procedure be followed even if you are not taking equipment apart but are just shutting down at the end of a training session.

1. Close the Liquid Propane valves at that Liquid Propane Supply cylinder.
2. When operating the vehicle Mock-Up, turn cooling water ON.

CAUTION

FOR THE VEHICLE MOCK-UP COOLING WATER MUST BE TURNED ON DURING TRAINER OPERATION

3. Using the Control Console light all connected Pilots. Verify that Pilots ignite.

WARNING

PRIOR TO PERFORMING THE NEXT STEP BE SURE THAT PILOT(S) HAVE IGNITED.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

WARNING

PERFORM THE NEXT STEP FOR EACH PILOT BOX ASSEMBLY THAT IS CONNECTED TO THE MULTI CONTROL SYSTEM.

4. **SEE ABOVE WARNINGS PRIOR TO PERFORMING THIS STEP!** While holding the Dead Man button in, turn "LOW FIRE" switch(s) on the Control Enclosure to the **ON** position.

WARNING

PRIOR TO PERFORMING THE NEXT STEP BE SURE THAT LOW FLAME FROM VAPOR BURNER ELEMENT IS VISIBLE.

WARNING

PERFORM THE NEXT STEP FOR EACH PILOT BOX ASSEMBLY THAT IS CONNECTED TO THE MULTI CONTROL SYSTEM.

5. **SEE ABOVE WARNINGS PRIOR TO PERFORMING THIS STEP!** Once all of the "LOW FIREs" have ignited, the "HIGH FIRE" switch(s) on the Control Enclosure to the **ON** position. Leave Pilot, "LOW" and "HIGH FIRE" flame(s) on for 2 minuets, at which time you should see the flame height reduce, indicating that the residual liquid gas has burned off.
6. When all the residual Liquid propane has burned off, release Dead Man button, turn "PILOT" and all high and low flame switches to the **"OFF"** position.
7. Close the Vapor Propane valves at the Vapor Propane Supply cylinder.

WARNING

PERFORM THE NEXT STEP FOR EACH PILOT BOX ASSEMBLY THAT IS CONNECTED TO THE MULTI CONTROL SYSTEM.

8. **SEE ABOVE WARNING PRIOR TO PERFORMING THIS STEP!** Using the Control Console light Pilot(s). Verify that Pilot(s) ignite.

Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this manual.

⚠ WARNING

PRIOR TO PERFORMING THE NEXT STEP BE SURE THAT PILOT(S) HAVE IGNITED.

⚠ WARNING

PERFORM THE NEXT STEP FOR EACH PILOT BOX ASSEMBLY THAT IS CONNECTED TO THE MULTI CONTROL SYSTEM.

9. **SEE ABOVE WARNINGS PRIOR TO PERFORMING THIS STEP!** While holding the Dead Man button in, turn "LOW FIRE" switch(s) on the Control Enclosure to the **ON** position. Leave Pilot and "LOW FIRE" flame on until the residual Vapor gas is burned off and Pilot and "LOW FIRE" flames self-extinguish.
10. When ALL flames have self-extinguished, release Dead Man button, turn "PILOT" and "LOW FIRE" switch(s) to the **OFF** position.
11. Turn the "POWER" key switch to the **OFF** position.
12. When operating the vehicle mock-up, leave cooling water on long enough to cool down the Pilot Box Assembly and vehicle mock-up, then turn cooling water off.
13. Disconnect the electric supply cord from the from power source and connection located on the right side of the Control Console and safely store.
14. Unhook the dead man switch and control cable from the control console and Inspect for damage. Safely store.
15. Remove the Control Console from the Multi Control System Stand and safely store.
16. Unhook ALL Liquid/Vapor propane gas hoses from the Propane Supply Tank, Multi Control System Stand, Pilot Box Assembly and mock-up (if applicable). Safely store all hoses.
17. Install ALL protective thread caps on the Liquid/Vapor Propane connections on the Propane Supply Tank, Multi Control System Stand, Pilot Box Assembly and mock-up (if required).
18. If operating the Automobile Mock-Up, disconnect the cooling water supply hose from the vehicle mock-up and safely store.
19. Safely store Multi Control Stand.
20. For the vehicle mock-up, remove Pilot Box Assembly from rear of vehicle mock-up.
21. Drain water from the Pilot Box assembly and Burn Pan by removing drain plugs, wash down as required. **ONLY USE WATER**. Drain any residual water.
22. Install wheels on Pilot Box Assembly and store all Equipment in a safe location.
23. Install wheels on 4' x 6' Fire Pan Assembly (if used) and store all Equipment in a safe location.

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⚠ CAUTION

WHEN NOT IN USE THE BURN PAN WATER BATH, PILOT BOX ASSEMBLY AND MOCK-UP SHOULD BE DRAINED OF WATER AND DURING COLD WEATHER PERIODS, WHEN THE ANTICIPATED AMBIENT TEMPERATURE IS GOING TO BE BELOW 32°F

WHEN TRAINING DURING THE ABOVE MENTIONED CONDITION, FILL BURN PAN WATER BATH, TOP OF THE PILOT BOX ASSEMBLY AND MOCK-UP JUST PRIOR TO OPERATION AND DRAIN IMMEDIATELY UPON THE COMPLETION OF TRAINING.

FAILURE TO FOLLOW THESE CAUTIONS MAY RESULT IN SERIOUS DAMAGE TO THE ASSOCIATED EQUIPMENT.

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4.0 CARE AND MAINTENANCE

4.1 GENERAL

This section provides Information to keep the trainer functional and operationally effective.

4.2 CARE AND MAINTENANCE

1. This device will be a very functional and operationally effective unit if cleaned and maintained properly.
2. When not in use be sure to place caps on all Acme fittings and junction box connections.
3. To clean be sure caps are on all Acme fittings and junction box connections. Clean surfaces with damp cloth only. *Never* use flammable liquids to clean surfaces.

WARNING

NEVER USE FLAMMABLE LIQUIDS TO CLEAN SURFACES.

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5.0 SPARE PARTS

5.1 General

This section provides a figure and parts list table for each of the Hose Line **FireTrainer® O-100** Multi Control System major assemblies.

5.2 Control Console Panel Assembly

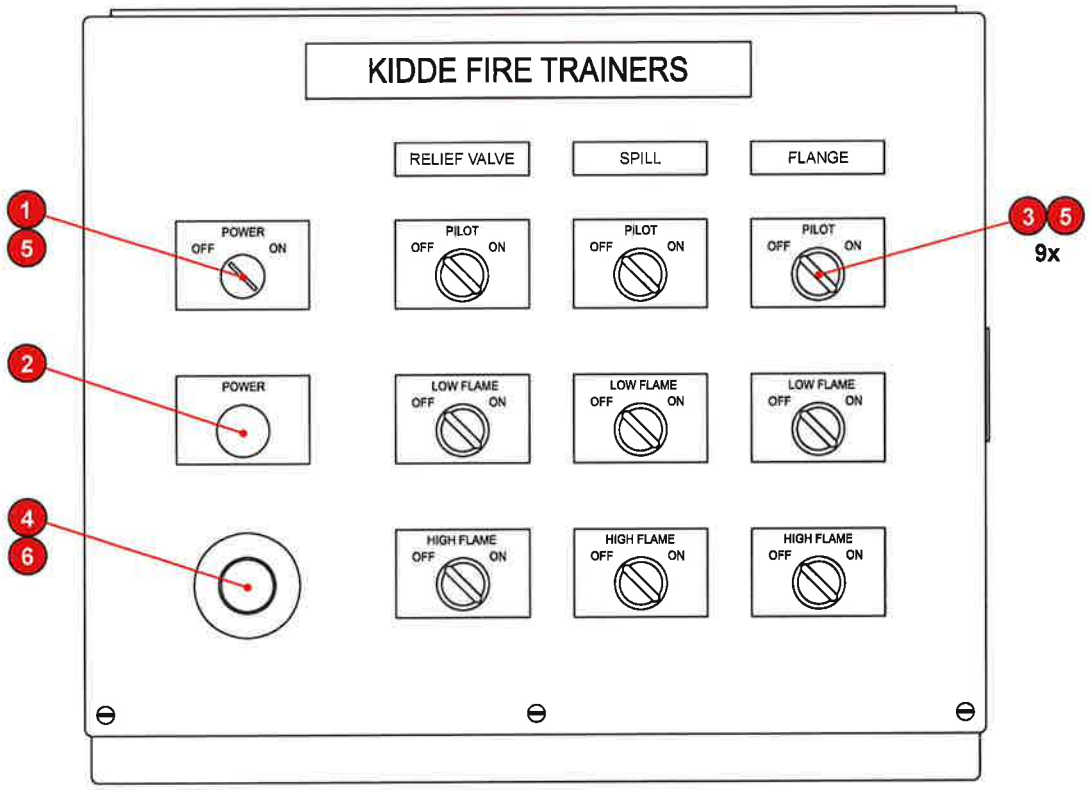

Figure 5-1 – Control Console Panel Assembly Details

Table 5-1. Control Console Panel Assembly Parts Lists			
Figure 5-1 Item No.	**Description**	**Part Number**	**Cage Code**
REF	**Control Console Assembly**	**4A1-006-750 Rev A**	**06003**
1	Switch, Sel, 2-Pos, MNTND, 1 N.O., Key	GCX1420	04YE1
2	Indicator, Light, Incndsnt, Green, 24V	ECX1052-24	04YE1
3	Switch, Sel, 2-Pos, MNTND, 1 N.O., Lvr	GCX1350	04YE1
4	Operator, Mushroom, Illum, 40mm, 120V	GCX1226-120	04YE1
5	Block, Contact, N.O., centsable	ECX1040-5	04YE1
6	Block, Contact, N.C., centsable	ECX1030-5	04YE1

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5.3 Control Console Assembly

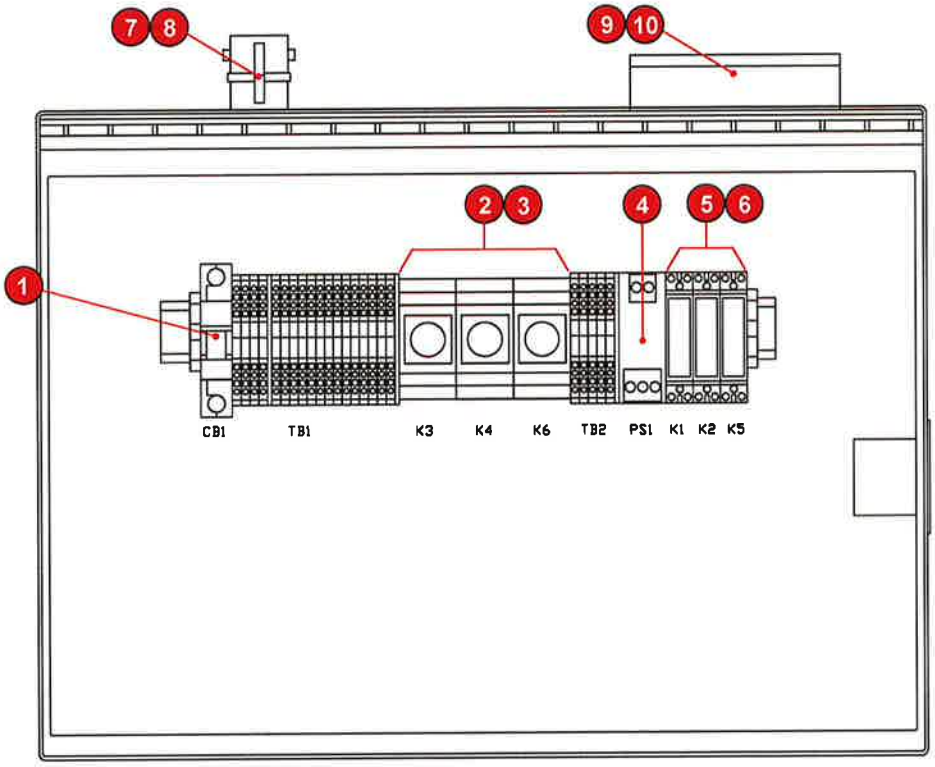

Figure 5-2 Control Console Assembly Details

Table 5-2. Control Console Assembly Parts Lists			
Figure 5-2 Item No.	**Description**	**Part Number**	**Cage Code**
REF	**Control Console Assembly**	**4A1-006-750**	**06003**
1	Circuit Breaker, 1-Pole, 6A	WMZS1D06	04YE1
2	Timer, On-Delay, 2PDT, 4-60Sec, 120V	ST7P-2A16T-ADC	04YE1
3	Socket, Relay, Din-Rail MNT, F/ST7P	TP88X2	04YE1
4	Power Supply, SW, 24VDC, 1A(24W) OP	PSP24-024C	04YE1
5	Relay, DPDT, 24VDC, 5A, Pilot Light	700-HK32Z24-4	01121
6	Socket, Relay, Mini, 2-Pole, 8-Blade	700-HN122	01121
7	Insert, FEM, Term, Screw, 4 CONT	09200032711	0G8R9
8	Cable Entry Protection, M20, 6-12mm	09200005082	0G8R9
9	Insert, Fem, Term, Screw, 4 CONT	09330242716	0G8R9
10	Housing, Bulkhead, MTG, SZ24, W/PL CVR	09300240302	0G8R9

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5.4 Pendant Assembly, Deadman

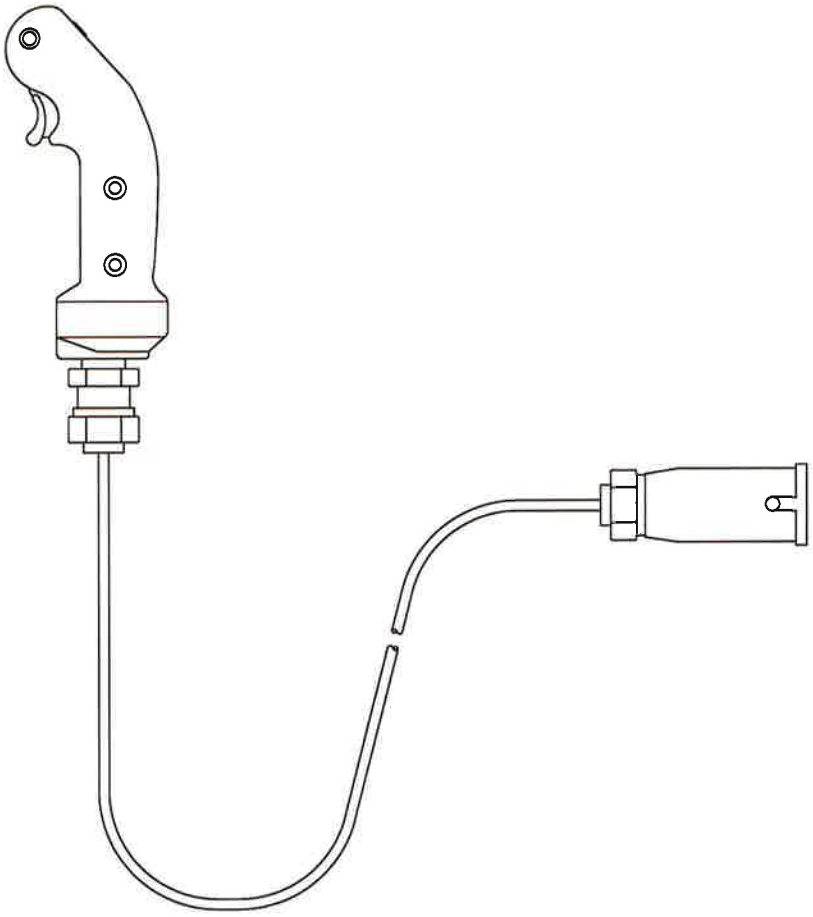

Figure 5–5 Pendant Assembly, Deadman

Table 5-5. Pendant Assembly, Deadman			
Figure 5-5 Item No.	**Description**	**Part Number**	**Cage Code**
REF	Pendant Assembly, Deadman	1R1-002-110	06003

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6.0 Drawings

6.1 INTRODUCTION

This section contains drawings that support the **HOSE LINE FIRETRAINER® O-100** Multi Control System.

Refer to Table 6-1 for a list of drawings that have been supplied in this section.

Table 6-1. Drawings

Drawing Number	Title
WD4A1-006-750 (Rev A)	Control Console Wiring Diagram

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DWG NO WD4A1-006-750 SH 1

REVISIONS			
REV	DRAWING CHANGE RECORD NO	DATE	APVD
A	ECO# 30441	08/14/06	DM/SH

Installation: TRIPLE 902 PAN FIRE CONTROL ENCL

Page	Page Title	Revision
1	REVISION STATUS OF PAGES	A
2	CONTROL WIRING	A
3	CONNECTOR WIRING	-

Legend of Common Reference Designators

AD	Agent Detection
B	Blower Motor
CAB	Combustion Air Blower
CB	Circuit Breaker
CGDS	Combustible Gas Drawn Sample
CGS	Combustible Gas Sensor
D	Diode
DS	Door Switch
ES	E-Stop
F	Flashover Effect
FHM	Fireplace Hardware Module
FP	Fireplace Mockups
FRCS	Fuel Regulator & Control Station
FS	Flame Safeguard
HOA	Hand-Off-Automatic Switch
I	Indicator
IG	Ignitor
JB	Junction Box
K	Relay
KS	Keyswitch
L	Solenoid Valve
LCP	Local Control Panel
LT	Lamp/Light
MC	Magnetic Contactor Enclosure
ME	Master I/O Rack Enclosure
MV	Motorized Valve
MVA	Motorized Valve Assy
OS	Operator Station
P	Plug
PND	Pendant, Control
POC	Proof of Closure
PRS	Pressure Switch
PS	Power Supply
PT	Pressure Transducer
R	Receptical
RBC	Remote Burner Control
S	Switch
SCP	Scenario Control Panel
SD	Smoke Damper
SG	Smoke Generator
SP	Service Panel
SPK	Speaker
SSOV	Safety Shutoff Valve
T	Transformer
TB	Terminal Block
TC	Thermocouple
TDR	Time Delay Relay
TS	Temperature Switch
VF	Ventilation Fan
VFC	Ventilation Fan Control
W	Cable
Z	Zener Diode

NOTES:

1. UNLESS OTHERWISE SPECIFIED, ALL WIRE IS TYPE THHN OR MTW OF SPECIFIED COLOR AND GAUGE.

PROPRIETARY INFORMATION OF Kidde Fire Trainers
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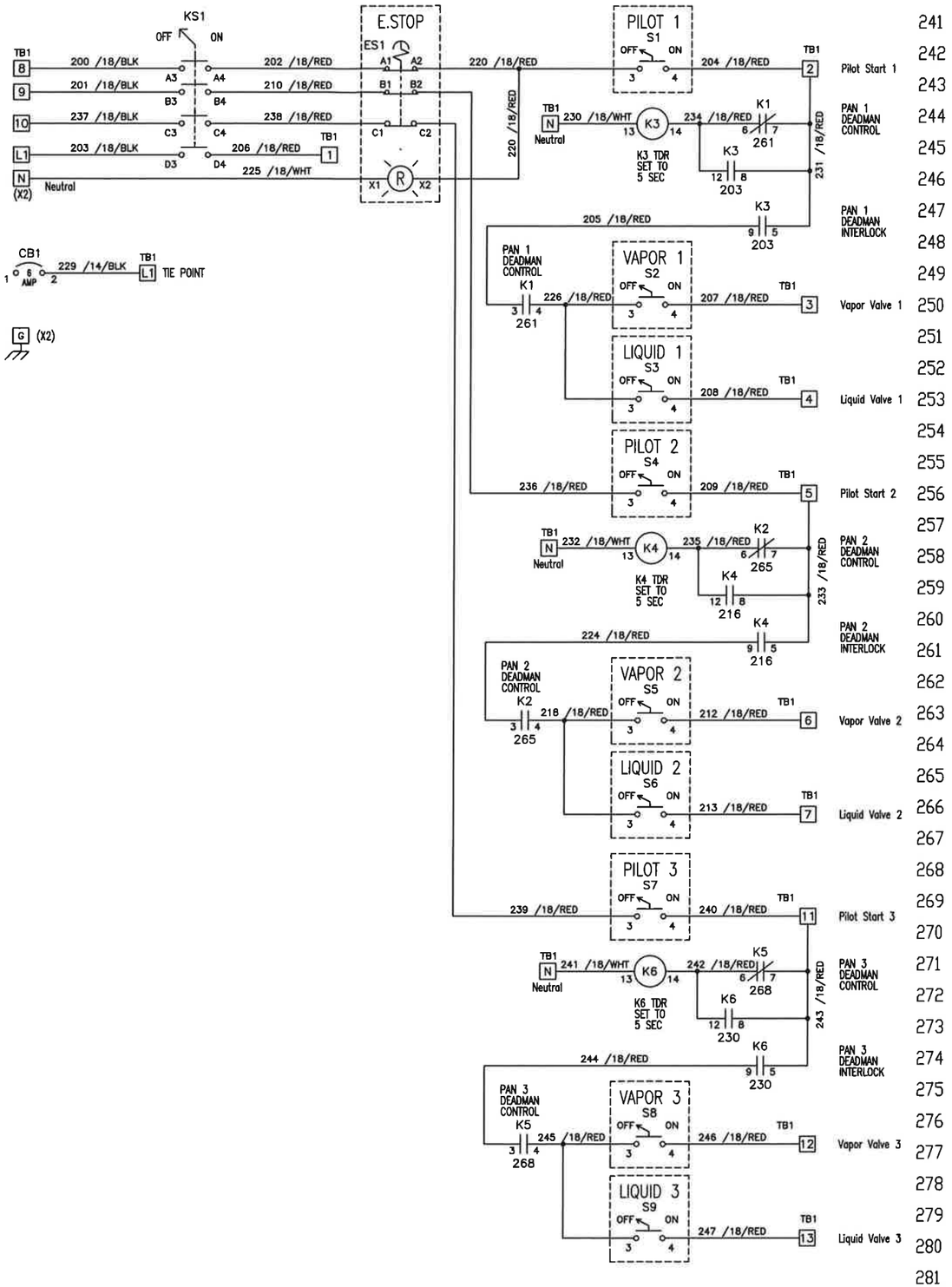
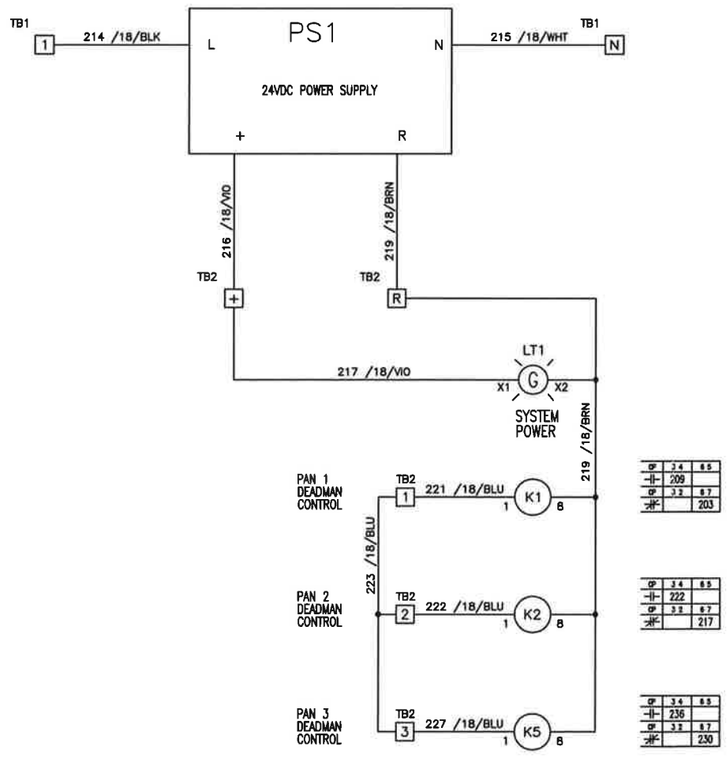
PROJECT: 902_PAN_3FIRE		INSTALLATION: WIRING_DIAG		Kidde Fire Trainers	
		LOCATION/S		TITLE: TRIPLE 902 PAN FIRE CONTROL ENCL WIRING DIAGRAM REVISION STATUS OF PAGES	
4A1-006-750	802 PAN	DWN	B AQUINO	02/17/05	SIZE D / CAGE CODE 06003 / DWG NO WD4A1-006-750
NEXT ASSY	USED ON	CHKD	S HUYNH	2/17/05	
APPLICATION		APVD	SL	2/17/05	SCALE NA / CAD FILE: WD4A8750A / SHEET 1 OF 3

DWG NO WD4A1-006-750 | SH 2

REVISIONS			
REV	DRAWING CHANGE RECORD NO	DATE	APVD
A	ECO# 30441	08/14/06	DM/SH





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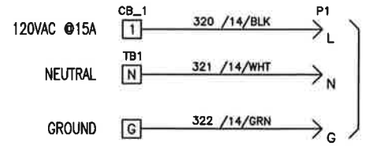
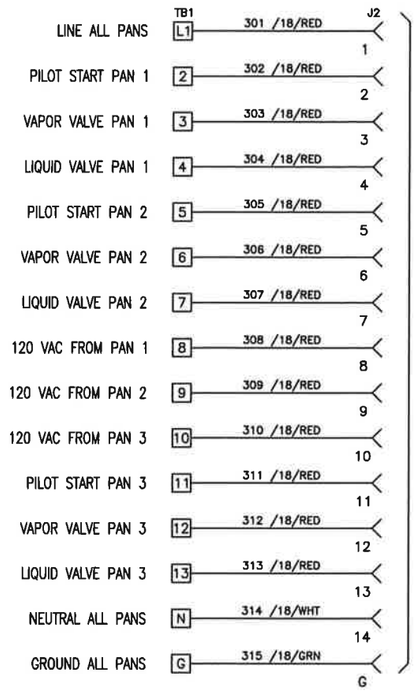
PROJECT	902_PAN_3FIRE	INSTALLATION	WIRING_DIAG	Kidde Fire Trainers	
		LOCATION/S		TITLE	TRIPLE 902 PAN FIRE CONTROL ENCL WIRING DIAGRAM CONTROL WIRING
4A1-006-750	802 PAN	DWN	B AQUINO	2/17/05	SIZE D / CAGE CODE 06003 / DWG NO WD4A1-006-750
NEXT ASSY	USED ON	CHKD	S HUYNH	2/17/05	
APPLICATION		APVD	SL	2/17/05	SCALE NA / CAD FILE: WD4A6750A / SHEET 2 OF 3

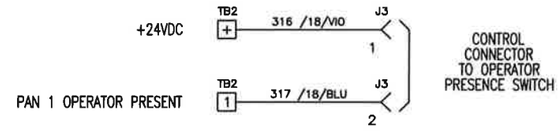
DWG NO	WD4A1-006-750	SH 3

REVISIONS

REV	DRAWING CHANGE RECORD NO	DATE	APVD







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PROJECT	INSTALLATION	Kidde Fire Trainers
902_PAN_3FIRE	WIRING_DIAG	TITLE: TRIPLE 902 PAN FIRE CONTROL ENCL WIRING DIAGRAM CONNECTOR WIRING
	LOCATION/S	

4A1-006-750	802 PAN	DWN	B AQUINO	02/17/05	SIZE D / CAGE CODE 06003 / DWG NO WD4A1-006-750
NEXT ASSY	USED ON	CHKD	S HUYNH	2/17/05	
APPLICATION		APVD	SL	2/17/05	SCALE NA / CAD FILE: WD4A6750A / SHEET 3 OF 3

O-100 Wireless Deadman
Data Sheets

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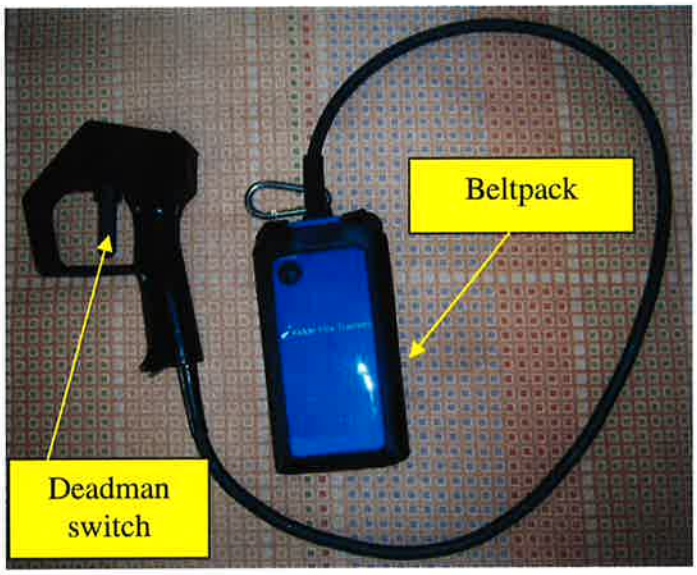
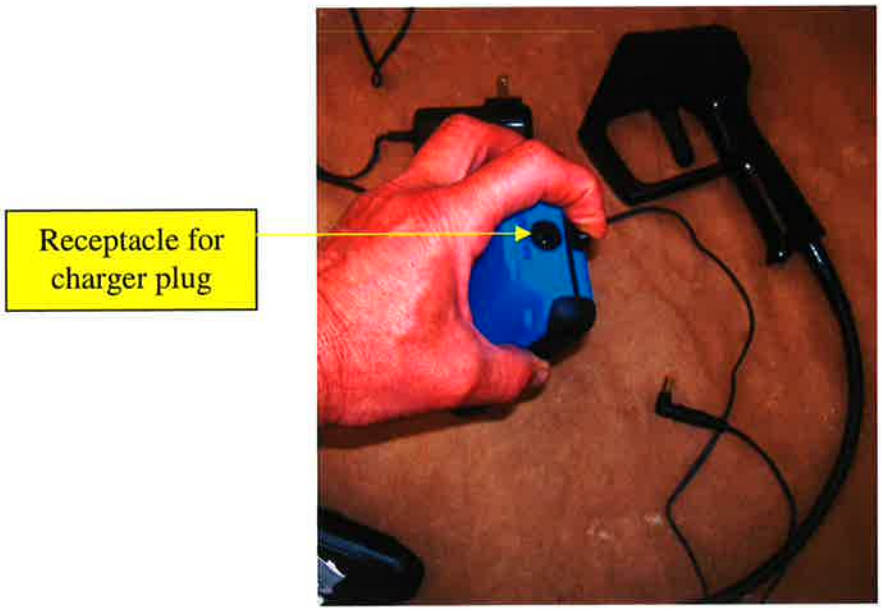
Fax +1 201.300.8101
www.kiddeft.com

Safety Officer Deadman Pendant



Recharging of Battery

- The battery should last up to 4 months
- Do not recharge until battery light comes on with a warning signal
- To recharge, plug wall charger into receptacle on bottom of unit.
- Recharging can take up to 4 hours



RANGER "DM" SERIES

AIRCRAFT REFUELING DEADMAN RADIO REMOTE CONTROLS

Model # FHSR-DM

Developed specifically for industrial applications requiring positive operator-to-machine contact. A wireless trigger-switch is used to send RF command signals to the pump. Operator is unencumbered by cables and wires normally used in 'deadman' fueling operations. System incorporates a secure frequency hopping, spread-spectrum radio link that is virtually immune to outside electrical or RF interference. Operations fail safe (shutdown) with any loss of signal from transmitter. Rechargeable transmitter delivers 30hrs of continuous use between charges (12/24VDC charger supplied).

IMPROVED SAFETY AND MOBILITY

*'Safety Timer' software available to limit switch closure duration prior to automatic shut-off. This feature requires periodic trigger pulse by operator to maintain receiver 'on' condition. Up to eleven other command functions available per customer requirements. Modular quick-connect components allow for easy installation and field service. Pre-wired solenoids are available for Pneumatic, Hydraulic, or Hose Reel control interface.

SYSTEMS FOR AC POWERED CABINETS

NO EXTRA DEADMAN CORDS OR REELS

TRANSMITTER SPECIFICATIONS

FREQUENCY OF OPERATION ..... 902-928MHz FHSS (433.92MHz, 869.7MHz & 2.4GHz radios available)
TRANSMITTER POWER 100mW
METHOD OF TRANSMISSION ..... 56 bit DES encryption key
ANTENNA ..... Internal 'on card' PCB mount (no visible antennas to damage)
NORMAL RANGE ..... Up to 500 feet (higher power available)
FCC APPROVAL # ..... KQLAC4490-100 (non-licensed)
CONSTRUCTION ..... Rugged weatherproof polymer enclosure
TEMPERATURE SPECIFICATION ..... Minus 20 F to 185 F
BATTERY LIFE ..... 30hrs of continuous use w/audible low battery
..... warning and flashing LED indicator
BATTERY TYPE ..... Rechargeable Lithium Ion cell with 12/24VDC charger
INTRINSICALLY SAFE FOR USE IN HAZARDOUS ENVIRONMENTS
ROHS LEAD-FREE COMPLIANT

HANDHELD DEADMAN & TRANSMITTER

RECEIVER SPECIFICATIONS

POWER SUPPLY ..... 12/24VDC (Optional 120VAC systems available)
CURRENT LOAD ..... Up to 4Amps per channel or 10Amps total
CONSTRUCTION ..... IP67 rated, impact resistant polycarbonate
ANTENNA ..... Internally protected and encased
..... (external antenna/coax available)
WIRING CONNECTIONS ..... Pre-wired to OEM requirements,
..... quick connect fittings supplied
PROGRAMMING MODES ..... Up to 12 programmable channels including:
..... LATCHED, MOMENTARY, SAFETY LATCHED,
..... and *SAFETY TIMER programming modes
..... (custom programming available)
SERIAL COMMUNICATION ..... Optional CAN bus interface available
ROHS LEAD-FREE COMPLIANT

PUMP MOUNTED RECEIVER